Symplectic Topology

A Field Guide

Prepared as an opinionated research-level overview,
with pointers into the literature

October 2026

Contents

0 Preface and conventions **1**
- 0.1 The shape of the subject . . . 1
- 0.2 How the document is organized . . . 2
- 0.3 A note on precision . . . 2
- 0.4 Standing conventions . . . 3
- 0.5 Prerequisites and notation . . . 5

1 Linear symplectic geometry and the Maslov index **6**
- 1.1 Symplectic vector spaces . . . 6
- 1.2 Compatible complex structures and symplectic vector bundles . . . 7
- 1.3 The symplectic group . . . 8
- 1.4 The Lagrangian Grassmannian and the Maslov class . . . 9
- 1.5 The Conley–Zehnder and Robbin–Salamon indices . . . 10
- 1.6 What to read . . . 13
- 1.7 Frontier and open problems . . . 14
- 1.8 Logical dependencies . . . 14

2 Symplectic manifolds **16**
- 2.1 Symplectic manifolds and examples . . . 16
- 2.2 Hamiltonian dynamics and the groups Symp and Ham . . . 18
- 2.3 Local theory: Moser's method and neighborhood theorems . . . 20
- 2.4 Lagrangian submanifolds . . . 22
- 2.5 Almost complex structures and Chern classes . . . 23
- 2.6 Symmetry: Hamiltonian group actions . . . 24
- 2.7 What to read . . . 27
- 2.8 Frontier and open problems . . . 28
- 2.9 Logical dependencies . . . 29

3 Constructions, existence, uniqueness **30**
- 3.1 Open symplectic manifolds with convex ends . . . 30
- 3.2 Closed constructions: blow-ups, fibrations, sums and surgeries . . . 33
- 3.3 Donaldson's theorems and Lefschetz fibrations . . . 36
- 3.4 The existence problem . . . 38
- 3.5 The landscape of closed symplectic 4-manifolds . . . 39
- 3.6 What to read . . . 42
- 3.7 Frontier and open problems . . . 43

3.8 Logical dependencies . . . 43

4 Contact geometry and flexibility 44
4.1 Contact manifolds . . . 44
4.2 Legendrian and transverse knots in dimension three . . . 46
4.3 Open books and contact surgery . . . 48
4.4 Tight versus overtwisted . . . 49
4.5 The h-principle . . . 52
4.6 Loose Legendrians and flexible Weinstein manifolds . . . 53
4.7 The rigid/flexible dichotomy . . . 55
4.8 What to read . . . 56
4.9 Frontier and open problems . . . 57
4.10 Logical dependencies . . . 57

5 Pseudoholomorphic curves: the foundations 58
5.1 J-holomorphic curves and their local theory . . . 58
5.2 Compactness and bubbling . . . 61
5.3 Fredholm theory and transversality . . . 63
5.4 Gromov–Witten invariants and quantum cohomology . . . 66
5.5 Virtual fundamental classes: the state of the foundations . . . 69
5.6 What to read . . . 71
5.7 Frontier and open problems . . . 72
5.8 Logical dependencies . . . 72

6 Rigidity: what holomorphic curves see 74
6.1 Nonsqueezing and symplectic capacities . . . 74
6.2 Four-dimensional embedding problems . . . 77
6.3 Lagrangian submanifolds and holomorphic disks . . . 80
6.4 Foliations by spheres and symplectomorphism groups . . . 82
6.5 Rational and ruled surfaces; Seiberg–Witten = Gromov . . . 83
6.6 What to read . . . 85
6.7 Frontier and open problems . . . 86
6.8 Logical dependencies . . . 87

7 Hamiltonian Floer theory 88
7.1 The Arnold conjecture . . . 88
7.2 Floer cohomology of a Hamiltonian . . . 90
7.3 PSS, products and what Floer cohomology cannot see . . . 92
7.4 Spectral invariants . . . 93
7.5 Hofer geometry . . . 94
7.6 Calabi quasimorphisms and symplectic quasi-states . . . 95
7.7 Barcodes . . . 96
7.8 C^0 symplectic topology . . . 97
7.9 Floer homotopy theory . . . 98
7.10 What to read . . . 99
7.11 Frontier and open problems . . . 100
7.12 Logical dependencies . . . 100

8 Lagrangian Floer theory and Fukaya categories **101**
8.1 Floer cohomology of a pair of Lagrangians . . . 101
8.2 Obstruction theory: curved A_∞-algebras and bounding cochains . . . 104
8.3 Fukaya categories . . . 106
8.4 Symplectic cohomology and wrapped Fukaya categories . . . 108
8.5 The nearby Lagrangian conjecture . . . 110
8.6 Homological mirror symmetry . . . 111
8.7 Microlocal sheaves . . . 112
8.8 Lagrangian cobordism . . . 113
8.9 What to read . . . 114
8.10 Frontier and open problems . . . 115
8.11 Logical dependencies . . . 116

9 Hamiltonian and Reeb dynamics **117**
9.1 Reeb orbits and the Weinstein conjecture . . . 117
9.2 Multiplicity, surfaces of section, and two-or-infinity . . . 119
9.3 Holomorphic curves in symplectizations: SFT and ECH . . . 122
9.4 Closing lemmas . . . 124
9.5 Hamiltonian diffeomorphisms: Conley, Hofer–Zehnder, pseudo-rotations . . . 125
9.6 Systolic inequalities and Viterbo's conjecture . . . 126
9.7 What to read . . . 128
9.8 Frontier and open problems . . . 128
9.9 Logical dependencies . . . 129

10 Frontiers and reading paths **130**
10.1 A map of the subject . . . 130
10.2 Reading paths . . . 131
10.3 Open problems . . . 133
10.4 Where the subject is going . . . 135
10.5 Logical dependencies . . . 136

References **137**

Chapter 0

Preface: what this is, and the conventions it uses

This is a field guide to symplectic topology, written for two readers at once. The first is a colleague who already works in the area and wants an opinionated map: which theorems are the load-bearing ones, which proofs are worth internalizing, which sources to send a student to, and where the frontier is in late 2026. The second is a graduate student with a good first-year background in differential and algebraic topology who wants to reach the point of stating, and ideally proving, research-level results, and who needs to know what to read and in what order. I have not tried to be complete, and I have almost never given proofs. What I have tried to do is to state every central definition completely, to state every central theorem with its hypotheses, to say honestly what is proved and what is merely believed, and to point into the literature at every turn.

Two warnings. First, the document is opinionated by design. When I say that a definition is the right one, that a proof is the one to learn, or that a book should be read in a particular order, I mean it as advice from someone who has made the mistakes, not as consensus. Second, it is a snapshot. The subject has moved fast in the last decade (foundations of virtual cycles, the integral Arnold conjecture, the simplicity conjecture, the disproof of Viterbo's conjecture, sheaf-theoretic Fukaya categories, closing lemmas), and some of what I call "the frontier" will be settled soon after this is written.

0.1 The shape of the subject

Symplectic topology studies the global properties of symplectic manifolds: smooth manifolds M^{2n} equipped with a closed nondegenerate 2-form ω. The definition comes from classical mechanics, where M is a phase space and ω is what turns an energy function H into a flow, and the oldest questions in the subject are about that flow: does it have periodic orbits, how many, and how are they constrained by the topology of M? Darboux's theorem says that symplectic manifolds have no local invariants, so everything interesting is global, and the central tension in the subject is between *flexibility* (symplectic objects often satisfy h-principles and behave like purely topological objects) and *rigidity* (sometimes they do not, and when they do not the obstruction is usually a holomorphic curve).

The modern subject has a clear birthday. Before 1985 there were the variational methods of Rabinowitz, Weinstein, Conley–Zehnder, Ekeland and Hofer, which proved the first cases of

Arnold's conjectures and Weinstein's conjecture by critical point theory for the action functional. In 1985 Gromov introduced pseudoholomorphic curves and proved, in one paper, nonsqueezing, the nonexistence of exact Lagrangians in $\mathbb{C}^n$, and the homotopy type of $\mathrm{Symp}(S^2 \times S^2)$. In 1988–89 Floer combined Gromov's curves with the variational picture to build what we now call Floer homology, and proved the Arnold conjecture for monotone manifolds. The 1990s produced Gromov–Witten invariants, quantum cohomology, Taubes's identification of Seiberg–Witten and Gromov invariants in dimension four, Donaldson's Lefschetz pencils, Hofer's metric, symplectic capacities, and Kontsevich's homological mirror symmetry conjecture. The 2000s produced Fukaya categories, symplectic field theory, embedded contact homology, Taubes's proof of the Weinstein conjecture in dimension three, and the first systematic uses of spectral invariants. The 2010s settled the foundations of virtual fundamental cycles by several independent routes, produced the flexible side of the subject in its modern form (loose Legendrians, flexible Weinstein manifolds, overtwistedness in all dimensions), and connected Fukaya categories to microlocal sheaf theory. The 2020s so far have given the Arnold conjecture over the integers, the solution of the simplicity conjecture, C^∞ closing lemmas, Birkhoff sections for generic Reeb flows, and a counterexample to Viterbo's conjecture. The subject is in good health.

0.2 How the document is organized

Chapter 1 is linear algebra and the index theory (Maslov, Conley–Zehnder) that grades every Floer theory; it fixes normalizations that everything else depends on. Chapter 2 is the basic theory of symplectic manifolds: examples, Moser's method and the neighborhood theorems, the groups Symp and Ham, almost complex structures, moment maps and reduction. Chapter 3 is about building closed and open symplectic manifolds and about the existence and uniqueness problems, with the four-dimensional landscape stated. Chapter 4 is contact topology together with the entire flexible side of the subject (h-principles, loose Legendrians, flexible Weinstein manifolds). Chapter 5 is the analytic core: pseudoholomorphic curves, compactness, transversality, Gromov–Witten invariants, quantum cohomology, and an honest account of the virtual foundations. Chapter 6 is what curves prove: nonsqueezing, capacities, embedding problems, Gromov's and McDuff's theorems, Taubes's SW = Gr. Chapter 7 is Hamiltonian Floer theory and its quantitative descendants (spectral invariants, Hofer geometry, quasimorphisms, barcodes, C^0 symplectic topology). Chapter 8 is Lagrangian Floer theory, Fukaya categories, symplectic cohomology, mirror symmetry and sheaves. Chapter 9 is dynamics: Weinstein's conjecture, SFT and ECH, closing lemmas, the Conley and Hofer–Zehnder conjectures, systolic inequalities. Chapter 10 collects the open problems and gives reading paths.

The dependencies are roughly linear through Chapter 5; after that, Chapters 6 to 9 can be read in any order, with Chapter 8 leaning on Chapter 7 for conventions and Chapter 9 leaning on Chapters 4 and 6. Every chapter ends with a "What to read" section, a "Frontier and open problems" section, and a short "Logical dependencies" paragraph.

0.3 A note on precision

This document was written with a second use in mind: as input to an autoformalization pipeline. That has shaped it in three ways that human readers will, I hope, also appreciate. Every numbered definition is meant to be complete (smoothness class, compactness, dimension, coefficients, connectedness all stated). Every numbered theorem is meant to be stated with its

hypotheses; where the fully general statement needs technology I did not want to develop, the numbered statement is the cleanest honest special case and the general case is described in a remark with attribution. And every chapter ends with a paragraph saying in which order its definitions must be introduced in order to *state* its main theorems, and which heavy foundations (Sobolev spaces, Fredholm theory, Gromov compactness) those statements presuppose. Informal statements are marked as such. I have tried to distinguish theorems (published or at least refereed), preprints, conjectures and folklore, and I have tried to get attributions and dates right; where I could not verify something I have said so.

0.4 Standing conventions

All manifolds are smooth (C^∞), Hausdorff, second countable, and, unless said otherwise, without boundary; "closed" means compact without boundary; "compact with boundary" is said explicitly. Symplectic manifolds have dimension $2n$. Sign conventions in symplectic topology are a notorious source of errors, and no two standard references agree. The conventions below are binding throughout the document, and each later chapter restates the ones it depends on. I chose them to minimize signs in the parts of the subject that are most active now (exact symplectic manifolds, symplectic cohomology, Fukaya categories), at the cost of one deviation from McDuff–Salamon that I flag in Warning 0.9.

Convention 0.1 (Standard structures on $\mathbb{R}^{2n} = \mathbb{C}^n$)**.** $\mathbb{R}^{2n}$ has coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ and the standard symplectic form

$$\omega_0 = \sum_{j=1}^{n} dx_j \wedge dy_j.$$

We identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ via $z_j = x_j + iy_j$; the standard complex structure J_0 is multiplication by i, so $J_0\partial_{x_j} = \partial_{y_j}$ and $J_0\partial_{y_j} = -\partial_{x_j}$. The standard primitive is $\lambda_0 = \frac{1}{2}\sum_j (x_j\, dy_j - y_j\, dx_j)$, with $d\lambda_0 = \omega_0$. In terms of the Euclidean inner product, $\omega_0(v, w) = \langle J_0 v, w\rangle$.

Convention 0.2 (Compatible and tame almost complex structures)**.** An almost complex structure J on (M, ω) is *ω-compatible* if $g_J(v, w) := \omega(v, Jw)$ is a Riemannian metric, and *ω-tame* if $\omega(v, Jv) > 0$ for all $v \neq 0$. With Convention 0.1, J_0 is ω_0-compatible and g_{J_0} is the Euclidean metric.

Convention 0.3 (Hamiltonian vector fields and Poisson brackets)**.** The Hamiltonian vector field of $H\colon M \to \mathbb{R}$ is defined by

$$\iota_{X_H}\omega = -dH, \qquad \text{equivalently} \qquad dH = \omega(\cdot, X_H).$$

On $(\mathbb{R}^{2n}, \omega_0)$ this gives $X_H = J_0\nabla H$, that is, $\dot{x}_j = -\partial H/\partial y_j$ and $\dot{y}_j = \partial H/\partial x_j$. The Poisson bracket is $\{F, G\} := \omega(X_F, X_G) = dG(X_F) = X_F(G)$, so that $\{F, \cdot\}$ is the derivation along the flow of F, and $X_{\{F,G\}} = [X_F, X_G]$.

Convention 0.4 (Exact symplectic manifolds, cotangent bundles, Liouville fields)**.** An exact symplectic manifold is a pair (M, λ) with $\omega = d\lambda$ symplectic. On a cotangent bundle T^*Q with coordinates (q, p), the canonical one-form is $\lambda_{\mathrm{can}} = \sum_j p_j\, dq_j$ (intrinsically, $\lambda_{\mathrm{can}}|_{(q,p)} = p \circ d\pi$), and $\omega_{\mathrm{can}} = d\lambda_{\mathrm{can}} = \sum_j dp_j \wedge dq_j$. Hamilton's equations on T^*Q then take the classical form $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$. A Liouville vector field for λ is the vector field Z with $\iota_Z\omega = \lambda$; it satisfies $\mathcal{L}_Z\omega = \omega$. The identification of $(T^*\mathbb{R}^n, \omega_{\mathrm{can}})$ with $(\mathbb{R}^{2n}, \omega_0)$ is $x = p$, $y = q$.

Convention 0.5 (Contact forms)**.** A contact form on a $(2n+1)$-manifold Y is a one-form α with $\alpha \wedge (d\alpha)^n$ nowhere zero; the contact structure is $\xi = \ker \alpha$, cooriented by α. The Reeb vector field R_α is defined by $\alpha(R_\alpha) = 1$ and $\iota_{R_\alpha} d\alpha = 0$. The symplectization is $(\mathbb{R} \times Y, d(e^r \alpha))$. A hypersurface in a symplectic manifold is of contact type if a Liouville vector field is defined near it and transverse to it; for a Liouville domain (Chapter 3) the Liouville field points outward along the boundary.

Convention 0.6 (Action functional and Floer's equation)**.** Let $H\colon S^1 \times M \to \mathbb{R}$, $S^1 = \mathbb{R}/\mathbb{Z}$, with $H_t = H(t, \cdot)$. For a contractible loop $x\colon S^1 \to M$ with a capping disk $\bar{x}\colon D^2 \to M$, the action is

$$\mathcal{A}_H(x, \bar{x}) = -\int_{D^2} \bar{x}^* \omega + \int_0^1 H_t(x(t))\, dt;$$

on an exact manifold, $\mathcal{A}_H(x) = -\int_{S^1} x^*\lambda + \int_0^1 H_t(x(t))\, dt$. Its critical points are the 1-periodic orbits of X_{H_t}. With Conventions 0.2 and 0.3 and the L^2 metric $\langle \xi, \eta \rangle = \int_0^1 \omega(\xi, J_t \eta)\, dt$, the gradient is $\nabla \mathcal{A}_H(x) = J_t(\dot{x} - X_{H_t}(x))$, so that the *negative* gradient flow equation for $u\colon \mathbb{R} \times S^1 \to M$ is Floer's equation

$$\partial_s u + J_t(u)\big(\partial_t u - X_{H_t}(u)\big) = 0,$$

and the energy of a solution asymptotic to $x_\pm$ as $s \to \pm\infty$ is $E(u) = \iint |\partial_s u|^2\, ds\, dt = \mathcal{A}_H(x_-) - \mathcal{A}_H(x_+)$. Action decreases along Floer cylinders.

Convention 0.7 (Gradings)**.** Hamiltonian Floer theory is written cohomologically. The Conley–Zehnder index is normalized (Chapter 1) so that for a symmetric matrix S with $\|S\| < 2\pi$ the path $t \mapsto \exp(tJ_0 S)$ has index $\frac{1}{2}\operatorname{sign}(S)$; a 1-periodic orbit x of H is placed in degree $\deg x = n - \mu_{\mathrm{CZ}}(x)$. For a C^2-small Morse function H this puts a critical point p in degree equal to its Morse index, so that $\mathrm{HF}^*(H) \cong H^*(M)$ with no shift.

Convention 0.8 (Terminology)**.** $\mathrm{Symp}(M, \omega)$ is the group of symplectomorphisms, Symp_0 its identity component, and $\mathrm{Ham}(M, \omega)$ the group of Hamiltonian diffeomorphisms, meaning time-one maps of time-dependent Hamiltonians, compactly supported if M is open. (M, ω) is *monotone* if $[\omega] = \kappa\, c_1(TM)$ on $\pi_2(M)$ for some $\kappa > 0$, *symplectically aspherical* if ω vanishes on $\pi_2(M)$, and *semipositive* in the sense of McDuff–Salamon (Chapter 2). A *Lagrangian* is an embedded submanifold $L^n \subset M^{2n}$ with $\omega|_L = 0$; immersed Lagrangians are always called immersed. Balls, ellipsoids and cylinders are labelled by capacity, not radius (Chapter 6): $B^{2n}(a) = \{\pi|z|^2 < a\}$. The Novikov field is written Λ (Chapter 5).

Warning 0.9 (Other conventions in the literature). McDuff–Salamon [MS17; MS12a] use $\iota_{X_H}\omega = +dH$ and $\omega_{\mathrm{can}} = -d\lambda_{\mathrm{can}}$ on cotangent bundles; with their choices, Hamilton's equations on $\mathbb{R}^{2n}$ read $\dot{x} = \partial H/\partial y$, $\dot{y} = -\partial H/\partial x$, and their action functional is $-\int \bar{x}^*\omega - \int H$. The conventions above agree with Seidel's "A biased view of symplectic cohomology" [Sei08a] and with most of the symplectic cohomology and Fukaya category literature. Hofer–Zehnder [HZ94] write $\omega_0 = \sum dy_j \wedge dx_j$ in some places. Conley–Zehnder index normalizations vary by sign and by a shift of n; Gutt [Gut14] is the careful comparison. When quoting a theorem from a source with other conventions I have translated it; when a sign in a statement is convention-dependent I have said so.

Remark 0.10 (Why these choices). The three desiderata "$J_0 = i$ is ω_0-compatible with the Euclidean metric", "$\omega = d\lambda$ with $\lambda_{\mathrm{can}} = p\, dq$" and "$\iota_{X_H}\omega = +dH$ gives the classical Hamilton equations

in the coordinates $(x, y) = (q, p)$" cannot all hold at once. I kept the first two, because the Liouville-manifold and Floer-theoretic literature does, and paid with the sign in Convention 0.3 and the identification $x = p,\ y = q$ in Convention 0.4. The sign $\iota_{X_H}\omega = -dH$ has the pleasant consequence that $\{F, \cdot\} = X_F$ and $X_{\{F,G\}} = [X_F, X_G]$ without a sign, and that Floer's equation is the negative gradient flow of the action functional as written in Convention 0.6, so that symplectic cohomology is the direct limit over Hamiltonians of increasing slope in the usual direction.

0.5 Prerequisites and notation

I assume the reader knows manifolds, differential forms, de Rham cohomology, vector bundles and characteristic classes, the basics of Riemannian geometry, singular (co)homology, and the elements of Sobolev spaces and Fredholm operators. Morse theory in Milnor's form is assumed in Chapter 7; the Morse-homology point of view is recalled there. Some complex geometry (Kähler manifolds, Chern classes via connections) helps in Chapters 2 and 5 but is not required. Fields of coefficients are written $\mathbb{F}$; $\mathbb{Z}/2$ coefficients are used by default where orientations would otherwise need discussion. The Novikov field Λ is defined in Chapter 5. Citations are alphabetic, and each chapter's reading list says what each source is good for.

Chapter 1

Linear symplectic geometry and the Maslov index

Every grading and Fredholm index in the Floer-theoretic chapters of this book is computed from the material of this chapter: the topology of the symplectic group $\mathrm{Sp}(2n)$ and of the Lagrangian Grassmannian $\mathrm{Lag}(n)$, and the integers (or half-integers) attached to loops and paths in them. The geometry is elementary—nothing here is harder than the polar decomposition—but the bookkeeping is not, and I have seen more wrong gradings in print than wrong theorems. The chapter therefore has two jobs. The first is to give the definitions that later chapters quote: compatible complex structures and c_1; the Maslov index of loops of symplectic matrices and of Lagrangian subspaces; the boundary Maslov index of a bundle pair; the Maslov homomorphism and minimal Maslov number of a Lagrangian; and the Conley–Zehnder and Robbin–Salamon indices. The second is to fix every normalization once, consistently with conventions C1–C3 and C8 of the preface, and to show how to check it.

The one idea to take away: a symplectic vector space is a complex vector space up to a contractible choice, so its homotopy theory is that of $\mathrm{U}(n)$, and every Maslov-type index is ultimately the winding number of a complex determinant.

1.1 Symplectic vector spaces

Definition 1.1 (Symplectic vector space)**.** A *symplectic vector space* is a pair (V, ω) where V is a finite-dimensional real vector space and $\omega\colon V \times V \to \mathbb{R}$ is bilinear, alternating ($\omega(v,v) = 0$ for all v) and nondegenerate ($\omega(v,w) = 0$ for all w implies $v = 0$). A *linear symplectomorphism* $(V,\omega) \to (V',\omega')$ is a linear isomorphism A with $A^*\omega' = \omega$. The model is $(\mathbb{R}^{2n}, \omega_0)$ with $\omega_0 = \sum_j dx_j \wedge dy_j$ and coordinates $(x_1,\dots,x_n,y_1,\dots,y_n)$ as in C1.

Proposition 1.2 (Linear Darboux theorem)**.** *If (V,ω) is a symplectic vector space then* $\dim V = 2n$ *is even and V has a* symplectic basis $e_1,\dots,e_n,f_1,\dots,f_n$*: $\omega(e_j,e_k) = \omega(f_j,f_k) = 0$ and $\omega(e_j,f_k) = \delta_{jk}$. Equivalently, the map $e_j \mapsto \partial_{x_j}$, $f_j \mapsto \partial_{y_j}$ is a linear symplectomorphism $(V,\omega) \to (\mathbb{R}^{2n},\omega_0)$.*

Proof idea. Choose e_1, f_1 with $\omega(e_1,f_1) = 1$; the ω-complement of their span is symplectic of dimension two less. Induct.

For a subspace $W \subset V$ put $W^\omega := \{v \in V : \omega(v,w) = 0 \text{ for all } w \in W\}$.

Definition 1.3. A linear subspace W of a symplectic vector space (V, ω) is *isotropic* if $W \subset W^\omega$, *coisotropic* if $W^\omega \subset W$, *Lagrangian* if $W = W^\omega$, and *symplectic* if $W \cap W^\omega = 0$. We write $\mathrm{Lag}(V, \omega)$ for the set of Lagrangian subspaces and $\mathrm{Lag}(n) \coloneqq \mathrm{Lag}(\mathbb{R}^{2n}, \omega_0)$.

Proposition 1.4. *Let (V, ω) be a symplectic vector space of dimension $2n$ and $W \subset V$ a subspace.*

1. $\dim W + \dim W^\omega = 2n$ *and* $(W^\omega)^\omega = W$. *Hence W is Lagrangian if and only if it is isotropic of dimension n.*
2. (Linear reduction.) *If W is coisotropic, then $\omega|_W$ descends to a symplectic form on W/W^ω, which has dimension $2\dim W - 2n$. If $\Lambda \subset V$ is Lagrangian, then $\big((\Lambda \cap W) + W^\omega\big)/W^\omega$ is Lagrangian in W/W^ω.*
3. *Every isotropic subspace is contained in a Lagrangian subspace. Every Lagrangian Λ has a Lagrangian complement Λ', and $v \mapsto \omega(v, \cdot)|_\Lambda$ is an isomorphism $\Lambda' \to \Lambda^*$.*

Proof idea. (1) W^ω is the kernel of the surjection $V \to W^*$, $v \mapsto \omega(v, \cdot)|_W$. (2) The kernel of $\omega|_W$ is $W \cap W^\omega = W^\omega$. (3) If I is isotropic, then I^ω is coisotropic with $(I^\omega)^\omega = I$, and the preimage in I^ω of a Lagrangian subspace of I^ω/I is Lagrangian in V. For a complement of Λ take $J\Lambda$ for any compatible J (Definition 1.7). This is the linear prototype of symplectic reduction (Chapter 2): reduce, solve the problem downstairs, pull back.

Proposition 1.5 (Graph chart on $\mathrm{Lag}(n)$)**.** *If $\Lambda \in \mathrm{Lag}(n)$ satisfies $\Lambda \cap i\mathbb{R}^n = 0$, where $i\mathbb{R}^n = \{x = 0\}$, then $\Lambda = \{(x, Sx) : x \in \mathbb{R}^n\}$ for a unique real $n \times n$ matrix S; conversely, the graph of a matrix S is Lagrangian if and only if $S = S^T$. Thus $\big\{\Lambda \in \mathrm{Lag}(n) : \Lambda \cap i\mathbb{R}^n = 0\big\}$ is identified with the space $\mathrm{Sym}(n)$ of real symmetric matrices. Translating these charts by $\mathrm{U}(n)$ makes $\mathrm{Lag}(n)$ a compact smooth manifold of dimension $n(n+1)/2$.*

The computation is one line: $\omega_0\big((x, Sx), (x', Sx')\big) = \langle x, Sx'\rangle - \langle Sx, x'\rangle$. When Λ is the graph of dF for a quadratic form $F(x) = \frac{1}{2}\langle x, Sx\rangle$, this is the linear shadow of the fact that graphs of closed 1-forms are Lagrangian (Chapter 2).

1.2 Compatible complex structures and symplectic vector bundles

Convention 1.6 (Matrices and the Hamiltonian vector field)**.** Under C1, multiplication by i is $i(x + iy) = -y + ix$, so

$$J_0 = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}, \qquad \omega_0(v, w) = \langle J_0 v, w\rangle = -\langle v, J_0 w\rangle = -v^T J_0 w, \qquad g_0(v, w) = \omega_0(v, J_0 w) = \langle v, w\rangle.$$

(Sources that write $\omega_0(v, w) = v^T J_0 w$ are using the opposite form or the opposite J_0.) A complex matrix $X + iY$ acts on $\mathbb{R}^{2n}$ as $\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}$. For $H \in C^\infty(\mathbb{R}^{2n})$, C3 says $dH = \omega_0(\cdot, X_H)$, while $dH(v) = g_0(v, \nabla H) = \omega_0(v, J_0 \nabla H)$; by nondegeneracy

$$X_H = J_0 \nabla H, \qquad \text{i.e.} \qquad \dot{x}_j = -\partial H/\partial y_j, \quad \dot{y}_j = \partial H/\partial x_j.$$

In particular $H = \frac{\theta}{2}|z|^2$ generates $e^{i\theta t}$: positive quadratic Hamiltonians rotate counterclockwise.

Definition 1.7 (Compatible and tame complex structures)**.** A *linear complex structure* on V is $J \in \mathrm{End}(V)$ with $J^2 = -\mathrm{id}$. It is *ω-tame* if $\omega(v, Jv) > 0$ for all $v \neq 0$, and *ω-compatible* if moreover $\omega(Jv, Jw) = \omega(v, w)$ for all v, w; equivalently (C2), $g_J := \omega(\cdot, J\cdot)$ is an inner product. Write $\mathcal{J}(V, \omega) \subset \mathcal{J}_\tau(V, \omega)$ for the spaces of compatible and of tame structures. For $J \in \mathcal{J}(V, \omega)$, the form $h_J := g_J - i\,\omega$ is a Hermitian inner product on (V, J), complex linear in the first variable; on $(\mathbb{C}^n, J_0)$ it is $h_{J_0}(z, w) = \sum_j z_j \bar{w}_j$, so $\omega_0 = -\operatorname{Im} h_{J_0}$.

Theorem 1.8. *For every symplectic vector space (V, ω) the spaces $\mathcal{J}(V, \omega)$ and $\mathcal{J}_\tau(V, \omega)$ are nonempty and contractible.*

Proof idea. The proof to learn is the *polar retraction*. Given any inner product g on V, define A by $\omega(v, w) = g(Av, w)$; A is g-skew, $-A^2$ is g-positive, and $J_g := (-A^2)^{-1/2}A$ is ω-compatible. The map $g \mapsto J_g$ is continuous and satisfies $J_{g_J} = J$ for $J \in \mathcal{J}(V, \omega)$, so $\mathcal{J}(V, \omega)$ is a retract of the convex cone of inner products. For tame structures, fix $J_0 \in \mathcal{J}(V, \omega)$ and write $J = J_0(1 + K)(1 - K)^{-1}$ (a Cayley transform; $J + J_0$ is invertible when J is tame). One checks that J is tame if and only if $KJ_0 = -J_0K$ and $\|K\|_{g_{J_0}} < 1$, and compatible if and only if in addition K is g_{J_0}-symmetric. Both are convex sets.

Definition 1.9 (Symplectic vector bundle)**.** A *symplectic vector bundle* over a paracompact Hausdorff space B is a real vector bundle $E \to B$ with a continuous section ω of $\Lambda^2 E^*$ that is a symplectic form on each fiber (smooth if B and E are smooth). If $A \subset B$, a *Lagrangian subbundle* of $E|_A$ is a subbundle F with $F_b \in \mathrm{Lag}(E_b, \omega_b)$ for all $b \in A$.

Corollary 1.10. *Let $(E, \omega) \to B$ be a symplectic vector bundle of rank $2n$.*

1. *The space of fiberwise ω-compatible complex structures J on E is nonempty and contractible. Hence $c_1(E, \omega) := c_1(E, J) \in H^2(B; \mathbb{Z})$ is independent of J.*
2. *On a fixed real vector bundle, $\omega \mapsto [J]$ induces a bijection between homotopy classes of fiberwise symplectic forms and homotopy classes of fiberwise complex structures. Isomorphism classes of rank-$2n$ symplectic bundles correspond bijectively to isomorphism classes of rank-n complex bundles.*
3. *If $F \subset E$ is a Lagrangian subbundle and J is compatible, then $E = F \oplus JF \cong F \otimes_{\mathbb{R}} \mathbb{C}$ as complex vector bundles.*

Proof idea. (1) Apply the polar retraction fiberwise to a bundle metric. (2) The space of compatible pairs (ω, J) fibers over both factors with contractible fibers: over ω the fiber is $\mathcal{J}(\omega)$, over J it is the convex set of forms compatible with J. (3) F and JF are g_J-orthogonal.

Part (1) defines $c_1(M, \omega) := c_1(TM, \omega)$ for symplectic manifolds (Chapter 2); part (2) is the linear input to Gromov's theorem that on open manifolds almost symplectic and almost complex structures are the same up to homotopy (Chapter 4); part (3) gives $TM|_L \cong TL \otimes \mathbb{C}$ along a Lagrangian L.

1.3 The symplectic group

Definition 1.11. $\mathrm{Sp}(2n) := \left\{A \in \mathbb{R}^{2n\times 2n} : A^*\omega_0 = \omega_0\right\} = \left\{A : A^T J_0 A = J_0\right\}$. Its Lie algebra is $\mathfrak{sp}(2n) = \left\{J_0 S : S = S^T\right\}$.

The two descriptions agree because $\omega_0(Av, Aw) = \langle A^T J_0 A v, w\rangle$, and $X^T J_0 + J_0 X = 0$ with $X = J_0 S$ is equivalent to $S = S^T$.

Proposition 1.12. *1.* $\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{Sp}(2n) \cap \mathrm{GL}(n,\mathbb{C}) = \mathrm{O}(2n) \cap \mathrm{GL}(n,\mathbb{C}) = \mathrm{U}(n)$*, where* $\mathrm{GL}(n,\mathbb{C}) \subset \mathrm{GL}(2n,\mathbb{R})$ *is the commutant of* J_0*, embedded as in Convention 1.6.*

2. *Every* $A \in \mathrm{Sp}(2n)$ *factors uniquely as* $A = PQ$ *with* $P = (AA^T)^{1/2}$ *symmetric, positive definite and symplectic, and* $Q \in \mathrm{U}(n)$*. The homotopy* $A_s = (AA^T)^{-s/2}A$*,* $s \in [0,1]$*, is a strong deformation retraction of* $\mathrm{Sp}(2n)$ *onto* $\mathrm{U}(n)$*, and* $\mathrm{Sp}(2n) \approx \mathrm{U}(n) \times \mathbb{R}^{n(n+1)}$*.*

3. *Consequently* $\mathrm{Sp}(2n)$ *is connected,* $\det A = 1$ *for all* $A \in \mathrm{Sp}(2n)$*, and* $\pi_1(\mathrm{Sp}(2n)) \cong \pi_1(\mathrm{U}(n)) \cong \mathbb{Z}$*, the last isomorphism induced by* $\det_{\mathbb{C}} \colon \mathrm{U}(n) \to S^1$*.*

($A^T J_0 A = J_0$ forces $\det A = \pm 1$, and connectedness gives $+1$.) So $\mathrm{Sp}(2n)$, $\mathrm{U}(n)$, $\mathrm{GL}(n,\mathbb{C})$ and $\mathcal{J}(\mathbb{R}^{2n}, \omega_0) \cong \mathrm{Sp}(2n)/\mathrm{U}(n)$ are all homotopy-theoretically "the same", which is the precise sense of the slogan in the introduction.

Proposition 1.13 (Spectrum)**.** *Every* $A \in \mathrm{Sp}(2n)$ *is conjugate to* $(A^T)^{-1}$*, so its characteristic polynomial satisfies* $p(\lambda) = \lambda^{2n} p(1/\lambda)$*. Hence* λ*,* $\bar\lambda$*,* λ^{-1}*,* $\bar\lambda^{-1}$ *are eigenvalues of the same algebraic multiplicity, and the eigenvalues* ± 1 *have even multiplicity. We call* A nondegenerate *if* $\det(I - A) \neq 0$*. For* $n = 1$*,* $A \neq \pm I$ *is* elliptic *if* $|\mathrm{tr}\, A| < 2$ *(eigenvalues* $e^{\pm i\theta}$*,* $\theta \notin \pi\mathbb{Z}$*),* parabolic *if* $|\mathrm{tr}\, A| = 2$*, and* positive *(resp.* negative*)* hyperbolic *if* $\mathrm{tr}\, A > 2$ *(resp.* $\mathrm{tr}\, A < -2$*).*

In higher dimensions an eigenvalue on the unit circle carries a Krein sign (the signature of $(v,w) \mapsto \omega(\bar v, w)$, suitably normalized, on its generalized eigenspace), and the determinant on $\mathrm{U}(n)$ extends to $\mathrm{Sp}(2n)$ by counting these.

Theorem 1.14 (The map ρ)**.** *There is a unique family of continuous maps* $\rho \colon \mathrm{Sp}(2n) \to S^1$*,* $n \geq 1$*, such that:* (naturality) $\rho(TAT^{-1}) = \rho(A)$ *for* $T \in \mathrm{Sp}(2n)$*;* (product) $\rho(A \oplus B) = \rho(A)\rho(B)$ *under* $\mathbb{C}^m \oplus \mathbb{C}^k = \mathbb{C}^{m+k}$*;* (determinant) $\rho(X + iY) = \det(X + iY)$ *on* $\mathrm{U}(n)$*;* (normalization) $\rho(A) \in \{\pm 1\}$ *if* A *has no eigenvalue on* S^1*.*

The map ρ was constructed by Salamon–Zehnder [SZ92]; see [MS17, Sect. 2.2] for this characterization and the explicit formula in terms of Krein signatures. The map ρ is not a homomorphism, but it restricts to $\det_{\mathbb{C}}$ on $\mathrm{U}(n)$, and that is all that matters on π_1.

Definition 1.15 (Maslov index of a symplectic loop)**.** For a continuous loop $\Phi \colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$, $\mu(\Phi) := \deg(\rho \circ \Phi) \in \mathbb{Z}$. Equivalently $\mu(\Phi) = \deg(\det_{\mathbb{C}} \circ r \circ \Phi)$, where $r(A) = (AA^T)^{-1/2}A$ is the polar retraction. Then $\mu \colon \pi_1(\mathrm{Sp}(2n)) \to \mathbb{Z}$ is an isomorphism, additive under pointwise products and direct sums. **Normalization:** the loop $t \mapsto \exp(2\pi t J_0)$ on $\mathbb{R}^2$, direct sum the identity on $\mathbb{R}^{2n-2}$ (that is, $e^{2\pi i t}$ on the first factor of $\mathbb{C}^n$), has $\mu = 1$ and generates $\pi_1(\mathrm{Sp}(2n))$.

1.4 The Lagrangian Grassmannian and the Maslov class

Proposition 1.16. $\mathrm{U}(n)$ *acts transitively on* $\mathrm{Lag}(n)$ *with stabilizer of* $\mathbb{R}^n$ *equal to* $\mathrm{O}(n)$*, so* $\mathrm{Lag}(n) \cong \mathrm{U}(n)/\mathrm{O}(n)$ *is a compact connected manifold of dimension* $n(n+1)/2$*. The map* $\det^2 \colon \mathrm{Lag}(n) \to S^1$*,* $U\mathbb{R}^n \mapsto \det_{\mathbb{C}}(U)^2$*, is well defined and induces an isomorphism* $\pi_1(\mathrm{Lag}(n)) \cong \mathbb{Z}$*.*

Proof idea. A g_0-orthonormal basis of $\Lambda \in \mathrm{Lag}(n)$ is a unitary basis of $\mathbb{C}^n$, since $\Lambda \perp J_0\Lambda$. The fibration $\mathrm{SU}(n)/\mathrm{SO}(n) \to \mathrm{Lag}(n) \xrightarrow{\det^2} S^1$ has simply connected fiber.

Definition 1.17 (Maslov index of a Lagrangian loop; Maslov class)**.** For a continuous loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Lag}(n)$, $\mu(\Lambda) \coloneqq \deg(\det^2 \circ \Lambda)$. The *Maslov class* $\mu \in H^1(\mathrm{Lag}(n);\mathbb{Z})$ is the pullback under $\det^2$ of the positive generator of $H^1(S^1;\mathbb{Z})$ [Arn67]. For a symplectic vector space (V,ω), use any linear symplectomorphism to $\mathbb{R}^{2n}$; the result is independent of the choice because $\mathrm{Sp}(2n)$ is connected. **Normalization:** the loop $t \mapsto e^{i\pi t}\mathbb{R} \subset \mathbb{C}$, $t \in [0,1]$, has $\mu = 1$. The index is additive under direct sums, and $\mu(\Phi\Lambda) = \mu(\Lambda) + 2\mu(\Phi)$ for a symplectic loop Φ acting pointwise.

Remark 1.18 (Arnold's Maslov cycle). The class goes back to Maslov's work on semiclassical asymptotics [Mas65]; the topological formulation is Arnold's [Arn67]. Fix $K \in \mathrm{Lag}(n)$. The *Maslov cycle* $\Sigma(K) = \{\Lambda : \Lambda \cap K \neq 0\}$ is stratified by $\Sigma_k = \{\dim \Lambda \cap K = k\}$, of codimension $k(k+1)/2$; the top stratum Σ_1 is co-oriented, and $\mu(\Lambda)$ is the intersection number of a loop with $\Sigma(K)$, for every K. This is why Maslov indices are local: one counts crossings with signs. The Robbin–Salamon index below is the same count for paths.

Definition 1.19 (Boundary Maslov index of a bundle pair)**.** Let Σ be a compact connected oriented surface with nonempty boundary, with boundary components $C_1,\dots,C_k$ oriented as the boundary. A *bundle pair* (E,F) is a symplectic vector bundle $E \to \Sigma$ of rank $2n$ with a Lagrangian subbundle $F \subset E|_{\partial\Sigma}$. Choose a symplectic trivialization $\Phi\colon E \to \Sigma \times \mathbb{R}^{2n}$ (one exists since Σ retracts onto a graph and $\mathrm{Sp}(2n)$ is connected) and orientation-preserving parametrizations $\gamma_j\colon \mathbb{R}/\mathbb{Z} \to C_j$. Define

$$\mu(E,F) \coloneqq \sum_{j=1}^{k} \mu\big(t \mapsto \Phi_{\gamma_j(t)}(F_{\gamma_j(t)})\big) \in \mathbb{Z}.$$

Changing Φ by $g\colon \Sigma \to \mathrm{Sp}(2n)$ changes the sum by $2\sum_j \mu(g\circ\gamma_j) = 0$, because $\partial\Sigma$ is null-homologous in Σ; so $\mu(E,F)$ is well defined. For closed Σ one sets $\mu(E) \coloneqq 2\langle c_1(E), [\Sigma]\rangle$, and then μ is additive under gluing surfaces along boundary circles.

The bundle-pair index is characterized by isomorphism invariance, additivity under direct sums, the gluing property, and the normalization $\mu(D^2\times\mathbb{C}, F) = k$ for $F_{e^{2\pi i t}} = e^{i\pi k t}\mathbb{R}$ [MS17, Sect. 2.7]. It is the topological term in the Riemann–Roch theorem for real Cauchy–Riemann operators on bundle pairs, index $= n\chi(\Sigma) + \mu(E,F)$, stated in Chapter 5.

Definition 1.20 (Maslov homomorphism; minimal Maslov number)**.** Let (M^{2n},ω) be a symplectic manifold and $L \subset M$ a connected Lagrangian. For a continuous map $u\colon (D^2,\partial D^2) \to (M,L)$, set $\mu_L(u) \coloneqq \mu\big(u^*TM, (u|_{\partial D^2})^*TL\big)$. This depends only on $[u] \in \pi_2(M,L)$ and defines a homomorphism $\mu_L\colon \pi_2(M,L) \to \mathbb{Z}$. The *minimal Maslov number* $N_L \in \{1,2,\dots\}\cup\{\infty\}$ is the positive generator of $\mu_L(\pi_2(M,L)) \subset \mathbb{Z}$, with $N_L = \infty$ if $\mu_L = 0$. The Lagrangian L is *monotone* if there is $\kappa > 0$ with $\omega(\beta) = \kappa\,\mu_L(\beta)$ for all $\beta \in \pi_2(M,L)$.

Example 1.21. For the unit disk $u(z) = z$ in $\mathbb{C}$ with boundary on $L = S^1$, the tangent lines are $T_{e^{2\pi i t}}S^1 = ie^{2\pi i t}\mathbb{R}$, so $\det^2 = -e^{4\pi i t}$ and $\mu_L(u) = 2$. Hence the Clifford torus $(S^1)^n \subset \mathbb{C}^n$ is monotone with $N_L = 2$. In general, if $v\colon S^2 \to M$ then $\mu_L(u\#v) = \mu_L(u) + 2c_1(v)$: on spherical classes μ_L is $2c_1$. If L is orientable then μ_L is even. As a sanity check against Chapter 5: the Fredholm index $n\chi(D^2) + \mu = 1 + 2 = 3$ at the unit disk is $\dim \mathrm{PSL}(2,\mathbb{R})$, the reparametrizations.

1.5 The Conley–Zehnder and Robbin–Salamon indices

Let $\mathcal{SP}(n)$ be the set of continuous paths $\Psi\colon [0,1] \to \mathrm{Sp}(2n)$ with $\Psi(0) = I$ and $\det(I - \Psi(1)) \neq 0$, with the compact-open topology. The set $\mathrm{Sp}^*(2n) = \{A : \det(I-A) \neq 0\}$ has two

components $\mathrm{Sp}^{\pm}(2n)$, distinguished by the sign of $\det(I-A)$ and containing $W_+ \coloneqq -I$ and $W_- \coloneqq \mathrm{diag}(2,-1,\dots,-1,\frac{1}{2},-1,\dots,-1)$ respectively; and every loop in $\mathrm{Sp}^{\pm}(2n)$ is contractible in $\mathrm{Sp}(2n)$ [SZ92; Sal99].

Definition 1.22 (Conley–Zehnder index; Conley–Zehnder 1984, Salamon–Zehnder 1992)**.** For $\Psi \in \mathcal{SP}(n)$, extend Ψ to $\widetilde{\Psi}\colon [0,2] \to \mathrm{Sp}(2n)$ so that $\widetilde{\Psi}([1,2]) \subset \mathrm{Sp}^*(2n)$ and $\widetilde{\Psi}(2) \in \{W_+, W_-\}$. Since $\rho(I)^2 = \rho(W_\pm)^2 = 1$, $\rho^2 \circ \widetilde{\Psi}$ is a loop, and $\mu_{\mathrm{CZ}}(\Psi) \coloneqq \deg(\rho^2 \circ \widetilde{\Psi}) \in \mathbb{Z}$, independent of the extension.

The index was introduced by Conley–Zehnder [CZ84] in their Morse-type index theory for periodic solutions, which grew out of their proof of the Arnold conjecture for tori [CZ83]; the form above is Salamon–Zehnder's [SZ92]. The definition is unusable for computation; the axioms, proved in [SZ92; RS93] (see also [Sal99]), are what one actually uses.

Theorem 1.23 (Axioms)**.** *The map $\mu_{\mathrm{CZ}}\colon \mathcal{SP}(n) \to \mathbb{Z}$ has the following properties.*

1. (Homotopy) *μ_{CZ} is constant on path components of $\mathcal{SP}(n)$.*
2. (Naturality) *$\mu_{\mathrm{CZ}}(\Phi\Psi\Phi^{-1}) = \mu_{\mathrm{CZ}}(\Psi)$ for every path $\Phi\colon [0,1] \to \mathrm{Sp}(2n)$.*
3. (Loop) *$\mu_{\mathrm{CZ}}(\Phi\Psi) = \mu_{\mathrm{CZ}}(\Psi) + 2\mu(\Phi)$ for every loop Φ with $\Phi(0) = \Phi(1) = I$.*
4. (Signature) *If $S = S^T$ is nondegenerate with $\|S\| < 2\pi$, then $\mu_{\mathrm{CZ}}\big(t \mapsto \exp(tJ_0S)\big) = \frac{1}{2}\operatorname{sgn}(S)$, where* sgn *is the signature.*
5. (Product) *$\mu_{\mathrm{CZ}}(\Psi_1 \oplus \Psi_2) = \mu_{\mathrm{CZ}}(\Psi_1) + \mu_{\mathrm{CZ}}(\Psi_2)$, under $\mathbb{C}^{n_1} \oplus \mathbb{C}^{n_2} = \mathbb{C}^{n_1+n_2}$.*
6. (Determinant) *$(-1)^{n-\mu_{\mathrm{CZ}}(\Psi)} = \operatorname{sgn}\det\big(I - \Psi(1)\big)$.*
7. (Inverse) *$\mu_{\mathrm{CZ}}(\Psi^{-1}) = -\mu_{\mathrm{CZ}}(\Psi)$, where $\Psi^{-1}(t) \coloneqq \Psi(t)^{-1}$.*

Homotopy, Loop and Signature already determine μ_{CZ} uniquely [Sal99; AD14; Gut14].

Proof idea. By the facts about $\mathrm{Sp}^{\pm}(2n)$ above, the path components of $\mathcal{SP}(n)$ form two $\mathbb{Z}$-torsors under multiplication by loops, one for each sign of $\det(I-\Psi(1))$. The Signature axiom applied to $S = \varepsilon I$ (index n) and to $S = \varepsilon\,\mathrm{diag}(-1,1,\dots,1)$ (index $n-1$) pins down one value in each torsor.

Remark 1.24 (Why this normalization). By Convention 1.6, $X_H = J_0\nabla H$ under C3. At a critical point p of H with Hessian S in Darboux coordinates the linearized flow solves $\dot{\xi} = J_0 S\xi$, so $d\phi_t^H(p) = \exp(tJ_0S)$. If H is a Morse function, C^2-small enough that its only 1-periodic orbits are its critical points and $\|S\| < 2\pi$ at each of them, then with constant cappings

$$\mu_{\mathrm{CZ}}(p) = \tfrac{1}{2}\operatorname{sgn}(S) = \tfrac{1}{2}\big((2n - \operatorname{ind}_p H) - \operatorname{ind}_p H\big) = n - \operatorname{ind}_p(H),$$

so the grading $\deg = n - \mu_{\mathrm{CZ}}$ of C8 puts p in degree $\operatorname{ind}_p(H)$, as $\mathrm{HF}^*(H) \cong H^*(M)$ requires. The Determinant axiom checks out: $\operatorname{sgn}\det(I - e^{J_0S}) = \operatorname{sgn}\det S = (-1)^{\operatorname{ind}_p H}$. Under the McDuff–Salamon sign $\iota_{X_H}\omega = +dH$ the linearized flow is $\exp(-tJ_0S)$ and $\mu_{\mathrm{CZ}}(p) = \operatorname{ind}_p(H) - n$; sources with that sign therefore grade homologically by $\mu_{\mathrm{CZ}} + n$, or by μ_{CZ} with $\mathrm{HF}_*(H) \cong H_{*+n}(M)$.

Example 1.25 (The case $n = 1$)**.** For $\theta \in \mathbb{R} \setminus 2\pi\mathbb{Z}$,

$$\mu_{\mathrm{CZ}}\big(t \mapsto \exp(t\theta J_0)\big) = 2\lfloor \theta/2\pi \rfloor + 1,$$

which equals $2\lceil \theta/2\pi \rceil - 1$ and is an odd function of θ; for $\theta \in (0, 2\pi)$ this is the Signature axiom with $S = \theta I$ (a local minimum, $\mu_{\mathrm{CZ}} = 1$), and the general case follows from the Loop axiom. For

$a \neq 0$, the hyperbolic path $\exp\big(tJ_0\,\mathrm{diag}(a,-a)\big) = \begin{pmatrix}\cosh at & \sinh at\\ \sinh at & \cosh at\end{pmatrix}$ (a saddle) has $\mu_{\mathrm{CZ}} = 0$, and multiplying by $\exp(2\pi k t J_0)$ gives $2k$. By the Determinant axiom, for $n = 1$ the index is even if and only if $\Psi(1)$ is positive hyperbolic; elliptic and negative hyperbolic endpoints (and endpoints with eigenvalue -1) give odd indices. These are exactly the formulas used for Reeb orbits in embedded contact homology, with $\theta/2\pi$ the rotation number [Hut14].

Definition 1.26 (Robbin–Salamon index; Robbin–Salamon 1993)**.** Let (V,ω) be a symplectic vector space, $K \in \mathrm{Lag}(V,\omega)$, and $\Lambda\colon [a,b] \to \mathrm{Lag}(V,\omega)$ a C^1 path. For $t \in [a,b]$ choose a Lagrangian complement W of $\Lambda(t)$ and write $\Lambda(s) = \{v + A_s v : v \in \Lambda(t)\}$ for s near t, with $A_s\colon \Lambda(t) \to W$ linear. The *crossing form* is the quadratic form $\Gamma(\Lambda,K,t)(v) \coloneqq \frac{d}{ds}\big|_{s=t}\omega(v,A_s v)$ on $\Lambda(t)\cap K$; it is independent of W. A *crossing* is a t with $\Lambda(t)\cap K \neq 0$; it is *regular* if Γ is nondegenerate. If all crossings are regular (hence finite in number),

$$\mu_{\mathrm{RS}}(\Lambda,K) \coloneqq \tfrac{1}{2}\operatorname{sgn}\Gamma(\Lambda,K,a) + \sum_{a<t<b}\operatorname{sgn}\Gamma(\Lambda,K,t) + \tfrac{1}{2}\operatorname{sgn}\Gamma(\Lambda,K,b).$$

For a path $\Psi\colon [0,1]\to \mathrm{Sp}(2n)$ set $\mu_{\mathrm{RS}}(\Psi) \coloneqq \mu_{\mathrm{RS}}(\mathrm{Gr}\,\Psi,\Delta)$, computed in $(\mathbb{R}^{2n}\times\mathbb{R}^{2n}, (-\omega_0)\oplus\omega_0)$; writing $\dot\Psi = J_0 S\Psi$, its crossings are the t with $\det(I-\Psi(t)) = 0$, with crossing form $v \mapsto \langle v, S(t)v\rangle$ on $\ker(I-\Psi(t))$.

Theorem 1.27 (Robbin–Salamon 1993)**.** *μ_{RS} extends uniquely to continuous paths so as to be invariant under homotopy with fixed endpoints. It takes values in $\frac{1}{2}\mathbb{Z}$, is additive under catenation, vanishes if* $\dim\Lambda(t)\cap K$ *is constant, equals $\mu(\Lambda)$ for loops (for every K), and satisfies* (localization) *$\mu_{\mathrm{RS}}(\Lambda,\mathbb{R}^n) = \frac{1}{2}\operatorname{sgn}A(b) - \frac{1}{2}\operatorname{sgn}A(a)$ if $\Lambda(t) = \{(x,A(t)x)\}$ with $A(t)$ symmetric. For symplectic paths, $\mu_{\mathrm{RS}}(\Psi) = \mu_{\mathrm{CZ}}(\Psi)$ if $\Psi \in \mathcal{SP}(n)$, and $\mu_{\mathrm{RS}}\big(t\mapsto \exp(tJ_0S)\big) = \frac{1}{2}\operatorname{sgn}S$ for every symmetric S with $\|S\| < 2\pi$, degenerate or not.*

For example, the shear $\begin{pmatrix}1 & 0\\ t & 1\end{pmatrix} = \exp\big(tJ_0\,\mathrm{diag}(1,0)\big)$, the flow of $H = \frac{1}{2}x^2$, has $\mu_{\mathrm{RS}} = \frac{1}{2}$. Half-integers are the price of degenerate endpoints, and they are what Morse–Bott and S^1-equivariant Floer theories need. The comparison of the many definitions of μ_{CZ}, and of the generalized index for arbitrary paths, is carried out in [Gut14]; the rotation formula in Example 1.25 is the start of the iteration theory of [Lon02].

Definition 1.28 (Conley–Zehnder index of a periodic orbit)**.** Let (M^{2n},ω) be a symplectic manifold, $H \in C^\infty(S^1\times M)$ with $S^1 = \mathbb{R}/\mathbb{Z}$, ϕ_t the flow of X_{H_t} (C3), and x a 1-periodic orbit, which is *nondegenerate* if $\det\big(I - d\phi_1(x(0))\big) \neq 0$. A *capping disk* is a smooth $\bar x\colon D^2 \to M$ with $\bar x(e^{2\pi i t}) = x(t)$. Choose a symplectic trivialization $\tau_z\colon (\mathbb{R}^{2n},\omega_0)\to (T_{\bar x(z)}M,\omega)$, $z\in D^2$, and set $\Psi(t) \coloneqq \tau_{e^{2\pi i t}}^{-1}\circ d\phi_t(x(0))\circ\tau_1$. For nondegenerate x, $\mu_{\mathrm{CZ}}(x,\bar x) \coloneqq \mu_{\mathrm{CZ}}(\Psi)$.

Proposition 1.29. *In the setting of Definition 1.28:*

1. *$\mu_{\mathrm{CZ}}(x,\bar x)$ is independent of τ and depends only on the homotopy class of $\bar x$ relative to the boundary.*
2. *For $A\in\pi_2(M)$, $\mu_{\mathrm{CZ}}(x,\bar x\#A) = \mu_{\mathrm{CZ}}(x,\bar x) + 2c_1(A)$. Hence, if $N_M \in \{1,2,\dots\}\cup\{\infty\}$ denotes the positive generator of $c_1(\pi_2(M)) \subset \mathbb{Z}$ (the* minimal Chern number*; $N_M = \infty$ if $c_1|_{\pi_2(M)} = 0$), the class $\mu_{\mathrm{CZ}}(x) \in \mathbb{Z}/2N_M$ is independent of the capping (with $\mathbb{Z}/2\infty \coloneqq \mathbb{Z}$).*
3. *With C8,* $\deg(x,\bar x) = n - \mu_{\mathrm{CZ}}(x,\bar x)$*; for a C^2-small Morse function and constant cappings this is the Morse index.*

Proof idea. (1) Two trivializations over D^2 differ by $g\colon D^2 \to \mathrm{Sp}(2n)$, and the boundary loop of g has Maslov index 0; use Naturality and Loop. (2) A clutching computation shows that the trivializations over $\bar{x}$ and $\bar{x}\#A$ differ along x by a loop of Maslov index $-c_1(A)$; apply the Loop axiom. (3) is Remark 1.24.

In the monotone case (C9), $\mathbb{Z}/2N_M$ is precisely the grading group of HF^* in Chapter 7; if $c_1|_{\pi_2(M)} = 0$ the grading is integral. For noncontractible orbits, and for Reeb orbits (Chapter 9), one needs other trivialization data, for instance a trivialization of the canonical bundle when $2c_1 = 0$ (Chapter 8).

Warning 1.30 (Sign traps). Every one of the following choices flips a sign or shifts by n, and published papers mix them.

(a) $\iota_{X_H}\omega = -dH$ (C3) versus $+dH$ (McDuff–Salamon and many others). This replaces the linearized flow at a critical point by $\exp(-tJ_0S)$ and $n - \mathrm{ind}_p$ by $\mathrm{ind}_p - n$.

(b) The normalization $\exp(tJ_0S) \mapsto \frac{1}{2}\operatorname{sgn} S$ versus $\exp(-tJ_0S) \mapsto \frac{1}{2}\operatorname{sgn} S$; equivalently, whether J_0 means multiplication by i or by $-i$ in the author's coordinates (it is the former under C1, and the ordering $(x_1, y_1, \dots)$ is harmless as long as J_0 is still i).

(c) The Floer grading $\deg = n - \mu_{\mathrm{CZ}}$ (C8), $\deg = \mu_{\mathrm{CZ}} + n$, $\deg = \mu_{\mathrm{CZ}}$, or $\deg = -\mu_{\mathrm{CZ}}$.

(d) Homological versus cohomological conventions: the generators are the same, the differential is transposed, and the comparison with $H_*(M)$ or $H^*(M)$ passes through Poincaré duality $k \leftrightarrow 2n - k$.

(e) The sign of the capping shift $\pm 2c_1(A)$ in Proposition 1.29.

Gutt's paper [Gut14] is the careful comparison. This document uses C8 throughout, together with $X_H = J_0\nabla H$ and the normalization $\mu_{\mathrm{CZ}}\big(t \mapsto \exp(tJ_0S)\big) = \frac{1}{2}\operatorname{sgn} S$, so that $\deg = n - \mu_{\mathrm{CZ}}$. My advice is to test any formula you import on a C^2-small Morse function before trusting it.

Remark 1.31 (Other indices). Hörmander's index of four Lagrangians [Hör71] measures the dependence of $\mu_{\mathrm{RS}}(\Lambda, K)$ on K; it is expressible through Kashiwara's triple index $\tau(\Lambda_1, \Lambda_2, \Lambda_3)$, the signature of the quadratic form $\omega(v_1, v_2) + \omega(v_2, v_3) + \omega(v_3, v_1)$ on $\Lambda_1 \oplus \Lambda_2 \oplus \Lambda_3$ [LV80], which is also the correction term in Wall's non-additivity of the signature [Wal69]. Cappell–Lee–Miller [CLM94] reconcile the various definitions. For a strip $u\colon \mathbb{R} \times [0,1] \to M$ with $u(\cdot, j) \in L_j$ between transverse intersection points, trivializing u^*TM turns TL_0 and TL_1 into two paths of Lagrangians, and the Robbin–Salamon index of the pair is the Maslov index of u, introduced in this setting by Viterbo [Vit87b] and Floer [Flo88a]; it is the relative grading of Lagrangian Floer theory, and absolute gradings come from graded Lagrangians [Sei00] (Chapter 8).

1.6 What to read

Cannas da Silva, Lectures on Symplectic Geometry *[Can01]*

The gentlest start: Part I (Lecture 1, symplectic linear algebra), Section 8.2 on Lagrangian subspaces, and Part V (Lectures 12–13) on compatible structures. It does not treat the Maslov index; read it in a week and move on.

McDuff–Salamon, Introduction to Symplectic Topology *[MS17]*

Chapter 2 is the canonical account. Read Sections 2.1–2.3 (the symplectic group, the map ρ, the Lagrangian Maslov index) and 2.5–2.7 (complex structures, bundles, c_1 and the bundle-pair

Maslov index with its axioms). Section 2.4 on affine nonsqueezing can wait until Chapter 6. Remember they use $\iota_{X_H}\omega = dH$.

Salamon, Lectures on Floer homology *[Sal99]*
Section 2 is the quickest route to the Conley–Zehnder index, with the axioms stated cleanly. Check its sign convention for X_H against C3 before importing formulas.

Robbin–Salamon [RS93] and Salamon–Zehnder [SZ92]
The original sources for crossing forms and for ρ. Read Robbin–Salamon in full; it is short and is the right way to think. From Salamon–Zehnder read the index sections; the rest constructs graded Floer homology for symplectically aspherical manifolds, which Chapter 7 covers.

Gutt, Generalized Conley–Zehnder index *[Gut14]*
Keep it open while reading any paper with gradings: the equivalence of the definitions, normal forms, and the minimal axioms.

Arnold [Arn67]
For the Maslov class and cycle, and for a sense of where the subject came from.

Audin–Damian, Morse Theory and Floer Homology *[AD14]*
Chapter 7, with its appendix constructing ρ, is the most detailed elementary treatment in a Floer context. Check their conventions against ours before quoting.

For later
McDuff–Salamon [MS12a, App. C] for the boundary Maslov index in Riemann–Roch; Long [Lon02] for iteration formulas; Cappell–Lee–Miller [CLM94] for triple indices.

1.7 Frontier and open problems

Honestly, there are no open problems in linear symplectic geometry: the classification of symplectic matrices up to conjugacy, the topology of $\mathrm{Sp}(2n)$ and $\mathrm{Lag}(n)$, and the uniqueness of the indices are all settled. The practical frontier is elsewhere.

First, conventions. The inconsistency described in Warning 1.30 is a real cost: gradings in symplectic cohomology, wrapped Floer theory and the closed–open maps are regularly off by n or by a sign, and errors surface when results from different groups are combined. A single convention-tagged reference, checkable by machine, would be valuable; this document is an attempt in that direction.

Second, gradings in symplectic field theory and embedded contact homology (Chapter 9) use Conley–Zehnder indices of Reeb orbits relative to trivializations of the contact structure, their behavior under iteration (Example 1.25 is the two-dimensional case), and Robbin–Salamon indices in Morse–Bott situations. Getting these right in families is where index theory is still actively used and occasionally miscomputed.

Third, Floer homotopy theory (Chapters 7 and 8) needs more than integers: it needs index bundles and the stable Lagrangian Grassmannian U/O as spaces, not just $\pi_1(\mathrm{U}/\mathrm{O}) = \mathbb{Z}$.

1.8 Logical dependencies

The order is: Definitions 1.1 and 1.3 (with Propositions 1.4 and 1.5); Convention 1.6; Definition 1.7, then Theorem 1.8, Definition 1.9 and Corollary 1.10 (which requires the Chern class of complex

bundles); Definition 1.11 and Proposition 1.12; Theorem 1.14, then Definition 1.15; Proposition 1.16, then Definitions 1.17, 1.19 and 1.20; finally Definition 1.22 (using ρ), Theorem 1.23, Definition 1.26, Theorem 1.27 and Definition 1.28 (using Definition 1.15 and c_1). No analysis is presupposed: only linear algebra, the degree of maps $S^1 \to S^1$, π_1 and π_2 of pairs, and Chern classes.

Chapter 2

Symplectic manifolds: local theory, global invariants, symmetry

This chapter fixes the objects that every later chapter manipulates: symplectic manifolds and their basic examples, with the normalizations used throughout the book; Hamiltonian vector fields and the groups $\mathrm{Symp} \supset \mathrm{Symp}_0 \supset \mathrm{Ham}$, together with the flux and Calabi homomorphisms that tell them apart; the neighborhood theorems; Lagrangian submanifolds; compatible almost complex structures and the Chern-class conditions (aspherical, monotone, Calabi–Yau, semipositive) that later theorems take as hypotheses; and Hamiltonian group actions, which supply the most computable examples and form a branch of the subject in their own right.

The take-away is a dichotomy that organizes the rest of the book. Everything *local* is flexible and is settled by one trick, Moser's: Darboux, Weinstein and Gotay say that neighborhoods are determined by linear algebra. Everything *global* is subtle, but the soft global invariants—$[\omega]$, c_1, the symplectic cone—say very little, and the invariants that matter come from holomorphic curves and Floer theory (Chapter 5). Hamiltonian torus actions are the exception that proves the rule: there, convexity, Duistermaat–Heckman and Delzant give complete answers with no analysis at all. I use the conventions of the preface (Chapter 0) and restate the ones I depend on.

2.1 Symplectic manifolds and examples

Definition 2.1 (Symplectic manifold)**.** A *symplectic manifold* is a pair (M, ω) where M is a C^∞ manifold without boundary and $\omega \in \Omega^2(M)$ is closed ($d\omega = 0$) and nondegenerate (for every $x \in M$ the map $T_xM \to T_x^*M$, $v \mapsto \iota_v\omega$, is bijective). Then $\dim M = 2n$ is even and ω^n is nowhere zero; we orient M by ω^n, and call $\omega^n/n!$ the *Liouville volume form*. A *symplectomorphism* $\psi\colon (M_1, \omega_1) \to (M_2, \omega_2)$ is a diffeomorphism with $\psi^*\omega_2 = \omega_1$. We write $M^- = (M, -\omega)$, and $M_1 \times M_2$ carries $\mathrm{pr}_1^*\omega_1 + \mathrm{pr}_2^*\omega_2$.

If M is closed then $[\omega]^n = [\omega^n] \neq 0$, so $[\omega]^k \neq 0$ in $H^{2k}(M;\mathbb{R})$ for $0 \le k \le n$; thus S^{2n}, $n \ge 2$, is not symplectic. Together with the existence of an almost complex structure (Section 2.5) this is essentially the only general obstruction known in dimension ≥ 6; whether it is the only one is open (Chapter 3).

Example 2.2 (Standard examples)**.** 1. $(\mathbb{R}^{2n}, \omega_0)$, $\omega_0 = \sum_j dx_j \wedge dy_j$, identified with $\mathbb{C}^n$ by $z_j = x_j + iy_j$; $J_0 = i$ is ω_0-compatible with g_{J_0} Euclidean, and $\lambda_0 = \frac{1}{2}\sum_j (x_j\,dy_j - y_j\,dx_j)$ is

the standard primitive. Here $\omega_0^n/n! = dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n$ is Lebesgue measure with the complex orientation, which differs from that of $dx_1 \wedge \cdots \wedge dx_n \wedge dy_1 \wedge \cdots \wedge dy_n$ by $(-1)^{n(n-1)/2}$. Following the preface, balls are labelled by capacity: $B^{2n}(a) = \{\pi|z|^2 < a\}$, so the open ball of radius r is $B^{2n}(\pi r^2)$ and has volume $\pi^n r^{2n}/n!$.

2. *Surfaces.* On a surface a 2-form is symplectic iff it is an area form. By Moser's theorem (Theorem 2.21) a closed connected symplectic surface is determined up to symplectomorphism by its genus and its total area.

3. *Products* $M_1 \times M_2$ and M^- as in Definition 2.1.

Definition 2.3 (Cotangent bundles)**.** Let Q be an n-manifold and $\pi\colon T^*Q \to Q$ the projection. The *canonical* 1*-form* is $(\lambda_{\mathrm{can}})_{(q,p)}(v) := p\big(d\pi_{(q,p)}v\big)$ for $v \in T_{(q,p)}T^*Q$, and $\omega_{\mathrm{can}} := d\lambda_{\mathrm{can}}$. In coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$ induced by coordinates on Q, $\lambda_{\mathrm{can}} = \sum_j p_j\, dq_j$ and $\omega_{\mathrm{can}} = \sum_j dp_j \wedge dq_j$. Viewing a 1-form σ on Q as a section $s_\sigma\colon Q \to T^*Q$, λ_{can} is the unique 1-form with $s_\sigma^*\lambda_{\mathrm{can}} = \sigma$ for all σ.

The map $(q,p) \mapsto (x,y) = (p,q)$ is a symplectomorphism $(T^*\mathbb{R}^n, \omega_{\mathrm{can}}) \to (\mathbb{R}^{2n}, \omega_0)$; so is $(x,y) = (q,-p)$, which sends the zero section to $\mathbb{R}^n \subset \mathbb{C}^n$. McDuff–Salamon use $\omega_{\mathrm{can}} = -d\lambda_{\mathrm{can}}$; since $(q,p) \mapsto (q,-p)$ intertwines the two forms and fixes the zero section, nothing in Sections 2.3 and 2.4 depends on the choice.

Definition 2.4 (Kähler manifolds)**.** A *Kähler manifold* is a triple (M, J, ω) with J an integrable almost complex structure, ω a symplectic form, and J ω-compatible: $g_J(v,w) = \omega(v, Jw)$ is a Riemannian metric, equivalently $\omega(v,w) = g_J(Jv, w)$. In local holomorphic coordinates $\omega = \frac{i}{2}\sum_{j,k} h_{j\bar{k}}\, dz_j \wedge d\bar{z}_k$ with $(h_{j\bar{k}})$ positive definite Hermitian, and locally $\omega = \frac{i}{2}\partial\bar{\partial}\varphi$ for a strictly plurisubharmonic φ; for instance $\omega_0 = \frac{i}{2}\partial\bar{\partial}|z|^2$.

So Kähler manifolds are symplectic by definition, and complex submanifolds of Kähler manifolds are Kähler. The normalization of $\mathbb{CP}^n$ matters for capacities and embedding theorems (Chapter 6), so I fix it once.

Convention 2.5 (Fubini–Study form)**.** ω_{FS} is the Kähler form on $\mathbb{CP}^n$ with $\mathrm{pr}^*\omega_{FS} = \frac{i}{2}\partial\bar{\partial}\log|z|^2$ on $\mathbb{C}^{n+1} \setminus \{0\}$, pr the quotient map. This is π times the integral form $\frac{i}{2\pi}\partial\bar{\partial}\log|z|^2$ common in algebraic geometry.

Proposition 2.6 (Normalization of $\mathbb{CP}^n$)**.** *Let $h \in H^2(\mathbb{CP}^n; \mathbb{Z})$ be the generator with $\langle h, [\mathbb{CP}^1]\rangle = 1$.*

1. *Every line has area π; $[\omega_{FS}] = \pi h$; $c_1(T\mathbb{CP}^n) = (n+1)h$, so $[\omega_{FS}] = \frac{\pi}{n+1}c_1$.*
2. *$\mathrm{pr}^*\omega_{FS}|_{S^{2n+1}} = \omega_0|_{S^{2n+1}}$ on the unit sphere $S^{2n+1} \subset \mathbb{C}^{n+1}$.*
3. *The map $u \mapsto [\sqrt{1-|u|^2} : u_1 : \cdots : u_n]$ is a symplectomorphism from the open ball of radius 1, $(B^{2n}(\pi), \omega_0)$, onto $\mathbb{CP}^n \setminus \mathbb{CP}^{n-1}$, where $\mathbb{CP}^{n-1} = \{z_0 = 0\}$. In particular $\mathrm{vol}(\mathbb{CP}^n, \omega_{FS}) = \pi^n/n!$.*

Proof idea. On TS^{2n+1} one has $\bar{\partial}|z|^2 = -\partial|z|^2$, which kills the second term of $\partial\bar{\partial}\log|z|^2 = \partial\bar{\partial}|z|^2/|z|^2 - \partial|z|^2 \wedge \bar{\partial}|z|^2/|z|^4$; this gives (2). The map in (3) is pr composed with the section $u \mapsto (\sqrt{1-|u|^2}, u)$ of $S^{2n+1} \to \mathbb{CP}^n$, whose image has real z_0-coordinate, so by (2) it pulls ω_{FS} back to ω_0. For $n = 1$ this gives area π; $c_1 = (n+1)h$ is the Euler sequence.

Smooth projective varieties $X \subset \mathbb{CP}^N$ with $\omega_{FS}|_X$ are therefore symplectic, and for most of the subject's history they were the only closed examples anyone could compute with.

Example 2.7 (Coadjoint orbits)**.** Let G be a Lie group and $\mathcal{O} \subset \mathfrak{g}^*$ an orbit of the coadjoint action $\langle \mathrm{Ad}^*_g \eta, \xi\rangle = \langle \eta, \mathrm{Ad}_{g^{-1}}\xi\rangle$. Its tangent space at η consists of the vectors $\mathrm{ad}^*_\xi \eta := \frac{d}{dt}|_{t=0}\mathrm{Ad}^*_{\exp t\xi}\eta = -\eta \circ \mathrm{ad}_\xi$, and the *Kirillov–Kostant–Souriau form*

$$\omega^{KKS}_\eta(\mathrm{ad}^*_\xi \eta, \mathrm{ad}^*_\zeta \eta) := -\langle \eta, [\xi, \zeta]\rangle$$

is well defined, closed, nondegenerate and G-invariant (Kirillov, Kostant, Souriau, 1960s–1970). The sign is the one for which, with $\iota_{X_H}\omega = -dH$, the inclusion $\mathcal{O} \hookrightarrow \mathfrak{g}^*$ is a moment map (Example 2.41); with McDuff–Salamon's sign it is $+\langle \eta, [\xi, \zeta]\rangle$. For $G = \mathrm{U}(n+1)$ the orbits are flag manifolds, including $\mathbb{CP}^n$ and Grassmannians.

Example 2.8 (Kodaira–Thurston manifold)**.** Let H be the Heisenberg group, $\mathbb{R}^3$ with product $(a,b,c)\cdot(x,y,z) = (a+x, b+y, c+z+ay)$, so that dx, dy and $dz - x\,dy$ are left invariant, and let $\Gamma = H_{\mathbb{Z}} \times \mathbb{Z}$, $H_{\mathbb{Z}} = \mathbb{Z}^3 \subset H$. Then $KT := \Gamma\backslash(H \times \mathbb{R})$ is a closed 4-manifold (a T^2-bundle over T^2) and the left-invariant form $\omega = dx \wedge dt + dy \wedge (dz - x\,dy)$ descends to a symplectic form. Since $\Gamma^{\mathrm{ab}} \cong \mathbb{Z}^3$, $b_1(KT) = 3$ is odd, so KT admits no Kähler structure. Thurston [Thu76] gave this as the first closed symplectic non-Kähler manifold; Kodaira [Kod64] had already found it as a complex surface, so it is complex and symplectic but not Kähler.

Remark 2.9 (Kähler versus symplectic). Hodge theory forces a closed Kähler manifold to have even odd-degree Betti numbers, to satisfy hard Lefschetz ($[\omega]^k\cup\colon H^{n-k}(M;\mathbb{R}) \to H^{n+k}(M;\mathbb{R})$ is an isomorphism), to be formal (Deligne–Griffiths–Morgan–Sullivan 1975), and to have a fundamental group in the restricted class of Kähler groups. Symplectic manifolds satisfy none of these in general: KT fails the first two; McDuff [McD84] blew up $\mathbb{CP}^5$ along an embedding of KT to get a simply connected example with b_3 odd; and every finitely presented group is the fundamental group of a closed symplectic 4-manifold (Gompf [Gom95]; Chapter 3). My advice is to think of Kähler geometry as a source of examples and of integrable J as a computational convenience, never as a guide to what is true.

2.2 Hamiltonian dynamics and the groups Symp and Ham

Convention 2.10 (Hamiltonian vector fields and Poisson brackets)**.** For $H \in C^\infty(M)$, X_H is defined by $\iota_{X_H}\omega = -dH$, i.e. $dH = \omega(\cdot, X_H)$. On $\mathbb{R}^{2n}$, $\dot{x}_j = -\partial H/\partial y_j$, $\dot{y}_j = \partial H/\partial x_j$; for every ω-compatible J, $X_H = J\nabla_{g_J} H$. On T^*Q with Definition 2.3, $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$. Example: $H = \frac{1}{2}|z|^2$ generates $\phi^t(z) = e^{it}z$. The Poisson bracket is $\{F,G\} := \omega(X_F, X_G) = dG(X_F) = X_F(G)$; on $\mathbb{R}^{2n}$, $\{F,G\} = \sum_j (\partial_{x_j}F\,\partial_{y_j}G - \partial_{y_j}F\,\partial_{x_j}G)$, so on T^*Q we get $\{p_j, q_k\} = \delta_{jk}$, the negative of the physicists' bracket. McDuff–Salamon use $\iota_{X_H}\omega = +dH$; their X_H is our X_{-H}, and moment maps, Calabi invariants and flux-type formulas change sign accordingly.

Proposition 2.11 (Basic identities)**.** *Let X be a vector field on (M,ω) and $F, G, H \in C^\infty(M)$.*

1. *(Cartan) $\mathcal{L}_X\omega = d\iota_X\omega$. Hence the flow of X preserves ω iff $\iota_X\omega$ is closed (X* symplectic*), and X is Hamiltonian iff $\iota_X\omega$ is exact; $X \mapsto [\iota_X\omega]$ identifies symplectic modulo Hamiltonian vector fields with $H^1(M;\mathbb{R})$.*
2. *H is constant along the flow of X_H, and $\frac{d}{dt}(G\circ\phi^t_H) = \{H,G\}\circ\phi^t_H$.*
3. *$\{\cdot,\cdot\}$ is antisymmetric, satisfies the Leibniz rule $\{F,GH\} = \{F,G\}H + G\{F,H\}$ and the Jacobi identity $\{F,\{G,H\}\} + \{G,\{H,F\}\} + \{H,\{F,G\}\} = 0$, and $X_{\{F,G\}} = [X_F, X_G]$, where $[X,Y]f = X(Yf) - Y(Xf)$.*

The flat torus T^{2n} with the translations ∂_{x_j} is the basic example of symplectic, non-Hamiltonian flows: $\iota_{\partial_{x_1}}\omega_0 = dy_1$ is closed and not exact.

Definition 2.12 (Hamiltonian isotopies and Ham)**.** A *time-dependent Hamiltonian* is a smooth $H\colon [0,1]\times M \to \mathbb{R}$, $H_t = H(t,\cdot)$, compactly supported if M is not closed. Its flow ϕ_H^t is defined by $\partial_t \phi_H^t = X_{H_t} \circ \phi_H^t$, $\phi_H^0 = \mathrm{id}$, and $\{\phi_H^t\}$ is a *Hamiltonian isotopy*. $\mathrm{Ham}(M,\omega) \coloneqq \{\phi_H^1\}$. $\mathrm{Symp}(M,\omega)$ carries the C^∞ topology and $\mathrm{Symp}_0(M,\omega)$ is its identity component; for open M one usually works with the group $\mathrm{Symp}_c(M,\omega)$ of compactly supported symplectomorphisms and the direct-limit topology. On closed M, H is *normalized* if $\int_M H_t\,\omega^n = 0$ for all t.

Since $\phi_H^t\phi_K^t = \phi_{H\#K}^t$ with $(H\#K)_t = H_t + K_t\circ(\phi_H^t)^{-1}$, $(\phi_H^t)^{-1} = \phi_{\bar H}^t$ with $\bar H_t = -H_t\circ\phi_H^t$, and $\psi\phi_H^t\psi^{-1} = \phi_{H\circ\psi^{-1}}^t$ for $\psi \in \mathrm{Symp}$, $\mathrm{Ham}(M,\omega)$ is a path-connected normal subgroup of $\mathrm{Symp}(M,\omega)$ contained in $\mathrm{Symp}_0(M,\omega)$. The universal cover $\widetilde{\mathrm{Ham}}$ consists of homotopy classes of Hamiltonian isotopies rel endpoints; it is the natural domain of Floer-theoretic invariants (Chapter 7).

Definition 2.13 (Flux)**.** Let M be closed. For a smooth path $\{\psi_t\}_{t\in[0,1]}$ in $\mathrm{Symp}_0(M,\omega)$ with $\psi_0 = \mathrm{id}$, let X_t be defined by $\partial_t\psi_t = X_t\circ\psi_t$. Then

$$\mathrm{Flux}(\{\psi_t\}) \coloneqq \int_0^1 [\iota_{X_t}\omega]\,dt \in H^1(M;\mathbb{R}).$$

The *flux group* is $\Gamma_\omega \coloneqq \mathrm{Flux}(\pi_1(\mathrm{Symp}_0(M,\omega),\mathrm{id})) \subset H^1(M;\mathbb{R})$.

Proposition 2.14 (Properties of flux)**.** *Let M be closed.*

1. *For every smooth loop $\gamma\colon S^1\to M$, $\langle \mathrm{Flux}(\{\psi_t\}),[\gamma]\rangle = \int_{[0,1]\times S^1}\Psi^*\omega$, where $\Psi(t,s)=\psi_t(\gamma(s))$ and $[0,1]\times S^1$ is oriented by $dt\wedge ds$: flux is the area swept by 1-cycles.*
2. $\mathrm{Flux}(\{\psi_t\})$ *depends only on the homotopy class of $\{\psi_t\}$ with fixed endpoints, and defines a surjective homomorphism* $\mathrm{Flux}\colon \widetilde{\mathrm{Symp}}_0(M,\omega)\to H^1(M;\mathbb{R})$ *vanishing on classes of Hamiltonian isotopies. Γ_ω is a countable subgroup.*

Theorem 2.15 (Banyaga 1978)**.** *Let (M,ω) be a closed connected symplectic manifold.*

1. *A path $\{\psi_t\}$ in Symp_0 from id with $\mathrm{Flux}(\{\psi_t\})=0$ is homotopic with fixed endpoints to a Hamiltonian isotopy. Hence $\mathrm{Ham}(M,\omega)$ is the image in $\mathrm{Symp}_0(M,\omega)$ of $\ker(\mathrm{Flux}\colon \widetilde{\mathrm{Symp}}_0\to H^1(M;\mathbb{R}))$, and Flux induces an isomorphism $\mathrm{Symp}_0/\mathrm{Ham}\cong H^1(M;\mathbb{R})/\Gamma_\omega$.*
2. *$\mathrm{Ham}(M,\omega)$ is a simple group, and $\mathrm{Ham}(M,\omega) = [\mathrm{Symp}_0(M,\omega),\mathrm{Symp}_0(M,\omega)]$.*
3. *$\mathrm{Ham}(M,\omega) = \mathrm{Symp}_0(M,\omega)$ if and only if $H^1(M;\mathbb{R})=0$.*

Proof idea. (1) Compose $\{\psi_t\}$ with a contractible loop of symplectic diffeomorphisms, built from flows of symplectic vector fields representing a basis of $H^1(M;\mathbb{R})$, so that the flux of the new path restricted to $[0,t]$ vanishes for every t; then each $\iota_{X_t}\omega$ is exact. (2) combines fragmentation (factoring into maps supported in Darboux balls), a perfectness theorem for compactly supported Hamiltonian diffeomorphisms of $\mathbb{R}^{2n}$ with zero Calabi invariant, and Epstein's simplicity criterion [Ban78; Ban97]. (3) follows from (1): Γ_ω is countable.

Simplicity means there is no homomorphism $\mathrm{Ham}(M,\omega)\to\mathbb{R}$ on closed M; the substitute is the Calabi *quasimorphism* of Entov–Polterovich [EP03], built from Floer theory (Chapter 7).

Theorem 2.16 (Flux conjecture; Ono 2006)**.** *For every closed symplectic manifold (M,ω) the flux group Γ_ω is discrete in $H^1(M;\mathbb{R})$. Equivalently,* $\mathrm{Ham}(M,\omega)$ *is closed in* $\mathrm{Symp}_0(M,\omega)$ *in the C^1 topology.*

Lalonde–McDuff–Polterovich [LMP98] proved it for spherically rational (M,ω) and when c_1 vanishes on $\pi_2(M)$ or N_M is large; Ono [Ono06] proved it in general using Floer–Novikov cohomology for symplectic isotopies, the nonzero-flux analogue of Hamiltonian Floer theory. The C^0 version—is $\mathrm{Ham}(M,\omega)$ closed in $\mathrm{Symp}_0(M,\omega)$ in the C^0 topology?—is open (Problem 2.51).

Definition 2.17 (Calabi homomorphism)**.** Let (M^{2n},ω) be connected and non-compact with $\omega = d\lambda$ exact, and let $\mathrm{Ham}_c(M,\omega)$ be the group of time-one maps of compactly supported Hamiltonians $H\colon [0,1]\times M\to\mathbb{R}$. Set

$$\mathrm{Cal}(\phi_H^1) := \int_0^1 \int_M H_t\,\omega^n\,dt.$$

Theorem 2.18 (Calabi 1970; Banyaga 1978)**.** *In the setting of Definition 2.17,* $\mathrm{Cal}(\phi)$ *is independent of the compactly supported Hamiltonian generating ϕ,* $\mathrm{Cal}\colon \mathrm{Ham}_c(M,\omega)\to\mathbb{R}$ *is a surjective homomorphism, and* $\ker \mathrm{Cal}$ *is a simple group.*

Calabi [Cal70] introduced the invariant; Banyaga [Ban78] proved simplicity. Normalizations vary (ω^n versus $\omega^n/n!$, and the sign flips under McDuff–Salamon's convention). Without exactness Cal is defined on $\widetilde{\mathrm{Ham}}_c$ and descends to Ham_c only modulo a countable subgroup [MS17, Ch. 10]. The Calabi homomorphism on the disk is the protagonist of the simplicity conjecture (Chapter 7).

Remark 2.19 (Hamiltonian diffeomorphisms as exact Lagrangians). For $\psi\in\mathrm{Symp}(M,\omega)$ the graph $\Gamma_\psi=\{(x,\psi(x))\}$ is Lagrangian in $M\times M^-$, and $\mathrm{Fix}(\psi)\cong\Gamma_\psi\cap\Delta$. If $\psi=\phi_H^1$, then Γ_ψ is Hamiltonian isotopic to Δ via $\mathrm{id}\times\phi_H^t$. By Weinstein's theorem (Theorem 2.26) a neighborhood of Δ is a neighborhood of the zero section in $T^*\Delta\cong T^*M$, so a ψ that is C^1-close to id has $\Gamma_\psi=\Gamma_{\sigma_\psi}$ for a unique small closed 1-form σ_ψ. This gives charts for Symp modeled on closed 1-forms (so Symp is locally contractible), and for such ψ one has $\psi\in\mathrm{Ham}$ iff $[\sigma_\psi]\in\Gamma_\omega$. Thus Theorem 2.16 says precisely that near the identity, Hamiltonian diffeomorphisms are those whose graphs are graphs of *exact* forms. This is the bridge from Hamiltonian to Lagrangian Floer theory: with suitable conventions $\mathrm{HF}(\psi)\cong\mathrm{HF}(\Delta,\Gamma_\psi)$ (Chapter 8).

2.3 Local theory: Moser's method and neighborhood theorems

Theorem 2.20 (Moser stability)**.** *Let M be closed and $\{\omega_t\}_{t\in[0,1]}$ a smooth family of symplectic forms with $[\omega_t]\in H^2(M;\mathbb{R})$ independent of t. Then there is an isotopy $\{\psi_t\}$ of M with $\psi_0=\mathrm{id}$ and $\psi_t^*\omega_t=\omega_0$ for all t.*

Proof idea. Write $\partial_t\omega_t = d\sigma_t$ with σ_t smooth in t and define X_t by $\iota_{X_t}\omega_t=-\sigma_t$. For the flow ψ_t of X_t, $\partial_t(\psi_t^*\omega_t)=\psi_t^*(d\iota_{X_t}\omega_t+d\sigma_t)=0$. Every result in this section is this argument plus a way of producing σ_t.

The same argument for volume forms is Moser's original theorem [Mos65]:

Theorem 2.21 (Moser 1965)**.** *Let M be a closed connected oriented manifold and σ_0,σ_1 positive volume forms with $\int_M\sigma_0=\int_M\sigma_1$. Then there is $\psi\in\mathrm{Diff}_0(M)$ with $\psi^*\sigma_1=\sigma_0$. Consequently two closed connected symplectic surfaces are symplectomorphic iff they have the same genus and the same total area.*

Lemma 2.22 (Relative Moser)**.** *Let $N \subset M$ be a compact submanifold and ω_0, ω_1 closed 2-forms on M which, at every $x \in N$, are equal and nondegenerate on T_xM. Then there are open neighborhoods U_0, U_1 of N and a diffeomorphism $\psi\colon U_0 \to U_1$ with $\psi|_N = \mathrm{id}$ and $\psi^*\omega_1 = \omega_0$.*

Theorem 2.23 (Darboux 1882)**.** *Every point of a symplectic manifold (M^{2n}, ω) has a neighborhood U and a diffeomorphism $\phi\colon U \to \phi(U) \subset \mathbb{R}^{2n}$ with $\phi^*\omega_0 = \omega$.*

Darboux [Dar82] proved the normal form for Pfaffian systems; in symplectic language, apply Lemma 2.22 with N a point. Next, the submanifolds.

Definition 2.24 (Symplectic, isotropic, coisotropic, Lagrangian)**.** Let $N \subset (M, \omega)$ be an embedded submanifold and $T_xN^\omega := \{v \in T_xM : \omega(v, w) = 0\ \forall w \in T_xN\}$, of dimension $2n - \dim N$. N is *symplectic* if $T_xN \cap T_xN^\omega = 0$, *isotropic* if $T_xN \subset T_xN^\omega$, *coisotropic* if $T_xN^\omega \subset T_xN$, for all $x \in N$; it is *Lagrangian* if it is isotropic and $\dim N = n$, i.e. $\omega|_N = 0$ and $\dim N = n$; immersed Lagrangians are always called immersed. For N symplectic, $TN^\omega \to N$ is a symplectic vector bundle, the *symplectic normal bundle*; for N isotropic, TN^ω/TN is a symplectic vector bundle, and $TM|_N/TN^\omega \cong T^*N$ via ω.

Proposition 2.25 (Characteristic foliation)**.** *If $C \subset M$ is coisotropic of codimension k, then $TC^\omega = \ker(\omega|_C)$ is an integrable distribution of rank k with isotropic leaves. For a hypersurface Y ($k = 1$) it is the* characteristic line field *$\mathcal{L}_Y$. If $Y = H^{-1}(c)$ for a regular value c of $H \in C^\infty(M)$, then $\mathcal{L}_Y = \mathbb{R}X_H|_Y$; hence the closed characteristics of Y are exactly the periodic orbits of X_H on Y up to parametrization, and they depend only on Y, not on H.*

This is what makes the existence of periodic orbits on a given energy surface a question about the hypersurface (Chapter 9). A compact hypersurface Y is of *contact type* if a Liouville vector field Z, $\mathcal{L}_Z\omega = \omega$, is defined near Y and transverse to it; then $\alpha = \iota_Z\omega|_Y$ is a contact form with $d\alpha = \omega|_Y$, and its Reeb field spans $\mathcal{L}_Y$. Contact forms are developed in Chapter 4 and Liouville domains in Chapter 3.

Theorem 2.26 (Lagrangian neighborhood theorem; Weinstein 1971)**.** *Let $L \subset (M, \omega)$ be a compact Lagrangian. There are a neighborhood U of the zero section 0_L in T^*L, a neighborhood V of L in M, and a diffeomorphism $\phi\colon U \to V$ with $\phi^*\omega = \omega_{\mathrm{can}} = d\lambda_{\mathrm{can}}$ and $\phi(0_L(q)) = q$ for $q \in L$.*

Proof idea. For $J \in \mathcal{J}(M, \omega)$, $J(TL)$ is a complement of TL, and g_J identifies $T^*L \cong TL \cong J(TL)$; composing with an exponential map gives a diffeomorphism matching ω_{can} and ω along 0_L. Apply Lemma 2.22.

Theorem 2.27 (Symplectic, isotropic and coisotropic neighborhood theorems)**.** *For $j = 0, 1$ let N_j be a compact submanifold of (M_j, ω_j).*

1. *If the N_j are symplectic and $\phi\colon (N_0, \omega_0|_{N_0}) \to (N_1, \omega_1|_{N_1})$ is a symplectomorphism covered by an isomorphism of symplectic normal bundles $TN_0^{\omega_0} \to TN_1^{\omega_1}$, then ϕ extends to a symplectomorphism between neighborhoods of N_0 and N_1.*
2. *(Weinstein [Wei81a]) If the N_j are isotropic and $\phi\colon N_0 \to N_1$ is a diffeomorphism covered by an isomorphism of symplectic vector bundles $TN_0^{\omega_0}/TN_0 \to TN_1^{\omega_1}/TN_1$, then ϕ extends to a symplectomorphism between neighborhoods.*
3. *(Gotay [Got82]) If the N_j are coisotropic and $\phi\colon N_0 \to N_1$ is a diffeomorphism with $\phi^*(\omega_1|_{N_1}) = \omega_0|_{N_0}$, then ϕ extends to a symplectomorphism between neighborhoods. Moreover, every manifold with a closed 2-form of constant rank embeds coisotropically in a symplectic manifold.*

Part (1) goes back to Weinstein's 1977 lectures [Wei77]; all proofs construct a diffeomorphism with the right linear data along N and then apply Lemma 2.22. In particular, a neighborhood of a compact hypersurface is determined by $\omega|_Y$, and a neighborhood of a closed connected symplectic surface in a 4-manifold by its genus, area and self-intersection number.

Remark 2.28 (No local invariants). The contrast with Riemannian geometry is total: a neighborhood of a point, of a compact symplectic, isotropic, coisotropic or Lagrangian submanifold is determined by bundle data. The soft global invariants of a closed (M, ω) are the class $[\omega]$, the *symplectic cone* $C_M = \{[\omega] : \omega \text{ symplectic}\} \subset H^2(M; \mathbb{R})$, the Chern classes of Section 2.5, and the deformation class of ω. There are three equivalence relations on symplectic forms: symplectomorphic, *isotopic* (joined by a path of cohomologous symplectic forms), and *deformation equivalent* (joined by a path of symplectic forms, up to diffeomorphism). Theorem 2.20 shows that isotopic forms are symplectomorphic by an isotopy; McDuff [McD87] found cohomologous, deformation-equivalent forms on $S^2 \times S^2 \times T^2$ that are not isotopic (Chapter 3). Everything finer—Gromov width, Gromov–Witten invariants, Floer theory—comes from holomorphic curves (Chapters 5 and 6).

2.4 Lagrangian submanifolds

Weinstein's *symplectic creed*, "everything is a Lagrangian submanifold" [Wei81b], is the best one-line summary of how the subject thinks: symplectomorphisms, generating functions, reductions and correspondences are all Lagrangians somewhere, and Floer theory is built on that.

Definition 2.29 (Exact Lagrangians)**.** A Lagrangian L in an exact symplectic manifold $(M, d\lambda)$ is *exact* if $\lambda|_L$ is exact. Two Lagrangians L_0, L_1 are *Hamiltonian isotopic* if $L_1 = \phi(L_0)$ for some $\phi \in \mathrm{Ham}(M, \omega)$ (compactly supported if M is open).

Proposition 2.30 (Graphs of 1-forms)**.** *Let Q be closed, $\sigma \in \Omega^1(Q)$ and $\Gamma_\sigma = s_\sigma(Q) \subset (T^*Q, \omega_{\mathrm{can}})$.*

1. *Γ_σ is Lagrangian iff $d\sigma = 0$.*
2. *If $\sigma = df$, then $\Gamma_{df} = \phi(0_Q)$, where ϕ is the time-one map of a compactly supported Hamiltonian equal to $-f \circ \pi$ near $\{(q, t\, df_q) : q \in Q,\ t \in [0,1]\}$.*
3. *Compactly supported Hamiltonian diffeomorphisms of T^*Q preserve exactness of Lagrangians. Consequently, for closed σ, Γ_σ is Hamiltonian isotopic to 0_Q iff $[\sigma] = 0$ in $H^1(Q; \mathbb{R})$; if $[\sigma] \neq 0$ it is Lagrangian isotopic to 0_Q (through $\Gamma_{t\sigma}$) but not Hamiltonian isotopic.*

Proof idea. (1) $s_\sigma^* \omega_{\mathrm{can}} = d\sigma$. (2) By Convention 2.10, $X_{-f\circ\pi}$ is the vertical field $(q,p) \mapsto df_q$. (3) If $\partial_t \phi_t = X_{H_t} \circ \phi_t$ then $\partial_t \phi_t^* \lambda = \phi_t^* d(\lambda(X_{H_t}) - H_t)$, and $s_\sigma^* \lambda_{\mathrm{can}} = \sigma$.

Example 2.31 (Lagrangians)**.**

1. The zero section 0_Q and the fibers T_q^*Q; more generally the *conormal bundle* $N^*K = \{(q,p) : q \in K,\ p|_{T_qK} = 0\}$ of a submanifold $K \subset Q$, on which λ_{can} vanishes identically.
2. *Fixed points of antisymplectic involutions.* If $\tau^2 = \mathrm{id}$ and $\tau^*\omega = -\omega$, every component of $\mathrm{Fix}(\tau)$ is a Lagrangian (the ± 1-eigenspaces of $d\tau$ at a fixed point are isotropic and complementary). Complex conjugation gives $\mathbb{R}^n \subset \mathbb{C}^n$ and $\mathbb{RP}^n \subset \mathbb{CP}^n$, and the real locus of any smooth projective variety defined over $\mathbb{R}$.
3. *Tori.* The product tori $T(r_1, \ldots, r_n) = \{|z_j| = r_j\} \subset \mathbb{C}^n$, $r_j > 0$, and the *Clifford torus* $\{|z_0| = \cdots = |z_n|\} \subset \mathbb{CP}^n$; both are orbits of Hamiltonian torus actions (Section 2.6).

4. *Graphs.* For a diffeomorphism $\psi\colon M_1 \to M_2$, the graph $\{(x, \psi(x))\}$ is Lagrangian in $M_1 \times M_2^-$ iff ψ is a symplectomorphism; in particular the diagonal $\Delta \subset M \times M^-$.

Theorem 2.32 (Arnold–Liouville)**.** *Let (M^{2n}, ω) be symplectic, $F = (F_1, \dots, F_n)\colon M \to \mathbb{R}^n$ smooth with $\{F_i, F_j\} = 0$ for all i, j, $c \in \mathbb{R}^n$ a regular value of F, and L a compact connected component of $F^{-1}(c)$. Then:*

1. *L is a Lagrangian submanifold; the flows of $X_{F_1}, \dots, X_{F_n}$ commute and define a transitive action of $\mathbb{R}^n$ on L whose stabilizer is a lattice, so L is diffeomorphic to T^n and each X_{F_i} is a linear flow on L.*
2. *(Action–angle coordinates) There are a neighborhood U of L that is a union of components of fibers of F, an open set $D \subset \mathbb{R}^n$, and a diffeomorphism $\Phi\colon U \to D \times T^n$, $T^n = \mathbb{R}^n/2\pi\mathbb{Z}^n$, such that $\omega = \Phi^*\big(\sum_j da_j \wedge d\theta_j\big)$ and $F \circ \Phi^{-1}(a, \theta) = h(a)$ for a diffeomorphism h of D onto an open subset of $\mathbb{R}^n$.*

In these coordinates (cotangent coordinates on T^*T^n with $p = a$, $q = \theta$) the flow of $X_{G\circ a}$ is $\dot a = 0$, $\dot\theta = \partial G/\partial a$, and the actions are $a_j = \frac{1}{2\pi}\int_{\gamma_j} \lambda$ for a primitive λ of ω on U and a basis γ_j of $H_1(L;\mathbb{Z})$, up to constants. The theorem goes back to Liouville (1855) for integration by quadratures, Mineur (1936) for action–angle variables and Arnold (1963) for the invariant formulation; read it in [Arn78]. The global theory of such *Lagrangian torus fibrations* has two obstructions, monodromy and a Chern class, found by Duistermaat [Dui80], and the base carries an integral affine structure. Moment maps of toric manifolds (Theorem 2.48) are the singular fibrations with the simplest singularities; almost toric fibrations [LS10] and SYZ fibrations are treated in Chapter 8.

2.5 Almost complex structures and Chern classes

Proposition 2.33 (Compatible and tame structures)**.** *Let $\mathcal{J}(M, \omega)$ (resp. $\mathcal{J}_\tau(M, \omega)$) be the space of smooth ω-compatible (resp. ω-tame) almost complex structures, with the C^∞ topology. Both are nonempty and contractible, and $\mathcal{J} \subset \mathcal{J}_\tau$. For $J \in \mathcal{J}(M, \omega)$, $g_J = \omega(\cdot, J\cdot)$ is a J-invariant metric and $h_J = g_J - i\omega$ is a Hermitian metric on (TM, J), complex linear in the first variable (Chapter 1).*

Proof idea. $\mathcal{J}$ is the space of sections of a bundle with fiber $\mathrm{Sp}(2n)/\mathrm{U}(n)$, which is contractible (Chapter 1); polar decomposition gives a canonical retraction of the space of metrics onto $\mathcal{J}$. For tame structures one shows that the fibers are contractible, an argument going back to Gromov [Gro85]; see [MS17].

Definition 2.34 (Chern classes, minimal Chern number)**.** $c_k(M, \omega) \coloneqq c_k(TM, J) \in H^{2k}(M;\mathbb{Z})$ for any $J \in \mathcal{J}_\tau(M, \omega)$; by Proposition 2.33 this is independent of J and invariant under deformation of ω. The *canonical class* is $K_M = -c_1(M, \omega)$. For $A \in \pi_2(M)$ write $c_1(A)$ and $\omega(A)$ for the pairings with the Hurewicz image of A. The *minimal Chern number* $N_M \in \{1, 2, \dots\} \cup \{\infty\}$ is the positive generator of $c_1(\pi_2(M)) \subset \mathbb{Z}$, with $N_M = \infty$ if c_1 vanishes on $\pi_2(M)$, as in Chapter 1 (McDuff–Salamon write $N_M = 0$ in that case).

Definition 2.35 (Positivity conditions)**.** A symplectic manifold (M^{2n}, ω) is

1. *symplectically aspherical* if $\omega(A) = 0$ for all $A \in \pi_2(M)$;
2. *monotone* if there is $\kappa > 0$ with $\omega(A) = \kappa\, c_1(A)$ for all $A \in \pi_2(M)$, and *negatively monotone* if this holds with $\kappa < 0$;

3. *symplectic Calabi–Yau* if $c_1(M,\omega) = 0$ in $H^2(M;\mathbb{Z})$;
4. *semipositive* [MS12a, Def. 6.4.1] if for every $A \in \pi_2(M)$, $\omega(A) > 0$ and $c_1(A) \geq 3-n$ imply $c_1(A) \geq 0$.

Warning 2.36 (Variants). These names are not used uniformly. Many papers on Hofer geometry and the Conley conjecture, following Schwarz [Sch00], call (M,ω) symplectically aspherical only if *both* ω and c_1 vanish on $\pi_2(M)$; I always say so explicitly. With (2) as stated, a symplectically aspherical manifold with $c_1|_{\pi_2} = 0$ is (vacuously) monotone; some authors exclude it, and some require $[\omega] = \kappa c_1$ in $H^2(M;\mathbb{R})$, which is equivalent for simply connected M. "Calabi–Yau" is used for $c_1 = 0$ in $H^2(M;\mathbb{Z})$ (my default), in $H^2(M;\mathbb{R})$, or on $\pi_2(M)$; the last is what $\mathbb{Z}$-gradings of Hamiltonian Floer cohomology of contractible orbits need (Chapter 7), while $\mathbb{Z}$-graded Fukaya categories need $2c_1 = 0$ plus a choice of grading [Sei00] (Chapter 8). Hofer–Salamon [HS95] call semipositive manifolds weakly monotone.

Proposition 2.37. *(M^{2n},ω) is semipositive if $n \leq 3$, or if it is monotone, or if $N_M \geq n-2$ (in particular if c_1 vanishes on $\pi_2(M)$).*

Proof idea. If $n \leq 3$ then $c_1(A) \geq 3-n$ already gives $c_1(A) \geq 0$. If $c_1(A) < 0$ and $N_M \geq n-2$ then $c_1(A) \leq -N_M < 3-n$.

Example 2.38. $\mathbb{CP}^n$ is monotone with $\kappa = \pi/(n+1)$ and $N_{\mathbb{CP}^n} = n+1$ (Proposition 2.6). A smooth hypersurface $X_d \subset \mathbb{CP}^{n+1}$ of degree d with $\omega_{FS}|_{X_d}$ has $[\omega] = \pi h|_{X_d}$ and $c_1 = (n+2-d)h|_{X_d}$: it is monotone for $d < n+2$, Calabi–Yau for $d = n+2$, and negatively monotone for $d > n+2$. Manifolds with $\pi_2(M) = 0$ (tori, surfaces of positive genus, KT) and exact manifolds such as T^*Q are symplectically aspherical; $2c_1(T^*Q) = 0$, and $c_1(T^*Q) = 0$ if Q is orientable.

2.6 Symmetry: Hamiltonian group actions

Convention 2.39 (Fundamental vector fields; tori)**.** For a left action of a Lie group G on M and $\xi \in \mathfrak{g}$, $X_\xi(x) := \frac{d}{dt}|_{t=0} \exp(t\xi)\cdot x$; then $[X_\xi, X_\eta] = -X_{[\xi,\eta]}$. Tori are $T^k = \mathbb{R}^k/2\pi\mathbb{Z}^k$ with $\mathfrak{t} = \mathbb{R}^k$ and $\exp(\xi) = [\xi]$. Weights of torus representations lie in $\mathbb{Z}^k \subset (\mathbb{R}^k)^* = \mathfrak{t}^*$: $\exp(\xi)$ acts on the weight-w line $\mathbb{C}_w$ by $e^{i\langle w,\xi\rangle}$.

Definition 2.40 (Hamiltonian actions and moment maps)**.** A smooth action of G on (M,ω) by symplectomorphisms is *Hamiltonian* if there is a smooth *moment map* $\mu\colon M \to \mathfrak{g}^*$ such that

1. for every $\xi \in \mathfrak{g}$, $\mu^\xi := \langle \mu, \xi\rangle$ is a Hamiltonian for X_ξ in the sense of Convention 2.10: $\iota_{X_\xi}\omega = -d\mu^\xi$, i.e. $d\mu^\xi = \omega(\cdot, X_\xi)$;
2. μ is equivariant: $\mu(g\cdot x) = \mathrm{Ad}^*_g \mu(x)$.

Then $\{\mu^\xi, \mu^\eta\} = -\mu^{[\xi,\eta]}$ and $\omega(X_\xi, X_\eta) = -\langle \mu, [\xi,\eta]\rangle$; for connected G this infinitesimal identity is equivalent to (2). For a torus, μ is invariant and, if M is connected, unique up to adding a constant in $\mathfrak{t}^*$.

A symplectic action of a compact connected G is Hamiltonian if $H^1(M;\mathbb{R}) = 0$ or $\mathfrak{g}$ is semisimple: $\xi \mapsto [\iota_{X_\xi}\omega] \in H^1(M;\mathbb{R})$ vanishes on $[\mathfrak{g},\mathfrak{g}]$ by Proposition 2.11, and averaging over G makes any choice equivariant. Translations of T^{2n} are symplectic and not Hamiltonian.

Example 2.41 (Moment maps with our signs)**.** 1. *Linear actions.* For $\rho\colon G \to \mathrm{Sp}(V,\omega)$, $\mu^\xi(v) = \frac{1}{2}\omega(v, \rho_*(\xi)v)$. For $\mathrm{U}(n)$ on $\mathbb{C}^n$ this is $\mu^\xi(z) = \frac{1}{2}\omega_0(z,\xi z) = -\frac{i}{2}z^*\xi z$ for skew-Hermitian ξ. For T^n acting by $(e^{i\theta_1}z_1,\dots,e^{i\theta_n}z_n)$, $\mu(z) = \frac{1}{2}(|z_1|^2,\dots,|z_n|^2)$, with image the closed positive orthant; for the diagonal circle, $\mu = \frac{1}{2}|z|^2$. (McDuff–Salamon's convention gives $-\frac{1}{2}|z|^2$.)

2. *Exact actions.* If $\omega = d\lambda$ and G preserves λ, then $\mu^\xi = \lambda(X_\xi)$. For the cotangent lift of an action on Q this is $\mu^\xi(q,p) = p(\xi_Q(q))$, ξ_Q the fundamental field on Q: linear momentum p for translations of $\mathbb{R}^3$ and angular momentum $q\times p$ for $\mathrm{SO}(3)$, with the physically correct signs.

3. *Coadjoint orbits.* With Example 2.7, the inclusion $\mathcal{O} \hookrightarrow \mathfrak{g}^*$ is an equivariant moment map for the coadjoint action.

Proposition 2.42 (Noether)**.** *Let G act in a Hamiltonian way with moment map μ, and let $H \in C^\infty(M)$ be G-invariant. Then every μ^ξ is constant along the flow of X_H. Conversely, if G is connected and every μ^ξ is constant along the flow of X_H, then H is G-invariant.*

Indeed $X_H(\mu^\xi) = \{H,\mu^\xi\} = -X_\xi(H)$. Noether's 1918 theorem is the origin of the moment map.

Theorem 2.43 (Marsden–Weinstein–Meyer reduction)**.** *Let a Lie group G act on (M,ω) in a Hamiltonian way with equivariant moment map μ, and suppose $\mu^{-1}(0) \neq \emptyset$ and G acts freely and properly on $\mu^{-1}(0)$ (properness is automatic if G is compact). Then 0 is a regular value of μ, $\mu^{-1}(0)$ is a coisotropic submanifold whose characteristic leaves are the G-orbits, $M/\!\!/G := \mu^{-1}(0)/G$ is a manifold of dimension $\dim M - 2\dim G$, and there is a unique symplectic form ω_{red} on it with $p^*\omega_{\mathrm{red}} = \iota^*\omega$, where $\iota\colon \mu^{-1}(0) \hookrightarrow M$ is the inclusion and $p\colon \mu^{-1}(0) \to M/\!\!/G$ the projection.*

Proof idea. $\langle d\mu_x(v),\xi\rangle = \omega(v, X_\xi(x))$, so $\operatorname{im} d\mu_x$ is the annihilator of the stabilizer algebra and $\ker d\mu_x = T_x(Gx)^\omega$; orbits in $\mu^{-1}(0)$ are isotropic because $\omega(X_\xi, X_\eta) = -\langle\mu,[\xi,\eta]\rangle$.

This is due to Marsden–Weinstein [MW74] and Meyer [Mey73]. Reduction at $\eta \in \mathfrak{g}^*$ is $\mu^{-1}(\eta)/G_\eta$, which is the reduction at 0 of $M \times \mathcal{O}_\eta^-$ ("shifting trick"); for a torus one simply replaces μ by $\mu - \eta$. Without freeness $\mu^{-1}(0)/G$ is a stratified symplectic space (Sjamaar–Lerman [SL91]). For closed M and compact connected G acting freely on $\mu^{-1}(0)$, Kirwan proved that $H^*_G(M;\mathbb{Q}) \to H^*(M/\!\!/G;\mathbb{Q})$ is surjective [Kir84b], via the equivariant perfection of $\|\mu\|^2$; equivariant cohomology is the natural language for everything below [AB84], but I only point to it.

Example 2.44 ($\mathbb{CP}^n$ as a reduced space)**.** The diagonal circle acting on $\mathbb{C}^{n+1}$ has moment map $\mu(z) = \frac{1}{2}|z|^2 - \frac{1}{2}$. Then $\mu^{-1}(0) = S^{2n+1}$, and by Proposition 2.6(2) the reduced space is $(\mathbb{CP}^n, \omega_{FS})$ with the normalization of Convention 2.5. The residual action of T^n on $[z_0 : z_1 : \cdots : z_n]$ by rotating $z_1,\dots,z_n$ has moment map $\mu([z]) = \frac{1}{2|z|^2}(|z_1|^2,\dots,|z_n|^2)$, with image the simplex $\Delta_n = \{x \in \mathbb{R}^n : x_j \geq 0,\ \sum_j x_j \leq \frac{1}{2}\}$; the Clifford torus is the fiber over its barycenter.

Theorem 2.45 (Convexity; Atiyah 1982, Guillemin–Sternberg 1982)**.** *Let (M,ω) be closed and connected, with a Hamiltonian action of a torus T and moment map $\mu\colon M \to \mathfrak{t}^*$. Then:*

1. *$\mu^{-1}(c)$ is connected for every $c \in \mu(M)$;*
2. *the fixed point set M^T has finitely many connected components $C_1,\dots,C_N$, μ is constant on each, $\mu(C_j) = \{c_j\}$, and $\mu(M)$ is the convex polytope* $\operatorname{conv}\{c_1,\dots,c_N\}$.

The proof [Ati82; GS82a] is Morse–Bott theory: every component μ^ξ is a Morse–Bott function with even indices, hence has connected level sets. For a compact connected nonabelian G, Kirwan [Kir84a] proved that $\mu(M) \cap \mathfrak{t}_+^*$ is a convex polytope, $\mathfrak{t}_+^*$ a closed positive Weyl chamber.

Theorem 2.46 (Duistermaat–Heckman 1982)**.** *Let (M^{2n}, ω) be closed and connected, with a Hamiltonian action of $T = T^k$ and moment map μ.*

1. *If the action is effective, the measure $\mu_*(\omega^n/n!)$ on $\mathfrak{t}^* = \mathbb{R}^k$ has a density with respect to Lebesgue measure which, on each connected component of the set of regular values of μ in $\mu(M)$, is a polynomial of degree at most $n - k$.*
2. *Let $U \subset \mathfrak{t}^*$ be a convex open set of regular values with T acting freely on $\mu^{-1}(U)$, $c_0 \in U$, $P = \mu^{-1}(c_0)$ and $B = P/T$. Under the identification of $\mu^{-1}(c)/T$ with B given by an equivariant trivialization $\mu^{-1}(U) \cong P \times U$ over U,*

$$[\omega_c] = [\omega_{c_0}] + \langle c - c_0, [F] \rangle \in H^2(B; \mathbb{R}) \qquad (c \in U),$$

   where $F \in \Omega^2(B; \mathfrak{t})$ is the curvature of any connection $\alpha \in \Omega^1(P; \mathfrak{t})$ normalized by $\alpha(X_\xi) = \xi$; $\frac{1}{2\pi}[F]$ is an integral class.
3. *(Localization) If M^T is finite and $\xi \in \mathfrak{t}$ satisfies $\langle w_j(p), \xi \rangle \neq 0$ for all $p \in M^T$ and all j, then*

$$\int_M e^{-\langle \mu, \xi \rangle} \frac{\omega^n}{n!} = (2\pi)^n \sum_{p \in M^T} \frac{e^{-\langle \mu(p), \xi \rangle}}{\prod_{j=1}^n \langle w_j(p), \xi \rangle},$$

   where $w_1(p), \dots, w_n(p) \in \mathbb{Z}^k$ are the weights of T on (T_pM, J_p) for any T-invariant $J \in \mathcal{J}(M, \omega)$.

The signs in (2) and (3) are those forced by Convention 2.10 and Convention 2.39; check on $\mathbb{CP}^1$ with Example 2.44, where $\mu_*(\omega) = 2\pi\, dx$ on $[0, \frac{1}{2}]$. Duistermaat–Heckman [DH82] proved (1)–(3); (3) is the first instance of the localization theorem in equivariant cohomology of Berline–Vergne [BV82] and Atiyah–Bott [AB84], and the template for virtual localization in Gromov–Witten theory (Chapter 5).

Definition 2.47 (Delzant polytopes; symplectic toric manifolds)**.** A *Delzant polytope* is a compact convex polytope $\Delta \subset \mathbb{R}^n$ such that each vertex v lies on exactly n edges, and these have the form $\{v + tu_j : 0 \le t \le \ell_j\}$, $j = 1, \dots, n$, with $u_1, \dots, u_n$ a $\mathbb{Z}$-basis of $\mathbb{Z}^n$. A *symplectic toric manifold* is a closed connected (M^{2n}, ω) with an effective Hamiltonian action of T^n and a chosen moment map μ.

Theorem 2.48 (Delzant 1988)**.** *The image $\mu(M)$ of a symplectic toric manifold is a Delzant polytope. Two symplectic toric manifolds (M, ω, μ), (M', ω', μ') with $\mu(M) = \mu'(M')$ are isomorphic, i.e. there is a T^n-equivariant symplectomorphism $\phi\colon M \to M'$ with $\mu' \circ \phi = \mu$. Every Delzant polytope is the image of a symplectic toric manifold.*

At a fixed point p the edges of $\mu(M)$ point along the weights $w_j(p)$, which form a $\mathbb{Z}$-basis by effectiveness; existence is reduction of $\mathbb{C}^d$, d the number of facets, by a subtorus of T^d [Del88]. With our normalization $\operatorname{vol}(M) = (2\pi)^n \operatorname{vol}_{\mathrm{Leb}}(\mu(M))$; for $\mathbb{CP}^n$ the polytope is Δ_n and the volume is $\pi^n/n!$, as it must be by Proposition 2.6. Every symplectic toric manifold is Kähler (it is a smooth projective toric variety), and its Lagrangian torus fibers are the laboratory for Lagrangian Floer theory (Chapter 8).

Proposition 2.49 (Symplectic cut; Lerman 1995)**.** *Let $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ act on (M, ω) with moment map μ, and let ϵ be a regular value such that S^1 acts freely on $\mu^{-1}(\epsilon)$. Let S^1 act on $M \times \mathbb{C}$ by $e^{i\theta}(x, w) = (e^{i\theta}x, e^{i\theta}w)$, with moment map $\mu(x) + \frac{1}{2}|w|^2$. Its reduction at ϵ, $\overline{M}_{\mu\le\epsilon}$, is a symplectic manifold containing $\{\mu < \epsilon\}$ as an open dense subset, via $x \mapsto [x, \sqrt{2(\epsilon - \mu(x))}]$, with complement the reduced space $\mu^{-1}(\epsilon)/S^1$, a codimension-2 symplectic submanifold.*

Cutting $\mathbb{C}^n$ at $\frac{1}{2}|z|^2 \le \frac{1}{2}$ gives $\mathbb{CP}^n = B^{2n}(\pi) \sqcup \mathbb{CP}^{n-1}$; cutting at $\frac{1}{2}|z|^2 \ge \epsilon$ (use the weight -1 on $\mathbb{C}$) is the symplectic blow-up of $\mathbb{C}^n$ at 0 whose exceptional lines have area $2\pi\epsilon$ (Chapter 3); on polytopes, cuts are intersections with rational half-spaces [Ler95].

Remark 2.50 (Further results, as pointers). Guillemin–Sternberg conjectured that "quantization commutes with reduction" [GS82b]; Meinrenken [Mei98] proved it in general using symplectic cutting. Karshon [Kar99] classified Hamiltonian S^1-actions on closed 4-manifolds by decorated graphs; all of them admit invariant Kähler structures. In higher dimensions this fails: Tolman [Tol98] found a Hamiltonian T^2-action on a closed 6-manifold, with isolated fixed points, admitting no invariant Kähler structure, and Woodward [Woo98] a multiplicity-free Hamiltonian action with the same property. Holomorphic curves still see symmetry: every closed Hamiltonian S^1-manifold is uniruled (McDuff [McD09a]). A circle action on a closed Kähler manifold that preserves the Kähler structure is Hamiltonian if and only if it has a fixed point (Frankel [Fra59]); for symplectic actions this holds in dimension 4 (McDuff [McD88]), fails in dimension 6 with fixed tori [McD88], holds for semifree actions with isolated fixed points (Tolman–Weitsman [TW00]), and fails in general even with isolated fixed points (Tolman [Tol17], a 6-manifold with 32 fixed points).

2.7 What to read

McDuff–Salamon, Introduction to Symplectic Topology *[MS17]*
: The reference for this chapter. Read Chapter 3 (Moser, Darboux, neighborhood theorems) and Chapter 4 (almost complex structures, c_1) carefully, Chapter 5 for moment maps, convexity and Duistermaat–Heckman, Chapter 10 for flux, Calabi and Banyaga, and Chapter 1 for the mechanics background. Their signs differ from ours ($\iota_{X_H}\omega = +dH$, $\omega_{\text{can}} = -d\lambda_{\text{can}}$); translate before copying a formula.

Cannas da Silva, Lectures on Symplectic Geometry *[Can01]*
: The best first pass: short chapters, the right examples, and a clean treatment of moment maps, reduction and Delzant's construction. Check her signs too.

Arnold, Mathematical Methods of Classical Mechanics *[Arn78]*
: For why any of this exists: Hamiltonian mechanics, Noether, integrable systems and action–angle variables (Part III and the appendices). Skim the first two parts if you know Lagrangian mechanics.

Audin, Torus Actions on Symplectic Manifolds *[Aud04]; Guillemin,* Moment Maps and Combinatorial Invariants of Hamiltonian T^n-spaces *[Gui94]*
: For toric geometry and convexity: Audin for topology and examples, Guillemin for Delzant, Duistermaat–Heckman and the combinatorics.

Polterovich, The Geometry of the Group of Symplectic Diffeomorphisms *[Pol01]*
: Read the opening chapters for Ham and its Lie algebra from the point of view that leads to Hofer geometry and quasimorphisms (Chapter 7).

Banyaga, The Structure of Classical Diffeomorphism Groups *[Ban97]*
For the simplicity theorems; read after Polterovich, and only if you need the proofs.

Kirwan, Cohomology of Quotients in Symplectic and Algebraic Geometry *[Kir84b]*
For the link between symplectic reduction and GIT and for Kirwan surjectivity.

Weinstein's papers [Wei71; Wei77; Wei81b]
The 1971 paper is the source of the Lagrangian neighborhood theorem and still reads well; the 1981 survey is where the creed is stated, and it is the best short statement of the Lagrangian philosophy.

Original papers
Moser [Mos65] (three pages of ideas); Atiyah [Ati82] and Guillemin–Sternberg [GS82a] on convexity; Duistermaat–Heckman [DH82]; Delzant [Del88]; Marsden–Weinstein [MW74]; Banyaga [Ban78]; Ono [Ono06] once you know Floer theory.

2.8 Frontier and open problems

Problem 2.51 (C^0 flux conjecture)**.** *Let (M,ω) be closed. Is* $\mathrm{Ham}(M,\omega)$ *closed in* $\mathrm{Symp}_0(M,\omega)$ *with respect to the C^0 topology?*

Open in general. Lalonde–McDuff–Polterovich [LMP98] proved it in some cases and Buhovsky [Buh14] extended these, also proving C^0-continuity of flux on smooth symplectic isotopies; see Chapter 7 for the C^0 theory.

Problem 2.52 (Hamiltonian S^1-manifolds in dimension 6)**.** *Classify closed 6-dimensional symplectic manifolds with Hamiltonian S^1-actions with isolated fixed points, and decide which of them admit Kähler structures.*

Karshon's dimension-4 classification has no analogue in dimension 6. Without the fixed-point hypothesis Kähler structures need not exist for trivial reasons ($KT\times S^2$, rotating S^2); with isolated fixed points M is simply connected with cohomology in even degrees, Tolman's example [Tol98] shows that invariant Kähler structures need not exist, and McDuff [McD09b] built several Fano 3-folds, including the Mukai–Umemura 3-fold, in purely symplectic terms and showed that certain such manifolds are determined by their fixed-point data. I do not know whether every closed 6-dimensional Hamiltonian S^1-manifold with isolated fixed points admits some, not necessarily invariant, Kähler structure; Goertsches–Konstantis–Zoller showed that the manifolds underlying Tolman's and Woodward's examples are diffeomorphic to Eschenburg's twisted flag manifold, so they do admit non-invariant Kähler structures [GKZ20].

Problem 2.53 (Complexity one and higher)**.** *Classify effective Hamiltonian T^{n-k}-actions on closed M^{2n} of complexity $k\geq 1$.*

For complexity one, the *tall* case (no reduced space is a point) is completely classified by Karshon–Tolman [KT03; KT14]; beyond that, the classification is not complete.

Question 2.54 (Uniqueness of toric structures)**.** *For which closed symplectic manifolds is the toric structure, if it exists, unique up to equivariant symplectomorphism and reparametrization of T^n?*

McDuff [McD11b], using an idea of Borisov, proved that a closed symplectic manifold has at most finitely many toric structures, and that the product of two projective spaces of complex dimension at least 2 with a product form has exactly one; blow-ups and bundles give examples with several.

Problem 2.55 (Lagrangian fibrations)**.** *Which closed symplectic manifolds admit Lagrangian torus fibrations with singularities, and which singularities are needed?*

In dimension 4 closed almost toric manifolds are classified up to diffeomorphism [LS10]; in higher dimensions this is the SYZ program (Chapter 8).

The symplectic cone and the gap between isotopy, deformation and symplectomorphism (Remark 2.28) are discussed in Chapter 3.

2.9 Logical dependencies

The definitions enter in the order Definitions 2.1 and 2.3, Convention 2.10, Definitions 2.12, 2.13, 2.17, 2.24 and 2.29; then Proposition 2.33 and Definition 2.34, which need Chern classes of complex vector bundles (Chapter 1), before Definition 2.35. Group actions need Convention 2.39 and Definition 2.40 before Theorems 2.43, 2.45, 2.46 and 2.48. Definition 2.4 and Convention 2.5 need only basic complex geometry. Every statement in this chapter is smooth differential topology: it presupposes de Rham cohomology, flows of time-dependent vector fields, the C^∞ topology on diffeomorphism groups (for Symp_0, $\widetilde{\mathrm{Symp}}_0$ and Theorem 2.16), Lie groups and principal bundles with connections (Theorem 2.46), and Lebesgue measure. No elliptic analysis is needed to state anything here, although Ono's proof of Theorem 2.16 uses Floer–Novikov theory.

Chapter 3

Constructions, and the existence and uniqueness problems

This chapter is about where symplectic manifolds come from, and how one decides whether a given manifold carries one. The first family of constructions produces *open* manifolds with a convex end—Liouville, Weinstein and Stein manifolds—the habitat of exact Lagrangians, symplectic cohomology and wrapped Fukaya categories. The second produces *closed* manifolds, by cut-and-paste (blow-ups, fibrations, symplectic sums, knot and Luttinger surgery) or by Donaldson's approximately holomorphic techniques.

What to take away. In dimension four we have both a rich supply of constructions and a powerful obstruction theory (Taubes' Seiberg–Witten theory), and the landscape of closed symplectic 4-manifolds can be mapped in outline by the symplectic Kodaira dimension. In dimension ≥ 6 we have the constructions but no obstructions beyond the obvious ones. Donaldson's theorems say that every closed symplectic manifold is "approximately algebraic"; in dimension four this turns symplectic topology into the combinatorics of factorizations in mapping class groups. That reduction is a theorem, but in my view it has so far been far better at producing examples than at classifying them. The analytic technology behind the obstructions (J-curves, SW = Gr) is in Chapter 6; here I state consequences.

3.1 Open symplectic manifolds with convex ends

3.1.1 Liouville domains and their completions

Recall convention C5: on an exact symplectic manifold $(M, \omega = d\lambda)$ the *Liouville vector field* Z is defined by $\iota_Z \omega = \lambda$. Cartan's formula gives $\mathcal{L}_Z \omega = d\iota_Z \omega = \omega$ and $\mathcal{L}_Z \lambda = \lambda$, so the flow satisfies $(\phi_Z^t)^* \lambda = e^t \lambda$ wherever it is defined. Note also $\lambda(Z) = \omega(Z, Z) = 0$.

Definition 3.1 (Liouville domains and manifolds)**.**

1. A *Liouville domain* (W, λ) is a compact $2n$-manifold W with boundary together with a 1-form λ such that $\omega = d\lambda$ is symplectic and the Liouville field Z points strictly outward along ∂W.
2. A *Liouville manifold of finite type* is an exact symplectic manifold (V, λ), without boundary, whose Liouville field Z is complete, and which contains a compact codimension-0 sub-

manifold W with smooth boundary such that Z points outward along ∂W and the map $[0,\infty)\times\partial W \to V$, $(r,y)\mapsto \phi_Z^r(y)$, is a diffeomorphism onto $V\setminus \operatorname{int} W$.

Proposition 3.2 (Boundary and completion)**.** *Let (W,λ) be a Liouville domain of dimension $2n$, oriented by ω^n, and let $\alpha := \lambda|_{\partial W}$.*

1. *α is a contact form on ∂W, positive for the boundary orientation: $\alpha\wedge(d\alpha)^{n-1} = \frac{1}{n}(\iota_Z\omega^n)|_{\partial W} > 0$.*
2. *The backward flow of Z defines a collar $(-\epsilon,0]\times\partial W \hookrightarrow W$, $(r,y)\mapsto\phi_Z^r(y)$, on which $\lambda = e^r\alpha$.*
3. *The* completion $\widehat{W} := W\cup_{\partial W}\big([0,\infty)\times\partial W\big)$*, with $\widehat{\lambda}=\lambda$ on W and $\widehat{\lambda}=e^r\alpha$ on the end, is a Liouville manifold of finite type; every Liouville manifold of finite type is the completion of the domain W in Definition 3.1.*
4. *If $(\lambda_t)_{t\in[0,1]}$ is a smooth family of Liouville forms on W (each making W a Liouville domain), then $(\widehat{W},\widehat{\lambda}_0)$ and $(\widehat{W},\widehat{\lambda}_1)$ are exact symplectomorphic.*

The end is the positive half of the symplectization of $(\partial W,\alpha)$ (C6). This is the right category for Floer theory on open manifolds, because the convex end gives a maximum principle for J-holomorphic curves (Chapter 5); it is the setting of symplectic cohomology and wrapped Floer theory (Chapter 8). For a short, careful treatment of completions read the early sections of Seidel's *A biased view* [Sei08a].

3.1.2 Weinstein structures

Definition 3.3 (Weinstein domains and manifolds; Eliashberg–Gromov 1991)**.** A *Weinstein domain* (W,λ,ϕ) is a Liouville domain (W,λ) together with a smooth function $\phi\colon W\to\mathbb{R}$ such that ∂W is a regular level set on which ϕ attains its maximum, ϕ is Morse (more generally, *generalized Morse*: each critical point is nondegenerate or a birth–death point), and Z is *gradient-like* for ϕ: there is a Riemannian metric and a constant $\delta>0$ with $d\phi(Z)\geq\delta\big(|Z|^2+|d\phi|^2\big)$. A *Weinstein manifold* (V,λ,ϕ) is defined in the same way with V without boundary, Z complete, ϕ exhausting (proper and bounded below), and δ a positive function; it has *finite type* if ϕ has finitely many critical points, and it is then the completion of a Weinstein domain. A *Weinstein homotopy* is a smooth family of such structures (with ϕ_t generalized Morse).

The notion, due to Eliashberg–Gromov [EG91], formalizes Weinstein's symplectic handlebodies [Wei91] and Eliashberg's Stein handlebodies [Eli90b]; birth–death points are allowed because they are unavoidable in generic one-parameter families [CE12].

Proposition 3.4 (Isotropic skeleton)**.** *Let (W,λ,ϕ) be a Weinstein domain of dimension $2n$. For each critical point p of ϕ, the stable manifold $W_p^s=\{x:\phi_Z^t(x)\to p \text{ as } t\to+\infty\}$ is an isotropic submanifold, of dimension $\operatorname{ind}_p(\phi)$; hence every critical point has index $\leq n$. The* skeleton

$$\operatorname{Skel}(W) := \bigcap_{t>0}\phi_Z^{-t}(W) = \bigcup_{p\in\operatorname{Crit}(\phi)} W_p^s$$

is compact, and W retracts into any neighborhood of it by the backward flow. If ϕ is Morse, W is homotopy equivalent to a CW complex with one k-cell for each critical point of index k, hence of dimension $\leq n$.

Proof idea. If $v,w\in T_xW_p^s$ then $\omega(v,w)=e^{-t}\omega(d\phi_Z^t v, d\phi_Z^t w)\to 0$ as $t\to+\infty$, because $(\phi_Z^t)^*\omega = e^t\omega$ while $d\phi_Z^t v$ stays bounded. The rest is Morse theory.

For a general Liouville domain the skeleton can be wild, even of codimension one, and Liouville is genuinely weaker than Weinstein: McDuff showed that $[0,1] \times S^*\Sigma$, for Σ a closed surface of genus ≥ 2, carries a Liouville structure [McD91b]; its boundary is disconnected, whereas a Weinstein domain of dimension ≥ 4 has connected boundary, since its handles have index $\leq n \leq 2n-2$.

Theorem 3.5 (Weinstein handle attachment; Weinstein 1991)**.** *Let (W, λ, ϕ) be a Weinstein domain of dimension $2n$ with contact boundary $(\partial W, \xi = \ker \alpha)$. Let $1 \leq k \leq n$ and let $\Lambda \subset \partial W$ be an embedded isotropic $(k-1)$-sphere, together with a trivialization of its* conformal symplectic normal bundle $\mathrm{CSN}(\Lambda) := \big((T\Lambda)^{d\alpha} \cap \xi\big)/T\Lambda$*, a bundle of rank $2(n-k)$ with the conformal symplectic structure induced by $d\alpha$. Then one can attach a $2n$-dimensional k-handle $D^k \times D^{2n-k}$ to W along a neighborhood of Λ so that the result W' is a Weinstein domain whose structure agrees with that of W away from a neighborhood of Λ, with exactly one critical point in $W' \setminus W$, of index k, whose stable manifold meets ∂W in Λ. The new boundary $\partial W'$ is obtained from ∂W by contact surgery along Λ.*

When $k = n$, Λ is Legendrian, $\mathrm{CSN}(\Lambda) = 0$, and no framing data are needed: this is *Legendrian (or Weinstein) surgery*. In dimension 4 it is the attachment of a 2-handle along a Legendrian knot with framing $\mathrm{tb}(\Lambda) - 1$ relative to the Seifert framing; learn to draw these from Gompf–Stipsicz Chapter 11 [GS99]. Conversely, cutting a Weinstein domain with Morse ϕ along regular levels exhibits it as built from finitely many standard balls by successive Weinstein handle attachments [CE12]. A Weinstein structure is *subcritical* if ϕ has no critical point of index n. (This is not invariant under Weinstein homotopy, since cancelling pairs of critical points of index $n-1$ and n can always be created; the meaningful notion is "homotopic to a subcritical structure".) All the symplectic rigidity of a Weinstein manifold lives in its index-n handles: subcritical Weinstein structures of dimension $2n > 4$ are flexible, hence determined up to homotopy by formal data (Chapter 4), and have vanishing symplectic cohomology [Cie02a]; whereas McLean showed that for every $n \geq 4$ there are infinitely many pairwise non-symplectomorphic finite-type Weinstein (indeed Stein) structures on $\mathbb{R}^{2n}$ [McL09].

Remark 3.6 (Where flexibility lives). Eliashberg's existence theorem for Stein structures [Eli90b], loose Legendrians, flexible Weinstein manifolds and their h-principles, and the rigid/flexible dichotomy are treated in Chapter 4.

3.1.3 Stein manifolds

Convention 3.7 (Sign of the Stein symplectic form)**.** For a complex manifold (V, J) and $\phi\colon V \to \mathbb{R}$ put $d^{\mathbb{C}}\phi := d\phi \circ J$, $\lambda_\phi := -d^{\mathbb{C}}\phi$ and $\omega_\phi := -dd^{\mathbb{C}}\phi = d\lambda_\phi = 2i\,\partial\bar\partial\phi$. With C1–C2, for $\phi_0 = \frac{1}{4}|z|^2$ on $\mathbb{C}^n$ one gets $\lambda_{\phi_0} = \lambda_0$, $\omega_{\phi_0} = \omega_0$, and Liouville field $\frac{1}{2}\sum_j (x_j\partial_{x_j} + y_j\partial_{y_j})$. This is the convention of Cieliebak–Eliashberg [CE12]. Complex geometers who set $d^c := i(\bar\partial - \partial)$ have $d^c\phi = -d^{\mathbb{C}}\phi$ and write the same form as $dd^c\phi$; with any convention, the test is that $\omega_\phi(v, Jv) > 0$ for strictly plurisubharmonic ϕ.

Definition 3.8 (Stein structures)**.** A *Stein structure* on a manifold V of real dimension $2n$ without boundary is a pair (J, ϕ) of an integrable complex structure J and an exhausting function $\phi\colon V \to \mathbb{R}$ which is strictly J-plurisubharmonic, i.e. $\omega_\phi(v, Jv) > 0$ for all nonzero $v \in TV$. A *Stein domain* is a sublevel set $\{\phi \leq c\}$ with c a regular value.

Then ω_ϕ is a symplectic form compatible with J (it is J-invariant, being of type $(1,1)$, and tames J), and the Liouville field X_ϕ of λ_ϕ is the gradient of ϕ for the metric $g_\phi = \omega_\phi(\cdot, J\cdot)$: indeed

$\omega_\phi(\nabla\phi, u) = d\phi(-Ju) = \lambda_\phi(u)$. After replacing ϕ by $f \circ \phi$ with $f' > 0$, $f'' \geq 0$, $f'(x) \to \infty$ as $x \to \infty$, the Liouville field becomes complete; if ϕ is (generalized) Morse we obtain the *associated Weinstein structure* $\mathfrak{W}(J, \phi) = (\omega_\phi, X_\phi, \phi)$ [EG91; CE12].

Theorem 3.9 (Stein = Weinstein up to homotopy; Cieliebak–Eliashberg 2012)**.** *Let V be a manifold of dimension $2n$ (or a compact cobordism, with structures required to have the boundary components as regular levels). Then the map $\mathfrak{W}$ from the space of Stein structures on V to the space of Weinstein structures on V induces a bijection on π_0 and a surjection on π_1. Concretely: every Weinstein structure on V is Weinstein homotopic to $\mathfrak{W}(J, \phi)$ for some Stein structure (J, ϕ), and two Stein structures are Stein homotopic if and only if their associated Weinstein structures are Weinstein homotopic.*

This is the main theorem of [CE12] (for domains with the C^∞ topology; manifolds need a finer topology); they conjecture that $\mathfrak{W}$ is a homotopy equivalence. My advice: think in Weinstein terms (handles, skeleta, Morse theory), compute in Stein or algebraic terms (affine varieties, Lefschetz fibrations).

Example 3.10 (Basic Weinstein manifolds)**.**

1. $\mathbb{C}^n$ with λ_0 and $\phi_0 = \frac{1}{4}|z|^2$: one critical point, of index 0; skeleton a point.
2. T^*Q for Q closed, with $\lambda_{\text{can}} = \sum_j p_j\, dq_j$ (C5): $Z = \sum_j p_j \partial_{p_j}$ is gradient-like for the Morse–Bott function $|p|^2$. After the standard perturbation of λ_{can} by an exact form, built from a Morse function f on Q, one obtains a Weinstein structure whose critical points are those of f, with the same indices, and whose skeleton is the zero section [CE12].
3. Smooth affine varieties: if $V \subset \mathbb{C}^N$ is a closed complex submanifold, $\phi = \frac{1}{4}|z - z_0|^2|_V$ is an exhausting strictly plurisubharmonic function, Morse for generic z_0, with finitely many critical points. The completion of the resulting Weinstein domain $\{\phi \leq c\}$, $c \gg 0$, is independent of the embedding up to exact symplectomorphism ([EG91]; [Sei08a, Sect. 4b]).
4. Milnor fibers and plumbings: the affine quadric $\left\{\sum_{j=0}^n z_j^2 = 1\right\}$ is exact symplectomorphic to T^*S^n, and the A_k Milnor fiber $\left\{z_0^{k+1} + \sum_{j=1}^n z_j^2 = 1\right\}$ (after completion) to a plumbing of k copies of T^*S^n along a chain, via matching cycles [Sei08b, Sect. 16]. Plumbings of cotangent bundles along trees, and Milnor fibers of isolated hypersurface singularities, are Weinstein.
5. Products of Weinstein manifolds are Weinstein; the *stabilization* $V \times \mathbb{C}$ is subcritical.

3.2 Closed constructions: blow-ups, fibrations, sums and surgeries

3.2.1 Blowing up and down

Definition 3.11 (Symplectic blow-up)**.** Let (M, ω) be a symplectic $2n$-manifold, $n \geq 2$. Write $B_r := \{z \in \mathbb{C}^n : |z| < r\}$ for the open ball of *radius* r (in the capacity labelling of the preface, $B_r = B^{2n}(\pi r^2)$). Let $\iota\colon (B_{\rho+\epsilon}, \omega_0) \to (M, \omega)$ be a symplectic embedding. The *blow-up* $\widetilde{M}_\iota$ is obtained by removing $\iota(B_\rho)$ and collapsing $\iota(\partial B_\rho)$ along the fibers of the Hopf fibration $\partial B_\rho \to \mathbb{CP}^{n-1}$ (the characteristic foliation of the sphere). The image $E \cong \mathbb{CP}^{n-1}$ of the sphere is the *exceptional divisor*.

Proposition 3.12 (Blow-up bookkeeping)**.** *In the situation of Definition 3.11, $\widetilde{M}_\iota$ carries a symplectic form $\widetilde{\omega}$ equal to ω outside $\iota(B_{\rho+\epsilon})$, for which E is a symplectic submanifold with normal bundle the*

tautological bundle $\mathcal{O}(-1)$ *and each projective line in* E *has area* $\pi\rho^2$*. Let* $\beta\colon \widetilde{M}_\iota \to M$ *be the (smooth) blow-down map and* $e := \mathrm{PD}[E]$*, so that* $H^2(\widetilde{M}_\iota;\mathbb{Z}) = \beta^* H^2(M;\mathbb{Z}) \oplus \mathbb{Z}e$*. Then*

$$[\widetilde{\omega}] = \beta^*[\omega] - \pi\rho^2 e, \qquad c_1(T\widetilde{M}_\iota) = \beta^* c_1(TM) - (n-1)e, \qquad \mathrm{vol}(\widetilde{M}_\iota) = \mathrm{vol}(M) - \frac{\pi^n \rho^{2n}}{n!}.$$

In dimension 4*:* E *is a symplectic sphere with* $E \cdot E = -1$ *and area* $\pi\rho^2$*, and* $K_{\widetilde{\omega}} = \beta^* K_\omega + e$*.*

Symplectic blow-up goes back to Gromov and was developed by McDuff [McD84], who used it to build the first simply connected closed symplectic manifolds with no Kähler structure (blowing up $\mathbb{CP}^5$ along an embedded Kodaira–Thurston manifold). The cleanest description is as a symplectic cut along the boundary of the ball [GS89; Ler95]. The symplectomorphism type of $(\widetilde{M}_\iota, \widetilde{\omega})$ depends only on the isotopy class of ι through symplectic embeddings [McD91a; McD93]; in dimension four, for embeddings of disjoint balls into $\mathbb{CP}^2$ the relevant embedding spaces are connected [McD98], so the blow-up depends only on the radii. Conversely, a symplectic sphere of square -1 in a 4-manifold has a standard neighborhood (Chapter 2) and can be blown down to a ball of radius ρ, $\pi\rho^2$ being its area. The dictionary between ball embeddings and symplectic forms on blow-ups is exploited in Chapter 6.

Definition 3.13 (Minimal; rational; ruled)**.** Let (M,ω) be a closed symplectic 4-manifold. An *exceptional sphere* is a symplectically embedded 2-sphere S with $S \cdot S = -1$; (M,ω) is *minimal* if it contains none. M is *rational* if it is diffeomorphic to $\mathbb{CP}^2$, $S^2 \times S^2$ or $\mathbb{CP}^2 \# k\overline{\mathbb{CP}}^2$ ($k \geq 1$), and *ruled* if it is diffeomorphic to the total space of an oriented S^2-bundle over a closed oriented surface or to a blow-up of one.

Theorem 3.14 (Minimal models; McDuff 1990)**.** *Let* (M,ω) *be a closed symplectic* 4*-manifold. Every family of pairwise disjoint exceptional spheres is finite, and blowing down a maximal such family yields a minimal symplectic* 4*-manifold* (N,τ)*, a* minimal model*, of which* (M,ω) *is a symplectic blow-up. If* M *is not rational or ruled, the exceptional classes are finite in number and pairwise orthogonal, and the minimal model is unique up to symplectomorphism.*

See [McD90; McD91a] and the survey [Li19]. By Seiberg–Witten theory, (M,ω) is symplectically minimal if and only if M contains no smoothly embedded sphere of square -1, so minimality is a smooth notion [Li19].

3.2.2 Symplectic fibrations and Thurston's construction

Theorem 3.15 (Thurston 1976)**.** *Let* $\pi\colon M \to B$ *be a smooth fiber bundle over a closed connected symplectic manifold* (B,β)*, with* M *closed and with connected fiber* F*. Suppose the structure group reduces to* $\mathrm{Symp}(F,\sigma)$ *for a symplectic form* σ *on* F*, so that each fiber* F_b *carries a symplectic form* σ_b*, and suppose some* $a \in H^2(M;\mathbb{R})$ *restricts to* $[\sigma_b]$ *on a fiber. Then for every sufficiently large* $K > 0$ *there is a symplectic form* ω_K *on* M *with* $[\omega_K] = a + K\pi^*[\beta]$ *and* $\omega_K|_{F_b} = \sigma_b$ *for all* $b \in B$*.*

Proof idea. Represent a by a closed form τ restricting to σ_b on the fibers (possible using the symplectic structure group and a partition of unity on B); then $\tau + K\pi^*\beta$ is nondegenerate for K large, being nondegenerate on fibers and on a horizontal complement.

Corollary 3.16 (Surface bundles over surfaces)**.** *Let* $\pi\colon M^4 \to B$ *be an oriented bundle with fiber a closed oriented surface* Σ_g *over a closed oriented surface. If* $[\Sigma_g] \neq 0$ *in* $H_2(M;\mathbb{R})$*, which is automatic when* $g \neq 1$*, then* M *carries a symplectic form for which every fiber is symplectic.*

(By Moser's theorem, the structure group of an oriented surface bundle reduces to area-preserving maps; when $g \neq 1$ the Euler class of the vertical tangent bundle evaluates to $2-2g \neq 0$ on the fiber.) The Kodaira–Thurston manifold $S^1 \times N_A$, where N_A is the mapping torus of $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ acting on T^2, is a T^2-bundle over T^2 with homologically essential fiber; so it is symplectic, while $b_1 = 3$, so it admits no Kähler structure [Thu76]. Likewise $S^1 \times N$ is symplectic whenever the closed 3-manifold N fibers over S^1; the converse is Theorem 3.46 below.

3.2.3 The symplectic sum

Theorem 3.17 (Symplectic normal connected sum; Gompf 1995, McCarthy–Wolfson 1994)**.** *Let (M_1, ω_1), (M_2, ω_2) be closed symplectic $2n$-manifolds, (V, ω_V) a closed symplectic $(2n-2)$-manifold, and $j_i \colon V \to M_i$ symplectic embeddings ($j_i^* \omega_i = \omega_V$) whose normal bundles satisfy $c_1(\nu_1) + c_1(\nu_2) = 0$ in $H^2(V; \mathbb{Z})$. Fix a fiber-orientation-reversing bundle isomorphism $\psi \colon \nu_1 \to \nu_2$ covering id_V. Then the manifold $X = M_1 \#_{V,\psi} M_2$, obtained by deleting tubular neighborhoods of $j_1(V)$, $j_2(V)$ and identifying their boundaries via ψ, carries a symplectic form that agrees with ω_i away from a neighborhood of $j_i(V)$. If $n = 2$, the hypothesis reads $[V_1]^2 + [V_2]^2 = 0$, and $\chi(X) = \chi(M_1) + \chi(M_2) - 2\chi(V)$, $\sigma(X) = \sigma(M_1) + \sigma(M_2)$.*

See [Gom95; MW94]. The typical use in dimension four is the *fiber sum* along square-zero tori. Let $E(1) = \mathbb{CP}^2 \# 9\overline{\mathbb{CP}}^2$, the rational elliptic surface (a pencil of cubics blown up at its nine base points), with fiber torus F, and set $E(n) = E(n-1) \#_F E(1)$. Then $E(n)$ is simply connected with $\chi = 12n$, $\sigma = -8n$, $b^+ = 2n-1$, canonical class $K = (n-2)\,\mathrm{PD}[F]$, and $E(2)$ is the K3 surface. For the symplectic sum formulas for Gromov–Witten invariants, which compute $\mathrm{GW}(X)$ from relative invariants of (M_i, V), see Ionel–Parker [IP04] and Li–Ruan [LR01], and Chapter 5.

Theorem 3.18 (Fundamental groups; Gompf 1995)**.** *For every finitely presented group G and every $n \geq 2$ there is a closed symplectic $2n$-manifold M with $\pi_1(M) \cong G$.*

Proof idea. Roughly: for a presentation with g generators, start from $\Sigma_g \times T^2$ with a product form; represent the relators by tori of the form (curve in Σ_g)$\times S^1$, made disjoint, embedded and (after perturbing ω) symplectic; fiber sum each with $E(1)$ along F. Since $\pi_1(E(1) \setminus F) = 1$, each sum kills the image of π_1 of its torus, hence the relator, together with the meridian; one more sum along a fiber $\{\mathrm{pt}\} \times T^2$ kills the T^2 factor. Products with S^2 give higher dimensions [Gom95].

The contrast with Kähler geometry is stark: Kähler groups have even first Betti number and satisfy many further restrictions [Amo+96]. So closed symplectic 4-manifolds cannot be classified up to diffeomorphism, and every classification statement below rests on strong hypotheses.

3.2.4 Knot surgery

Definition 3.19 (Knot surgery; Fintushel–Stern)**.** Let X be a closed oriented 4-manifold containing an embedded torus T with $T \cdot T = 0$, so $\nu T \cong T^2 \times D^2$, and let $K \subset S^3$ be a knot with exterior $E_K = S^3 \setminus \nu K$. The *knot surgery* manifold is $X_K := (X \setminus \nu T) \cup_\varphi (S^1 \times E_K)$, where $\varphi \colon S^1 \times \partial E_K \to \partial \nu T = T^2 \times S^1$ is a diffeomorphism sending the class of a 0-framed longitude of K to the class of the meridian $\mathrm{pt} \times \partial D^2$ of T.

Theorem 3.20 (Fintushel–Stern 1998)**.** *Let X be a closed simply connected 4-manifold with $b^+(X) > 1$ containing a torus T that is a smooth fiber of a cusp neighborhood, with $\pi_1(X \setminus T) = 1$. Then for every*

knot $K \subset S^3$, X_K is homeomorphic to X and its Seiberg–Witten invariant, written as a symmetric Laurent polynomial, is

$$\mathrm{SW}_{X_K} = \mathrm{SW}_X \cdot \Delta_K(t), \qquad t = \exp(2[T]),$$

where Δ_K is the symmetrized Alexander polynomial. If X is symplectic, T is a symplectic submanifold and K is fibered, then X_K admits a symplectic structure; if Δ_K is not monic, X_K admits none.

Proof idea. X_K is the fiber sum of X with $S^1 \times M_K$ (M_K the 0-surgery on K) along T and $S^1\times$(meridian); when K is fibered, M_K fibers over S^1 and $S^1 \times M_K$ is symplectic. The product formula is a Seiberg–Witten gluing computation; non-monic Δ_K contradicts Taubes' theorem that the extremal coefficient of a symplectic manifold is ± 1 (Theorem 3.43) [FS98].

Corollary 3.21 (Exotic symplectic K3 surfaces)**.** *For the torus knots $K_k = T(2, 2k+1)$, $k \geq 1$, the manifolds $E(2)_{K_k}$ are closed symplectic 4-manifolds, homeomorphic to K3, and pairwise non-diffeomorphic (their Seiberg–Witten polynomials have different numbers of nonzero coefficients).*

For knot surgery, rational blow-down and how far Seiberg–Witten invariants go towards a classification, read Fintushel–Stern's *Six lectures* [FS09].

3.2.5 Luttinger surgery

Definition 3.22 (Luttinger surgery)**.** Let $L \subset (M^4, \omega)$ be an embedded Lagrangian torus, $\gamma \subset L$ an embedded closed curve and $k \in \mathbb{Z}$. A Weinstein neighborhood $U \cong T^2 \times D^2$ of L provides the *Lagrangian push-off* $\gamma' \subset \partial U$ of γ (a parallel copy in a nearby Lagrangian torus $T^2 \times \{\mathrm{pt}\}$) and a meridian $\mu \subset \partial U$. Define $M_{L,\gamma,k} := (M \setminus U) \cup_\varphi (T^2 \times D^2)$, where $\varphi \colon T^2 \times \partial D^2 \to \partial U$ is a diffeomorphism with $\varphi_*[\mathrm{pt} \times \partial D^2] = [\mu] + k[\gamma']$.

Luttinger showed that $M_{L,\gamma,k}$ carries a symplectic form agreeing with ω outside U [Lut95]; Auroux–Donaldson–Katzarkov developed its properties systematically and used it to give a topological explanation of Moishezon's non-isotopic singular plane curves [ADK03]. Luttinger surgery preserves χ and σ but can change π_1, which makes it the standard tool for building small exotic symplectic 4-manifolds (Section 3.5); Luttinger's own application, to Lagrangian tori in $\mathbb{R}^4$, is in Chapter 6.

3.3 Donaldson's theorems and Lefschetz fibrations

Donaldson's idea, informally: choose a compatible J and a Hermitian line bundle $L \to M$ with $c_1(L)$ an integral lift of $[\omega]$ and a connection of curvature $-2\pi i\,\omega$. For $k \gg 0$, in the rescaled metric $k\,g_J$ the manifold looks locally like $\mathbb{C}^n$ and $L^{\otimes k}$ has many approximately holomorphic sections (concentrated Gaussians). Sections s_k with $|\bar\partial s_k| = O(k^{-1/2})$ that are *uniformly transverse* to zero ($|\partial s_k| \geq \eta > 0$ wherever $|s_k| < \eta$, in rescaled norms) have symplectic zero sets. Achieving uniform transversality is the heart of the matter: a quantitative Sard theorem plus local perturbations whose sum is controlled globally. Donaldson's 1996 paper [Don96] is unusually readable and the right first original paper here.

Theorem 3.23 (Symplectic hypersurfaces; Donaldson 1996)**.** *Let (M, ω) be a closed symplectic $2n$-manifold and $h \in H^2(M;\mathbb{Z})$ an integral class whose image in $H^2(M;\mathbb{R})$ is $[\omega]$. Then there is k_0 such that for every $k \geq k_0$ there exists a closed symplectic submanifold $V_k \subset M$ of dimension $2n-2$, Poincaré dual to kh, such that the inclusion induces isomorphisms $H_j(V_k;\mathbb{Z}) \to H_j(M;\mathbb{Z})$ for $j < n-1$ and a surjection for $j = n-1$.*

The integrality hypothesis is harmless for existence questions: perturb ω to have rational class and rescale. Auroux extended the method to zero sets of approximately holomorphic sections of $E \otimes L^{\otimes k}$ for Hermitian vector bundles E, giving symplectic submanifolds of higher codimension, and proved that for k large the submanifolds obtained this way are unique up to symplectic isotopy [Aur97].

Definition 3.24 (Lefschetz pencils)**.** Let M be a closed oriented $2n$-manifold, $n \geq 2$. A *Lefschetz pencil* on M is a pair (B, f) where $B \subset M$ is a nonempty closed codimension-4 submanifold and $f : M \setminus B \to \mathbb{CP}^1$ is a smooth map such that: (i) each $b \in B$ has orientation-preserving complex coordinates in which $B = \{z_1 = z_2 = 0\}$ and $f = [z_1 : z_2]$; (ii) f has finitely many critical points, each with orientation-preserving complex coordinates in which $f(z) = z_1^2 + \cdots + z_n^2$ in an affine chart of $\mathbb{CP}^1$; (iii) distinct critical points have distinct critical values. It is *symplectic* for ω if B and the fibers $\overline{f^{-1}(t)} = f^{-1}(t) \cup B$ are symplectic away from the critical points and the coordinate charts in (i), (ii) can be chosen so that ω tames the standard complex structure there.

Theorem 3.25 (Lefschetz pencils; Donaldson 1999)**.** *Under the hypotheses of Theorem 3.23, for all sufficiently large k the manifold M admits a symplectic Lefschetz pencil whose fibers are Poincaré dual to kh.*

See [Don99]. Blowing up the base locus (in dimension four, $(kh)^2$ points) turns the pencil into a Lefschetz fibration over $\mathbb{CP}^1$ with exceptional sections. So, after finitely many blow-ups, every closed symplectic 4-manifold fibers over S^2 with Lefschetz singularities.

Definition 3.26 (Lefschetz fibrations in dimension 4)**.** A *Lefschetz fibration* is a smooth map $f : X \to \Sigma$ from a compact connected oriented 4-manifold onto a compact connected oriented surface, with $f^{-1}(\partial\Sigma) = \partial X$, connected fibers, and finitely many critical points, all in the interior, each with orientation-preserving complex coordinates in which $f(z_1, z_2) = z_1^2 + z_2^2$, and with distinct critical values.

Near a critical value the singular fiber is obtained from a regular fiber Σ_g by collapsing a simple closed curve, the *vanishing cycle*, and the monodromy around it is a right-handed (positive) Dehn twist along that curve (Picard–Lefschetz). Choosing a base point and an ordered system of arcs to the critical values, a Lefschetz fibration over S^2 with m singular fibers yields a *positive factorization* $t_{c_m} \cdots t_{c_1} = 1$ in the mapping class group $\mathrm{Mod}(\Sigma_g) = \pi_0 \mathrm{Diff}^+(\Sigma_g)$. Changing the arcs changes the factorization by *Hurwitz moves*, replacing two consecutive factors $\varphi_i \varphi_{i+1}$ by $\varphi_{i+1} \cdot (\varphi_{i+1}^{-1} \varphi_i \varphi_{i+1})$, and changing the identification of the fiber changes it by global conjugation.

Theorem 3.27 (Monodromy classification; Kas 1980)**.** *For $g \geq 2$, the monodromy construction gives a bijection between isomorphism classes of genus-g Lefschetz fibrations over S^2 and positive factorizations of 1 in $\mathrm{Mod}(\Sigma_g)$ modulo Hurwitz moves and global conjugation.*

See [Kas80] and Gompf–Stipsicz Chapter 8 [GS99]. For the fibration obtained from a pencil with k base points, the monodromy lifts to the mapping class group $\mathrm{Mod}(\Sigma_{g,k})$ of the fiber with k boundary components, and the relation becomes $t_{c_m} \cdots t_{c_1} = t_{\delta_1} \cdots t_{\delta_k}$, the product of boundary-parallel twists.

Theorem 3.28 (Lefschetz fibrations are symplectic; Gompf)**.** *Let $f : X \to \Sigma$ be a Lefschetz fibration on a closed oriented 4-manifold whose fiber class $[F]$ is nonzero in $H_2(X; \mathbb{R})$; this holds automatically if $g \neq 1$. Then X carries a symplectic form, inducing its orientation, for which every regular fiber is symplectic; any given finite collection of sections can be made symplectic as well. Every closed oriented 4-manifold admitting a Lefschetz pencil carries a symplectic form for which the fibers are symplectic.*

This is [GS99, Thm. 10.2.18]; it is a Thurston-type argument (Theorem 3.15) with care near the critical points. Combining with Theorem 3.25:

Corollary 3.29 (Topological characterization in dimension 4)**.** *A closed oriented 4-manifold admits a symplectic form inducing its orientation if and only if it admits a Lefschetz pencil.*

Remark 3.30 (The slogan and its limits). Closed symplectic 4-manifolds, up to blow-up and the choice of k, "are" positive factorizations of boundary multitwists in mapping class groups. The trouble is that we cannot decide Hurwitz equivalence, and the invariants we can extract from factorizations are mostly classical. The best structural result is Auroux's *stable classification* [Aur05]: for each g there is a universal genus-g fibration f_g^0 such that two Lefschetz fibrations over S^2 with the same Euler characteristic and signature, the same numbers of reducible singular fibers of each type, and sections of the same self-intersection become isomorphic after repeated fiber sums with f_g^0. On the holomorphic side, Siebert–Tian proved that every genus-2 Lefschetz fibration without reducible fibers and with transitive monodromy is holomorphic [ST05].

The other combinatorial model is Auroux's: for $[\omega]$ integral and $k \gg 0$, a closed symplectic 4-manifold is, roughly, a branched covering of $\mathbb{CP}^2$ whose branch curve is a symplectic curve with cusps and (possibly negative) nodes, canonical for large k up to isotopy and creation or cancellation of pairs of nodes [Aur00; AS08]; the braid monodromy of the branch curve encodes the manifold. Here the *symplectic isotopy problem* for singular plane curves enters, and Luttinger surgery explains non-isotopy [ADK03]. In the smooth case, Siebert–Tian proved that every symplectic surface in $\mathbb{CP}^2$ of degree ≤ 17 is symplectically isotopic to a complex curve [ST05].

Theorem 3.31 (Stein surfaces as Lefschetz fibrations; Loi–Piergallini 2001, Akbulut–Ozbagci 2001)**.** *A compact oriented 4-manifold W with boundary admits a Stein domain structure inducing its orientation if and only if it admits a* positive allowable Lefschetz fibration *over D^2: a Lefschetz fibration $W \to D^2$ (in the sense of Definition 3.26, W having corners) whose regular fiber is a compact surface with nonempty boundary and all of whose vanishing cycles are homologically nontrivial in the fiber.*

See [LP01; AO01]. This is the relative version of the closed story, and it restricts on the boundary to an open book supporting the induced contact structure; for open books and Giroux's correspondence see Chapter 4.

Donaldson's philosophy is that symplectic manifolds are "approximately algebraic": constructions of projective geometry that use only sections of high powers of an ample bundle have symplectic analogues. Gompf pushed the converse into higher dimensions with *hyperpencils*, maps to $\mathbb{CP}^{n-1}$ generalizing linear systems of curves, whose existence characterizes symplectic manifolds topologically in dimensions ≤ 6 [Gom04].

3.4 The existence problem

Proposition 3.32 (Necessary conditions)**.** *If a closed $2n$-manifold M carries a symplectic form ω, then (i) M admits an almost complex structure (unique up to homotopy among those compatible with ω; Chapter 2), and (ii) the class $a = [\omega] \in H^2(M;\mathbb{R})$ satisfies $a^n \neq 0$.*

For open manifolds these are essentially the only constraints:

Theorem 3.33 (Gromov 1969)**.** *Let M be a manifold of dimension $2n$ with no closed components, and σ a nondegenerate 2-form on M. Then for every $a \in H^2(M;\mathbb{R})$ there is a symplectic form ω with $[\omega] = a$ that is homotopic to σ through nondegenerate 2-forms.*

This is an h-principle [Gro69; Gro86; EM02]; see Chapter 4. For closed manifolds the first test is (i), which in dimension four is computable.

Theorem 3.34 (Almost complex 4-manifolds; Wu)**.** *A closed oriented 4-manifold X admits an almost complex structure inducing its orientation if and only if there is $c \in H^2(X;\mathbb{Z})$ with $c \equiv w_2(X) \pmod 2$ and $c^2 = 2\chi(X) + 3\sigma(X)$; such a c is then $c_1(TX, J)$ for some J.*

See [GS99, Thm. 1.4.15]. Since a characteristic class satisfies $c^2 \equiv \sigma \pmod 8$, a consequence is that $1 - b_1(X) + b^+(X)$ is even. Beyond (i) and (ii), dimension four has a genuinely new obstruction.

Theorem 3.35 (Taubes 1994)**.** *Let (X, ω) be a closed symplectic 4-manifold, oriented by ω, with $b^+(X) > 1$. Then the Seiberg–Witten invariant of the canonical spin^c structure $\mathfrak{s}_\omega$, whose first Chern class is $c_1(TX, J) = -K_\omega$, is ± 1.*

See [Tau94]; the proof (perturbing the Seiberg–Witten equations by a large multiple of ω) is sketched in Chapter 6.

Corollary 3.36 (Connected sums are not symplectic)**.** *If $X = X_1 \# X_2$ with X_1, X_2 closed and $b^+(X_1), b^+(X_2) \geq 1$, then X admits no symplectic form inducing its orientation. In particular $\mathbb{CP}^2\#\mathbb{CP}^2\#\mathbb{CP}^2$ is almost complex (take $c = (3,3,1)$: $c^2 = 19 = 2 \cdot 5 + 3 \cdot 3$) but admits no symplectic form with either orientation.*

(Seiberg–Witten invariants of such connected sums vanish; with the reversed orientation the manifold is negative definite, so no class has positive square.) In dimension ≥ 6 nothing of the kind is known.

Problem 3.37 (Existence in higher dimensions)**.** *Let M be a closed $2n$-manifold, $n \geq 3$, that admits an almost complex structure and a class $a \in H^2(M;\mathbb{R})$ with $a^n \neq 0$. Does M admit a symplectic form? (Stronger versions: in the class a; in a prescribed homotopy class of almost complex structures.)*

To my knowledge no closed manifold of dimension ≥ 6 satisfying both conditions is known not to be symplectic; Salamon calls the problem "completely open" [Sal13]. The classical non-examples fail (ii): S^6 has $H^2 = 0$, and on $S^2 \times S^4$ the generator of H^2 squares to zero. Nor need closed symplectic manifolds be Kähler: besides the Kodaira–Thurston manifold, there are simply connected examples [McD84], also in dimension four [Gom95].

3.5 The landscape of closed symplectic 4-manifolds

Throughout, (M, ω) is a closed symplectic 4-manifold oriented by ω, $K = K_\omega = -c_1(TM, J)$ its canonical class for any ω-compatible J, and K^2, $K \cdot [\omega]$ are evaluated on $[M]$. Recall $K^2 = 2\chi + 3\sigma$ (Wu).

Definition 3.38 (Symplectic Kodaira dimension; T.-J. Li)**.** For (M,ω) minimal set

$$\kappa^s(M,\omega)=\begin{cases}-\infty & \text{if } K\cdot[\omega]<0 \text{ or } K^2<0,\\ 0 & \text{if } K\cdot[\omega]=0 \text{ and } K^2=0,\\ 1 & \text{if } K\cdot[\omega]>0 \text{ and } K^2=0,\\ 2 & \text{if } K\cdot[\omega]>0 \text{ and } K^2>0.\end{cases}$$

For non-minimal (M,ω), κ^s is that of any minimal model (Theorem 3.14).

Theorem 3.39 (Well-definedness; Li 2006)**.** *The definition is exhaustive: no minimal (M,ω) has $K\cdot[\omega]=0$ and $K^2>0$. The value κ^s does not depend on the minimal model or on ω, so it is an invariant of the oriented diffeomorphism type of M. If ω is a Kähler form for a complex structure on M, then κ^s equals the holomorphic Kodaira dimension. A minimal (M,ω) has $\kappa^s=0$ if and only if K is torsion.*

The proofs rest on Taubes' theorems and on wall-crossing (Li–Liu, Liu) [Li06b; Li06c; Li19].

Theorem 3.40 ($\kappa^s=-\infty$: rational and ruled manifolds; McDuff 1990, Liu 1996, Ohta–Ono 1996)**.** *For a closed symplectic 4-manifold (M,ω) the following are equivalent: (a) $\kappa^s(M)=-\infty$; (b) M is rational or ruled; (c) M contains a symplectically embedded 2-sphere of nonnegative self-intersection. If (M,ω) is minimal, these hold if and only if M is diffeomorphic to $\mathbb{CP}^2$ or to an S^2-bundle over a closed oriented surface.*

See [McD90; Liu96; OO96; Li19]. McDuff's (c)⇒(b) is the prototype J-curve classification theorem; see Chapter 6 and Wendl [Wen18].

Theorem 3.41 (Symplectic structures on rational and ruled manifolds)**.** *Let M be a rational or ruled 4-manifold.*

1. *Any two cohomologous symplectic forms on M are diffeomorphic (Lalonde–McDuff 1996, Li–Liu 1995).*
2. *The symplectic canonical class is unique up to orientation-preserving diffeomorphism, and the cone of classes represented by symplectic forms is explicitly determined by the set of classes of smoothly embedded (-1)-spheres (Li–Liu 2001).*
3. *Every symplectic form on $\mathbb{CP}^2$ is diffeomorphic to $c\,\omega_{\mathrm{FS}}$ for some $c>0$ (Gromov 1985, Taubes 1995).*

See [LM96b; LM96a; LL95b; LL95a; LL01; Gro85; Tau95b], and for isotopy (rather than diffeomorphism) of cohomologous forms that are deformation equivalent, [McD98]. So the moduli space of symplectic structures on such M is the symplectic cone modulo the diffeomorphism group's action on cohomology [Sal13].

Theorem 3.42 ($\kappa^s=0$: symplectic Calabi–Yau surfaces; Li 2006, Bauer 2008)**.** *Let (M,ω) be minimal with $\kappa^s=0$. Then M has the rational homology and rational intersection form of the K3 surface, of the Enriques surface, or of a T^2-bundle over T^2. In particular $\chi(M)\in\{0,12,24\}$, $\sigma(M)\in\{-16,-8,0\}$, $b^+(M)\le 3$ and $b_1(M)\le 4$. If moreover M is simply connected, it is homeomorphic to K3.*

See [Li06a; Bau08; Li19]. The key step is $b^+\le 3$, obtained by showing that a refined (Pin(2)-equivariant) Seiberg–Witten invariant of K is even when $b^+>3$, contradicting Theorem 3.35. All known examples are K3, Enriques, and the symplectic T^2-bundles over T^2 (classified by Geiges [Gei92]). The class $\kappa^s=1$ contains the elliptic surfaces $E(n)$, $n\ge 3$, and Gompf's manifolds realizing arbitrary finitely presented fundamental groups [Li19], so no classification is possible there; $\kappa^s=2$ ("general type") contains all the small exotic manifolds discussed below.

Theorem 3.43 (Taubes' constraints)**.** *Let (M, ω) be a closed symplectic 4-manifold with $b^+ > 1$.*

1. *If $\mathfrak{s}$ is a spin^c structure with $\mathrm{SW}(\mathfrak{s}) \neq 0$, then $|c_1(\mathfrak{s}) \cdot [\omega]| \leq K \cdot [\omega]$, with equality only if $c_1(\mathfrak{s}) = \pm K$ (Taubes 1995).*
2. *$\mathrm{PD}(K)$ is represented by an embedded, possibly empty or disconnected, symplectic surface; in particular $K \cdot [\omega] \geq 0$ (Taubes 1996).*
3. *If (M, ω) is minimal, then $K^2 \geq 0$ (Taubes 1996).*

See [Tau95a; Tau96a; Tau00]; part (2) is "SW $\Rightarrow$ Gr", and (3) follows from (2) and adjunction, since a component of negative square would be an exceptional sphere. Consequently minimal symplectic 4-manifolds with $b^+ > 1$ have $\kappa^s \geq 0$, and $\kappa^s = -\infty$ forces $b^+ = 1$.

Theorem 3.44 (Symplectic Thom conjecture; Ozsváth–Szabó 2000)**.** *Let Σ be a closed connected symplectic surface in a closed symplectic 4-manifold X. Then every closed connected oriented smoothly embedded surface $\Sigma' \subset X$ with $[\Sigma'] = [\Sigma] \in H_2(X; \mathbb{Z})$ has genus $g(\Sigma') \geq g(\Sigma)$.*

See [OS00]; the case of complex curves in $\mathbb{CP}^2$ is the Thom conjecture of Kronheimer–Mrowka [KM94b].

Proposition 3.45 (Symplectic homotopy $\mathbb{CP}^2$'s)**.** *Let (X, ω) be a closed symplectic 4-manifold homeomorphic to $\mathbb{CP}^2$, and $h \in H^2(X; \mathbb{Z})$ the generator with $h \cdot [\omega] > 0$. Then $K = \pm 3h$. If $K = -3h$, i.e. $K \cdot [\omega] < 0$, then (X, ω) is symplectomorphic to $(\mathbb{CP}^2, c\,\omega_{\mathrm{FS}})$ for some $c > 0$. If $K = 3h$, then (X, ω) is minimal with $\kappa^s = 2$ and $K^2 = 9 = 3\chi(X)$, and X is not diffeomorphic to $\mathbb{CP}^2$.*

Proof idea. $K^2 = 2\chi + 3\sigma = 9$ and $b_2 = 1$ force $K = \pm 3h$; $b^- = 0$ gives minimality. If $K \cdot [\omega] < 0$, Theorem 3.40 gives $X \cong \mathbb{CP}^2$, and Theorem 3.41(3) finishes. If $K = 3h$ the canonical class of ω is not that of any symplectic form on $\mathbb{CP}^2$.

Whether the second case occurs is open (Problem 3.47). Kähler surfaces with these numbers exist—the fake projective planes—but they have infinite fundamental group.

Uniqueness and deformation. Two symplectic forms ω_0, ω_1 on M are *deformation equivalent* if ω_0 and $\varphi^*\omega_1$ are joined by a path of symplectic forms for some diffeomorphism φ. McMullen–Taubes found the first closed 4-manifolds with deformation-inequivalent symplectic forms, built from 3-manifolds with inequivalent fibrations over S^1 [MT99]; Smith found simply connected ones with arbitrarily many deformation classes, distinguished by the divisibility of c_1 [Smi00]; Vidussi found homotopy K3 surfaces with arbitrarily many [Vid01]. Earlier, Ruan had distinguished deformation classes on $X \times S^2 \cong Y \times S^2$, for non-diffeomorphic 4-manifolds X, Y, by Gromov–Witten invariants [Rua94]. The symplectic cone of a 4-manifold with $b^+ = 1$ is determined by Li–Liu [LL01]; on K3, with its standard orientation, every class of positive square is represented by a symplectic (indeed hyperkähler) form [Sal13]. For products with a circle the picture is complete:

Theorem 3.46 (Thurston; Friedl–Vidussi 2011)**.** *Let N be a closed oriented 3-manifold. Then $S^1 \times N$ admits a symplectic form if and only if N fibers over S^1.*

The forward direction, conjectured by Taubes, is [FV11]; the converse is Theorem 3.15.

Exotica. A closed symplectic 4-manifold homeomorphic but not diffeomorphic to M is an *exotic symplectic* M. After Corollary 3.21 the race went to small b_2, via rational blow-down and Luttinger surgery: J. Park's exotic $\mathbb{CP}^2\#7\overline{\mathbb{CP}}^2$ [Par05]; exotic $\mathbb{CP}^2\#3\overline{\mathbb{CP}}^2$ by Akhmedov–Park [AP08] and Baldridge–Kirk [BK08]; and exotic $\mathbb{CP}^2\#2\overline{\mathbb{CP}}^2$, symplectic and (infinitely many) non-symplectic, by Akhmedov–Park [AP10b]. As of this writing $\mathbb{CP}^2\#2\overline{\mathbb{CP}}^2$ is the smallest manifold known to carry an exotic symplectic structure, and no exotic smooth structure of any kind is known on $\mathbb{CP}^2$, $S^2 \times S^2$ or $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. (A 2010 preprint claiming an exotic $S^2 \times S^2$ [AP10a] has never appeared in print; I do not count it.)

3.6 What to read

Gompf–Stipsicz, 4-Manifolds and Kirby Calculus *[GS99]*
The four-manifold bible. Chapter 8 (elliptic and Lefschetz fibrations, monodromy), Chapter 10 (symplectic 4-manifolds: Gompf sums, Lefschetz fibrations are symplectic, fundamental groups) and Chapter 11 (Stein surfaces and Legendrian handle attachment). Learn to draw Kirby diagrams from Part 2 before you need them; you will.

McDuff–Salamon, Introduction to Symplectic Topology*, 3rd ed. [MS17]*
Chapters 6–7 (symplectic fibrations and Thurston's theorem; blow-ups and sums) for careful proofs; Chapter 13 (existence and uniqueness, Taubes–Seiberg–Witten theory, symplectic four-manifolds) for an overview; Chapter 14 for open problems. Mind their sign conventions (C3, C5).

Cieliebak–Eliashberg, From Stein to Weinstein and Back *[CE12]*
The reference. Read the Weinstein chapters and the main theorems first; consult the J-convexity chapters as needed. Their survey [CE14] is the shortcut to the main statements. Supplement with Seidel [Sei08a] for completions.

Donaldson 1996 and 1999 [Don96; Don99]
Read the 1996 paper in full: it is short, readable and contains the whole idea. Then Auroux–Smith's lectures [AS08] on pencils, branched covers and their invariants, and Auroux's stable classification [Aur05] and problem list [Aur06] for what the combinatorial approach can and cannot do.

Gompf 1995 [Gom95]
Still the best place to see what the symplectic sum can do, including the fundamental group construction.

Fintushel–Stern, Six lectures on four 4-manifolds *[FS09]*
Knot surgery, rational blow-down, and the use of Seiberg–Witten invariants as a working tool. The fastest route to the exotica literature.

T.-J. Li's surveys [Li06c; Li19]; Salamon [Sal13]
Li for the Kodaira-dimension landscape and its open problems; Salamon for uniqueness questions, with careful statements of what is known in examples.

Wendl, Holomorphic Curves in Low Dimensions *[Wen18]*
For the J-curve proof of McDuff's classification of rational and ruled manifolds (see Chapter 6).

3.7 Frontier and open problems

- Existence in dimension ≥ 6 (Problem 3.37): wide open; there is no higher-dimensional substitute for Seiberg–Witten theory.
- Exotic $\mathbb{CP}^2$'s:

  Problem 3.47. *Is every closed symplectic 4-manifold homeomorphic to $\mathbb{CP}^2$ symplectomorphic to $(\mathbb{CP}^2, c\,\omega_{\mathrm{FS}})$? Equivalently (Proposition 3.45), is there none with $K \cdot [\omega] > 0$?*

  More broadly, no exotic smooth structure is known on $\mathbb{CP}^2$, $S^2 \times S^2$ or $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$.
- Calabi–Yau surfaces:

  Conjecture 3.48 (Li)**.** *A minimal closed symplectic 4-manifold with $\kappa^s = 0$ is diffeomorphic to K3, to the Enriques surface, or to a T^2-bundle over T^2.*

  Known up to rational homology (Theorem 3.42). Even whether every degree-4 symplectic hypersurface in $\mathbb{CP}^3$ is diffeomorphic to K3 is open [Li19].
- Geography of general type:

  Conjecture 3.49 (Symplectic Bogomolov–Miyaoka–Yau)**.** *Every minimal closed symplectic 4-manifold with $\kappa^s = 2$ satisfies $K^2 \leq 3\chi$.*

  True for Kähler surfaces (Miyaoka, Yau); as of Li's survey [Li19], no non-Kähler symplectic example on the line $K^2 = 3\chi$ was known.
- Uniqueness: for $M = T^4$ or K3 and $a \in H^2(M;\mathbb{R})$ with $a^2 > 0$, is the space of symplectic forms in the class a connected? This is long-standing for T^4 [Sal13]. More basically, it is open whether there is any closed 4-manifold and class a for which the space of symplectic forms representing a is disconnected [Sal13]. Donaldson's "four-six" conjecture (closed symplectic 4-manifolds X, Y are diffeomorphic iff $X \times S^2$, $Y \times S^2$ with product forms are deformation equivalent) is wide open [Sal13].
- The symplectic isotopy problem: is every connected symplectic surface in $\mathbb{CP}^2$ symplectically isotopic to a complex curve? Known for degree ≤ 17 [ST05]; open beyond.

3.8 Logical dependencies

To state the open-manifold results one needs, in order: exact symplectic forms and Liouville fields (C5), Definition 3.1, contact forms (C6) for Proposition 3.2, generalized Morse functions and gradient-like fields for Definition 3.3, and integrable complex structures and plurisubharmonicity for Definition 3.8 and Theorem 3.9. The closed constructions need the symplectic neighborhood theorem and c_1 (Chapter 2), symplectic ball embeddings (Definition 3.11), and fiber bundles with structure group $\mathrm{Symp}(F, \sigma)$. Donaldson's and Gompf's theorems need only Poincaré duality, Dehn twists and $\mathrm{Mod}(\Sigma_g)$ to be stated. Theorem 3.20, Theorem 3.35, Definition 3.38 (via Theorem 3.39) and everything in Section 3.5 presuppose Seiberg–Witten invariants (elliptic moduli spaces, Fredholm theory, transversality, orientations) as a black box; McDuff's theorems presuppose Gromov compactness and positivity of intersections (Chapter 5).

Chapter 4

Contact geometry, and the flexible side of the subject

Contact geometry enters symplectic topology through three doors. It is the geometry of the boundary of a Liouville domain and of the ends of a Liouville manifold (Chapter 3); it is the home of Reeb dynamics and of the holomorphic curve theories of symplectizations (Chapter 9); and—the reason it gets its own chapter—it is where the subject first learned to tell rigidity from flexibility. Bennequin's 1983 proof that the standard contact structure on $\mathbb{R}^3$ has no overtwisted disk, and Eliashberg's 1989 theorem that overtwisted structures are classified by homotopy theory, set a pattern that has organized the field ever since: each geometric category comes in a flexible version, governed by an h-principle and invisible to holomorphic curves, and a rigid version, invisible to the h-principle and detected by holomorphic curves.

What to take away: the definitions and normal forms of contact topology, which you must know cold; in dimension three, the tight/overtwisted dichotomy, Legendrian invariants, open books and fillability; the h-principle as a technique, with a precise list of what satisfies one; and the post-2012 picture, in which Murphy's loose Legendrians propagated flexibility to Weinstein manifolds, to Lagrangians with boundary and, via Borman–Eliashberg–Murphy, to overtwisted contact structures in all dimensions. The slogan "flexible objects are invisible to curves" is a theorem in one direction and false in the other.

4.1 Contact manifolds

Definition 4.1 (Contact forms and structures)**.** Let Y be a smooth manifold of dimension $2n+1$, $n \geq 1$. A *contact form* is a 1-form α on Y such that $\alpha \wedge (d\alpha)^n$ vanishes nowhere. A *(cooriented) contact structure* is a hyperplane field $\xi \subset TY$ of the form $\xi = \ker \alpha$ for a contact form α, together with the coorientation in which α is positive; contact forms α, α' define the same cooriented contact structure iff $\alpha' = f\alpha$ with $f : Y \to (0, \infty)$. Equivalently, $\ker \alpha$ is contact iff $d\alpha|_\xi$ is nondegenerate; then $(\xi, d\alpha|_\xi)$ is a symplectic vector bundle whose conformal class depends only on ξ, and $c_1(\xi)$ denotes its first Chern class. Since $(f\alpha) \wedge (d(f\alpha))^n = f^{n+1}\alpha \wedge (d\alpha)^n$, ξ orients Y; if Y is already oriented, ξ is *positive* when the orientations agree. A *contactomorphism* $(Y, \xi) \to (Y', \xi')$ is a diffeomorphism φ with $d\varphi(\xi) = \xi'$, preserving coorientations; $\mathrm{Cont}(Y, \xi)$ is the group of contactomorphisms of (Y, ξ).

In this chapter contact structures are cooriented and, on oriented 3-manifolds, positive.

Remark 4.2 (Non-coorientable structures). In general a contact structure need only be *locally* the kernel of a contact form; the space $\mathbb{P}(T^*Q)$ of unoriented contact elements of a manifold Q is the basic non-coorientable example, with coorienting double cover the unit cotangent bundle. Since $(-\alpha)\wedge(-d\alpha)^n = (-1)^{n+1}\alpha\wedge(d\alpha)^n$, contact manifolds of dimension $4k+3$ are orientable, while in dimension $4k+1$ a coorientation is the same as an orientation.

Definition 4.3 (Reeb vector field)**.** The *Reeb vector field* of a contact form α is the unique vector field R_α with $\alpha(R_\alpha)=1$ and $\iota_{R_\alpha}d\alpha=0$ (C6). Its flow preserves α. It depends on α, not only on $\ker\alpha$; its dynamics is the subject of Chapter 9.

Convention 4.4 (The standard contact space)**.** On $\mathbb{R}^{2n+1}$ with coordinates $(x_1,\dots,x_n,y_1,\dots,y_n,z)$ we use

$$\alpha_0 = dz - \textstyle\sum_{j=1}^n y_j\,dx_j, \qquad \xi_0=\ker\alpha_0, \qquad R_{\alpha_0}=\partial_z.$$

This choice matches C1 and C5: $d\alpha_0=\sum_j dx_j\wedge dy_j$ is the pull-back of ω_0 under $(x,y,z)\mapsto(x,y)$, and with $q=x$, $p=y$ the form α_0 is $dz-\lambda_{\text{can}}$ on the 1-jet space $J^1\mathbb{R}^n = T^*\mathbb{R}^n\times\mathbb{R}_z$, so that 1-jet graphs $j^1f=\{(q,df(q),f(q))\}$ are Legendrian. Other normal forms are related to it by explicit contactomorphisms: $(x,y,z)\mapsto(x,y,z+\sum_j x_jy_j)$ pulls α_0 back to $dz+\sum_j x_j\,dy_j$, and $(x,y,z)\mapsto(x,y,z+\frac12\sum_j x_jy_j)$ pulls it back to $dz+\lambda_0$ with λ_0 as in C1. In dimension three, $\ker(dz-y\,dx)$ is the convention of most of the Legendrian knot literature [Etn05; EN20].

Example 4.5 (The standard sphere)**.** Let $n\geq 2$ and $S^{2n-1}\subset\mathbb{C}^n$ the unit sphere. The radial field $Z=\frac12\sum_j(x_j\partial_{x_j}+y_j\partial_{y_j})$ satisfies $\iota_Z\omega_0=\lambda_0$, so it is Liouville; it is transverse to S^{2n-1}, so $\alpha_{\text{st}}=\lambda_0|_{S^{2n-1}}$ is a contact form (Proposition 4.9), and $\xi_{\text{st}}=\ker\alpha_{\text{st}}=TS^{2n-1}\cap J_0TS^{2n-1}$ is the field of complex tangencies. In coordinates $z_j=r_je^{i\theta_j}$ one has $\lambda_0=\frac12\sum_j r_j^2\,d\theta_j$, so on $\{\sum_j r_j^2=1\}$

$$R_{\alpha_{\text{st}}}=2\textstyle\sum_j\partial_{\theta_j}, \qquad \text{with flow } z\mapsto e^{2it}z.$$

All Reeb orbits are closed: they are the Hopf circles $S^{2n-1}\cap\mathbb{C}v$, each of minimal period π, which is also its action $\int\alpha_{\text{st}}=\pi$, the area of the unit disk and the Gromov width of the unit ball (Chapter 6). For every $p\in S^{2n-1}$, $(S^{2n-1}\setminus\{p\},\xi_{\text{st}})$ is contactomorphic to $(\mathbb{R}^{2n-1},\xi_0)$ [Gei08].

Theorem 4.6 (Darboux theorem for contact forms)**.** *Let α be a contact form on Y^{2n+1} and $p\in Y$. There are coordinates (x,y,z) on a neighbourhood of p, centred at p, in which $\alpha=dz-\sum_j y_j\,dx_j$.*

Theorem 4.7 (Gray stability; Gray 1959)**.** *Let Y be a closed manifold and $(\xi_t)_{t\in[0,1]}$ a smooth family of contact structures on Y. Then there is a smooth isotopy $(\psi_t)_{t\in[0,1]}$ of Y with $\psi_0=\mathrm{id}$ and $d\psi_t(\xi_0)=\xi_t$ for all t.*

Proof idea. Pick forms α_t and look for ψ_t generated by $X_t\in\xi_t$ with $\psi_t^*\alpha_t=\mu_t\alpha_0$. Differentiating gives $\dot\alpha_t+\iota_{X_t}d\alpha_t=h_t\alpha_t$; evaluating on R_{α_t} forces $h_t=\dot\alpha_t(R_{\alpha_t})$, and then $X_t\in\xi_t$ is uniquely determined by nondegeneracy of $d\alpha_t|_{\xi_t}$. Compactness lets one integrate [Gra59; Gei08].

Unlike Moser's theorem (Chapter 2), Gray needs no cohomological hypothesis: contact structures have no deformation moduli. But Gray says nothing about whether contact structures that are merely *homotopic as plane fields* are isotopic, and that is where the rigidity of the subject lives. There is no Gray theorem for contact forms.

Definition 4.8 (Symplectization; contact-type hypersurfaces)**.** The *symplectization* of $(Y,\xi=\ker\alpha)$ is $(\mathbb{R}\times Y, d(e^r\alpha))$, with Liouville field ∂_r. For $g\colon Y\to\mathbb{R}$ the map $(r,y)\mapsto(r+g(y),y)$ is a

symplectomorphism from the symplectization of $e^g\alpha$ to that of α, so the symplectization is an invariant of (Y,ξ). Following C6, a compact hypersurface Y in a symplectic manifold (M^{2n+2},ω) is of *contact type* if there is a Liouville vector field Z ($\mathcal{L}_Z\omega=\omega$) defined near Y and transverse to it.

Proposition 4.9 (Collars)**.** *Let $Y\subset(M,\omega)$ be a compact hypersurface of contact type with Liouville field Z, and put $\lambda=\iota_Z\omega$. Then $\alpha=\lambda|_{TY}$ is a contact form; for small $\varepsilon>0$ the map $(r,y)\mapsto\phi_r^Z(y)$ is a symplectomorphism from $((-\varepsilon,\varepsilon)\times Y, d(e^r\alpha))$ onto a neighbourhood of Y; and the characteristic line field $\ker(\omega|_{TY})$ is spanned by R_α. If $Y=\partial W$ and Z points out of W, the contact orientation of Y is its boundary orientation.*

I find it clarifying to say that contact topology *is* the symplectic topology of conical objects: contactomorphisms of Y are the symplectomorphisms of $\mathbb{R}\times Y$ commuting with translation in r (the Liouville flow), Legendrians Λ correspond to the conical Lagrangians $\mathbb{R}\times\Lambda$, and contact Hamiltonians to Hamiltonians homogeneous of degree one. Every Floer-type invariant of a contact manifold is built from holomorphic curves in a symplectization or a completed filling (Chapters 8 and 9).

Definition 4.10 (Isotropic and Legendrian submanifolds)**.** A submanifold $\Lambda\subset(Y^{2n+1},\xi)$ is *isotropic* if $T\Lambda\subset\xi$. Then $d\alpha|_{T\Lambda}=0$, so $T\Lambda$ is isotropic in $(\xi,d\alpha|_\xi)$ and $\dim\Lambda\le n$; Λ is *Legendrian* if $\dim\Lambda=n$. A *Legendrian isotopy* is a smooth isotopy through Legendrian submanifolds; for closed Λ it extends to an ambient contact isotopy.

Theorem 4.11 (Legendrian neighbourhood theorem)**.** *Let Λ be a closed Legendrian submanifold of (Y,ξ). There are a neighbourhood U of Λ and a contactomorphism from $(U,\xi|_U)$ onto a neighbourhood of the zero section $\{p=0,\,z=0\}$ in $(J^1\Lambda=T^*\Lambda\times\mathbb{R}_z,\ \ker(dz-\lambda_{\mathrm{can}}))$ mapping Λ to the zero section. More generally, the germ of ξ along a closed isotropic submanifold Λ is determined by its conformal symplectic normal bundle $(T\Lambda)^{\perp_{d\alpha}}/T\Lambda\subset\xi|_\Lambda/T\Lambda$.*

Legendrians C^1-close to the zero section of $J^1\Lambda$ are the graphs j^1f. The *front projection* $J^1\Lambda\to\Lambda\times\mathbb{R}_z$ maps Legendrians to singular hypersurfaces from which they are recovered by $p=\partial z/\partial q$, so Legendrians can be drawn.

Definition 4.12 (Almost contact structures)**.** An *almost contact structure* on Y^{2n+1} is a pair (ξ,J) of a cooriented hyperplane field ξ and a complex structure J on the bundle ξ; equivalently, a reduction of the structure group of TY to $\mathrm{U}(n)\times 1$. A contact structure determines an almost contact structure, unique up to homotopy, by choosing J compatible with $d\alpha|_\xi$. On an oriented 3-manifold, homotopy classes of almost contact structures are homotopy classes of cooriented 2-plane fields.

4.2 Legendrian and transverse knots in dimension three

In $(\mathbb{R}^3,\xi_0=\ker(dz-y\,dx))$ a Legendrian knot L has two useful projections: the *front* (x,z), a closed curve with semicubical cusps and no vertical tangencies from which L is recovered by $y=dz/dx$; and the *Lagrangian projection* (x,y), an immersed curve enclosing zero signed area whose double points are the Reeb chords of L. Every knot type has Legendrian representatives.

Definition 4.13 (Classical invariants)**.** Let L be an oriented Legendrian knot in $(\mathbb{R}^3,\xi_0)$. The *Thurston–Bennequin invariant* is $\mathrm{tb}(L)=\mathrm{lk}(L,L+\varepsilon\partial_z)$ for small $\varepsilon>0$. The *rotation number* $\mathrm{rot}(L)$ is

the winding number of $\dot{L}$ with respect to the trivialization $(\partial_x + y\partial_z,\ \partial_y)$ of ξ_0, which is positively oriented for $d\alpha_0|_{\xi_0}$. For a generic front F of L,

$$\mathrm{tb}(L) = w(F) - \tfrac{1}{2}\#\{\text{cusps}\}, \qquad \mathrm{rot}(L) = \tfrac{1}{2}\big(\#\{\text{down cusps}\} - \#\{\text{up cusps}\}\big),$$

where $w(F)$ is the writhe and a cusp is *down* (*up*) if L traverses it in the direction of decreasing (increasing) z. For a null-homologous Legendrian knot L in a contact 3-manifold (Y, ξ) with Seifert surface Σ, $\mathrm{tb}(L)$ is the twisting of the framing of L by ξ relative to the framing by Σ, and $\mathrm{rot}(L, \Sigma)$ is the winding number of $\dot{L}$ relative to a trivialization of $\xi|_\Sigma$; it depends on Σ only through its relative homology class.

Reversing the orientation of L negates rot and preserves tb. *Stabilization*, adding a zigzag to the front, changes $(\mathrm{tb}, \mathrm{rot})$ by $(-1, \pm 1)$, so tb is unbounded below in every knot type. It is bounded above:

Theorem 4.14 (Bennequin's inequality; Bennequin 1983)**.** *For every oriented Legendrian knot $L \subset (\mathbb{R}^3, \xi_0)$ with Seifert surface Σ, $\mathrm{tb}(L) + |\mathrm{rot}(L)| \le -\chi(\Sigma)$. Hence $\mathrm{tb}(L) + |\mathrm{rot}(L)| \le 2g(K) - 1$, where $g(K)$ is the Seifert genus of the knot type K of L.*

Bennequin [Ben83] proved the transverse form (below) using braid foliations of Seifert surfaces; Eliashberg's proof, via the characteristic foliation of Σ, is the one that generalizes (Theorem 4.24). A knot type is *Legendrian simple* if its Legendrian representatives are classified up to Legendrian isotopy by $(\mathrm{tb}, \mathrm{rot})$; the unknot [EF09], torus knots and the figure-eight knot [EH01a] are simple. Not all are:

Theorem 4.15 (Chekanov 2002)**.** *There exist Legendrian knots $L_1, L_2 \subset (\mathbb{R}^3, \xi_0)$, both of smooth knot type $m(5_2)$ and both with $\mathrm{tb} = 1$ and $\mathrm{rot} = 0$, which are not Legendrian isotopic.*

The invariant is the *Chekanov–Eliashberg DGA* [Che02; Eli98]. For L with generic Lagrangian projection, $\mathcal{A}(L)$ is the unital noncommutative algebra over $\mathbb{Z}/2$ freely generated by the Reeb chords of L, graded in $\mathbb{Z}/2\,\mathrm{rot}(L)$ by a Maslov-type index; ∂a counts immersed polygons in the (x, y)-plane with boundary on the projection of L, convex corners at chords, one positive corner at a and negative corners $b_1, \dots, b_k$ in order, each contributing $b_1 \cdots b_k$. Then $\partial^2 = 0$, $\deg \partial = -1$, and the stable tame isomorphism type of $(\mathcal{A}(L), \partial)$ is a Legendrian isotopy invariant; Chekanov's knots are distinguished by linearized homologies. Eliashberg identified the polygons with rigid holomorphic disks in $\mathbb{R} \times \mathbb{R}^3$ with boundary on $\mathbb{R} \times L$, a piece of symplectic field theory; Ekholm–Etnyre–Sullivan built the analytic theory in $\mathbb{R}^{2n+1}$ and found infinite families of non-isotopic Legendrians with the same classical invariants [EES05a; EES05b; EES07]. In my opinion the DGA in $\mathbb{R}^3$ is the best first exposure to SFT: every curve can be seen. That augmentations correspond to constructible sheaves [Ng+20] belongs to Chapter 8; SFT foundations to Chapter 9.

A knot T is *(positively) transverse* if $\alpha(\dot{T}) > 0$. Its *self-linking number* $\mathrm{sl}(T, \Sigma)$ is the linking number of T with its push-off along a nowhere-zero section of $\xi|_\Sigma$. A Legendrian L has transverse push-offs $L_\pm$ with $\mathrm{sl}(L_\pm) = \mathrm{tb}(L) \mp \mathrm{rot}(L)$, and transverse knots up to transverse isotopy correspond to Legendrian knots up to Legendrian isotopy and negative stabilization [Etn05]. Bennequin's original inequality is $\mathrm{sl}(T, \Sigma) \le -\chi(\Sigma)$, proved after showing that every transverse link in $\mathbb{R}^3$ is transversely isotopic to a closed braid.

4.3 Open books and contact surgery

Definition 4.16 (Open books; supported contact structures)**.** An *open book decomposition* of a closed oriented manifold Y^{2n+1} is a pair (B, π) of a closed codimension-two submanifold $B \subset Y$ with trivial normal bundle (the *binding*) and a smooth fibre bundle $\pi\colon Y \setminus B \to S^1$ which, on a neighbourhood $B \times D^2$ of $B = B \times \{0\}$, is $(b, re^{i\theta}) \mapsto \theta$. The *pages* are the closures $P_\theta = \overline{\pi^{-1}(\theta)}$, compact with $\partial P_\theta = B$, cooriented by $\pi^* d\theta$ and oriented compatibly with Y; B is oriented as ∂P_θ. A contact structure ξ on Y is *supported* by (B, π) if $\xi = \ker \alpha$ for a contact form α such that $\alpha|_{TB}$ is a positive contact form on B and $d\alpha$ restricts to a positive symplectic form on the interior of every page.

An open book is determined up to diffeomorphism by its abstract data: a page P and a *monodromy* $\phi \in \mathrm{Diff}(P)$ equal to the identity near ∂P, up to isotopy rel boundary. In dimension three, a *positive stabilization* replaces (P, ϕ) by $(P \cup h,\ \phi \circ \tau_c)$, where h is a 1-handle attached along ∂P, $c \subset P \cup h$ is a simple closed curve meeting the cocore of h transversely once, and τ_c is the right-handed Dehn twist along c.

Theorem 4.17 (Thurston–Winkelnkemper 1975)**.** *Every open book decomposition of a closed oriented 3-manifold supports a positive contact structure.*

Since every closed oriented 3-manifold has an open book (Alexander), this reproves Martinet's 1971 theorem that every closed oriented 3-manifold is contact [Mar71; TW75]; Lutz twists along transverse knots then reach every homotopy class of plane fields [Gei08].

Theorem 4.18 (Giroux correspondence in dimension three; Giroux 2002)**.** *Let Y be a closed oriented 3-manifold.*

(a) Every open book on Y supports a positive contact structure, and any two contact structures supported by the same open book are isotopic.

(b) Every positive contact structure on Y is supported by some open book.

(c) Two open books supporting isotopic contact structures become isotopic after finitely many positive stabilizations of each.

Hence "supports" induces a bijection between open books on Y modulo isotopy and positive stabilization, and positive contact structures on Y modulo isotopy.

Giroux's ICM address [Gir02] gave the strategy; (a) and (b) are written out in [Etn06]. For the hard half (c) I point students to Breen–Honda–Huang's proof in all dimensions via convex hypersurface theory [BHH23] and to Licata–Scharitzer–Vértesi's three-dimensional proof via contact Heegaard splittings [LSV24], both preprints. In my view this is the most useful structural theorem of contact 3-manifold theory, because it turns contact structures into mapping classes: a contact 3-manifold is Stein fillable iff it is supported by *some* open book whose monodromy is a product of right-handed Dehn twists [Gir02; LP01; AO01]; see [OS04].

Remark 4.19 (Higher dimensions). In dimension $2n + 1 \geq 5$ the right objects are *Weinstein open books*: the page is a Weinstein domain P^{2n} and the monodromy an exact symplectomorphism equal to the identity near ∂P. Positive stabilization attaches a Weinstein n-handle to P along the Legendrian boundary of a properly embedded Lagrangian disk $D \subset P$ and composes the monodromy with the positive Dehn–Seidel twist along the Lagrangian sphere formed by D and the core. Giroux announced in 2002 that every closed contact manifold is supported by a

Weinstein open book, unique up to positive stabilization; existence via convex hypersurfaces is in [Bre+19], the full correspondence in [BHH23] (preprints). Every Weinstein domain admits a Weinstein Lefschetz fibration [GP17] (Chapter 3).

Theorem 4.20 (Weinstein handles; Weinstein 1991, Eliashberg 1990)**.** *Let (W^{2n}, λ) be a Liouville domain with contact boundary (Y, ξ), let $1 \le k \le n$, and let $\Lambda \subset Y$ be an isotropic $(k-1)$-sphere together with a trivialization of its conformal symplectic normal bundle (no datum if $k = n$). Then the manifold W' obtained by attaching a k-handle along Λ, with the framing determined by these data, carries a Liouville form extending λ whose Liouville field is gradient-like for a function with exactly one new critical point, of index k, whose stable manifold is an isotropic disk with boundary Λ; W' is Weinstein if W is. If $2n = 4$ and Λ is null-homologous in Y, the 2-handle is attached with framing $\mathrm{tb}(\Lambda) - 1$ relative to the Seifert framing.*

The new boundary is obtained from Y by *contact surgery* along Λ; for $k = n$ this is *Legendrian surgery* [Wei91; Eli90b]. Legendrian surgery preserves weak, strong, exact and Stein fillability, and on closed 3-manifolds it preserves tightness [Wan15]. Ding–Geiges contact (± 1)-surgery is the three-dimensional calculus: every closed contact 3-manifold is obtained from (S^3, ξ_{st}) by contact (± 1)-surgery along a Legendrian link [DG04].

4.4 Tight versus overtwisted

4.4.1 Dimension three

Definition 4.21 (Overtwisted disks; tightness)**.** Let (Y, ξ) be a contact 3-manifold. An *overtwisted disk* is an embedded closed 2-disk $D \subset Y$ with $T_pD = \xi_p$ for every $p \in \partial D$. ξ is *overtwisted* if it contains an overtwisted disk, and *tight* otherwise.

Thus ∂D is a Legendrian unknot with $\mathrm{tb} = 0$, precisely what Theorem 4.14 forbids in $\mathbb{R}^3$. The model is $\ker(\cos r\, dz + r \sin r\, d\theta)$ on $\mathbb{R}^3$, with overtwisted disk $\{z = 0,\ r \le \pi\}$.

Theorem 4.22 (Eliashberg 1989)**.** *Let Y be a closed connected oriented 3-manifold, $\Delta \subset Y$ an embedded closed disk, and ζ a germ of positive contact structure along Δ for which Δ is an overtwisted disk. Let $\mathrm{Cont}(Y; \zeta)$ be the space of positive contact structures on Y that agree with ζ near Δ, and $\mathrm{Distr}(Y; \zeta)$ the space of cooriented 2-plane fields that agree with ζ near Δ, both with the C^∞ topology. Then the inclusion $\mathrm{Cont}(Y; \zeta) \hookrightarrow \mathrm{Distr}(Y; \zeta)$ is a weak homotopy equivalence. Consequently every homotopy class of cooriented 2-plane fields on Y contains an overtwisted contact structure, and two overtwisted contact structures on Y are isotopic iff they are homotopic as plane fields.*

Remark 4.23 (The fixed disk matters). Vogel proved two results attributed to Chekanov: the contact mapping class group $\pi_0 \mathrm{Cont}(S^3, \xi)$ of an overtwisted ξ on S^3 is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ for exactly one isotopy class (the only one admitting non-loose unknots) and $\mathbb{Z}/2$ for all others, and the identity component of $\mathrm{Cont}(S^3, \xi)$ need not act transitively on boundaries of overtwisted disks [Vog18]. Without a fixed disk, rigidity survives on π_0.

Theorem 4.24 (Bennequin 1983; Eliashberg 1992)**.** *(a) $(\mathbb{R}^3, \xi_0)$ and (S^3, ξ_{st}) are tight.*

(b) Every positive tight contact structure on S^3 is isotopic to ξ_{st}, and every tight contact structure on $\mathbb{R}^3$ is contactomorphic to ξ_0.

(c) *Let (Y, ξ) be tight. Then* $\mathrm{tb}(L)+|\mathrm{rot}(L,\Sigma)| \le -\chi(\Sigma)$ *for every oriented null-homologous Legendrian knot L with Seifert surface Σ;* $|\langle c_1(\xi), [S]\rangle| \le -\chi(S)$ *for every closed oriented embedded surface $S \neq S^2$; and* $\langle c_1(\xi), [S]\rangle = 0$ *for every embedded sphere S.*

Part (a) is Bennequin's [Ben83]; (b) and (c) are Eliashberg's [Eli92], whose method—simplify characteristic foliations of surfaces, then use uniqueness of tight structures on the ball with fixed boundary germ—is the template for everything after. The modern tools are Giroux's convex surfaces [Gir00], whose germs are encoded by dividing sets, and Honda's bypasses [Hon00]. Representative classifications: if $p > q > 0$ are coprime and $-p/q = [r_0, \dots, r_k]$ is the continued fraction with all $r_i \le -2$, then $L(p,q)$ carries exactly $|(r_0+1)(r_1+1)\cdots(r_k+1)|$ tight contact structures up to isotopy, all Stein fillable [Hon00; Gir00]—so $p-1$ on $L(p,1)$; on $T^3 = \mathbb{R}^3/\mathbb{Z}^3$ every tight structure is contactomorphic to exactly one $\xi_m = \ker(\cos(2\pi mz)\,dx + \sin(2\pi mz)\,dy)$, $m \ge 1$ [Kan97; Gir00].

Theorem 4.25 (Colin–Giroux–Honda 2009)**.** *Let Y be a closed oriented 3-manifold. Only finitely many homotopy classes of plane fields on Y contain tight contact structures. If Y is atoroidal, i.e. contains no incompressible embedded torus, then Y carries only finitely many tight contact structures up to isotopy.*

Conversely, closed irreducible toroidal 3-manifolds carry infinitely many (Colin [Col01]; Honda–Kazez–Matić [HKM02]), so for irreducible Y finiteness is equivalent to atoroidality [CGH09]. Existence fails in general:

Theorem 4.26 (Etnyre–Honda 2001)**.** *The Poincaré homology sphere with reversed orientation, $-\Sigma(2,3,5)$, carries no positive tight contact structure.*

This was the first closed 3-manifold known to carry none [EH01b]; Lisca–Stipsicz later determined exactly which closed Seifert fibred 3-manifolds carry positive tight structures [LS09].

Definition 4.27 (Symplectic fillings)**.** Let (Y^3, ξ) be a closed contact 3-manifold and (W, ω) a compact symplectic 4-manifold, oriented by ω^2, with $\partial W = Y$ as oriented manifolds. Then (W, ω) is

(i) a *weak filling* if $\omega|_\xi > 0$, i.e. ω restricts to each plane ξ_y as a positive area form for the orientation of ξ by $d\alpha|_\xi$;

(ii) a *strong filling* if there is a Liouville field Z defined near ∂W, pointing outward, with $\xi = \ker(\iota_Z\omega|_{TY})$;

(iii) an *exact filling* if such a Z exists on all of W, i.e. $(W, \iota_Z\omega)$ is a Liouville domain with contact boundary (Y, ξ);

(iv) a *Stein filling* if W carries a Weinstein (equivalently, up to deformation, Stein) domain structure with contact boundary (Y, ξ) (Chapter 3).

Definitions (ii)–(iv) make sense in all dimensions; the correct higher-dimensional notion of weak filling is due to Massot–Niederkrüger–Wendl [MNW13].

Theorem 4.28 (The fillability hierarchy)**.** *For closed contact 3-manifolds, Stein fillable $\Rightarrow$ exactly fillable $\Rightarrow$ strongly fillable $\Rightarrow$ weakly fillable $\Rightarrow$ tight. The last implication is due to Gromov and Eliashberg [Gro85; Eli90a]. Moreover:*

(a) *there are tight contact structures with no weak filling, e.g. on the Seifert fibred space over S^2 with invariants $(-\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ [EH02];*

(b) *the structures ξ_m, $m \geq 2$, on T^3 are weakly but not strongly fillable [Eli96];*

(c) *there are strongly fillable contact 3-manifolds without Stein fillings [Ghi05], and exactly fillable ones without Stein fillings [Bow12].*

Theorem 4.29 (Fillings of S^3 and $L(p,1)$; Gromov 1985; Eliashberg, McDuff 1990)**.** *Fillings are scarce:*

(a) *Every weak symplectic filling of (S^3, ξ_{st}) is diffeomorphic to a blow-up $B^4 \# k\,\overline{\mathbb{CP}}^2$ of the ball for some $k \geq 0$.*

(b) *Let $p \geq 2$ and let ξ_{st} on $L(p,1) = S^3/(\mathbb{Z}/p)$, the quotient by the diagonal action, be induced from ξ_{st}. Every strong symplectic filling of $(L(p,1), \xi_{\mathrm{st}})$ is diffeomorphic to a blow-up of the disk bundle D_{-p} of Euler number $-p$ over S^2 if $p \neq 4$; if $p = 4$, to a blow-up of D_{-4} or of the rational homology ball $\mathbb{CP}^2 \setminus \nu(C)$, C a smooth conic.*

Proof idea. Cap the filling off by a concave neighbourhood of a symplectic sphere of self-intersection $+1$ (for S^3) or $+p$ (for $L(p,1)$), and apply McDuff's classification of closed symplectic 4-manifolds containing such a sphere [McD90; Wen18]; Eliashberg's alternative for (a) fills W by holomorphic disks [Eli90a].

Remark 4.30 (Floer-theoretic contact invariants). Ozsváth–Szabó's contact class $c(\xi) \in \widehat{\mathrm{HF}}(-Y)$ vanishes if ξ is overtwisted and is nonzero if ξ is Stein fillable [OS05]; Kronheimer–Mrowka's monopole invariant [KM97] and the ECH contact class (Chapter 9) play the same role. These are the main tools for proving tightness and non-fillability today; Ghiggini's examples in Theorem 4.28(c) are detected by $c(\xi)$.

4.4.2 Higher dimensions

For decades there was no agreed notion of overtwistedness in dimension ≥ 5. Niederkrüger's *plastikstufe* [Nie06] was the first candidate: for a closed $(n-1)$-manifold S, an embedding of $S \times D^2$ into Y^{2n+1} whose characteristic foliation is singular along $S \times \{0\}$, with Legendrian leaves $S \times$(rays) and closed Legendrian leaf $S \times \partial D^2$. Holomorphic disks with boundary on it obstruct semipositive fillings [Nie06; MNW13]. The definitive notion is due to Borman–Eliashberg–Murphy.

Definition 4.31 (Overtwisted, all dimensions; Borman–Eliashberg–Murphy 2015)**.** Borman, Eliashberg and Murphy construct an explicit germ $(D^{2n}_{\mathrm{OT}}, \xi_{\mathrm{OT}})$ of contact structure along a $2n$-disk [BEM15, Sect. 3]. A contact manifold (Y^{2n+1}, ξ) is *overtwisted* if it admits a contact embedding of this germ, and *tight* otherwise. For $n = 1$ this agrees with Definition 4.21.

Theorem 4.32 (Borman–Eliashberg–Murphy 2015)**.** *Let Y be a closed manifold of dimension $2n+1 \geq 3$.*

(a) *Every almost contact structure on Y is homotopic, through almost contact structures, to an overtwisted contact structure.*

(b) *Two overtwisted contact structures on Y that are homotopic as almost contact structures are isotopic.*

More precisely, the inclusion of the space of contact structures that agree with ξ_{OT} near a fixed embedded D_{OT} into the corresponding space of almost contact structures is a weak homotopy equivalence; relative versions hold.

Note the logic: the existence of contact structures on closed manifolds, a question that sounds rigid, was settled by flexibility. Before [BEM15] existence was known in dimension three

(Martinet, Lutz), for simply connected 5-manifolds [Gei91], and for all closed almost contact 5-manifolds [CPP15].

Theorem 4.33 (Casals–Murphy–Presas 2019)**.** *Let (Y^{2n+1}, ξ) be a contact manifold, $n \geq 2$. The following are equivalent: (i) ξ is overtwisted; (ii) the standard Legendrian unknot in a Darboux ball of Y is loose in Y (Definition 4.42); (iii) Y contains a small plastikstufe with spherical core and trivial rotation, in the sense of [CMP19]. If Y is closed, these are also equivalent to: (iv) ξ is supported by a negatively stabilized Weinstein open book.*

By (iii), overtwisted manifolds have no semipositive fillings, and their Reeb flows always have contractible closed orbits: Hofer in dimension three [Hof93b], Albers–Hofer for manifolds containing a plastikstufe [AH09] (Chapter 9). Tightness in high dimensions is far less structured than in dimension three: for $n \geq 3$, every $(2n+1)$-manifold carrying a tight contact structure carries a tight non-fillable one in the same almost contact class, and for $n = 2$ the same holds when $c_1 = 0$ [Bow+26].

4.5 The h-principle

Definition 4.34 (Differential relations; h-principle)**.** Let $X \to V$ be a smooth fibre bundle and $J^r X \to V$ its bundle of r-jets of sections. A *differential relation* of order r is a subset $\mathcal{R} \subset J^r X$, *open* if it is open. A *formal solution* is a continuous section $F\colon V \to J^r X$ with $F(V) \subset \mathcal{R}$; a *(genuine) solution* is a smooth section f of X with $j^r f(V) \subset \mathcal{R}$, and F is *holonomic* if $F = j^r f$. $\mathcal{R}$ satisfies the *h-principle* if every formal solution is homotopic through formal solutions to a holonomic one, and the *parametric h-principle* if $f \mapsto j^r f$ is a weak homotopy equivalence from the space of solutions (C^∞ topology) to the space of formal solutions (C^0 topology).

The engine of modern h-principle proofs, to be learned before anything in Gromov's book, is:

Theorem 4.35 (Holonomic approximation; Eliashberg–Mishachev 2001)**.** *Fix metrics on V and X. Let $A \subset V$ be a polyhedron with $\dim A < \dim V$, and F a section of $J^r X$ over a neighbourhood of A. For all $\varepsilon, \delta > 0$ there are a diffeotopy $h^\tau\colon V \to V$, $\tau \in [0,1]$, with $\operatorname{dist}_{C^0}(h^\tau, \mathrm{id}) < \delta$, and a holonomic section $\widetilde{F} = j^r f$ over a neighbourhood U of $h^1(A)$ such that $\operatorname{dist}_{C^0}(\widetilde{F}, F) < \varepsilon$ on U.*

One cannot approximate a formal section by holonomic ones on all of V, but one can after wiggling a positive-codimension skeleton [EM02]. Since an open manifold retracts by isotopy into a neighbourhood of a positive-codimension polyhedron, this gives:

Theorem 4.36 (Gromov 1969)**.** *Let V be an open manifold (no component is closed), $X \to V$ a natural fibre bundle (one to which diffeomorphisms of V lift, e.g. $\Lambda^k T^*V$), and $\mathcal{R} \subset J^r X$ an open differential relation invariant under the induced action of $\mathrm{Diff}(V)$. Then $\mathcal{R}$ satisfies the parametric h-principle.*

Theorem 4.37 (Symplectic forms on open manifolds; Gromov 1969)**.** *Let V be an open $2n$-manifold and $a \in H^2(V; \mathbb{R})$. The inclusion of the space of symplectic forms ω on V with $[\omega] = a$ into the space of nondegenerate 2-forms on V is a weak homotopy equivalence. In particular V admits a symplectic form in the class a iff it admits an almost complex structure.*

Similarly the inclusion of contact structures into almost contact structures on an open manifold is a weak homotopy equivalence [Gro69; EM02]. On closed manifolds both statements fail (Chapter 3): S^6 is almost complex but not symplectic, and Taubes's theory shows that $\mathbb{CP}^2 \# \mathbb{CP}^2 \# \mathbb{CP}^2$ is not symplectic either.

Definition 4.38 (Formal Lagrangian immersions)**.** Let (M^{2n}, ω) be symplectic and L an n-manifold. A *formal Lagrangian immersion* is a pair (f, F) of a continuous map $f : L \to M$ and a fibrewise injective bundle map $F : TL \to f^*TM$ with $F(T_qL)$ Lagrangian for every $q \in L$.

Theorem 4.39 (Lagrangian immersions; Gromov, Lees 1976)**.** *(a) Let $(M, \omega = d\lambda)$ be exact. The map $f \mapsto (f, df)$ from the space of exact Lagrangian immersions $f : L \to M$ (those with $f^*\lambda$ exact) to the space of formal Lagrangian immersions is a weak homotopy equivalence.*

(b) A closed n-manifold L admits a Lagrangian immersion into $(\mathbb{C}^n, \omega_0)$ iff the complex vector bundle $TL \otimes \mathbb{C}$ is trivial. For instance S^n, T^n and all closed orientable 3-manifolds do.

For (b): a formal Lagrangian immersion into $\mathbb{C}^n$ is the same as a complex trivialization of $TL \otimes \mathbb{C}$, up to the deformation retraction $\mathrm{GL}(n, \mathbb{C})/\mathrm{GL}(n, \mathbb{R}) \simeq \mathrm{U}(n)/\mathrm{O}(n)$ [Lee76; Gro86].

Warning 4.40 (Immersions versus embeddings). Theorem 4.39 is false for embeddings: there is no closed exact Lagrangian submanifold of $\mathbb{C}^n$ (Gromov 1985, Chapter 6), whereas S^n has an exact Lagrangian immersion with one double point. Gromov's theorem, more than any other, marks the line between flexibility and rigidity, and it is proved with holomorphic curves.

4.6 Loose Legendrians and flexible Weinstein manifolds

Definition 4.41 (Formal Legendrian embeddings)**.** Let (Y^{2n+1}, ξ) be a contact manifold and Λ a closed n-manifold. A *formal Legendrian embedding* is a pair $(f, (F_s)_{s\in[0,1]})$ of a smooth embedding $f : \Lambda \to Y$ and a homotopy of fibrewise injective bundle maps $F_s : T\Lambda \to f^*TY$ with $F_0 = df$ and $F_1(T\Lambda)$ a Lagrangian subbundle of $(f^*\xi, d\alpha|_\xi)$. A *formal Legendrian isotopy* is a continuous path of formal Legendrian embeddings. A Legendrian embedding is formal with $F_s \equiv df$.

Definition 4.42 (Loose charts and loose Legendrians; Murphy 2012)**.** Let $n \geq 2$. Let $C \subset (\mathbb{R}^3, \ker(dz - y_1 dx_1))$ be a cube of side length 1 and $\Lambda_0 \subset C$ a properly embedded Legendrian arc whose front is a single zigzag. For $\rho > 0$ let $V_\rho = \{(q, p) \in T^*\mathbb{R}^{n-1} : |q| < \rho,\ |p| < \rho\}$ and $Z_\rho = V_\rho \cap \{p = 0\}$, and give $\mathbb{R}^3 \times T^*\mathbb{R}^{n-1} = \mathbb{R}^{2n+1}$ the form $dz - y_1 dx_1 - \sum_{j=2}^n p_j\, dq_j$, which is α_0 of Convention 4.4. A *loose chart* for a Legendrian $\Lambda \subset (Y^{2n+1}, \xi)$ is an open set $U \subset Y$ such that $(U, U \cap \Lambda)$ is contactomorphic to $(C \times V_\rho, \Lambda_0 \times Z_\rho)$ for some $\rho > 1$. Λ is *loose* if it has a loose chart. A Legendrian link is loose if each component is loose in the complement of the others.

The condition $\rho > 1$, a lower bound on the width of the chart in the $T^*\mathbb{R}^{n-1}$ directions relative to the zigzag, is essential.

Theorem 4.43 (Murphy 2012)**.** *Let (Y^{2n+1}, ξ) be a contact manifold with $n \geq 2$, and Λ a closed n-manifold. If two loose Legendrian embeddings $\Lambda \to Y$ are formally Legendrian isotopic, then they are Legendrian isotopic. Every formal Legendrian isotopy class contains loose Legendrian embeddings, and they are C^0-dense in it.*

The analogue in dimension three is false (Theorem 4.15). In $\mathbb{R}^{2n+1}$, $n \geq 2$, each formal class of Legendrian embeddings thus contains a unique loose Legendrian up to isotopy, and the standard unknot is not loose—it has non-trivial Legendrian contact homology, whereas that of a loose Legendrian vanishes [Mur12]. Read the proof: it starts from a regular homotopy through Legendrian immersions, supplied by the h-principle for immersions, and uses the loose chart to remove self-intersections by an explicit front construction.

Loose boundaries make exact Lagrangian *embeddings* with boundary flexible. Eliashberg–Murphy prove an h-principle for exact Lagrangian embeddings with concave Legendrian boundary ("Lagrangian caps") whose boundary is loose [EM13]; in particular:

Theorem 4.44 (Lagrangian caps; Eliashberg–Murphy 2013)**.** *Let $2n \geq 6$ and $B \subset (\mathbb{R}^{2n}, \omega_0)$ the closed unit ball. There is a properly embedded Lagrangian n-disk $D \subset \mathbb{R}^{2n} \setminus \operatorname{Int} B$, attached transversely to ∂B along a loose Legendrian sphere $\partial D \subset (\partial B, \xi_{\mathrm{st}})$.*

Ekholm–Eliashberg–Murphy–Smith deduce an h-principle for exact Lagrangian immersions into $\mathbb{C}^n$ with the minimal number of double points; for example every closed orientable 3-manifold has an exact Lagrangian immersion into $\mathbb{C}^3$ with exactly one transverse double point [Ekh+13], which is optimal by Warning 4.40. Flexible Lagrangians with Legendrian boundary in Weinstein domains satisfy an existence and uniqueness h-principle up to ambient symplectomorphism [EGL20].

Definition 4.45 (Flexible Weinstein structures; Cieliebak–Eliashberg)**.** Let $(W^{2n}, \omega, X, \phi)$ be a Weinstein cobordism (Chapter 3) with $2n \geq 6$. It is *elementary* if there are no X-trajectories between distinct critical points of ϕ; an elementary Weinstein cobordism is *flexible* if the attaching spheres of its index-n critical points form a loose Legendrian link in $\partial_- W$. A Weinstein cobordism, domain or manifold is *flexible* if it can be cut along regular level sets of ϕ into elementary flexible cobordisms (countably many for manifolds). Subcritical structures, those with no critical points of index n, are flexible.

Theorem 4.46 (h-principle for flexible Weinstein structures; Cieliebak–Eliashberg 2012)**.** *Let W be a compact domain or a manifold of dimension $2n \geq 6$, let η be a nonempty homotopy class of nondegenerate 2-forms on W, and let $\mathfrak{M}_n$ denote the space of generalized Morse functions on W (having ∂W as maximal regular level set, resp. exhausting) all of whose critical points have index $\leq n$.*

(a) Every $\phi \in \mathfrak{M}_n$ is the function of a flexible Weinstein structure (ω, X, ϕ) with $\omega \in \eta$.

(b) If (ω_0, X_0, ϕ_0) and (ω_1, X_1, ϕ_1) are flexible Weinstein structures with $\omega_0, \omega_1 \in \eta$, then every path (ϕ_t) in $\mathfrak{M}_n$ from ϕ_0 to ϕ_1 lifts to a path (ω_t, X_t, ϕ_t) of flexible Weinstein structures.

Corollary 4.47 (Cieliebak–Eliashberg 2012)**.** *Let $2n \geq 6$. Two flexible Weinstein structures on the same domain or manifold whose symplectic forms are homotopic as nondegenerate 2-forms (equivalently, whose almost complex structures are homotopic) are Weinstein homotopic; in particular their completions are symplectomorphic.*

The Stein precursor is:

Theorem 4.48 (Eliashberg 1990)**.** *Let V be a smooth manifold of dimension $2n \geq 6$, J an almost complex structure on V, and $\phi \colon V \to \mathbb{R}$ an exhausting Morse function without critical points of index $> n$. Then there is an integrable complex structure J' on V, homotopic to J, for which ϕ is strictly plurisubharmonic; in particular (V, J') is Stein. Hence V^{2n}, $2n \geq 6$, admits a Stein structure iff it admits an almost complex structure and an exhausting Morse function without critical points of index $> n$.*

Remark 4.49 (Dimension four). Theorem 4.48 fails for $2n = 4$. $S^2 \times \mathbb{R}^2$ is almost complex and has an exhausting Morse function with critical points of index 0 and 2, but it is not Stein, because a Stein surface contains no homologically essential embedded sphere of nonnegative self-intersection [LM97]. What survives is Eliashberg's criterion (Theorem 4.20): a handlebody with 0-, 1- and 2-handles is Stein if the 2-handles are attached along Legendrian knots in $\#_k(S^1 \times S^2, \xi_{\mathrm{st}})$ with

framing tb -1. Gompf turned this into a diagrammatic calculus and showed that the interior of every such handlebody is *homeomorphic* to a Stein surface [Gom98]. Gauge theory, not flexibility, governs dimension four; see [OS04].

Subcritical Weinstein manifolds are split: each is deformation equivalent to $V \times \mathbb{C}$ for a Stein manifold V [Cie02b]. Since the flexible structure in a formal class is unique, the others are "exotic": Seidel–Smith found the first exotic structures on Euclidean spaces convex at infinity [SS05], McLean infinitely many pairwise non-symplectomorphic finite-type Stein structures on $\mathbb{R}^{2n}$, $n \geq 4$ [McL09], and Abouzaid–Seidel infinitely many finite-type Weinstein structures on every affine variety of real dimension > 4 [AS10b]. Lazarev glues Weinstein domains along flexible subdomains to build "maximal" Weinstein and contact structures containing any finite collection with the same topology [Laz20b], and shows that flexible fillings of a contact manifold with $c_1 = 0$ all have the same integral cohomology [Laz20a].

4.7 The rigid/flexible dichotomy

Theorem 4.50 (Flexible Weinstein domains have vanishing symplectic cohomology)**.** *Let W be a flexible Weinstein domain of dimension $2n \geq 6$. Then $\mathrm{SH}^*(W) = 0$, with coefficients in any field.*

This is due to Bourgeois–Ekholm–Eliashberg, via their surgery formula expressing SH of a Weinstein domain in terms of the Legendrian contact homology of its critical attaching spheres, which vanishes for loose ones [BEE12]; Murphy–Siegel give an alternative proof via an embedding h-principle [MS18]. In the same vein, overtwisted contact manifolds have vanishing contact homology: negatively stabilized open books have exact unit [BK10], and Theorem 4.33(iv) says every closed overtwisted manifold has one (modulo the foundations of contact homology, Chapter 9).

So flexible objects are invisible to holomorphic curves. For a closed manifold Q of dimension $n \geq 3$, T^*Q and the flexible Weinstein structure in its formal class are almost symplectomorphic, but $\mathrm{SH}^*(T^*Q)$ is the homology of the free loop space by Viterbo's theorem (Chapter 8) while the flexible one has $\mathrm{SH} = 0$. The converse slogan is false. Murphy–Siegel call a Weinstein domain *subflexible* if it is Weinstein deformation equivalent to a sublevel set of a flexible Weinstein manifold; Viterbo transfer forces $\mathrm{SH} = 0$, yet some subflexible domains are not flexible, as a twisted version of symplectic cohomology detects, and every flexible Weinstein manifold can be homotoped to have a non-flexible sublevel set [MS18]. Lazarev has since explained flexibility categorically, as a geometric shadow of Thomason's correspondence between split-generating subcategories and subgroups of the Grothendieck group [Laz22]. Arborealization, Álvarez-Gavela–Eliashberg–Nadler's theorem that polarized Weinstein manifolds have, after Weinstein homotopy, positive arboreal skeleta [ÁEN20], uses flexibility to simplify skeleta, with the payoff in microlocal sheaf theory (Chapter 8).

The list I keep in my head:

Flexible: an h-principle holds	**Rigid: no h-principle**
Symplectic and contact structures on open manifolds (Theorem 4.37)	Symplectic structures on closed manifolds (Chapter 3)
Overtwisted contact structures (Theorems 4.22 and 4.32)	Tight contact structures (Theorems 4.24 and 4.25)
Lagrangian and Legendrian immersions (Theorem 4.39)	Exact Lagrangian embeddings (Warning 4.40)
Loose Legendrians in dimension ≥ 5 (Theorem 4.43); Lagrangian caps; flexible Lagrangians	Legendrian knots in dimension three; non-loose Legendrians, e.g. the standard unknot
Flexible, in particular subcritical, Weinstein structures in dimension ≥ 6 (Theorem 4.46)	Weinstein structures in general (T^*Q, exotic $\mathbb{R}^{2n}$); Stein surfaces
Isosymplectic embeddings in codimension ≥ 4 [Gro86; EM02]	Symplectic embeddings in codimension 0: non-squeezing, ellipsoid embeddings (Chapter 6)

4.8 What to read

Etnyre, Introductory lectures on contact geometry *[Etn03]*
Start here: thirty pages, the right pictures.

Geiges, An Introduction to Contact Topology *[Gei08]*
The reference. Chapters 2 (Darboux, Gray, neighbourhood theorems) and 3 (knots) are essential, the rest (Martinet and Lutz, fillings, contact surgery) on demand; its proofs of the local theorems are the ones to learn.

Etnyre, Legendrian and transversal knots *[Etn05]; Etnyre–Ng [EN20]*
The first for classical invariants and Legendrian simplicity, the second for the Chekanov–Eliashberg DGA in $\mathbb{R}^3$.

Etnyre, Lectures on open book decompositions and contact structures *[Etn06]; Giroux's ICM address [Gir02]*
Etnyre for the proofs, Giroux for the vision.

Ozbagci–Stipsicz, Surgery on Contact 3-Manifolds and Stein Surfaces *[OS04]*
Kirby diagrams of Stein surfaces and contact surgery in practice.

Eliashberg 1989 and 1992 [Eli89; Eli92]; Giroux 2000 [Gir00]; Honda 2000 [Hon00]
Still the best sources for the 3-dimensional techniques: Eliashberg 1992 for characteristic foliations, then Honda for bypasses.

Eliashberg–Mishachev, Introduction to the h-Principle *[EM02]*
Read this before Gromov [Gro86]: holonomic approximation and its applications to open manifolds and to Lagrangian immersions; convex integration can wait.

Cieliebak–Eliashberg, From Stein to Weinstein and Back *[CE12]*
The definitive Weinstein reference. Read Chapter 1 and the Weinstein half (Weinstein homotopies, loose Legendrians, flexibility); consult the J-convexity chapters only for the Stein side. The survey [CE14] is a 40-page extract.

Murphy [Mur12]; Borman–Eliashberg–Murphy [BEM15]
Murphy is readable by a second-year student; everyone should read the introduction of BEM and its overtwisted disk.

Eliashberg, Recent advances in symplectic flexibility *[Eli15]*
THE survey of post-2012 flexibility; read it right after this chapter.

Massot–Niederkrüger–Wendl [MNW13]; Wendl [Wen18]
Fillings in high dimensions; holomorphic-curve proofs of Theorem 4.29.

4.9 Frontier and open problems

Resolved, though partly only in preprints: the higher-dimensional Giroux correspondence [BHH23]; the existence of infinitely many exotic Weinstein structures on $\mathbb{R}^{2n}$ for all $n \geq 3$ ([McL09] for $n \geq 4$; [AS10b] in general); and the splitting of subcritical Weinstein manifolds [Cie02b], a preprint whose argument is widely accepted.

Question 4.51 (Tight structures on hyperbolic 3-manifolds)**.** *Does every closed orientable hyperbolic 3-manifold carry a positive tight contact structure? More ambitiously, classify tight contact structures on hyperbolic 3-manifolds (there are finitely many by Theorem 4.25).*

Manifolds with co-orientable taut foliations carry weakly fillable structures by Eliashberg–Thurston perturbation [ET98], so the interesting cases are hyperbolic manifolds without them, such as hyperbolic L-spaces.

Problem 4.52 (Recognizing flexibility)**.** *Characterize flexible Weinstein domains of dimension ≥ 6 among all Weinstein domains by Floer-theoretic invariants.*

$\mathrm{SH} = 0$ is necessary (Theorem 4.50) but not sufficient [MS18]. My guess is the wrapped Fukaya category with all twistings, but the evidence is thin.

Question 4.53 (Intermediate regimes)**.** *Is there a natural class of Weinstein domains, containing the subflexible ones, that satisfies an h-principle relative to algebraic data, such as a twisted symplectic cohomology or the Grothendieck group of the wrapped Fukaya category [Laz22]?*

Finally, the nearby Lagrangian conjecture—the rigid counterpart of Theorem 4.39 for exact Lagrangians in T^*Q—is discussed in Chapter 8.

4.10 Logical dependencies

Introduce Definitions 4.1 and 4.3, Convention 4.4, Definitions 4.8, 4.10 and 4.12 first; these state Theorems 4.6, 4.7 and 4.11. Definition 4.13 (linking numbers, Seifert surfaces) states Theorems 4.14 and 4.15; Definition 4.16 states Theorems 4.17 and 4.18; Definition 4.21 states Theorems 4.22, 4.24 and 4.25. Definition 4.27 needs Liouville and Weinstein domains (Chapter 3). Definition 4.31 imports the model germ of [BEM15]. Definition 4.34 (jet bundles, C^0 and C^∞ topologies, weak homotopy equivalence) precedes Theorems 4.35 to 4.37 and 4.39; Definitions 4.41 and 4.42 precede Theorem 4.43 and, with Weinstein cobordisms, Definition 4.45 and Theorem 4.46. Only Theorem 4.50 and Remark 4.30 presuppose Floer theory and hence compactness and transversality for holomorphic curves.

Chapter 5

Pseudoholomorphic curves: the foundations

Gromov's 1985 paper [Gro85] is the watershed of the subject: before it, symplectic topology had normal forms and conjectures; after it, a machine for producing invariants. Choose an auxiliary almost complex structure J tamed by ω and study maps from Riemann surfaces solving the nonlinear Cauchy–Riemann equation of J. The symplectic area of such a curve is topological, which gives compactness; the equation is elliptic, which makes moduli spaces finite dimensional and, for generic J, smooth of computable dimension; and the space of tame J is contractible, so whatever one can count is an invariant of (M,ω). Chapters 6–9 are variations on this theme with other domains, boundary conditions and targets.

This chapter covers the closed theory. What to take away: the energy identity is why everything works; multiply covered curves are why everything is hard; in the semipositive genus-zero world everything can be done by hand with pseudocycles, and that theory should be learned thoroughly; beyond it one needs virtual techniques, whose foundations, after a contentious decade, are now settled by several independent routes.

5.1 J-holomorphic curves and their local theory

Convention 5.1 (Cauchy–Riemann operator)**.** For a Riemann surface (Σ, j), an almost complex manifold (M,J) and a C^1 map $u\colon \Sigma \to M$ we write $\bar\partial_J u := \tfrac12\big(du + J(u)\circ du \circ j\big)$, a section of the bundle $\Lambda^{0,1}T^*\Sigma \otimes_J u^*TM$ of J-antilinear 1-forms ($\eta\circ j = -J\eta$). In a local holomorphic coordinate $s+it$ on Σ, the equation $\bar\partial_J u = 0$ reads $\partial_s u + J(u)\,\partial_t u = 0$. With C1 and $J = J_0$ on $\mathbb{C}^n$ this is the usual Cauchy–Riemann equation. On the cylinder $\mathbb{R}\times S^1$ with coordinate $s+it$ it is Floer's equation of C7 with $H = 0$.

Definition 5.2 (J-holomorphic curve)**.** Let (M,ω) be a symplectic manifold of dimension $2n$. Write $\mathcal{J}_\tau(M,\omega) \supset \mathcal{J}(M,\omega)$ for the spaces of smooth ω-tame and ω-compatible almost complex structures (C2), with the C^∞ topology. Let (Σ, j) be a Riemann surface, connected unless stated otherwise. A *J-holomorphic curve* is a smooth map $u\colon \Sigma \to M$ with $\bar\partial_J u = 0$, i.e. $du\circ j = J(u)\circ du$. For $p > 2$, a map $u \in W^{1,p}_{\mathrm{loc}}(\Sigma, M)$ (continuous by Sobolev embedding) satisfying the equation almost everywhere is a J-holomorphic curve of class $W^{1,p}_{\mathrm{loc}}$. If Σ is closed, u represents $[u] := u_*[\Sigma] \in H_2(M;\mathbb{Z})$. A *$J$-holomorphic sphere* is a J-holomorphic curve with domain $S^2 = \mathbb{CP}^1$ carrying its standard complex structure.

Later chapters also use punctured domains $\dot\Sigma = \Sigma \setminus \Gamma$ with cylindrical ends (Floer cylinders, Chapter 7; curves in symplectizations, Chapter 9) and compact surfaces with boundary and $u(\partial\Sigma) \subset L$ for a Lagrangian L (Chapter 8). A Lagrangian is totally real for every tame J (if $v, Jv \in T_xL$ then $\omega(v, Jv) = 0$), which is what makes that boundary problem Fredholm. The interior local theory is the same in all cases.

Definition 5.3 (Energy)**.** For $J \in \mathcal{J}_\tau(M,\omega)$ let $g_J(v,w) := \frac{1}{2}\big(\omega(v,Jw) + \omega(w,Jv)\big)$. This is a Riemannian metric for which J is orthogonal, and it equals $\omega(\cdot, J\cdot)$ when J is compatible. For a smooth $u\colon \Sigma \to M$ and a measurable $S \subset \Sigma$, the *energy* is $E(u;S) := \frac{1}{2}\int_S |du|^2_{g_J}\,\mathrm{dvol}_\Sigma$, where the norm and the area form are computed with any metric on Σ in the conformal class of j. The integrand is conformally invariant, so it does not matter which. We write $E(u) := E(u;\Sigma)$.

Lemma 5.4 (Energy identity; Gromov 1985)**.** *Let $J \in \mathcal{J}_\tau(M,\omega)$ and let $u\colon \Sigma \to M$ be J-holomorphic. Then pointwise $\frac{1}{2}|du|^2_{g_J}\,\mathrm{dvol}_\Sigma = u^*\omega$. Hence $E(u;S) = \int_S u^*\omega$ for every measurable S, and $E(u) = \langle [\omega], [u]\rangle$ if Σ is closed. If $J \in \mathcal{J}(M,\omega)$, then every smooth u satisfies $E(u;S) = \int_S u^*\omega + \int_S |\bar\partial_J u|^2_{g_J}\,\mathrm{dvol}_\Sigma$.*

Proof idea. In a conformal coordinate we have $\partial_t u = J\partial_s u$, so $u^*\omega = \omega(\partial_s u, J\partial_s u)\,ds\wedge dt = |\partial_s u|^2\,ds\wedge dt$. Since J is g_J-orthogonal, $|\partial_t u| = |\partial_s u|$.

So closed J-curves in a class A have energy $\omega(A)$, independent of J and of the curve; for compatible J they are g_J-area minimizing in their homology class, Gromov's original point of view. I recommend working with *tame* structures, as McDuff–Salamon do [MS12a]: tameness is open in J and (on compact M) in ω, so $\mathcal{J}_\tau(M,\omega)$ is open in the Fréchet manifold of all almost complex structures and a path ω_t can be accompanied by a path J_t, which is what deformation invariance needs. Both $\mathcal{J}_\tau$ and $\mathcal{J}$ are contractible (Chapter 2). All one loses is the inequality $E(u) \geq \int u^*\omega$ for non-holomorphic u.

Theorem 5.5 (Elliptic regularity)**.** *Let $\ell \geq 1$ be an integer, J an almost complex structure of class C^ℓ on M, and $p > 2$. If $u \in W^{1,p}_{\mathrm{loc}}(\Sigma, M)$ is J-holomorphic, then $u \in W^{\ell+1,p}_{\mathrm{loc}}$; in particular u is smooth if J is. Moreover, if $J_\nu \to J$ in C^∞, $u_\nu\colon \Sigma \to M$ are J_ν-holomorphic with images in a fixed compact set, and $\sup_\nu \|du_\nu\|_{L^\infty(K)} < \infty$ for every compact $K \subset \Sigma$, then a subsequence converges in C^∞_{loc} to a J-holomorphic curve.*

See [MS12a, App. B, Thm. 4.1.1]. *Gradient* bounds give compactness; energy bounds alone do not (Section 5.2).

Theorem 5.6 (Carleman similarity principle)**.** *Let $p > 2$, let $B_\varepsilon \subset \mathbb{C}$ be the open ε-disk, and let $J \in W^{1,p}(B_\varepsilon, \mathbb{R}^{2n\times 2n})$ with $J(z)^2 = -\mathbb{1}$, $C \in L^p(B_\varepsilon, \mathbb{R}^{2n\times 2n})$ and $u \in W^{1,p}(B_\varepsilon, \mathbb{R}^{2n})$ satisfy $\partial_s u + J\partial_t u + Cu = 0$ and $u(0) = 0$. Then there exist $\delta \in (0,\varepsilon)$, $\Phi \in W^{1,p}(B_\delta, \mathrm{GL}(2n,\mathbb{R}))$ and a holomorphic $\sigma\colon B_\delta \to \mathbb{C}^n$ with $\sigma(0) = 0$, $\Phi(z)^{-1}J(z)\Phi(z) = J_0$ and $u = \Phi\sigma$ on B_δ.*

So solutions of linear Cauchy–Riemann-type equations behave like holomorphic functions up to a continuous change of frame. Applied to the difference of two curves, or to du, this gives the structure theory [MS12a, §§2.3–2.4].

Corollary 5.7 (Unique continuation and isolated singularities)**.** *Let J be smooth and Σ connected.*

1. *If two J-holomorphic curves $u, v\colon \Sigma \to M$ agree to infinite order at a point (in particular, if they agree on a nonempty open set), then $u \equiv v$.*
2. *If u is nonconstant, then its critical set $\{z : du(z) = 0\}$ and every fibre $u^{-1}(x)$ are discrete.*

3. *If Σ_1, Σ_2 are closed and connected and $u_i \colon \Sigma_i \to M$ are J-holomorphic with $u_1(\Sigma_1) \neq u_2(\Sigma_2)$, then the set $\{(z_1, z_2) : u_1(z_1) = u_2(z_2)\}$ is finite.*

Two quantitative facts make the analysis global.

Lemma 5.8 (Mean value inequality)**.** *Let $r > 0$ and $c > 0$, and let $f \colon B_r(0) \to [0, \infty)$ be of class C^2 with $\Delta f \geq -c f^2$ and $\int_{B_r(0)} f < \pi/(8c)$. Then $f(0) \leq \frac{8}{\pi r^2} \int_{B_r(0)} f$.*

This is [MS12a, §4.3] (Lemma 4.3.2 in the first edition). The energy density $\frac{1}{2}|du|^2$ of a J-curve satisfies $\Delta f \geq -c f^2$ with $c = c(M, \omega, J)$, so small energy on a disk forces a gradient bound at its center.

Lemma 5.9 (Monotonicity)**.** *Let (M, ω) be closed, $J \in \mathcal{J}_\tau(M, \omega)$, and let $B_r(x)$ denote open g_J-balls. There are constants $r_0 > 0$ and $c_0 > 0$ with the following property. Let $x \in M$ and $0 < r \leq r_0$, let (Σ, j) be a compact Riemann surface with (possibly empty) boundary, and let $u \colon \Sigma \to M$ be J-holomorphic with $x \in u(\Sigma)$ and $u(\partial\Sigma) \cap B_r(x) = \emptyset$. Then $E\big(u; u^{-1}(B_r(x))\big) \geq c_0 r^2$. The same conclusion holds in $(\mathbb{C}^n, \omega_0, J_0)$ with the Euclidean metric, with $r_0 = \infty$ and $c_0 = \pi$.*

The sharp Euclidean constant (from the monotonicity formula for minimal surfaces) is the inequality behind nonsqueezing (Chapter 6). Sikorav's chapter [Sik94] is the cleanest source for monotonicity and the isoperimetric inequality.

Proposition 5.10 (Energy quantization)**.** *Let (M, ω) be closed and $\mathcal{K} \subset \mathcal{J}_\tau(M, \omega)$ compact. There is $\hbar = \hbar(\mathcal{K}) > 0$ such that $E(u) \geq \hbar$ for every $J \in \mathcal{K}$ and every nonconstant J-holomorphic curve $u \colon \Sigma \to M$ with Σ closed (of any genus).*

Proof idea. A closed curve whose image lies in a small ball, where ω is exact, has $E(u) = \int u^* d\lambda = 0$ and is constant. Otherwise its image leaves the ball, and Lemma 5.9 gives $E(u) \geq c_0 r_0^2$, uniformly over $\mathcal{K}$.

Theorem 5.11 (Removal of singularities)**.** *Let (M, ω) be symplectic, $J \in \mathcal{J}_\tau(M, \omega)$, $B \subset \mathbb{C}$ the closed unit disk, and $u \colon B \setminus \{0\} \to M$ a J-holomorphic curve with $E(u) < \infty$ and $u(B \setminus \{0\})$ contained in a compact subset of M. Then u extends to a smooth J-holomorphic curve $B \to M$.*

This is [MS12a, Thm. 4.1.2]. A finite-energy J-plane in a closed M therefore extends to a sphere: this is why bubbles are spheres.

Definition 5.12 (Simple and multiply covered curves)**.** Let Σ be closed and connected, and let $u \colon \Sigma \to M$ be a nonconstant J-holomorphic curve. Then u is *multiply covered* if $u = v \circ \phi$, where $\phi \colon \Sigma \to \Sigma'$ is a holomorphic branched covering of degree ≥ 2 onto a closed Riemann surface and $v \colon \Sigma' \to M$ is J-holomorphic. It is *simple* otherwise. A point $z \in \Sigma$ is an *injective point* if $du(z) \neq 0$ and $u^{-1}(u(z)) = \{z\}$, and u is *somewhere injective* if it has one. Constant maps are neither simple nor somewhere injective.

Proposition 5.13 (Simple curves; McDuff–Salamon 2012)**.** *Let J be smooth, Σ closed and connected, and $u \colon \Sigma \to M$ a nonconstant J-holomorphic curve.*

1. *There are a simple J-holomorphic curve $v \colon \Sigma' \to M$ and a holomorphic branched covering $\phi \colon \Sigma \to \Sigma'$ with $u = v \circ \phi$ and $[u] = \deg(\phi)\,[v]$.*
2. *u is simple if and only if it is somewhere injective. In that case the complement of the set of injective points is finite.*

See [MS12a, Prop. 2.5.1, App. E]; finiteness of the non-injective set uses the local structure theory of Micallef–White [MW95]. A reparametrization fixing a simple u fixes every injective point, so it is the identity.

Warning 5.14. For disks with Lagrangian boundary conditions the dichotomy of Proposition 5.13 fails: a nonconstant disk need not factor through a simple one. What survives is that its image contains that of a simple disk and, suitably interpreted, decomposes into simple pieces (Kwon–Oh [KO00], Lazzarini [Laz00]). Much of the extra difficulty of Lagrangian Floer theory (Chapter 8) starts here.

5.2 Compactness and bubbling

An energy bound controls $\|du\|_{L^2}$, but compactness needs L^p with $p > 2$. The culprit is conformal invariance: energy is invariant under the noncompact group $G := \mathrm{PSL}(2,\mathbb{C})$ and can concentrate at points. The smooth conics $\{xy = \varepsilon z^2\} \subset \mathbb{CP}^2$ converge as $\varepsilon \to 0$ to the line pair $\{xy = 0\}$; parametrized as spheres, the maps converge away from one point, where a second line *bubbles off*. The limit is a tree of two spheres, and no energy is lost. In general, if $|du_\nu(z_\nu)| \to \infty$, rescaling by the gradient produces a nonconstant finite-energy J-plane, hence (Theorem 5.11) a sphere of energy $\geq \hbar$ (Proposition 5.10). The rescaling centers are chosen with:

Lemma 5.15 (Hofer's lemma)**.** *Let (X, d) be a metric space, $f : X \to [0,\infty)$ continuous, $x_0 \in X$ and $\delta_0 > 0$, and assume the closed ball $\bar{B}_{2\delta_0}(x_0)$ is complete. Then there are $\xi \in X$ and $\delta \in (0, \delta_0]$ with $d(\xi, x_0) \leq 2\delta_0$, $\sup_{B_\delta(\xi)} f \leq 2f(\xi)$ and $\delta f(\xi) \geq \delta_0 f(x_0)$.*

There are finitely many bubble points, and one iterates. The hard step is to show that no energy disappears in the necks between bubbles and that the limit is connected; this uses the isoperimetric inequality and exponential decay of energy on long cylinders of small energy [MS12a, §§4.4, 4.7]. The limits are Kontsevich's stable maps [Kon95a].

Definition 5.16 (Genus-zero stable maps)**.** Let (M, J) be an almost complex manifold, $A \in H_2(M;\mathbb{Z})$ and $k \geq 0$. A *k-pointed genus-zero stable J-map in class A* consists of

- a finite tree T (vertices α, edges written $\alpha E \beta$);
- J-holomorphic spheres $u_\alpha : S^2 \to M$ for $\alpha \in T$, with $\sum_\alpha [u_\alpha] = A$;
- nodal points $z_{\alpha\beta} \in S^2$ for each ordered pair with $\alpha E \beta$;
- marked points given by a labelling $i \mapsto \alpha_i \in T$ and points $z_i \in S^2$ $(1 \leq i \leq k)$.

These data must satisfy the following conditions: $u_\alpha(z_{\alpha\beta}) = u_\beta(z_{\beta\alpha})$; for each α the special points $\{z_{\alpha\beta}\}_\beta \cup \{z_i : \alpha_i = \alpha\}$ are pairwise distinct; and (*stability*) every α with u_α constant carries at least three special points. Two stable maps are *equivalent* if there are a tree isomorphism $f : T \to T'$ and Möbius maps $\phi_\alpha \in G$ with $u'_{f(\alpha)} \circ \phi_\alpha = u_\alpha$, $z'_{f(\alpha)f(\beta)} = \phi_\alpha(z_{\alpha\beta})$, $\alpha'_i = f(\alpha_i)$ and $z'_i = \phi_{\alpha_i}(z_i)$. The set of equivalence classes is $\overline{\mathcal{M}}_{0,k}(M, A; J)$. The equivalence classes with T a single vertex form the open subset $\mathcal{M}_{0,k}(M, A; J)$.

Definition 5.17 (Gromov convergence, genus zero)**.** A sequence of k-pointed genus-zero stable J_ν-maps *Gromov converges* to a stable J-map $(\{u_\alpha\}, \{z_{\alpha\beta}\}, \{\alpha_i, z_i\})$ modelled on T if, after passing to equivalent representatives, there are for each $\alpha \in T$ Möbius maps $\phi_{\alpha,\nu}$ such that the following hold.

1. (*Map*) $u_\nu \circ \phi_{\alpha,\nu} \to u_\alpha$ in C^∞_{loc} on S^2 minus the nodal points of α (for stable maps u_ν, compose the appropriate component of u_ν).
2. (*Energy*) For each edge $\alpha E \beta$, the energy of $u_\nu \circ \phi_{\alpha,\nu}$ on the ε-disk around $z_{\alpha\beta}$ converges, as $\nu \to \infty$ and then $\varepsilon \to 0$, to the total energy of the branch of T at β.
3. (*Rescaling*) For $\alpha E \beta$, $\phi^{-1}_{\alpha,\nu} \circ \phi_{\beta,\nu}$ converges to the constant $z_{\alpha\beta}$ uniformly on compact subsets of $S^2 \setminus \{z_{\beta\alpha}\}$.
4. (*Marked points*) $\phi^{-1}_{\alpha_i,\nu}(z_{i,\nu}) \to z_i$.

This is the definition of [MS12a, §5.2], compressed. It implies $E(u_\nu) \to \sum_\alpha E(u_\alpha)$ and Hausdorff convergence of the images.

Theorem 5.18 (Gromov compactness, genus zero)**.** *Let (M,ω) be a closed symplectic manifold, and let $J_\nu \to J$ in the C^∞ topology with $J_\nu, J \in \mathcal{J}_\tau(M,\omega)$. Every sequence of k-pointed genus-zero stable J_ν-maps with $\sup_\nu E(u_\nu) < \infty$ has a subsequence that Gromov converges to a k-pointed genus-zero stable J-map. In particular this applies to a sequence of J_ν-holomorphic spheres u_ν with pairwise distinct marked points $z_{1,\nu}, \dots, z_{k,\nu}$, provided each u_ν is nonconstant or $k \geq 3$.*

Gromov sketched this in [Gro85]. Complete proofs, with varying notions of convergence, are due to Parker–Wolfson [PW93], Ye [Ye94], Pansu [Pan94] and Hummel [Hum97]. The statement above, with the energy and tree structure built in, is [MS12a, Thm. 5.3.1, §5.5].

Theorem 5.19 (Topology of the moduli space)**.** *Let (M,ω) be closed, $J \in \mathcal{J}_\tau(M,\omega)$, $A \in H_2(M;\mathbb{Z})$ and $k \geq 0$. The set $\overline{\mathcal{M}}_{0,k}(M,A;J)$ carries a natural topology, the* Gromov topology*, with the following properties: a sequence converges in it if and only if it Gromov converges; and $\overline{\mathcal{M}}_{0,k}(M,A;J)$ is compact and metrizable.*

See [MS12a, Thm. 5.6.6]. By Theorem 5.18 the parametrized space $\bigcup_{t\in[0,1]} \{t\} \times \overline{\mathcal{M}}_{0,k}(M,A;J_t)$ of a smooth path J_t is also compact, which is what cobordism arguments need. The finiteness statement behind Novikov rings is a direct consequence.

Corollary 5.20. *Let (M,ω) be closed, $\mathcal{K} \subset \mathcal{J}_\tau(M,\omega)$ compact and $C > 0$. Only finitely many classes $A \in H_2(M;\mathbb{Z})$ with $\omega(A) \leq C$ are represented by genus-zero stable J-maps with $J \in \mathcal{K}$.*

Definition 5.21 (Stable maps of higher genus)**.** A *nodal Riemann surface of arithmetic genus g with k marked points* is a closed, possibly disconnected Riemann surface $\tilde{\Sigma}$ (the normalization), together with a finite set of unordered pairs $\{n_i, n_i'\}$ of points (the nodes) and k ordered marked points. All of these points are distinct. The quotient $\Sigma = \tilde{\Sigma}/(n_i \sim n_i')$ must be connected, and g is its arithmetic genus, $g = \sum_\alpha g_\alpha + \#\text{nodes} - \#\text{components} + 1$. A *stable J-map* of genus g with k marked points in class A is such a Σ together with a continuous $u\colon \Sigma \to M$ that is J-holomorphic on each component of $\tilde{\Sigma}$, with $u_*[\Sigma] = A$. Stability requires that every component Σ_α on which u is constant satisfies $2g_\alpha + (\text{number of special points on } \Sigma_\alpha) \geq 3$; equivalently, the automorphism group of $(\Sigma, \text{marked points}, u)$ is finite. Equivalence is isomorphism of nodal curves preserving the marked points and intertwining the maps. The set of equivalence classes is $\overline{\mathcal{M}}_{g,k}(M,A;J)$.

Remark 5.22 (Higher genus, boundaries, punctures). For all g, k and closed M, $\overline{\mathcal{M}}_{g,k}(M,A;J)$ is compact and Hausdorff in the Gromov topology; domains vary in Deligne–Mumford space, and collapsing necks produce nodes just as bubbling does (Hummel [Hum97], Ye [Ye94], Fukaya–Ono

[FO99]). With boundary on a closed Lagrangian L one adds disk bubbling; for monotone L each disk bubble has Maslov index at least the minimal Maslov number, and these bubbles are the source of the obstructions of Chapter 8. For punctured curves in symplectizations the limits are holomorphic buildings [Bou+03] (Chapter 9).

5.3 Fredholm theory and transversality

Definition 5.23 (Real Cauchy–Riemann operator)**.** Let $E \to \Sigma$ be a complex vector bundle over a Riemann surface. A *real linear Cauchy–Riemann operator* on E is a real-linear first order differential operator $D\colon \Omega^0(\Sigma, E) \to \Omega^{0,1}(\Sigma, E)$ with $D(f\xi) = fD\xi + (\bar\partial f)\xi$ for $f \in C^\infty(\Sigma, \mathbb{R})$. Here $(\bar\partial f)\xi := \frac{1}{2}\big(\xi\, df + (J\xi)\, df \circ j\big)$, where J is the complex structure on E. Every such operator is $\bar\partial^\nabla + a$ for a Hermitian connection ∇ and a zeroth-order term $a \in \Omega^0(\mathrm{Hom}_{\mathbb{R}}(E, \Lambda^{0,1}T^*\Sigma \otimes_{\mathbb{C}} E))$, which in general is not complex linear.

Theorem 5.24 (Riemann–Roch)**.** *Let Σ be a closed Riemann surface, $E \to \Sigma$ a complex vector bundle of rank n, D a real linear Cauchy–Riemann operator on E with smooth coefficients, $k \geq 1$ and $p > 1$. Then $D\colon W^{k,p}(\Sigma, E) \to W^{k-1,p}(\Sigma, \Lambda^{0,1} \otimes E)$ is Fredholm of index $n\chi(\Sigma) + 2\langle c_1(E), [\Sigma]\rangle$. If Σ is compact with boundary and $F \subset E|_{\partial\Sigma}$ is a totally real subbundle, then D restricted to sections with boundary values in F is Fredholm of index $n\chi(\Sigma) + \mu(E, F)$, where $\mu(E, F)$ is the boundary Maslov index of Chapter 1. If $n = 1$ and Σ is closed, D is injective when $c_1(E) < 0$ and surjective when $c_1(E) + \chi(\Sigma) > 0$.*

See [MS12a, Thm. C.1.10]. The $n = 1$ statements follow from the similarity principle as for holomorphic line bundles; they are the engine of automatic transversality.

Now fix $p > 2$, a closed Riemann surface (Σ, j) and $A \in H_2(M; \mathbb{Z})$. The space $\mathcal{B}$ of $W^{1,p}$ maps $\Sigma \to M$ representing A is a smooth separable Banach manifold with $T_u\mathcal{B} = W^{1,p}(\Sigma, u^*TM)$. Over it lies the Banach bundle $\mathcal{E}$ with fibre $\mathcal{E}_u = L^p(\Sigma, \Lambda^{0,1}T^*\Sigma \otimes_J u^*TM)$, and $\bar\partial_J$ is a smooth section of $\mathcal{E}$. At a zero u its vertical differential

$$D_u\colon W^{1,p}(\Sigma, u^*TM) \to L^p(\Sigma, \Lambda^{0,1}T^*\Sigma \otimes_J u^*TM)$$

does not depend on the choice of connection, and it is a real linear Cauchy–Riemann operator on (u^*TM, J) [MS12a, §3.1]. By Theorem 5.24, $\operatorname{index} D_u = n\chi(\Sigma) + 2c_1(A)$, where $c_1(A) := \langle c_1(TM, J), A\rangle$.

Theorem 5.25 (Transversality for simple curves)**.** *Let (M, ω) be a closed symplectic $2n$-manifold, $A \in H_2(M; \mathbb{Z})$, and (Σ, j) a closed connected Riemann surface. Let $\mathcal{M}^*(A; \Sigma; J)$ be the space of simple J-holomorphic maps $(\Sigma, j) \to M$ in class A, and let $\mathcal{J}_{\mathrm{reg}}(A; \Sigma)$ be the set of $J \in \mathcal{J}_\tau(M, \omega)$ such that D_u is onto for every $u \in \mathcal{M}^*(A; \Sigma; J)$.*

1. *$\mathcal{J}_{\mathrm{reg}}(A; \Sigma)$ is comeagre in $\mathcal{J}_\tau(M, \omega)$, i.e. it contains a countable intersection of open dense subsets. The same holds with $\mathcal{J}(M, \omega)$ in place of $\mathcal{J}_\tau(M, \omega)$.*
2. *For $J \in \mathcal{J}_{\mathrm{reg}}(A; \Sigma)$, $\mathcal{M}^*(A; \Sigma; J)$ is a smooth manifold of dimension $n\chi(\Sigma) + 2c_1(A)$, possibly empty, with $T_u\mathcal{M}^* = \ker D_u$. It carries a canonical orientation.*
3. *For $J_0, J_1 \in \mathcal{J}_{\mathrm{reg}}(A; \Sigma)$, a comeagre set of smooth paths $\{J_t\}_{t\in[0,1]}$ in $\mathcal{J}_\tau(M, \omega)$ from J_0 to J_1 has the property that $\{(t, u) : u \in \mathcal{M}^*(A; \Sigma; J_t)\}$ is a smooth oriented manifold of dimension $n\chi(\Sigma) + 2c_1(A) + 1$ with oriented boundary $\mathcal{M}^*(A; \Sigma; J_1) \sqcup \big(-\mathcal{M}^*(A; \Sigma; J_0)\big)$.*

4. *Let $\Sigma = S^2$ and $A \neq 0$. Then $G = \mathrm{PSL}(2,\mathbb{C})$ acts freely and properly on $\mathcal{M}^*(A;J) := \mathcal{M}^*(A;S^2;J)$ by $\phi\cdot u = u\circ\phi^{-1}$. For $J \in \mathcal{J}_{\mathrm{reg}}(A;S^2)$ and $k \geq 0$, the quotient $\mathcal{M}^*_{0,k}(A;J) := \big(\mathcal{M}^*(A;J)\times Z_k\big)/G$, where $Z_k \subset (S^2)^k$ is the set of k-tuples of pairwise distinct points, is a smooth oriented manifold of dimension $2n + 2c_1(A) + 2k - 6$.*

Parts (i)–(iii) are proved in [MS12a, §3.1]; the orientation comes from deforming D_u to its complex linear part.

Proof idea. Work first with C^ℓ structures $J \in \mathcal{J}^\ell_\tau$, a Banach manifold. The *universal moduli space* $\big\{(u,J) : \bar\partial_J u = 0,\ u \text{ simple}\big\}$ is cut out transversally. If $\eta \in L^q$ annihilates the image of $(\xi, Y) \mapsto D_u\xi + \frac{1}{2}Y(u)\,du\circ j$, then η vanishes near an injective point, because Y can be chosen supported near the image of that point; by unique continuation $\eta \equiv 0$. This is the only place where simplicity is used. The projection to $\mathcal{J}^\ell_\tau$ is Fredholm with the index of D_u, so the Sard–Smale theorem [Sma65] makes its regular values comeagre. To pass to C^∞ one uses either Taubes' trick, which writes $\mathcal{J}_{\mathrm{reg}}$ as a countable intersection of sets $\mathcal{J}_{\mathrm{reg},K}$ (curves with derivative bounds K) that are open and dense in C^∞, or Floer's separable Banach space C^ε of smooth perturbations [Flo88c].

Example 5.26 (Why multiple covers cannot be perturbed away)**.** Let v be a simple J-sphere in class B with D_v onto, and $d \geq 2$. Degree-d rational maps $S^2 \to S^2$ form a manifold of dimension $4d+2$, and $(v',\phi) \mapsto v'\circ\phi$ has 6-dimensional fibres, so the d-fold covers of curves near v form a family of dimension $(2n + 2c_1(B)) + (4d+2) - 6$ in the moduli space of class dB, whose expected dimension is $2n + 2dc_1(B)$. The excess is

$$\big(2n + 2c_1(B) + 4d - 4\big) - \big(2n + 2dc_1(B)\big) = 2(d-1)\big(2 - c_1(B)\big),$$

positive whenever $c_1(B) \leq 1$. Then D_u is onto at *no* d-fold cover u of v, and since v persists under perturbation of J, no nearby J helps. If $n = 3$ and $c_1(B) = 0$ (a rigid sphere in a Calabi–Yau threefold), the covers form a $(4d-4)$-dimensional family in a moduli space of expected dimension 0 after dividing by G: the origin of the Aspinwall–Morrison contribution $1/d^3$ [AM93]. If $c_1(B) < 0$ (which needs $n \geq 4$ for v to exist generically), the covers lie in moduli spaces of *negative* expected dimension $2n + 2dc_1(B) - 6$ for d large, yet exist for every J near the original one.

There are three ways out, and each defines a part of the subject. *(a) Semipositivity.* Recall (Chapter 2) that a closed (M^{2n},ω) is *semipositive* if $c_1(A) \geq 0$ for every $A \in \pi_2(M)$ with $\omega(A) > 0$ and $c_1(A) \geq 3 - n$; monotone manifolds, symplectically aspherical manifolds and all manifolds of dimension ≤ 6 are semipositive. For generic J a simple sphere exists only if $2n + 2c_1(B) - 6 \geq 0$, so semipositivity rules out simple spheres with $c_1 < 0$; multiply covered and nodal configurations then have evaluation images of codimension ≥ 2, with one exception treated below. This is McDuff–Salamon's route [MS12a, Ch. 6]. *(b) Dimension four*, where positivity of intersections and automatic transversality replace genericity (Theorems 5.27 to 5.29). *(c) Virtual techniques* (Section 5.5).

Theorem 5.27 (Positivity of intersections; Gromov 1985, McDuff 1991)**.** *Let (M,J) be an almost complex 4-manifold with J smooth. Let $u_1 : \Sigma_1 \to M$ and $u_2 : \Sigma_2 \to M$ be simple J-holomorphic curves from closed connected Riemann surfaces with $u_1(\Sigma_1) \neq u_2(\Sigma_2)$. Then the set $\mathcal{I} = \{(z_1,z_2) : u_1(z_1) = u_2(z_2)\}$ is finite, and each $(z_1,z_2) \in \mathcal{I}$ contributes a local intersection index ≥ 1, with equality if and only if u_1 and u_2 are immersed at z_1, z_2 and intersect transversally there. Consequently $[u_1]\cdot[u_2] \geq \#\mathcal{I}$, with equality if and only if all intersections are transverse.*

See McDuff [McD91c; McD94], Micallef–White [MW95] and [MS12a, Thm. 2.6.3, App. E]. In practice: distinct J-curves with $[u_1]\cdot[u_2]=0$ are disjoint, and with $[u_1]\cdot[u_2]=1$ meet once, transversally.

Theorem 5.28 (Adjunction formula)**.** *Let (M,J) be an almost complex 4-manifold with J smooth, and $u\colon\Sigma\to M$ a simple J-holomorphic curve from a closed connected Riemann surface of genus g, with $A=[u]$. There is an integer $\delta(u)\geq 0$, the number of double points counted with the multiplicities of Theorem 5.27, with singular points contributing via their local δ-invariants, such that*

$$2\delta(u)=A\cdot A-c_1(A)+2-2g,$$

and $\delta(u)=0$ if and only if u is an embedding. Equivalently, $g\leq 1+\frac{1}{2}\big(A\cdot A-c_1(A)\big)$, with equality if and only if u is an embedding.

As a sanity check, take a line in $\mathbb{CP}^2$: $1-3+2=0$. A smooth cubic gives $9-9+0=0$, and a nodal rational cubic gives $9-9+2=2=2\delta$. See McDuff [McD91c] and [MS12a, Thm. 2.6.4, App. E].

Theorem 5.29 (Automatic transversality in dimension four)**.** *Let (M,J) be an almost complex 4-manifold with J smooth, and $u\colon\Sigma\to M$ an immersed J-holomorphic curve from a closed Riemann surface of genus g with $c_1([u])\geq 1$. Then u is regular: the moduli space of unparametrized J-curves of genus g (with varying domain complex structure) is, near $[u]$, a smooth manifold of dimension $2c_1([u])+2g-2$. This holds for every J, not just generic ones.*

This is Hofer–Lizan–Sikorav [HLS97]: split $u^*TM=T\Sigma\oplus N_u$ and apply the $n=1$ case of Theorem 5.24 to the normal operator. Wendl [Wen10] extended it to punctured and non-immersed, including multiply covered, curves; for closed curves his criterion reads $c_1([u])>Z(du)$, where $Z(du)\geq 0$ counts critical points with multiplicity. An embedded sphere of self-intersection m has $c_1=m+2$ (Theorem 5.28), so it is regular for every J once $m\geq -1$. That fact drives McDuff's classification of rational and ruled surfaces (Chapter 6).

Gluing is the converse of compactness: every nodal curve that is cut out transversally is a Gromov limit of smooth curves, in a unique way up to a gluing parameter.

Theorem 5.30 (Gluing two spheres)**.** *Let (M,ω) be closed and $J\in\mathcal{J}_\tau(M,\omega)$. Let u_0,u_1 be simple J-spheres in classes A_0,A_1 with $u_0(S^2)\neq u_1(S^2)$ and $u_0(\infty)=u_1(0)$. Assume D_{u_0},D_{u_1} are onto and that $(v_0,v_1)\mapsto(v_0(\infty),v_1(0))$, defined on $\mathcal{M}^*(A_0;J)\times\mathcal{M}^*(A_1;J)$, is transverse at (u_0,u_1) to the diagonal of $M\times M$. Put $A=A_0+A_1$ and $d=2n+2c_1(A)-6$. Then the class $[u_0,u_1]\in\overline{\mathcal{M}}_{0,0}(M,A;J)$ has a neighbourhood U in the Gromov topology with the following properties.*

- *$U_2\coloneqq U\cap\big\{\text{two-component stable maps}\big\}$ is a smooth manifold of dimension $d-2$, and $U\setminus U_2\subset\mathcal{M}^*_{0,0}(A;J)$.*
- *There is a homeomorphism from a neighbourhood of the zero section of the complex line bundle $L\to U_2$ with fibre $T_\infty S^2\otimes_{\mathbb{C}}T_0S^2$ onto U. It is the identity on the zero section and restricts to a diffeomorphism from the complement of the zero section onto $U\setminus U_2$.*

This is the content of McDuff–Salamon's gluing theorem [MS12a, Ch. 10, esp. §10.9], stated there for parametrized curves with an explicit gluing parameter; the form above follows by passing to quotients. The same analysis applies to every simple stable map with transverse edge evaluation, so for semipositive M and generic J the space $\overline{\mathcal{M}}_{0,k}(M,A;J)$ is, away from multiple covers, stratified with top stratum $\mathcal{M}^*_{0,k}(A;J)$ and other strata of codimension ≥ 2. Gluing is the part of the foundations students most often take on faith; Chapter 10 of [MS12a] is where to stop.

5.4 Gromov–Witten invariants and quantum cohomology

Definition 5.31 (Pseudocycle)**.** Let X be a smooth manifold. A *d-dimensional pseudocycle* is a smooth map $f\colon V \to X$ from an oriented d-manifold V without boundary (not necessarily compact) such that $f(V)$ has compact closure and the *limit set* $\Omega_f := \bigcap_{K\subset V \text{ compact}} \overline{f(V\setminus K)}$ is contained in the image of a smooth map $W \to X$ defined on a second countable manifold all of whose components have dimension $\le d-2$. Two pseudocycles of complementary dimension can be put in general position, and their intersection number is then a finite signed count that depends only on their bordism classes.

Bordism classes of d-dimensional pseudocycles in X correspond naturally to $H_d(X;\mathbb{Z})$ (Zinger [Zin08]; [MS12a, Ch. 6]), so a pseudocycle defines an integral class without triangulating the moduli space.

Convention 5.32 (Novikov ring)**.** For a closed symplectic manifold (M,ω) let $\Gamma := H_2^S(M)/(\ker\omega \cap \ker c_1)$, where $H_2^S(M)\subset H_2(M;\mathbb{Z})$ is the image of the Hurewicz map. This is the same group as the deck group $\pi_2(M)/(\ker\omega\cap\ker c_1)$ used in Floer theory. The *Novikov ring* is

$$\Lambda := \left\{ \textstyle\sum_{A\in\Gamma} \lambda_A q^A :\ \lambda_A\in\mathbb{Q},\ \#\{A : \lambda_A\neq 0,\ \omega(A)\le c\} < \infty \text{ for all } c\in\mathbb{R}\right\},$$

graded by $\deg q^A := 2c_1(A)$; Chapter 7 writes e^A for q^A and allows any coefficient field. Since ω is injective on $\ker c_1\subset\Gamma$, every nonzero homogeneous element is invertible (Λ is a graded field, the Novikov field of C9), and Λ is a field if ω is injective on Γ. If M is monotone with minimal Chern number N and $\omega\neq 0$ on $\pi_2(M)$, then $\Gamma\cong\mathbb{Z}$ and $\Lambda=\mathbb{Q}((q))$ with $\deg q = 2N$. The *universal Novikov field* over a field $K\supseteq\mathbb{Q}$ is $\Lambda_{\text{univ}} := \left\{\sum_{i\ge1} a_i T^{\lambda_i} : a_i\in K,\ \lambda_i\in\mathbb{R},\ \lambda_i\to+\infty\right\}$. It is ungraded, and $q^A\mapsto T^{\omega(A)}$ defines a ring map $\Lambda\to\Lambda_{\text{univ}}$ that forgets the grading. Throughout this document, Hamiltonian Floer theory and spectral invariants (Chapter 7) use Λ, whose grading is indispensable there. Lagrangian Floer theory and Fukaya categories (Chapter 8) use Λ_{univ} and its valuation ring $\Lambda_0=\{\lambda_i\ge 0\}$, since bounding cochains need a complete valued field. Authors using q^{-A} instead reverse every finiteness condition.

Let (M,ω) be closed and semipositive. Call $J\in\mathcal{J}_\tau(M,\omega)$ *GW-regular* if D_u is onto for every simple J-sphere u, and if for every tree T the edge evaluation maps of tuples of simple J-spheres with pairwise distinct images are transverse to the diagonals. The GW-regular structures form a comeagre set $\mathcal{J}_{\text{reg}}(M,\omega)$ [MS12a, Ch. 6]. Consider the condition

($\star$) $\dim M = 4$, or $c_1(B)\neq 0$ for every $B\in H_2^S(M)$ with $\omega(B)>0$.

It is a global form of the per-class hypothesis of [MS12a, Thm. 6.6.1]; it holds for monotone manifolds and in dimension four, and fails for Calabi–Yau threefolds, where Example 5.26 shows that the argument below breaks down.

Theorem 5.33 (Genus-zero GW pseudocycle; Ruan–Tian 1995, McDuff–Salamon)**.** *Let (M,ω) be a closed semipositive symplectic $2n$-manifold satisfying ($\star$), $A\in H_2(M;\mathbb{Z})\setminus\{0\}$, $k\ge1$, and $J\in\mathcal{J}_{\text{reg}}(M,\omega)$. Then $\mathrm{ev}_J\colon \mathcal{M}^*_{0,k}(A;J)\to M^k$, $[u,z_1,\dots,z_k]\mapsto(u(z_1),\dots,u(z_k))$, is a pseudocycle of dimension $2n+2c_1(A)+2k-6$. Its bordism class is independent of $J\in\mathcal{J}_{\text{reg}}(M,\omega)$, and it is unchanged under deformation of ω through symplectic forms satisfying the same hypotheses.*

For $a_1,\dots,a_k\in H^*(M;\mathbb{Z})$ one defines $\mathrm{GW}^M_{A,k}(a_1,\dots,a_k)\in\mathbb{Z}$ as the intersection number of ev_J with a product of pseudocycles Poincaré dual to the a_i, and as 0 unless $\sum_i \deg a_i =$

$2n+2c_1(A)+2k-6$. For the zero class one sets $\mathrm{GW}_{0,3}(a_1,a_2,a_3) := \int_M a_1\cup a_2\cup a_3$ and $\mathrm{GW}_{0,k} := 0$ for $k \neq 3$. Without $(\star)$, the covers of Example 5.26 with $c_1(B) = 0$ have evaluation images of full dimension, and one needs domain-dependent (inhomogeneous) perturbations of $\bar\partial_J$, as in Ruan–Tian [RT95]; McDuff–Salamon do this with domain-dependent almost complex structures via the graph construction [MS12a, §6.7]. The resulting invariants agree with the virtual ones of Remark 5.43.

Theorem 5.34 (Kontsevich–Manin axioms, genus zero)**.** *Let (M,ω) be as in Theorem 5.33 and let $a_i \in H^*(M;\mathbb{Z})$.*

1. *(*Effectivity*) If $\mathrm{GW}_{A,k}(a_1,\dots,a_k) \neq 0$, then for every $J \in \mathcal{J}_\tau(M,\omega)$ there is a genus-zero stable J-map in class A; in particular $A = 0$ or $\omega(A) > 0$.*
2. *(*Symmetry*) For a permutation σ, $\mathrm{GW}_{A,k}(a_{\sigma(1)},\dots,a_{\sigma(k)}) = \varepsilon\ \mathrm{GW}_{A,k}(a_1,\dots,a_k)$, where $\varepsilon = \pm 1$ is the Koszul sign of σ acting on the odd-degree a_i.*
3. *(*Grading*) $\mathrm{GW}_{A,k}(a_1,\dots,a_k) = 0$ unless $\sum_i \deg a_i = 2n + 2c_1(A) + 2k - 6$.*
4. *(*Fundamental class*) If $k \geq 1$ and $(A,k) \neq (0,3)$, then $\mathrm{GW}_{A,k}(a_1,\dots,a_{k-1},1) = 0$.*
5. *(*Divisor*) If $k \geq 2$, $(A,k) \neq (0,3)$ and $\deg a_k = 2$, then*
$$\mathrm{GW}_{A,k}(a_1,\dots,a_k) = \Big(\int_A a_k\Big)\ \mathrm{GW}_{A,k-1}(a_1,\dots,a_{k-1}).$$
6. *(*WDVV*) For $a,b,c,d \in H^{\mathrm{ev}}(M;\mathbb{Z})$, a basis $\{e_\nu\}$ of $H^*(M;\mathbb{Q})$ and the dual basis $\{e^\nu\}$ with $\int_M e_\nu \cup e^\mu = \delta_\nu^\mu$, and every A,*
$$\sum_{A_0+A_1=A}\ \sum_\nu \mathrm{GW}_{A_0,3}(a,b,e_\nu)\ \mathrm{GW}_{A_1,3}(e^\nu,c,d)$$
$$= \sum_{A_0+A_1=A}\ \sum_\nu \mathrm{GW}_{A_0,3}(a,c,e_\nu)\ \mathrm{GW}_{A_1,3}(e^\nu,b,d).$$
The sums are finite by Corollary 5.20.
7. *(*Deformation*) The invariants depend only on the deformation class of ω among forms satisfying the hypotheses.*

See [MS12a, Ch. 7], [KM94a] and [RT95]. WDVV is the shadow of the splitting axiom, which expresses invariants with a fixed cross-ratio of four marked points as sums over two-component trees; gluing (Theorem 5.30) proves it. The same argument gives WDVV with any number of extra insertions distributed over the two factors, with Koszul signs for odd classes; in all genera GW classes form a cohomological field theory [KM94a].

Example 5.35 (Lines)**.** In $\mathbb{CP}^n$ with L the line class, $\mathrm{GW}^{\mathbb{CP}^n}_{L,2}(\mathrm{pt},\mathrm{pt}) = 1$: through two points passes exactly one line. Indeed $2n + 2(n+1) + 4 - 6 = 4n = \deg\mathrm{pt} + \deg\mathrm{pt}$, the integrable structure J_0 is regular (every holomorphic sphere in $\mathbb{CP}^n$ is, since $T\mathbb{CP}^n$ is globally generated), and the line is cut out transversally with positive sign.

Theorem 5.36 (Kontsevich's formula; Kontsevich–Manin 1994, Ruan–Tian 1995)**.** *For $d \geq 1$ let $N_d := \mathrm{GW}^{\mathbb{CP}^2}_{dL,3d-1}(\mathrm{pt},\dots,\mathrm{pt})$. Then N_d is the number of irreducible rational plane curves of degree d*

through $3d-1$ *points in general position. Moreover* $N_1 = 1$ *and, for* $d \geq 2$,

$$N_d = \sum_{\substack{d_1+d_2=d \\ d_1,d_2\geq 1}} N_{d_1} N_{d_2}\, d_1^2 d_2 \left(d_2 \binom{3d-4}{3d_1-2} - d_1 \binom{3d-4}{3d_1-1} \right).$$

Hence $N_2 = 1$, $N_3 = 12$, $N_4 = 620$ *and* $N_5 = 87304$.

The recursion is WDVV for $h, h, \mathrm{pt}, \mathrm{pt}$ with the remaining $3d-4$ point insertions distributed over the two factors, plus the divisor axiom; for the enumerative statement see also [FP97]. It is the best five-minute advertisement for the subject, and $N_3 = 12$ (the nodal cubics in a pencil) is a good check of one's signs.

Definition 5.37 (Small quantum cohomology)**.** Let (M, ω) be as in Theorem 5.33. Set $\mathrm{QH}^*(M) := H^*(M;\mathbb{Q}) \otimes_{\mathbb{Q}} \Lambda$, graded by $\deg(a \otimes q^A) = \deg a + 2c_1(A)$, and extend $\langle x, y\rangle := \int_M x \cup y$ Λ-bilinearly. For $a, b \in H^*(M;\mathbb{Q})$ the *quantum product* $a * b \in \mathrm{QH}^*(M)$ is the unique element with

$$\langle a * b, c\rangle = \sum_{A \in H_2^S(M)} \mathrm{GW}_{A,3}(a, b, c)\, q^{[A]} \qquad \text{for all } c \in H^*(M;\mathbb{Q}),$$

where $[A] \in \Gamma$ is the image of A. The product is extended Λ-bilinearly.

Proposition 5.38. $(\mathrm{QH}^*(M), *)$ *is an associative, graded-commutative unital* Λ*-algebra:* $a * b = (-1)^{\deg a \deg b}\, b * a$*, and the unit is* $1 \in H^0(M)$*. It satisfies* $\langle a * b, c\rangle = \langle a, b * c\rangle$ *and* $a * b = a \cup b +$ (*terms with* $\omega(A) > 0$)*. Associativity is equivalent to WDVV.*

See [MS12a, Prop. 11.1.9]. The grading works because, by the grading axiom, $\deg q^A = 2c_1(A)$ exactly compensates the degree drop.

Example 5.39. Let $h \in H^2(\mathbb{CP}^n;\mathbb{Z})$ satisfy $\int_L h = 1$. Then $h^{*k} = h^k$ for $k \leq n$ by degree reasons, and $h * h^n = q$ because $\mathrm{GW}_{L,3}(h, h^n, h^n) = \mathrm{GW}_{L,2}(\mathrm{pt}, \mathrm{pt}) = 1$ by the divisor axiom. Hence

$$\mathrm{QH}^*(\mathbb{CP}^n) \cong \Lambda[h]/(h^{n+1} - q), \qquad \Lambda = \mathbb{Q}((q)),\ \deg q = 2(n+1).$$

Since $h^{n+1} - q$ is Eisenstein over $\mathbb{Q}((q))$, $\mathrm{QH}^*(\mathbb{CP}^n)$ is a field. For $M = S^2 \times S^2$ with $\omega = a\sigma \oplus b\sigma$, let $A = [S^2 \times \mathrm{pt}]$ and $B = [\mathrm{pt} \times S^2]$, and let $x, y \in H^2$ be dual to A and B ($\int_A x = \int_B y = 1$, $\int_B x = \int_A y = 0$). Then $\mathrm{QH}^*(S^2 \times S^2) \cong \Lambda[x,y]/(x^2 - q^A,\ y^2 - q^B)$ with $\deg q^A = \deg q^B = 4$; when $a = b$ one has $q^A = q^B$ in Γ. The computation uses only $\mathrm{GW}_{A,1}(\mathrm{pt}) = \mathrm{GW}_{B,1}(\mathrm{pt}) = 1$ and the divisor axiom.

Definition 5.40 (Semisimplicity)**.** A finite-dimensional commutative algebra Q over a field F is *semisimple* if it is isomorphic to a finite product of field extensions of F, equivalently if it has no nonzero nilpotents. When Λ is a field we call $\mathrm{QH}^*(M)$ semisimple if it is semisimple as a Λ-algebra.

For example, $\mathrm{QH}^*(\mathbb{CP}^n)$ is a field, and for monotone $S^2 \times S^2$ the algebra splits as $L \times L$ with $L = \Lambda[x]/(x^2 - q)$. Entov–Polterovich [EP03] use a version for the degree-zero part of quantum homology over the degree-zero part of the Novikov ring; a field summand yields a Calabi quasimorphism (Chapter 7).

A loop γ in $\mathrm{Ham}(M, \omega)$ based at id defines by clutching a Hamiltonian fibration $P_\gamma \to S^2$ with fibre M. Counting J-holomorphic sections of P_γ, weighted by q to the power of their section class relative to a reference class and recorded by their intersection with a fibre, gives the *Seidel element* $\mathcal{S}(\gamma) \in \mathrm{QH}^*(M)$.

Theorem 5.41 (Seidel 1997)**.** *Let (M,ω) be closed and monotone. Then $\gamma \mapsto \mathcal{S}(\gamma)$ induces a group homomorphism*

$$\mathcal{S}\colon \pi_1(\mathrm{Ham}(M,\omega)) \to \mathrm{QH}^*(M)^\times_{\mathrm{hom}}/\{q^A : A \in \Gamma\}$$

to the group of homogeneous invertible elements modulo the subgroup of powers of q. It satisfies $\mathcal{S}(\gamma_1\gamma_2) = \mathcal{S}(\gamma_1) * \mathcal{S}(\gamma_2)$ and $\mathcal{S}(const) = 1$.

Seidel [Sei97] works under a condition similar to weak monotonicity, with $\mathcal{S}$ defined on an extension of π_1; McDuff [McD00] extended it to all closed manifolds using virtual cycles. For $\gamma_t[z_0 : \cdots : z_n] = [e^{2\pi i t} z_0 : z_1 : \cdots : z_n]$, generating $\pi_1(\mathrm{PU}(n+1)) \cong \mathbb{Z}/(n+1)$, Seidel's computation [Sei97, §11] gives $\mathcal{S}(\gamma) = h$ up to a power of q. Since $h^{n+1} = q$ and $h^k \notin q^{\mathbb{Z}}$ for $0 < k \le n$, it follows that $\pi_1(\mathrm{PU}(n+1)) \to \pi_1(\mathrm{Ham}(\mathbb{CP}^n))$ is injective. For $(S^2 \times S^2, \lambda\sigma \oplus \sigma)$ with $\lambda > 1$, McDuff found an element of infinite order in $\pi_1(\mathrm{Ham})$ [McD87], which Seidel recovered from a circle action on the Hirzebruch surface F_2 [Sei97, §11]; McDuff–Tolman [MT06] computed $\mathcal{S}$ for circle actions on toric manifolds. Lalonde–McDuff–Polterovich [LMP99] used Seidel's method to show, for many M, that the map $H_*(M;\mathbb{Q}) \to H_{*+1}(M;\mathbb{Q})$ swept out by a Hamiltonian loop vanishes, so that Hamiltonian loops are topologically rigid. The equivalent splitting $H^*(P_\gamma) \cong H^*(M) \otimes H^*(S^2)$ was proved over $\mathbb{Q}$ for all closed M by McDuff [McD00] and over $\mathbb{Z}$ by Abouzaid–McLean–Smith [AMS21].

Remark 5.42 (GW invariants as symplectic invariants). Being deformation invariant, GW invariants distinguish deformation classes. Ruan [Rua94] showed that the Kähler forms on $V \times S^2$ and $W \times S^2$, with $V = \mathbb{CP}^2 \# 8\overline{\mathbb{CP}}^2$ and W the Barlow surface, are not deformation equivalent, although $V \times S^2$ and $W \times S^2$ are diffeomorphic (V and W are homeomorphic but not diffeomorphic). For smooth projective varieties the symplectic invariants agree with the algebraic ones defined via the virtual class of Li–Tian and Behrend–Fantechi [LT98a; BF97], by Li–Tian [LT99] and Siebert [Sie99]. In dimension four the right count is Taubes' Gr, which counts embedded curves (with special weights for multiply covered square-zero tori) and equals Seiberg–Witten [Tau96a; Tau00]; see Chapter 6.

Remark 5.43 (The general case). For every closed symplectic (M,ω) and all $g, k \ge 0$ and A with $A \neq 0$ or $2g+k \ge 3$, there is a virtual fundamental class of $\overline{\mathcal{M}}_{g,k}(M,A;J)$ of dimension $(1-g)(2n-6) + 2c_1(A) + 2k$. It is independent of J and yields rational invariants $\mathrm{GW}_{g,k,A}(a_1,\dots,a_k;\beta) \in \mathbb{Q}$, where β is a class pulled back from Deligne–Mumford space when $2g+k \ge 3$. These satisfy the Kontsevich–Manin axioms and reduce to the invariants above in the situation of Theorem 5.33. The first constructions were by Fukaya–Ono [FO99], Li–Tian [LT98b], Ruan [Rua99] and Siebert [Sie96] (higher-genus semipositive invariants: Ruan–Tian [RT97]); their status is discussed in Section 5.5. Virtual invariants are genuinely rational: in the quintic threefold the 2875 lines contribute $2875/8$ to the genus-zero invariant of $2L$.

5.5 Virtual fundamental classes: the state of the foundations

For general (M,ω) and any J, the space $\overline{\mathcal{M}} = \overline{\mathcal{M}}_{g,k}(M,A;J)$ is compact and Hausdorff, and near each point $[u]$ it is modelled on a *Kuranishi chart* $(V, E, \Gamma_u, s, \psi)$. Here Γ_u is the finite automorphism group of u; E is a finite-dimensional Γ_u-representation that covers $\operatorname{coker} D_u$; V is a Γ_u-manifold of solutions of $\bar\partial_J u \in E$ near u, including gluing parameters; $s\colon V \to E$ is an equivariant section; and ψ identifies $s^{-1}(0)/\Gamma_u$ with a neighbourhood of $[u]$. One always has $\dim V - \operatorname{rank} E = \mathrm{vdim}$, but s need not be transverse. One wants a class $[\overline{\mathcal{M}}]^{\mathrm{vir}}$ in $\check{H}_{\mathrm{vdim}}(\overline{\mathcal{M}};\mathbb{Q})$,

or at least its pushforward under evaluation and stabilization, independent of all choices. The obstacles: charts of different dimensions are related by embeddings rather than diffeomorphisms; isotropy forces multivalued perturbations and rational coefficients; and charts built from gluing parameters are not smooth in the naive sense, so one must choose a category. The last is the most often underestimated. The main approaches are these.

Kuranishi structures

Fukaya–Ono [FO99] built the theory and applied it to the Arnold conjecture; FOOO [Fuk+09] made it the basis of Lagrangian Floer theory. In 2012 McDuff and Wehrheim questioned several foundational points (the topology of the glued virtual neighbourhood, the definition of good coordinate systems, the treatment of isotropy). FOOO replied with a long technical account [Fuk+12], which became the monograph [Fuk+20]. McDuff–Wehrheim developed their own framework of Kuranishi atlases [MW17a; MW17b; MW18]. The debate was largely settled by 2017.

Polyfolds

Hofer–Wysocki–Zehnder [HWZ21] replace Banach manifolds by *polyfolds*, built on a new notion of smoothness (sc-calculus) in which gluing is smooth. The whole compactified moduli space becomes the zero set of a single sc-Fredholm section, which is perturbed by multisections. The application to Gromov–Witten theory is [HWZ17]; the SFT programme is surveyed in Fish–Hofer [FH18].

Implicit atlases

Pardon [Par16] avoids smooth structures on gluing altogether: thickened moduli spaces with obstruction bundles, and a sheaf-theoretic virtual class with $\mathbb{Q}$ coefficients built from topological transversality. The same method yields a construction of contact homology [Par19].

Global Kuranishi charts

Abouzaid–McLean–Smith [AMS21] construct, for genus-zero moduli spaces, a single chart $(G, \mathcal{T}, \mathcal{E}, s)$. Here G is a compact Lie group, $\mathcal{T}$ a G-manifold, $\mathcal{E} \to \mathcal{T}$ a G-vector bundle and s a section with $s^{-1}(0)/G \cong \overline{\mathcal{M}}$; domains are framed by projective embeddings coming from an integral symplectic form, and the virtual class becomes finite-dimensional equivariant topology. Hirschi–Swaminathan [HS24b] built global charts in all genera and proved a product formula; Abouzaid–McLean–Smith [AMS24] defined GW invariants in K-theory and Morava-local complex-oriented theories; Bai–Xu [BX22] combined global charts with Fukaya–Ono–Parker normally complex perturbations to get integral counts and the Arnold conjecture over $\mathbb{Z}$ (Chapter 7).

Other routes

Siebert [Sie96] and Li–Tian [LT98b] used global constructions in an algebraic spirit. Ionel–Parker's thin compactifications [IP19] need only a manifold-like top stratum with homologically thin complement. Cieliebak–Mohnke [CM07] defined genus-zero invariants of closed manifolds with rational $[\omega]$ without virtual techniques, stabilizing domains by extra marked points mapped to a Donaldson hypersurface.

My view: for closed Gromov–Witten theory and Hamiltonian Floer theory the foundations are settled, by several independent routes that agree wherever compared, and nobody should hesitate to use GW invariants of arbitrary closed symplectic manifolds. What remains concerns structure (chain-level operations, integrality, generalized cohomology) rather than existence. A

student should learn the semipositive pseudocycle theory thoroughly from McDuff–Salamon, then read the introduction and first sections of Pardon [Par16] and the global chart construction of [AMS21]. In 2026 global charts are the best entry point: they turn the virtual class into equivariant differential topology that a topologist can check. Kuranishi structures and polyfolds are references for their particular strengths: the A_∞ machinery of FOOO and the SFT-scale generality of polyfolds.

5.6 What to read

McDuff–Salamon, J-holomorphic Curves and Symplectic Topology *[MS12a]*
The reference, and the book whose numbering I have cited throughout. Read Ch. 2 (local theory, with App. E on singularities), Ch. 3 (Fredholm theory and transversality, with App. C on Riemann–Roch), Ch. 4 (compactness for spheres), Ch. 5–6 (stable maps, their moduli spaces, pseudocycles) and Ch. 7 (GW invariants), in that order. Chapter 10 (gluing) is long but should be read at least once. Section 6.7 (domain-dependent structures via the graph construction) is needed for the general semipositive case, Chapter 8 treats Hamiltonian perturbations, and Chapter 11 covers quantum cohomology and the Seidel representation. Chapter 9 overlaps with Chapter 6; consult Appendix B as needed.

Wendl, Lectures on Holomorphic Curves in Symplectic and Contact Geometry *[Wen14]*
The friendliest complete account of the local theory, Fredholm theory and compactness, with an eye toward punctured curves. For dimension four (positivity of intersections, automatic transversality, McDuff's theorems) read his book [Wen18].

Gromov [Gro85]
Read the introduction and the first section, to see how much was there from the start: compactness, nonsqueezing, $\mathrm{Symp}(S^2 \times S^2)$, the count of curves through points. Do not try to extract complete proofs.

Audin–Lafontaine (eds.) [AL94]
Sikorav's chapter [Sik94] for monotonicity and the local theory, McDuff's chapter [McD94] for singularities and positivity of intersections, and Pansu's [Pan94] for compactness. The book is dated but these chapters are not.

Hummel [Hum97]
Gromov compactness for closed curves of arbitrary genus with varying domains, done carefully.

McDuff, Lectures on Gromov invariants for symplectic 4-manifolds *[McD97]*
How the four-dimensional machinery is used; a bridge to Taubes.

Kontsevich–Manin [KM94a]; Ruan–Tian [RT95]
The axioms and the enumerative payoff; and the first complete symplectic construction of quantum cohomology for semipositive manifolds.

Fulton–Pandharipande [FP97]; Cox–Katz [CK99]
The algebro-geometric side: stable maps, Kontsevich's formula and mirror symmetry. A symplectic topologist who cannot compute with these is handicapped.

Pardon [Par16] (§§1–2); Abouzaid–McLean–Smith [AMS21] (introduction and the chart construction)
The modern entry points to virtual techniques.

FOOO [Fuk+20]; Hofer–Wysocki–Zehnder [HWZ21] (introduction only)
References, not first reads.

5.7 Frontier and open problems

Multiple covers and super-rigidity. Wendl [Wen23] proved that unbranched covers are generically regular and that simple index-zero curves are generically super-rigid in dimension greater than four. So GW invariants of symplectic Calabi–Yau threefolds reduce to local contributions of finitely many embedded curves. Doan–Walpuski [DW21] studied when an energy bound forces a genus bound (a Castelnuovo-type bound) and proved a new result of this kind in dimension six, in particular for symplectic Calabi–Yau 6-manifolds. Generic regularity of branched covers, and super-rigidity in dimension four, are only partially understood.

Gopakumar–Vafa. Integrality of the BPS numbers of symplectic 6-manifolds was proved by Ionel–Parker [IP18]. The finiteness part was proved by Doan–Ionel–Walpuski [DIW21], to appear in the *Annals*.

Integral and generalized invariants. Global charts give GW invariants in K-theory and Morava-local theories [AMS24] and integral counts [BX22]; a general theory of GW invariants in arbitrary complex-oriented theories, and its Floer analogues (Abouzaid–Blumberg, Rezchikov; Chapter 7), are under active development.

Open and real GW theory. Welschinger's invariants [Wel05] and Solomon's open GW invariants [Sol06] count real curves and disks. Their foundations and their relation to Floer theory are active areas (Chapter 8).

GW versus sheaf counts. The MNOP correspondence [Mau+06] between GW and Donaldson–Thomas/Pandharipande–Thomas invariants of threefolds was proved by Pardon [Par23] for threefolds with nef anticanonical bundle and primary insertions (preprint, 2023).

Question 5.44. *To what extent do Gromov–Witten invariants, including all their structure (descendants, higher genus, the action of $\pi_1(\mathrm{Ham})$), determine the deformation class of a symplectic form on a closed manifold, among forms whose compatible almost complex structures are homotopic?*

Ruan's examples (Remark 5.42) show that GW invariants see more than the diffeomorphism type and c_1. Beyond that, very little is known. In dimension four the analogous question for Taubes' invariants (is a symplectic form determined up to deformation and diffeomorphism by its canonical class?) is open in general [Sal13].

Problem 5.45 (Symplectic isotopy problem)**.** *Is every smooth connected symplectic surface in $(\mathbb{CP}^2, \omega_{\mathrm{FS}})$ symplectically isotopic to a complex algebraic curve?*

The answer is yes in degree ≤ 17 by Siebert–Tian [ST05], building on Gromov (degrees 1, 2) and later work. The general case is open. The method is pure J-holomorphic curve theory: deform J to the integrable structure while keeping the surface J-holomorphic, and control the singular degenerations.

5.8 Logical dependencies

To state the main results one needs, in order: tame and compatible J and c_1 (Chapter 2); J-holomorphic curves and energy (Definitions 5.2 and 5.3); simple curves (Definition 5.12); stable

maps and Gromov convergence (Definitions 5.16, 5.17 and 5.21); real Cauchy–Riemann operators and the boundary Maslov index (Definition 5.23, Chapter 1); pseudocycles (Definition 5.31); semipositivity; and the Novikov ring (Convention 5.32). The statements presuppose Sobolev spaces $W^{k,p}$ on surfaces, Banach manifolds and bundles, Fredholm operators and the Sard–Smale theorem (Theorem 5.25), the Gromov topology (Theorem 5.19) and Deligne–Mumford spaces (Remark 5.43). Theorems 5.33 and 5.34 presuppose Theorems 5.18, 5.25 and 5.30. Definition 5.37 and Theorem 5.41 presuppose the axioms. Virtual classes (Section 5.5) presuppose orbifold, Čech or sheaf-theoretic homology.

Chapter 6

Rigidity: what holomorphic curves see

This chapter collects the theorems that turned symplectic topology from a branch of Hamiltonian mechanics into a subject of its own. Almost all of them follow one template, which Gromov introduced in 1985 [Gro85]: find a moduli space of pseudoholomorphic curves that is *forced* to be nonempty by a deformation-invariant count (a Gromov–Witten invariant, a Seiberg–Witten invariant, or a cobordism argument), then extract geometry from the curves it contains, using the area–energy identity, monotonicity, and, in dimension four, positivity of intersections and the adjunction formula. I treat the foundations of Chapter 5 as a black box: Gromov compactness, transversality for somewhere injective curves, and the definition of genus-zero Gromov–Witten invariants of semipositive manifolds.

What to take away: (i) nonsqueezing and the theory of symplectic capacities, the quantitative face of rigidity; (ii) the embedding problem, which in dimension four is now astonishingly well understood (ellipsoids are governed by ECH capacities and by exceptional spheres in blow-ups of $\mathbb{CP}^2$) and in higher dimensions is still largely open; (iii) holomorphic disks as obstructions for Lagrangians, and the exotic tori they detect; (iv) Gromov's foliation technique in dimension four, which computes symplectomorphism groups and underlies McDuff's characterization of rational and ruled surfaces; (v) Taubes' theorem SW = Gr, the bridge to gauge theory that produces most of what we know about symplectic 4-manifolds.

Convention 6.1 (Normalizations of standard domains)**.** Throughout this chapter domains in $\mathbb{C}^n$ carry ω_0 (C1) and are normalized by *capacity*, not radius. For $a, a_1, \dots, a_n > 0$ set

$$E(a_1,\dots,a_n) = \left\{ z \in \mathbb{C}^n : \sum_{j=1}^n \pi|z_j|^2/a_j < 1 \right\}, \qquad B^{2n}(a) = E(a,\dots,a) = \left\{ z : \pi|z|^2 < a \right\},$$
$$Z^{2n}(a) = B^2(a) \times \mathbb{C}^{n-1}, \qquad P(a_1,\dots,a_n) = B^2(a_1) \times \cdots \times B^2(a_n).$$

All of these are open. Thus $\operatorname{vol} B^{2n}(a) = a^n/n!$, $\operatorname{vol} B^4(a) = a^2/2$ and $\operatorname{vol} E(a,b) = ab/2$, where $\operatorname{vol} = \int \omega^n/n!$. The ball "of radius r" of many sources (Gromov, Hofer–Zehnder, Guth, Hind–Kerman) is our $B^{2n}(\pi r^2)$, so statements quoted from them have πr^2 where we have a. These are the conventions of McDuff–Schlenk and Hutchings. A *symplectic embedding* $(M,\omega) \hookrightarrow (M',\omega')$ is a smooth embedding φ with $\varphi^*\omega' = \omega$; unless said otherwise $\dim M = \dim M'$.

6.1 Nonsqueezing and symplectic capacities

Theorem 6.2 (Nonsqueezing; Gromov 1985)**.** *Let $n \geq 1$ and $a, A > 0$. There is a symplectic embedding $B^{2n}(a) \hookrightarrow Z^{2n}(A)$ if and only if $a \leq A$.*

Proof idea. This is the proof to learn, in the form given in [MS12a, Ch. 9]. Suppose $\varphi\colon B^{2n}(a) \hookrightarrow Z^{2n}(A)$ and fix $a' < a$, $\epsilon > 0$. The image of the closed ball $\overline{B^{2n}(a')}$ is compact, so it lies in $B^2(A)\times[-R,R]^{2n-2}$, and this embeds symplectically into $M = S^2\times T^{2n-2}$ with $\omega = \sigma\oplus\omega_T$, where $\int_{S^2}\sigma = A+\epsilon$ and the flat torus is large. Choose an ω-compatible J on M that equals $\varphi_* J_0$ on $\varphi(B^{2n}(a'))$. The class $\beta = [S^2\times\mathrm{pt}]$ generates $\pi_2(M)\cong\mathbb{Z}$ and has minimal positive area, so no bubbling can occur in class β; for the product structure there is exactly one β-sphere through each point, so the genus-zero invariant counting β-spheres through a point is 1 (Chapter 5). Hence for *every* compatible J there is a J-sphere u in class β through $\varphi(0)$, of area $A+\epsilon$. The preimage $\varphi^{-1}(u)\cap B^{2n}(a')$ is a proper J_0-holomorphic curve through the centre of a Euclidean ball of capacity a', i.e. a minimal surface, and the monotonicity formula gives it area at least a'. So $a' \le A+\epsilon$ for all $a' < a$, $\epsilon > 0$.

The theorem exhibits the first symplectic invariant beyond volume, and the point of the proof is the mechanism, not the inequality: a count of curves in a closed model forces a curve through a prescribed point, and area controls everything. Ekeland–Hofer's variational proof [EH89] and Viterbo's generating functions give the same inequality from Hamiltonian dynamics; they are worth seeing, but they do not generalize to the four-dimensional results later in this chapter.

Definition 6.3 (Symplectic capacity)**.** Fix $n \ge 1$ and a class $\mathcal{C}$ of $2n$-dimensional symplectic manifolds, closed under $(M,\omega)\mapsto(M,\lambda\omega)$ for $\lambda\neq 0$ and containing $B^{2n}(1)$ and $Z^{2n}(1)$. A *symplectic capacity* on $\mathcal{C}$ is a map $c\colon\mathcal{C}\to[0,\infty]$ such that:

1. (Monotonicity) if $(M,\omega)\hookrightarrow(M',\omega')$ symplectically, then $c(M,\omega)\le c(M',\omega')$;
2. (Conformality) $c(M,\lambda\omega) = |\lambda|\,c(M,\omega)$ for all $\lambda\in\mathbb{R}\setminus\{0\}$;
3. (Nontriviality) $0 < c(B^{2n}(1))$ and $c(Z^{2n}(1)) < \infty$.

It is *normalized* if $c(B^{2n}(1)) = c(Z^{2n}(1)) = 1$. (In radius conventions this reads $c(B^{2n}(r)) = c(Z^{2n}(r)) = \pi r^2$.) A sequence of functions that satisfy (1) and (2), such as the c_k below, is also called a sequence of capacities.

The axioms are due to Ekeland–Hofer; I follow the survey [Cie+07]. Existence of a single normalized capacity is *equivalent* to nonsqueezing: if c is one and $B^{2n}(a)\hookrightarrow Z^{2n}(A)$, then $a = c(B^{2n}(a)) \le c(Z^{2n}(A)) = A$; conversely, nonsqueezing says exactly that the Gromov width below is normalized.

Definition 6.4 (Gromov width and cylindrical capacity)**.** For a $2n$-dimensional symplectic manifold (M,ω) set

$$c_G(M,\omega) = \sup\left\{a>0 : B^{2n}(a)\hookrightarrow(M,\omega)\right\}, \qquad c_Z(M,\omega) = \inf\left\{A>0 : (M,\omega)\hookrightarrow Z^{2n}(A)\right\},$$

with $\inf\emptyset = \infty$.

Proposition 6.5. *c_G and c_Z are normalized capacities on the class of all $2n$-dimensional symplectic manifolds, and every normalized capacity c on this class satisfies $c_G \le c \le c_Z$.*

Theorem 6.6 (C^0-rigidity; Eliashberg 1987, Gromov 1986)**.** *Let (M,ω) be a symplectic manifold and $\varphi_k\in\mathrm{Symp}(M,\omega)$ a sequence converging uniformly on compact subsets to a diffeomorphism φ of M. Then $\varphi^*\omega=\omega$. Thus $\mathrm{Symp}(M,\omega)$ is closed in $\mathrm{Diff}(M)$ for the C^0 (compact-open) topology.*

Proof idea. The proof via capacities is in [HZ94, Ch. 2]; Eliashberg's original [Eli87] uses wave fronts and Gromov's is in [Gro86]. A linear map $A \in \mathrm{GL}(2n,\mathbb{R})$ that preserves a normalized capacity of all ellipsoids satisfies $A^*\omega_0 = \pm\omega_0$. Monotonicity and uniform convergence give, for ellipsoids E in a Darboux chart and $\lambda < 1 < \mu$, the sandwich $c(\varphi(\lambda E)) \le c(E) \le c(\varphi(\mu E))$; rescaling at a point shows that $d\varphi(x)$ preserves capacities of ellipsoids, hence is $\pm$-symplectic. The sign is $+$: otherwise $\varphi \times \mathrm{id}_{\mathbb{R}^2}$, a C^0-limit of symplectomorphisms of $M \times \mathbb{R}^2$, would pull $\omega \oplus \omega_0$ back to $(-\omega) \oplus \omega_0$, which is neither $+$ nor $-$ of it.

This is the founding result of C^0 symplectic topology (Hamiltonian homeomorphisms, the simplicity conjecture), which is the business of Chapter 7.

Definition 6.7 (Hofer–Zehnder capacity)**.** Let (M,ω) be a symplectic manifold. Let $\mathcal{H}(M)$ be the set of smooth $H\colon M \to [0,\infty)$ such that H vanishes on a nonempty open set, and $H \equiv \max H$ outside a compact subset of $M \setminus \partial M$. Call H *admissible* if every nonconstant periodic orbit of X_H has period > 1. Then $c_{HZ}(M,\omega) = \sup\{\max H : H \in \mathcal{H}(M) \text{ admissible}\}$.

The sign convention for X_H (C3) is irrelevant here. Restricting to orbits that are contractible in M defines the larger π_1-sensitive variant c°_{HZ}.

Theorem 6.8 (Hofer–Zehnder 1990)**.** *c_{HZ} is a normalized capacity on the class of all $2n$-dimensional symplectic manifolds. If $U \subset \mathbb{R}^{2n}$ is a bounded convex open set with smooth boundary, then $c_{HZ}(U) = \min\left\{\int_\gamma \lambda_0\right\}$, the minimum over closed characteristics γ of ∂U oriented as Reeb orbits of $\lambda_0|_{\partial U}$.*

See [HZ90] and [HZ94, Ch. 3]. The real content of c_{HZ} is dynamical: if $c_{HZ}(U) < \infty$, then near any compact regular energy level in U of an autonomous Hamiltonian, almost every level carries a periodic orbit ("almost existence"), which is the route to the Weinstein conjecture in $\mathbb{R}^{2n}$ (Chapter 9).

Theorem 6.9 (Ekeland–Hofer capacities; 1989–1990)**.** *There are functions c_k^{EH}, $k \ge 1$, from subsets of $\mathbb{R}^{2n}$ to $[0,\infty]$ such that: $U \subset V \Rightarrow c_k^{EH}(U) \le c_k^{EH}(V)$; $c_k^{EH}(\psi(U)) = c_k^{EH}(U)$ for $\psi \in \mathrm{Symp}(\mathbb{R}^{2n},\omega_0)$; $c_k^{EH}(\sqrt{\lambda}\,U) = \lambda c_k^{EH}(U)$ for $\lambda > 0$; $c_1^{EH} \le c_2^{EH} \le \cdots$; and c_1^{EH} is normalized. Moreover*

$$c_k^{EH}(E(a_1,\dots,a_n)) = M_k(a_1,\dots,a_n), \qquad c_k^{EH}(P(a_1,\dots,a_n)) = k\min_j a_j,$$

where $M_k(a)$ denotes the k-th smallest element, counted with multiplicity, of the multiset of all ma_j with $m \in \mathbb{Z}_{\ge 1}$ and $1 \le j \le n$. On bounded convex domains $c_1^{EH} = c_{HZ}$.

They are built in [EH89; EH90] from S^1-equivariant minimax for the action functional on loops in $\mathbb{R}^{2n}$. For star-shaped U a symplectic embedding $U \hookrightarrow V \subset \mathbb{R}^{2n}$ agrees on compact subsets of U with a compactly supported symplectomorphism of $\mathbb{R}^{2n}$, so the c_k^{EH} are monotone under symplectic embeddings of star-shaped domains.

Example 6.10 (Polydisks in balls)**.** $P(a,\dots,a) \hookrightarrow B^{2n}(A)$ if and only if $na \le A$. The inclusion gives "if"; conversely $c_n^{EH}(P(a,\dots,a)) = na$ while $c_n^{EH}(B^{2n}(A)) = A$. The Gromov width, which equals a on the polydisk, cannot see this; higher capacities can.

Definition 6.11 (Displacement energy)**.** Let (M,ω) be symplectic. For $H \in C_c^\infty([0,1]\times M)$ put $\|H\| = \int_0^1 (\max_M H_t - \min_M H_t)\,dt$. For compact $K \subset M$ let $e(K) = \inf\{\|H\| : \varphi_H^1(K) \cap K = \emptyset\}$, and for arbitrary $A \subset M$ let $e(A) = \sup\{e(K) : K \subset A \text{ compact}\}$.

Theorem 6.12 (Energy–capacity inequalities)**.** *Let A be a subset of a symplectic manifold (M,ω).*

1. *(Hofer 1990, 1993; Hofer–Zehnder 1994) If $M=\mathbb{R}^{2n}$, then $c_G(A)\le c_{HZ}(A)\le e(A)$; in particular $e(B^{2n}(a))=a$.*
2. *(Lalonde–McDuff 1995) In general $c_G(A)\le 2e(A)$.*
3. *(Usher 2010) If M is closed, then $c^\circ_{HZ}(A)\le e(A)$; in particular $c_G(A)\le e(A)$.*

Hofer introduced the displacement energy and the metric in [Hof90]; for the inequalities see [Hof93a], [HZ94, §5.5], [LM95b], [Ush10]. Part (2) already shows that Hofer's metric on $\mathrm{Ham}(M,\omega)$ is nondegenerate for every (M,ω), which is where Chapter 7 picks up.

Capacities from Floer theory and beyond. Gutt–Hutchings [GH18] defined capacities c_k^{GH} on star-shaped domains in $\mathbb{R}^{2n}$ from positive S^1-equivariant symplectic homology. Their virtue is computability: for a *convex toric domain* $X_\Omega=\mu^{-1}(\Omega)$, where $\mu(z)=(\pi|z_1|^2,\dots,\pi|z_n|^2)$ and the reflection-symmetric extension of $\Omega\subset\mathbb{R}^n_{\ge0}$ is compact and convex,

$$c_k^{GH}(X_\Omega)=\min\Big\{\max_{w\in\Omega}\langle v,w\rangle:\ v\in\mathbb{Z}^n_{\ge0},\ \textstyle\sum_i v_i=k\Big\},$$

which recovers the formulas of Theorem 6.9. They conjectured $c_k^{GH}=c_k^{EH}$, and Gutt–Ramos prove this for all star-shaped domains in a 2024 preprint [GR24]. Siegel's higher capacities [Sie25], built from rational SFT with local tangency constraints, and the curve-theoretic capacities of McDuff–Siegel [MS24] give the strongest known obstructions to stabilized embeddings. Cieliebak–Mohnke's *Lagrangian capacity* (the supremum of the minimal area of disks on Lagrangian tori in the domain) takes the value $1/n$ on $B^{2n}(1)$ but 1 on $Z^{2n}(1)$ [CM18], a useful reminder that capacities need not agree on the ball and the cylinder. Capacities of Lagrangian products and their relation to billiards belong to Chapter 9.

Warning 6.13. It was long expected (the "strong Viterbo conjecture") that all normalized capacities agree on convex domains. This is false: Haim-Kislev–Ostrover's counterexample to Viterbo's volume–capacity conjecture [HO26] is a convex domain K with $c_{HZ}(K)^n>n!\,\mathrm{vol}(K)\ge c_G(K)^n$. See Chapter 9.

6.2 Four-dimensional embedding problems

In dimension four the embedding problem for ellipsoids, and more generally for "concave into convex" toric domains, is solved. Two technologies meet: exceptional spheres in blow-ups of $\mathbb{CP}^2$ (Gromov–Witten and Seiberg–Witten theory), and ECH capacities.

Theorem 6.14 (ECH capacities; Hutchings 2011)**.** *To every four-dimensional Liouville domain (X,ω) (Chapter 3) there is associated a sequence $0=c_0(X)<c_1(X)\le c_2(X)\le\cdots\le\infty$ such that:*

1. *(Monotonicity) a symplectic embedding $X\hookrightarrow X'$, not necessarily exact, implies $c_k(X)\le c_k(X')$ for all k;*
2. *(Conformality) $c_k(X,r\omega)=r\,c_k(X,\omega)$ for $r>0$;*
3. *(Disjoint unions) $c_k(\bigsqcup_{i=1}^m X_i)=\max\big\{\sum_i c_{k_i}(X_i):\sum_i k_i=k\big\}$;*
4. *(Ellipsoids) $c_k(\overline{E(a,b)})=N_k(a,b)$, the $(k+1)$-st smallest element, counted with repetitions, of $\big\{pa+qb:p,q\in\mathbb{Z}_{\ge0}\big\}$;*

5. *(Polydisks)* $c_k(\overline{P(a,b)}) = \min\{pa+qb : p,q\in\mathbb{Z}_{\geq 0},\ (p+1)(q+1)\geq k+1\}$.

For an open $U\subset\mathbb{C}^2$ one sets $c_k(U)=\sup c_k(X)$ over Liouville domains $X\subset U$.

They are defined in [Hut11a] by spectral invariants of embedded contact homology (Chapter 9); for this chapter only the axioms matter. For instance $c_k(B^4(a)) = da$, where $d\geq 0$ is the integer with $d(d+1)/2\leq k\leq d(d+3)/2$; and the disjoint union axiom with $k=2$ gives $c_2(B^4(a)\sqcup B^4(a)) = 2a \leq c_2(B^4(1)) = 1$, recovering Gromov's two-ball theorem.

Theorem 6.15 (Volume property; Cristofaro-Gardiner–Hutchings–Ramos 2015)**.** *If X is a four-dimensional Liouville domain with all $c_k(X)<\infty$, then $\lim_{k\to\infty} c_k(X)^2/k = 4\,\mathrm{vol}(X)$, where $\mathrm{vol}(X) = \frac{1}{2}\int_X \omega\wedge\omega$.*

Check with Convention 6.1: $c_k(B^4(a))\approx a\sqrt{2k}$ and $4\,\mathrm{vol}\,B^4(a) = 2a^2$ [CHR15].

Ball packings. The *packing number* $p_k(M,\omega)$ of a closed (or finite-volume) 4-manifold is the supremum of $k\,\mathrm{vol}(B^4(a))/\mathrm{vol}(M)$ over a such that k disjoint copies of $B^4(a)$ embed symplectically.

Theorem 6.16 (Gromov 1985; McDuff–Polterovich 1994; Biran 1997)**.** *For $M=B^4(1)$, equivalently $M=(\mathbb{CP}^2,\omega_{FS})$, the packing numbers $p_1,\dots,p_9$ are*

$$1,\ \tfrac{1}{2},\ \tfrac{3}{4},\ 1,\ \tfrac{20}{25},\ \tfrac{24}{25},\ \tfrac{63}{64},\ \tfrac{288}{289},\ 1,$$

and $p_k=1$ for all $k\geq 9$.

The case $k=2$ is Gromov's two-ball theorem [Gro85]; $k\leq 9$ is [MP94], with explicit packings by Traynor [Tra95]; full packings for $k\geq 9$ are Biran's [Bir97], who later proved *packing stability* (full packings by k equal balls for all large k) for every closed symplectic 4-manifold with rational $[\omega]$ [Bir99]; Buse–Hind extended this to all dimensions [BH13]. The mechanism, due to McDuff–Polterovich, is the dictionary between disjoint balls of capacities $a_1,\dots,a_k$ in $\mathbb{CP}^2(1)$ and symplectic forms on $\mathbb{CP}^2\#k\overline{\mathbb{CP}^2}$ in the class $\mathrm{PD}(L-\sum a_iE_i)$; obstructions come from exceptional classes $dL-\sum m_iE_i$, which are represented by (possibly nodal) J-holomorphic curves for every tame J (Section 6.5), forcing $d>\sum m_ia_i$. The discrepancy between these symplectic packings and their algebro-geometric analogues for $k\geq 10$ is the territory of Nagata's conjecture [Bir01].

Theorem 6.17 (Hofer's conjecture; McDuff 2011)**.** *Let $a,b,c,d>0$. There is a symplectic embedding $E(a,b)\hookrightarrow E(c,d)$ if and only if $N_k(a,b)\leq N_k(c,d)$ for all $k\geq 0$.*

Necessity is ECH monotonicity. For sufficiency [McD11a] combines McDuff's reduction [McD09d] of the problem, for rational ratios, to embedding a disjoint union of balls into a blow-up of $\mathbb{CP}^2$ (the ball sizes are read off from continued fraction expansions) with a combinatorial argument showing that the resulting exceptional-class conditions are equivalent to $N(a,b)\leq N(c,d)$. It is one of the most satisfying facts in the subject that a gauge-theoretic invariant (ECH, via Taubes' ECH = HM) and the Gromov–Witten theory of blow-ups give *identical* answers. The extreme case of a ball target is completely computed:

Theorem 6.18 (Fibonacci staircase; McDuff–Schlenk 2012)**.** *Let $c(a)=\inf\{\mu : E(1,a)\hookrightarrow B^4(\mu)\}$ for $a\geq 1$, so $c(a)\geq\sqrt{a}$ by volume. Let $\tau=(1+\sqrt{5})/2$, let $g_0=1$ and $g_n=f_{2n-1}$ for $n\geq 1$ (the odd-index Fibonacci numbers $1,2,5,13,34,\dots$), and set $a_n=(g_{n+1}/g_n)^2$, $b_n=g_{n+2}/g_n$, so that $1=a_0<b_0<a_1<b_1<\cdots\to\tau^4$. Then:*

1. *for each $n \geq 0$, $c(a) = a/\sqrt{a_n}$ on $[a_n, b_n]$ and $c(a) = \sqrt{a_{n+1}}$ on $[b_n, a_{n+1}]$;*
2. *$c(a) = (a+1)/3$ on $[\tau^4, 7]$;*
3. *there are finitely many disjoint closed intervals $I_j \subset [7, (17/6)^2]$ such that $c(a) = \sqrt{a}$ for $a > 7$, $a \notin \bigcup_j I_j$, and c is piecewise linear on each I_j with exactly one nonsmooth point in its interior;*
4. *$c(a) = \sqrt{a}$ for $a \geq (17/6)^2$.*

So the graph alternates between segments of lines through the origin and horizontal segments, with corners accumulating at τ^4, where it meets the volume curve [MS12b]; the first nontrivial value, $c(4) = 2$, is due to Opshtein [Ops07]. The lower bounds come from exceptional classes in blow-ups of $\mathbb{CP}^2$ whose coefficients are continued-fraction expansions of ratios of Fibonacci numbers, and McDuff–Schlenk check that the ECH bound reproduces c.

Theorem 6.19 (Concave into convex; Cristofaro-Gardiner 2019)**.** *Let X be a concave and X' a convex toric domain in $\mathbb{C}^2$. The interior of X embeds symplectically into X' if and only if $c_k(X) \leq c_k(X')$ for all k.*

Here a toric domain $X_\Omega \subset \mathbb{C}^2$ is *concave* if Ω is the closed region between the axes and the graph of a convex function $f\colon [0, A] \to [0, B]$ with $f(0) = B$, $f(A) = 0$; see [Cri19]. ECH capacities are not sharp in general: for some polydisk-into-ellipsoid problems the Ekeland–Hofer capacities do better, and finer ECH obstructions are needed [Hut16].

Infinite staircases. For a target X write $c_X(a) = \inf\{\mu : E(1, a) \hookrightarrow \mu X\}$. Frenkel–Müller showed that for the cube $P(1, 1)$ there is again an infinite staircase, governed by Pell numbers and accumulating at $3 + 2\sqrt{2}$ [FM15]; Usher found infinite staircases for $P(1, \beta)$ at a doubly indexed family of irrational β [Ush19]. Cristofaro-Gardiner–Holm–Mandini–Pires showed that for finite-type convex toric targets an infinite staircase can only accumulate at one explicit point, where c_X equals the volume bound, and conjectured that among rational convex toric domains exactly six families, all with reflexive moment polygons, carry infinite staircases [Cri+25a]. For the one-parameter family of one-point blow-ups of $\mathbb{CP}^2$ the picture is now complete [Ber+21; MM23; MMW24; MPW25]: for rational blow-up size, a staircase occurs only in the monotone case, while many irrational parameters carry staircases, and there is a necessary and sufficient numerical criterion. The general conjecture is open.

Higher dimensions. Here we know much less, and in my opinion this is the most important gap in the quantitative theory. Guth showed that for polydisks the only obstructions, up to a dimensional constant, are nonsqueezing and volume:

Theorem 6.20 (Guth 2008)**.** *For each n there is $C(n)$ such that if $a_1 \leq \cdots \leq a_n$, $a_1' \leq \cdots \leq a_n'$, $C(n)a_1 \leq a_1'$ and $C(n)a_1 \cdots a_n \leq a_1' \cdots a_n'$, then $P(a_1, \ldots, a_n) \hookrightarrow P(a_1', \ldots, a_n')$.*

In particular [Gut08] there are no "intermediate" capacities (Hofer's question: finite on $B^{2k}(1) \times \mathbb{C}^{n-k}$ but infinite on $B^{2k-2}(1) \times \mathbb{C}^{n-k+1}$) among capacities continuous under exhaustions; Pelayo–Vũ Ngọc removed the continuity assumption [PV15]. The sharp version of the first nontrivial case is:

Theorem 6.21 (Hind–Kerman 2014; Pelayo–Vũ Ngọc 2016)**.** *Let $n \geq 3$. There is a symplectic embedding $Z^{2n}(1) \hookrightarrow B^4(A) \times \mathbb{C}^{n-2}$ if $A \geq 3$, and there is none if $A < 3$.*

The obstruction [HK14] comes from degree-d planes in the completed complement of a thin ellipsoid in $\mathbb{CP}^2(A) \times \mathbb{C}^{n-2}$, whose existence is reduced to dimension four by a cobordism argument; the construction refines Guth's and is completed in [PV16]. For stabilized ellipsoids let $f(x) = \inf\{\mu : E(1,x) \times \mathbb{C}^N \hookrightarrow B^4(\mu) \times \mathbb{C}^N\}$, $N \geq 1$. Then $f = c$ on $[1, \tau^4]$, i.e. the Fibonacci staircase survives stabilization [CH18]; symplectic folding gives $f(x) \leq 3x/(x+1) < c(x)$ for $x > \tau^4$ [Hin15]; and equality $f(x) = 3x/(x+1)$ holds at the "ghost stairs" $x = h_{2n+3}/h_{2n+1}$, where $(h_j)_{j\geq 1} = (1, 3, 8, 21, 55, \dots)$ are the even-index Fibonacci numbers, e.g. $f(8) = 8/3$ and $f(55/8) = 55/21$ [CHM18]. Whether $f(x) = 3x/(x+1)$ for all $x > \tau^4$ is open (Question 6.39). Beyond such stabilized problems, even $E(1, a, b) \hookrightarrow B^6(\mu)$ is not understood; Schlenk's survey [Sch18] is the place to see the whole landscape.

6.3 Lagrangian submanifolds and holomorphic disks

Theorem 6.22 (Gromov 1985)**.** *Let $L \subset (\mathbb{C}^n, \omega_0)$ be a closed Lagrangian submanifold. Then there is a nonconstant J_0-holomorphic disk $u : (D, \partial D) \to (\mathbb{C}^n, L)$, smooth up to the boundary. Consequently $\int_{\partial D} u^*\lambda_0 = \int_D u^*\omega_0 > 0$, so $\lambda_0|_L$ is not exact: $\mathbb{C}^n$ contains no closed exact Lagrangian submanifold.*

Proof idea. Pick a vector $v \in \mathbb{C}^n$ so large that $L + v$ is disjoint from L, and consider maps $u : (D, \partial D) \to (\mathbb{C}^n, L)$, homotopic to constants, solving $\bar\partial u = t\, v\, d\bar z$, $t \in [0,1]$. At $t = 0$ the solutions, with one boundary marked point, are the constants, and evaluation at the marked point has degree one onto L; at $t = 1$ there are none, because the inhomogeneous term acts like a Hamiltonian perturbation that displaces L. So the one-parameter moduli space cannot be compact; since energy is bounded and $\mathbb{C}^n$ has no holomorphic spheres, Gromov compactness forces a nonconstant J_0-disk to bubble off. Perturbing J generically and passing to the limit gives the J_0 statement [Gro85]; see [ALP94] for a leisurely account.

Chekanov proved that a closed Lagrangian L in a geometrically bounded (M, ω) cannot be displaced with Hofer energy below the minimal area of nonconstant J-holomorphic spheres in M and disks on L, for any tame J [Che98]; in $\mathbb{C}^n$ this bounds the area of the smallest disk of Theorem 6.22 by $e(L)$.

Definition 6.23 (Minimal Maslov number)**.** Let $L \subset (M, \omega)$ be a Lagrangian with Maslov class $\mu_L : \pi_2(M, L) \to \mathbb{Z}$ (Chapter 1). The *minimal Maslov number* $N_L \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$ is the positive generator of $\mu_L(\pi_2(M, L))$, or ∞ if this image is zero. L is *monotone* if $\int \omega = \kappa\, \mu_L$ on $\pi_2(M, L)$ for some $\kappa > 0$. If L is orientable, all Maslov indices are even.

Theorem 6.24 (Audin's conjecture; Cieliebak–Mohnke 2018)**.** *Every Lagrangian torus $L \subset (\mathbb{C}^n, \omega_0)$ bounds a smooth disk $f : (D, \partial D) \to (\mathbb{C}^n, L)$ with $\int_D f^*\omega_0 > 0$ and Maslov index 2. In particular $N_L = 2$.*

Audin asked this in 1988. The case $n = 2$ is due to Viterbo [Vit90] and Polterovich [Pol91]; monotone tori were done by Oh for $n \leq 24$ [Oh96] and in general by Buhovsky [Buh10b] and by Fukaya–Oh–Ohta–Ono [Fuk+09], see also Damian [Dam12]; Fukaya outlined a loop-space approach [Fuk06]. The full conjecture was proved by Cieliebak–Mohnke [CM18] by stretching the neck around L (which admits a flat metric) and counting punctured spheres with a tangency constraint in $\mathbb{CP}^n$; their method gives more generally a disk of Maslov index 1 or 2 on every monotone Lagrangian $L \subset \mathbb{CP}^n$ admitting a metric of nonpositive curvature, of area at most $1/(n+1)$ times that of a line.

Exotic tori. The *product tori* $T(a_1,\dots,a_n) = \{\pi|z_j|^2 = a_j\ \forall j\} \subset \mathbb{C}^n$ are Lagrangian; $T(a,\dots,a)$ is monotone (the Clifford torus).

Theorem 6.25 (Chekanov 1996)**.** *For every $n \geq 2$ there is a monotone Lagrangian torus in $(\mathbb{R}^{2n}, \omega_0)$ that is not the image of any product torus under any symplectomorphism of $\mathbb{R}^{2n}$.*

Theorem 6.26 (Vianna 2014, 2016)**.** *For each Markov triple (a,b,c), i.e. positive integers with $a^2+b^2+c^2 = 3abc$, there is a monotone Lagrangian torus $T(a^2,b^2,c^2) \subset (\mathbb{CP}^2, \omega_{FS})$, and tori associated to distinct (unordered) triples are not Hamiltonian isotopic.*

Here $T(1,1,1)$ is the Clifford torus, $T(1,1,4)$ the Chekanov torus and $T(1,4,25)$ the first new one [Via14]; the infinite family is [Via16]. Each torus is a fibre of an almost toric fibration on $\mathbb{CP}^2$ [Sym03; LS10] associated with a degeneration of $\mathbb{CP}^2$ to the weighted projective plane $\mathbb{CP}(a^2,b^2,c^2)$, and tori for neighbouring Markov triples differ by a mutation. They are told apart by the number of families of Maslov-2 holomorphic disks, i.e. by their Landau–Ginzburg potentials (Chapter 8). Vianna extended this to the monotone del Pezzo surfaces $\mathbb{CP}^2\#k\overline{\mathbb{CP}^2}$, $k \neq 2$ [Via17], Chekanov–Schlenk constructed twist tori in higher dimensions [CS10], and Auroux found infinitely many monotone tori in $\mathbb{R}^6$, pairwise not symplectomorphic [Aur15].

Luttinger surgery and knotting.

Definition 6.27 (Luttinger surgery)**.** Let T be a Lagrangian torus in a symplectic 4-manifold (X,ω), $\gamma \subset T$ a co-oriented simple closed curve and $k \in \mathbb{Z}$. Identify a neighbourhood of T symplectically with $U_r = T^2 \times [-r,r]^2 \subset T^*T^2$, coordinates $(q_1,q_2) \in \mathbb{R}^2/\mathbb{Z}^2$ and (p_1,p_2), $\omega = \omega_{\text{can}} = dp_1 \wedge dq_1 + dp_2 \wedge dq_2$, with γ the q_1-axis, co-oriented by q_2. Fix a smooth $\chi\colon [-r,r] \to [0,1]$ with $\chi = 0$ on $[-r,-r/3]$, $\chi = 1$ on $[r/3,r]$ and $\int_{-r}^{r} t\chi'(t)\,dt = 0$, and define ϕ_k on $U_r \setminus U_{r/2}$ by $\phi_k(q,p) = (q_1 + k\chi(p_1), q_2, p_1, p_2)$ if $p_2 \geq r/2$ and $\phi_k = \text{id}$ otherwise. Then ϕ_k is a well-defined symplectomorphism (it preserves each $dp_j \wedge dq_j$), and $X(T,\gamma,k) = (X \setminus U_{r/2}) \cup_{\phi_k} U_r$ (we follow Auroux–Donaldson–Katzarkov).

Topologically this is $1/k$ Dehn surgery along T with respect to the Lagrangian framing: the meridian is sent to meridian plus $k\gamma$. The symplectic form on $X(T,\gamma,k)$ is well defined up to isotopy [ADK03]. Luttinger introduced the operation [Lut95] to show that some smooth isotopy classes of tori in $\mathbb{C}^2$ contain no Lagrangian torus: surgery on a hypothetical Lagrangian representative yields a symplectic manifold standard at infinity, hence symplectomorphic to $\mathbb{R}^4$ by Theorem 6.30, whereas for the knotted examples its topology would be wrong. Since then it has become a basic construction tool: it explains Moishezon's non-isotopic singular plane curves [ADK03] and produces small exotic symplectic 4-manifolds, e.g. a minimal symplectic manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2\#3\overline{\mathbb{CP}^2}$ [BK08].

Theorem 6.28 (Dimitroglou Rizell–Goodman–Ivrii 2016)**.** *Up to Lagrangian isotopy there is exactly one Lagrangian torus in each of $(\mathbb{R}^4,\omega_0)$, $(\mathbb{CP}^2,\omega_{FS})$ and $(S^2\times S^2, \sigma\oplus\sigma)$. Moreover every closed exact Lagrangian in T^*T^2 is Hamiltonian isotopic to the zero section.*

The proof [DGI16] stretches the neck along the torus and studies the resulting foliations by holomorphic spheres. Earlier unknotting theorems are of the same flavour: Lagrangian planes in $\mathbb{C}^2$ that are linear at infinity are compactly supported Hamiltonian isotopic to the linear plane [EP96]; Lagrangian spheres in $(S^2\times S^2, \sigma\oplus\sigma)$ are all Hamiltonian isotopic [Hin04], as are Lagrangian spheres in T^*S^2 to the zero section [Hin12]; Lagrangian tori in T^*T^2 homologous to

the zero section are smoothly isotopic to it [EP93]. The general problem of Lagrangian knots in four-manifolds was posed by Eliashberg–Polterovich [EP97], and Evans applied the same foliation techniques to Lagrangian spheres in del Pezzo surfaces [Eva10]. In higher dimensions monotone Lagrangian n-tori in $\mathbb{C}^n$, $n \geq 4$, that are homotopic through Lagrangian immersions are smoothly isotopic [DE14]. On the other hand knotted Lagrangian tori exist in suitable symplectic 4-manifolds [Vid06], and the Klein bottle admits no Lagrangian embedding into $\mathbb{C}^2$ [Nem09; She09]. Knotted Lagrangian spheres and Dehn twists are treated in Chapter 8.

Warning 6.29 (Lagrangian vs. Hamiltonian isotopy). By Theorem 6.28 the Chekanov and Clifford tori in $\mathbb{C}^2$ are Lagrangian isotopic, while by Theorem 6.25 they are not Hamiltonian isotopic. There is no contradiction: a Lagrangian isotopy is induced by a Hamiltonian isotopy iff its flux in $H^1(L;\mathbb{R})$ vanishes at all times, and no isotopy between these two tori has this property. Statements "up to Lagrangian isotopy" are essentially smooth topology plus holomorphic-curve input; statements "up to Hamiltonian isotopy" are the genuinely symplectic ones.

6.4 Foliations by spheres and symplectomorphism groups

Theorem 6.30 (Gromov 1985)**.** *Let σ be an area form on S^2.*

1. $\mathrm{Symp}(S^2 \times S^2, \sigma \oplus \sigma)$ *deformation retracts onto* $(\mathrm{SO}(3) \times \mathrm{SO}(3)) \rtimes \mathbb{Z}/2$*, the* $\mathbb{Z}/2$ *swapping the factors.*
2. $\mathrm{Symp}(\mathbb{CP}^2, \omega_{FS})$ *deformation retracts onto* $\mathrm{PU}(3)$*.*
3. $\mathrm{Symp}_c(\mathbb{R}^4, \omega_0)$*, with the C^∞ topology, is contractible.*
4. *If ω is a symplectic form on $\mathbb{R}^4$ that equals ω_0 outside a compact set, then $(\mathbb{R}^4, \omega)$ is symplectomorphic to $(\mathbb{R}^4, \omega_0)$.*

Proof idea. For $S^2 \times S^2$ and any tame J, the classes $A = [S^2 \times \mathrm{pt}]$ and $B = [\mathrm{pt} \times S^2]$ are represented through every point by a unique embedded J-sphere: the count is 1 for the product structure, there is no bubbling because A and B have minimal area, and positivity of intersections ($A \cdot A = B \cdot B = 0$, $A \cdot B = 1$) makes distinct curves in one class disjoint and curves in different classes meet transversally once. So each J yields two transverse foliations by J-spheres, hence a "J-holomorphic product structure". The space of tame J is contractible, and analysing the action of Symp on it with these foliations identifies Symp up to homotopy with the stabilizer of the product structure. For $\mathbb{CP}^2$ one uses the lines (one through any two points), and for $\mathbb{R}^4$ one compactifies to $S^2 \times S^2$ keeping the two spheres at infinity fixed [Gro85], [MS12a, §9.5].

Gromov observed that already for unequal areas the answer changes. Abreu and McDuff computed it completely [Abr98; AM00]:

Theorem 6.31 (Abreu–McDuff 2000)**.** *Let σ have area 1, let $\mu > 1$, $M_\mu = (S^2 \times S^2, \mu\sigma \oplus \sigma)$ and $G_\mu = \mathrm{Symp}(M_\mu)$. Then G_μ is connected and $\pi_1(G_\mu) \cong \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2$. If $\ell \in \mathbb{Z}_{\geq 1}$ and $\ell < \mu \leq \ell + 1$, then*

$$H^*(G_\mu;\mathbb{Q}) = \Lambda(a,x,y) \otimes \mathbb{Q}[w_\ell], \qquad H^*(BG_\mu;\mathbb{Q}) = \mathbb{Q}[A,X,Y] \big/ \big(A(X-Y)(4X-Y)\cdots(\ell^2 X - Y)\big),$$

with $\deg a = 1$, $\deg x = \deg y = 3$, $\deg w_\ell = 4\ell$, $\deg A = 2$, $\deg X = \deg Y = 4$.

The homotopy type jumps exactly when μ crosses an integer, because a new J-holomorphic sphere of self-intersection -2ℓ (the section of a Hirzebruch surface) appears; the space of

compatible J is stratified by which of these exist. Anjos and Anjos–Granja determined the full homotopy type for $1 < \mu \leq 2$ [Anj02; AG04], Lalonde–Pinsonnault related symplectomorphisms of one-point blow-ups to spaces of embedded balls [LP04], and there are computations for some multiple blow-ups of $\mathbb{CP}^2$ [AP13; Eva11; Anj+24]. McDuff's lectures [McD04] are the right entry point.

Beyond such small rational surfaces the mapping class group $\pi_0 \operatorname{Symp}$ is mostly detected by Floer theory (Chapter 8). Seidel showed that squared Dehn twists along Lagrangian spheres, which are smoothly isotopic to the identity, are often not symplectically so [Sei99; Sei00]; the braid group injects into $\pi_0 \operatorname{Symp}_c$ of A_m Milnor fibres [KS02]; for certain K3 surfaces the symplectic Torelli group is infinitely generated, via mirror symmetry [SS20b], and for many K3 surfaces $\pi_0 \operatorname{Symp}$ itself is infinitely generated, via Seiberg–Witten theory [Smi22]. Li–Li–Wu show that for rational surfaces with $c_1 \cdot [\omega] > 0$ the symplectic Torelli group is generated by Lagrangian Dehn twists (trivial in type $\mathbb{A}$, a sphere braid group in type $\mathbb{D}$; preprint 2022 [LLW22]). Finally, the contrast with Theorem 6.30(4) is instructive: for $n \geq 4$ there are infinitely many pairwise non-symplectomorphic finite-type Stein manifolds diffeomorphic to $\mathbb{R}^{2n}$ [McL09] (Chapter 4).

6.5 Rational and ruled surfaces; Seiberg–Witten = Gromov

Recall from Chapter 3: an *exceptional sphere* in a symplectic 4-manifold is a symplectically embedded 2-sphere of self-intersection -1; (M, ω) is *minimal* if it contains none; a *symplectic ruled surface* is the total space of a smooth S^2-bundle $\pi\colon M \to \Sigma$ over a closed oriented surface with a symplectic form that is nondegenerate on the fibres; M is *rational* if it is diffeomorphic to $S^2 \times S^2$ or to $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$, $k \geq 0$.

Theorem 6.32 (McDuff 1990)**.** *Let (M, ω) be a closed connected symplectic 4-manifold containing a symplectically embedded 2-sphere C with $C \cdot C = p \geq 0$.*

1. *If $p = 0$, there are finitely many pairwise disjoint exceptional spheres in $M \setminus C$ whose symplectic blow-down is a symplectic ruled surface in which C is a fibre.*
2. *If $p = 1$, there are finitely many pairwise disjoint exceptional spheres in $M \setminus C$ whose blow-down is symplectomorphic to $(\mathbb{CP}^2, \lambda\omega_{FS})$ for some $\lambda > 0$, by a symplectomorphism taking C to a line.*

In all cases $p \geq 0$, M is rational or diffeomorphic to a blow-up of an S^2-bundle over a surface. If $M \setminus C$ contains no exceptional sphere, no blow-down is needed.

Proof idea. Take $p = 0$ and J generic among those making C holomorphic. Adjunction gives $c_1([C]) = 2$, so spheres in class $[C]$ with one marked point form a 4-dimensional moduli space; they are embedded, automatically transverse [HLS97], and pairwise disjoint by positivity of intersections. For generic J every simple sphere has $c_1 \geq 1$, so a nodal limit has exactly two components with $c_1 = 1$; adjunction and positivity make them exceptional spheres meeting once transversally, and there are finitely many such singular fibres. So M carries a singular fibration by J-spheres, and blowing down one component of each singular fibre gives a genuine S^2-fibration. For $p = 1$ the curves through pairs of points behave like lines; for $p \geq 2$ blow up $p - 1$ points of C to reduce to $p = 1$ [McD90], [MS12a, Ch. 9], [Wen18].

Combined with Seiberg–Witten theory, which supplies the sphere, this yields an essentially complete classification of symplectic forms on these manifolds; the proofs are assembled in [Sal13].

Theorem 6.33 (Lalonde–McDuff, Li–Liu, McDuff, Taubes)**.** *1. Let M be an S^2-bundle over a closed oriented surface of genus $g \geq 1$, with fibre class F. A class $a \in H^2(M;\mathbb{R})$ is represented by a symplectic form iff $a^2 \neq 0$ and $\langle a, F\rangle \neq 0$; any two symplectic forms on M are deformation equivalent, and cohomologous ones are diffeomorphic.*

2. *Let $M_k = \mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$, $k \geq 0$, with classes $L, E_1, \dots, E_k \in H_2(M_k;\mathbb{Z})$ and $c_{\mathrm{st}} = \mathrm{PD}(3L - \sum_i E_i)$. A class $a \in H^2(M_k;\mathbb{R})$ is represented by a symplectic form ω with $c_1(\omega) = c_{\mathrm{st}}$ iff $a^2 > 0$ and $\langle a, E\rangle > 0$ for every $E \in H_2(M_k;\mathbb{Z})$ with $E \cdot E = -1$ and $\langle c_{\mathrm{st}}, E\rangle = 1$. Two cohomologous symplectic forms on M_k, or on $S^2 \times S^2$, are diffeomorphic by a diffeomorphism acting trivially on homology.*

See [LM96a; LM96b; LL95b; LL01; McD98]. In (2), necessity holds because Taubes' theory represents every such E by a symplectic sphere; sufficiency is Kähler geometry for $k \leq 8$ and Li–Liu's theorem for $k \geq 9$. In all these cases cohomologous symplectic forms that are homotopic through nondegenerate forms are isotopic (Lalonde–McDuff for ruled surfaces; McDuff and Biran for blow-ups of $\mathbb{CP}^2$). The engine behind "supplies the sphere" is the following.

Definition 6.34 (Taubes' Gromov invariant, informally precise)**.** Let (M, ω) be a closed symplectic 4-manifold with canonical class K, and $e \in H_2(M;\mathbb{Z})$. Put $d(e) = \frac{1}{2}(e \cdot e - K \cdot e)$. For generic tame J and a generic set Ω of $d(e)$ points, $\mathrm{Gr}(M, \omega)(e) \in \mathbb{Z}$ is the signed, weighted count of finite sets $\{(C_i, m_i)\}$ where the C_i are pairwise disjoint connected embedded J-curves, the m_i are positive integers with $m_i = 1$ unless C_i is a torus with $C_i \cdot C_i = 0$, $\sum_i m_i[C_i] = e$, and $\Omega \subset \bigcup_i C_i$. Curves with $m_i = 1$ count with the sign of their linearized operator; multiply covered square-zero tori count with weights defined by Taubes from the linearized operators of their unbranched covers. $\mathrm{Gr}(e) = 0$ if $d(e) < 0$ and $\mathrm{Gr}(0) = 1$.

By adjunction $2d(e)$ is the index of embedded curves in class e, so this is a count of expected-dimension-zero configurations. Disconnected curves are essential and the torus weights are what make the count invariant [Tau96b]; McDuff's lectures [McD97] explain why this is not the Gromov–Witten count. Write Γ_ω for the spin^c structure of ω ($c_1(\Gamma_\omega) = -K$) and $\Gamma_{\omega,e} = \Gamma_\omega \otimes E$ with $c_1(E) = \mathrm{PD}(e)$.

Theorem 6.35 (SW = Gr; Taubes 1996, 1999)**.** *Let (M, ω) be a closed symplectic 4-manifold with $b^+(M) \geq 2$, and give M the homological orientation induced by ω. Then $\mathrm{SW}(\Gamma_{\omega,e}) = \mathrm{Gr}(M, \omega)(e)$ for every $e \in H_2(M;\mathbb{Z})$. If $b^+(M) = 1$ the same holds with SW computed in the chamber determined by ω.*

The direction SW $\Rightarrow$ Gr (solutions degenerate to curves as the perturbation $r\omega$ grows) is [Tau96a], the converse [Tau99]; everything is collected in [Tau00].

Theorem 6.36 (Taubes 1994–1995)**.** *Let (M, ω) be a closed symplectic 4-manifold with $b^+(M) \geq 2$.*

1. $\mathrm{SW}(\Gamma_\omega) = 1$ *for the homological orientation of ω.*
2. *If $\mathrm{SW}(\Gamma_{\omega,e}) \neq 0$ then $[\omega] \cdot e \geq 0$, with equality only for $e = 0$. In particular, since $\mathrm{SW}(\Gamma_{\omega,K}) = \pm\mathrm{SW}(\Gamma_\omega)$ by charge conjugation, $K \cdot [\omega] \geq 0$ with equality only if $K = 0$.*
3. *If $\mathrm{SW}(\Gamma_{\omega,e}) \neq 0$ then e is represented by an embedded, possibly disconnected, ω-symplectic surface. In particular K is.*
4. *M is not diffeomorphic to a connected sum $M_1 \# M_2$ with $b^+(M_1), b^+(M_2) \geq 1$.*

See [Tau94; Tau95a; Tau95b]; (3) is the application of Theorem 6.35 and (4) combines (1) with the vanishing of SW for such connected sums. With $b^+ = 1$ there are analogous statements in the appropriate chamber [LL95a], which together with Theorem 6.32 give the constraints stated in Chapter 3: e.g. $K \cdot [\omega] < 0$ forces M rational or ruled [Liu96; OO96].

Corollary 6.37 (Uniqueness on $\mathbb{CP}^2$; Taubes 1995, Gromov, McDuff)**.** *Every symplectic form on the smooth manifold $\mathbb{CP}^2$ is diffeomorphic to $\lambda\omega_{FS}$ for some $\lambda > 0$.*

Proof idea. For any symplectic ω on $\mathbb{CP}^2$ the wall-crossing formula and Theorem 6.35 give $\mathrm{Gr}(L) \neq 0$ for the generator L with $\omega(L) > 0$; adjunction makes the curve an embedded sphere with $L \cdot L = 1$, and Theorem 6.32 finishes [Tau95b].

Theorem 6.38 (Symplectic Thom conjecture; Ozsváth–Szabó 2000)**.** *Let Σ be a closed connected symplectic surface in a closed symplectic 4-manifold (M, ω). Every closed connected oriented surface smoothly embedded in M and homologous to Σ has genus at least $g(\Sigma) = 1 + \frac{1}{2}(\Sigma \cdot \Sigma + K \cdot \Sigma)$.*

Kronheimer–Mrowka proved this for $\mathbb{CP}^2$ (the Thom conjecture) [KM94b], Morgan–Szabó–Taubes for $\Sigma \cdot \Sigma \geq 0$ [MST96] (with Li–Liu [LL95b] for the symplectic case), and Ozsváth–Szabó in general [OS00].

Super-rigidity. Gromov–Witten theory of symplectic Calabi–Yau 3-folds was long hostage to multiply covered curves. Wendl proved that for generic J on a closed symplectic manifold of dimension at least six, every simple J-curve of index zero is *super-rigid*: informally, all of its multiple covers, branched or not, have injective linearized operators, so they are isolated and as transverse as they can be [Wen23]; Doan–Walpuski gave an equivariant Brill–Noether framework for this [DW23]. Consequently the Gromov–Witten invariants of a symplectic Calabi–Yau 3-fold are sums of local contributions of finitely many embedded curves in each class. This is the backdrop for the symplectic work on the Gopakumar–Vafa conjectures: integrality was proved by Ionel–Parker [IP18] and finiteness by Doan–Ionel–Walpuski (to appear in Ann. of Math. [DIW21]); see also Doan–Walpuski's Castelnuovo-type genus bounds in dimension six [DW21].

6.6 What to read

McDuff–Salamon, Introduction to Symplectic Topology *[MS17]*
The chapters on symplectic capacities and on questions of existence and uniqueness give the statements of this chapter with minimal machinery; read them first.

McDuff–Salamon, J-holomorphic Curves and Symplectic Topology *[MS12a]*
Chapter 9 is the reference for the applications in Sections 6.1, 6.4 and 6.5: nonsqueezing via Gromov–Witten invariants, Gromov's and McDuff's four-dimensional theorems, $\mathrm{Symp}(S^2 \times S^2)$ and $\mathrm{Symp}(\mathbb{CP}^2)$. Read it after the foundations of Chapter 5.

Hofer–Zehnder, Symplectic Invariants and Hamiltonian Dynamics *[HZ94]*
Chapters 1–3 and 5: capacities, the construction of c_{HZ}, C^0-rigidity, and the displacement energy. The cleanest route to capacities without curves.

Cieliebak–Hofer–Latschev–Schlenk, Quantitative symplectic geometry *[Cie+07]*
The capacity zoo with precise axioms and a table of known values and open problems.

Schlenk, Symplectic embedding problems, old and new *[Sch18]*
The survey of Section 6.2; read it before any original paper. Schlenk's book [Sch05] is the reference for folding and wrapping constructions.

Hutchings [Hut11b; Hut14]
The PNAS note is the fastest overview of four-dimensional embeddings; the lecture notes explain ECH and its capacities. Then [Hut11a] and McDuff's survey [McD09c], followed by [McD11a; MS12b].

Wendl, Holomorphic Curves in Low Dimensions *[Wen18]*
The modern treatment of McDuff's theorem, with automatic transversality done properly. Read it alongside [McD90], [LM96a] and Salamon's survey [Sal13].

McDuff, Lectures on Gromov invariants *[McD97]; Taubes [Tau00]*
The lectures first; Taubes' volume is for those who need the analysis.

McDuff, Lectures on groups of symplectomorphisms *[McD04]*
Then [AM00].

Gromov [Gro85]
Read the long introduction to see how many theorems were announced at once; consult the proofs through the books above.

Audin–Lafontaine (eds.) [AL94]
The chapter [ALP94] is a good account of Gromov's disk theorem and of early Lagrangian rigidity.

Lagrangian tori
Chekanov–Schlenk [CS10], then Vianna [Via16]; the Floer-theoretic background is in Chapter 8.

6.7 Frontier and open problems

Question 6.39 (Stabilized ellipsoids)**.** *Is* $\inf\left\{\mu : E(1,x)\times\mathbb{C}^N \hookrightarrow B^4(\mu)\times\mathbb{C}^N\right\} = 3x/(x+1)$ *for all* $x > \tau^4$ *and* $N \geq 1$*? Is it independent of* N*? Known at the ghost stairs [CHM18] and some further values; see [MS24; Sie25].*

Problem 6.40 (Ellipsoids in dimension six)**.** *Determine when* $E(a_1,a_2,a_3) \hookrightarrow E(b_1,b_2,b_3)$*, even for a ball target. Beyond volume, the Ekeland–Hofer (equivalently Gutt–Hutchings) capacities and the higher capacities of Siegel and McDuff–Siegel, no complete set of obstructions is known in dimension* ≥ 6*.*

Conjecture 6.41 (Cristofaro-Gardiner–Holm–Mandini–Pires)**.** *A rational convex toric domain in* $\mathbb{C}^2$ *has an infinite staircase if and only if it belongs to one of the six families with reflexive moment polygon identified in [Cri+25a]. Proved for Hirzebruch surfaces [MPW25]; open in general.*

Question 6.42 (Monotone tori)**.** *Is every monotone Lagrangian torus in* $\mathbb{CP}^2$ *Hamiltonian isotopic to one of Vianna's tori* $T(a^2,b^2,c^2)$*? Is every monotone Lagrangian torus in* $\mathbb{C}^2$ *Hamiltonian isotopic to a Clifford or a Chekanov torus? Both are open as far as I know; Lagrangian isotopy is settled by Theorem 6.28.*

Question 6.43 (Symplectomorphism groups)**.** *For which closed symplectic* 4*-manifolds is* $\mathrm{Symp}(M, \omega)$ *homotopy equivalent to a CW complex of finite type? (Not for some K3 surfaces, where* π_0 *is infinitely generated [Smi22], as is its Torelli subgroup [SS20b].) What is* $\pi_0 \mathrm{Symp}$ *of a rational surface with* $c_1 \cdot [\omega] \leq 0$*, and is the symplectic Torelli group of K3 generated by squared Dehn twists?*

Conjecture 6.44 (Symplectic isotopy problem)**.** *Every connected symplectic surface in* $(\mathbb{CP}^2, \omega_{FS})$ *is symplectically isotopic to a complex curve. Known for degree* ≤ 2 *by Gromov's methods and for degree* ≤ 17 *by Siebert–Tian [ST05]; open in general.*

For the uniqueness of symplectic structures on manifolds homotopy equivalent to $\mathbb{CP}^2$ see Chapter 3; for C^0 questions, Chapter 7; for Viterbo's conjecture and its disproof, Chapter 9.

6.8 Logical dependencies

To state Theorem 6.2 one needs only Convention 6.1; the capacity statements need Definitions 6.3 and 6.4 and, for Theorems 6.8 and 6.12, Hamiltonian flows and Definitions 6.7 and 6.11. Theorems 6.14 and 6.15 presuppose Liouville domains (Chapter 3); their definition, not their statement, needs ECH and hence Seiberg–Witten Floer theory. Theorems 6.16 to 6.18 are elementary to state. Theorem 6.24 needs Definition 6.23, hence the Maslov index of Chapter 1. Theorems 6.32 and 6.33 need symplectic blow-down (Chapter 3); Theorems 6.35 and 6.36 need spin^c structures, Seiberg–Witten invariants with their chamber structure for $b^+ = 1$, and Definition 6.34, which itself presupposes Fredholm theory and Gromov compactness for embedded curves (Chapter 5).

Chapter 7

Hamiltonian Floer theory and its quantitative offspring

Floer's papers of 1988–89 [Flo88b; Flo88c; Flo88a; Flo89b] did two things at once: they proved a large part of Arnold's conjecture on fixed points of Hamiltonian diffeomorphisms, and they built an infinite-dimensional Morse theory for the action functional whose output, Floer cohomology, became the basic invariant of the subject. This chapter is about the closed-manifold Hamiltonian version of that theory and about what grew out of it. The organizing fact is a paradox. The Floer cohomology $\mathrm{HF}^*(H)$ of a Hamiltonian is isomorphic to the quantum cohomology of M, so it does not see the Hamiltonian diffeomorphism at all; but the *action filtration* of the Floer complex does see it, and nearly every quantitative result of the last twenty-five years (spectral invariants, Hofer geometry, Calabi quasimorphisms, symplectic quasi-states, barcodes, C^0 rigidity, the simplicity conjecture) exploits that filtration. My advice: learn the construction once, carefully, in the aspherical case; then treat everything else as filtered linear algebra on top of it.

I use the foundations of Chapter 5 (Gromov compactness, transversality, Gromov–Witten invariants, quantum cohomology, the Seidel representation, the status of virtual techniques), the Conley–Zehnder index of Chapter 1, normalized by $\mu_{\mathrm{CZ}}(t \mapsto e^{tJ_0S}) = \frac{1}{2}\operatorname{sign} S$ for small symmetric S, and the capacities of Chapter 6. Symplectic cohomology is the open-manifold version of this chapter and lives in Chapter 8; periodic orbits beyond fixed-point counts (Conley and Hofer–Zehnder conjectures, pseudo-rotations, barcode entropy) live in Chapter 9.

7.1 The Arnold conjecture

Throughout, (M, ω) is a closed connected symplectic manifold of dimension $2n$. A fixed point p of a diffeomorphism ϕ is *nondegenerate* if 1 is not an eigenvalue of $d\phi_p$; ϕ is *nondegenerate* if all its fixed points are. Nondegenerate fixed points are isolated, so a nondegenerate ϕ has finitely many.

Definition 7.1 (Topological lower bounds)**.** For a closed manifold M and a field $\mathbb{F}$ set $N_{\mathrm{crit}}(M) := \min\{\#\operatorname{Crit}(f) : f \in C^\infty(M)\}$, $N_{\mathrm{Morse}}(M) := \min\{\#\operatorname{Crit}(f) : f \in C^\infty(M) \text{ Morse}\}$, $b(M;\mathbb{F}) := \sum_k \dim_{\mathbb{F}} H^k(M;\mathbb{F})$, and let the *cup-length* $\mathrm{cl}_{\mathbb{F}}(M)$ be the largest $k \geq 0$ for which there are $a_1, \dots, a_k \in \bigoplus_{j>0} H^j(M;\mathbb{F})$ with $a_1 \cup \cdots \cup a_k \neq 0$.

Lusternik–Schnirelmann theory gives $N_{\mathrm{crit}}(M) \geq \mathrm{cl}_{\mathbb{F}}(M) + 1$; more precisely $\mathrm{cl}_{\mathbb{F}}(M) + 1 \leq \operatorname{cat}(M) + 1 \leq N_{\mathrm{crit}}(M) \leq \dim M + 1$, the last inequality by Takens [Tak68]. The Morse inequalities

give $N_{\mathrm{Morse}}(M) \geq b(M;\mathbb{F})$. By contrast, for a nondegenerate diffeomorphism isotopic to the identity the Lefschetz fixed point theorem gives only the lower bound $|\chi(M)|$. Arnold's conjecture, formulated in a 1965 note [Arn65] and in Appendix 9 of [Arn78] as a generalization of Poincaré's last geometric theorem, says that Hamiltonian diffeomorphisms behave like gradient flows rather than like arbitrary diffeomorphisms.

Conjecture 7.2 (Arnold)**.** *Let (M,ω) be a closed symplectic manifold and $\phi \in \mathrm{Ham}(M,\omega)$.*

(AC-crit) $\#\mathrm{Fix}(\phi) \geq N_{\mathrm{crit}}(M)$.

(AC-cl) $\#\mathrm{Fix}(\phi) \geq \mathrm{cl}_{\mathbb{F}}(M)+1$ *for every field* $\mathbb{F}$.

(AC-Morse) If ϕ is nondegenerate, $\#\mathrm{Fix}(\phi) \geq N_{\mathrm{Morse}}(M)$.

(AC-hom) If ϕ is nondegenerate, $\#\mathrm{Fix}(\phi) \geq b(M;\mathbb{F})$ *for every field* $\mathbb{F}$.

(AC-crit) implies (AC-cl) and (AC-Morse) implies (AC-hom). Only the last is now a theorem in general.

Remark 7.3 (Status of the degenerate versions). (AC-crit) is known for tori T^{2n} (Conley–Zehnder [CZ83], by a finite-dimensional reduction of the action functional; this was the breakthrough that made the conjecture credible), for closed surfaces (Eliashberg [Eli79], Floer [Flo86]), for $\mathbb{CP}^n$ (Fortune–Weinstein [FW85; For85]), and for closed M with $\omega|_{\pi_2(M)} = 0 = c_1|_{\pi_2(M)}$ (Rudyak [Rud99] proved $\#\mathrm{Fix}(\phi) \geq \mathrm{cat}(M)+1$ there, upgrading the cup-length estimates of Floer [Flo89a] and Hofer [Hof88] for symplectically aspherical manifolds, and Rudyak–Oprea [RO99] showed $\mathrm{cat}(M) = \dim M$ when $\omega|_{\pi_2(M)} = 0$). Schwarz [Sch98] proved an estimate in which cup products are replaced by quantum products. Beyond such cases both (AC-crit) and (AC-cl) are, to my knowledge, open. Theorem 7.34 below shows that any proof must use smoothness of ϕ in an essential way once $\dim M \geq 4$.

Theorem 7.4 (Homological Arnold conjecture; Floer, Hofer–Salamon, Ono, Fukaya–Ono, Liu–Tian, Ruan)**.** *Let (M^{2n},ω) be closed, $\mathbb{F}$ a field and $\phi \in \mathrm{Ham}(M,\omega)$ nondegenerate. Then $\#\mathrm{Fix}(\phi) \geq b(M;\mathbb{F})$ in each of the following cases: (i) (M,ω) is semipositive and $\mathbb{F}$ is arbitrary; (ii) (M,ω) is arbitrary and $\mathrm{char}\,\mathbb{F} = 0$.*

Proof idea. Write $\phi = \phi_H^1$ with H nondegenerate. The Floer complex of Definition 7.14 is a free module of rank $\#\mathcal{P}(H) \leq \#\mathrm{Fix}(\phi)$ over the graded field Λ, and by Theorem 7.15 its cohomology is $H^*(M;\mathbb{F}) \otimes \Lambda$, of rank $b(M;\mathbb{F})$. Cohomology of a complex over a graded field has rank at most that of the complex.

Remark 7.5 (History, and the integral story). Floer proved Theorem 7.4 for monotone manifolds [Flo89b], after first treating aspherical ones [Flo88c; Flo88a]; Hofer–Salamon [HS95] and Ono [Ono95] extended it to semipositive ones, over $\mathbb{Z}$. Case (ii) is due independently to Fukaya–Ono [FO99], Liu–Tian [LT98c] and Ruan [Rua99], and was reproved with Pardon's virtual cycles [Par16] and with polyfolds [FW22]. Positive characteristic is harder because virtual counts in the presence of finite isotropy are naturally rational. Abouzaid–Blumberg [AB21] proved (AC-hom) for every field of positive characteristic, using Morava K-theory and the canonical complex orientations of Floer moduli spaces. Bai–Xu [BX22] and Rezchikov [Rez22] then proved integral versions, building on the global charts of Abouzaid–McLean–Smith [AMS21]. These three papers were still preprints when I checked (October 2026). I have no doubts about the results; long foundational papers take a long time to referee.

Theorem 7.6 (Integral Arnold conjecture; Bai–Xu 2022, preprint)**.** *Let (M,ω) be closed, let $N = N_M \in \{1,2,\dots\}\cup\{\infty\}$ be its minimal Chern number (Chapter 2), with $\mathbb{Z}/2N := \mathbb{Z}$ when $N=\infty$, and for $i \in \mathbb{Z}/2N$ let τ_i be the minimal number of generators of the torsion subgroup of $\bigoplus_{j\equiv i\ (2N)} H_j(M;\mathbb{Z})$. If H is a nondegenerate Hamiltonian, then $\#\mathcal{P}(H) \geq \sum_k \operatorname{rank} H_k(M;\mathbb{Z}) + 2\sum_{i\in\mathbb{Z}/2N}\tau_i$.*

Rezchikov's formulation [Rez22] bounds $\#\operatorname{Fix}(\phi)$ below by the minimal rank of a cochain complex of free abelian groups with cohomology $H^*(M;\mathbb{Z})$. By the universal coefficient theorem either version implies (AC-hom) for all fields at once. It is strictly stronger when torsion of different primes sits in different degrees.

Remark 7.7 (An observation about (AC-Morse)). If M is simply connected then $c_1|_{\pi_2(M)} = 0$ forces $c_1(M) = 0$, and then $N = \infty$, the grading in Theorem 7.6 is a $\mathbb{Z}$-grading, and the bound reads $\sum_k (b_k + 2\tau(H_k(M;\mathbb{Z})))$. By Smale's theorem [Sma62] this is exactly $N_{\mathrm{Morse}}(M)$ when $\dim M \geq 6$. So, granting Theorem 7.6, (AC-Morse) holds for closed simply connected symplectic manifolds of dimension ≥ 6 with $c_1 = 0$. This is my own deduction; I have not seen it stated in the literature. Bai–Xu themselves only suggest that the general simply connected case might follow from the simple homotopy type of their integral complex. For general $N < \infty$ the reduction of the grading mod $2N$ loses information, and (AC-Morse) is open.

The Lagrangian analogue, $\#(L\cap\phi(L)) \geq b(L;\mathbb{F})$, is Floer's original setting [Flo88b]; together with the Arnold–Givental conjecture (for which a proof was posted in August 2026 [Bai+26]) it belongs to Chapter 8.

7.2 Floer cohomology of a Hamiltonian

Let $H\colon S^1\times M\to\mathbb{R}$ be smooth, $S^1 = \mathbb{R}/\mathbb{Z}$, $H_t := H(t,\cdot)$, with Hamiltonian vector field X_{H_t} defined by C3, flow ϕ_H^t and $\phi_H := \phi_H^1$. Let $\mathcal{P}(H)$ be the set of contractible 1-periodic orbits $x\colon S^1\to M$, $\dot{x} = X_{H_t}(x)$. H is *nondegenerate* if ϕ_H is; then $\mathcal{P}(H)$ is finite.

Definition 7.8 (Capped orbits and the Novikov field)**.** A *capping* of $x\in\mathcal{P}(H)$ is a smooth $\bar{x}\colon D^2\to M$ with $\bar{x}(e^{2\pi i t}) = x(t)$. Let $H_2^S(M)\subset H_2(M;\mathbb{Z})$ be the image of the Hurewicz map and $\Gamma := H_2^S(M)/(\ker[\omega]\cap\ker c_1)$, a finitely generated free abelian group. Cappings $\bar{x},\bar{x}'$ of x are *equivalent* if the sphere $\bar{x}\#(-\bar{x}')$ obtained by gluing them along x is zero in Γ. A *capped orbit* $\tilde{x} = [x,\bar{x}]$ is an equivalence class, and $\widetilde{\mathcal{P}}(H)$ is the set of capped orbits. Γ acts freely on $\widetilde{\mathcal{P}}(H)$ by $[x,\bar{x}]\#A := [x,\bar{x}\#A]$, with quotient $\mathcal{P}(H)$. For a field $\mathbb{F}$ the *Novikov field* is

$$\Lambda := \left\{ \textstyle\sum_{A\in\Gamma}\lambda_A e^A :\ \lambda_A\in\mathbb{F},\ \#\{A : \lambda_A\neq 0,\ \omega(A) < C\} < \infty \ \text{ for every } C\in\mathbb{R}\right\},$$

graded by $\deg e^A := 2c_1(A)$. It is a graded field (every nonzero homogeneous element is invertible), and $\Lambda = \mathbb{F}$ when $\Gamma = 0$.

If Chapter 5 works with a different Novikov field, for instance a universal one in a formal variable T, the substitution $e^A\mapsto T^{\omega(A)}$ compares the two at the cost of the $\mathbb{Z}$-grading. Nothing below depends on the choice.

Definition 7.9 (Action, index, spectrum)**.** For $\tilde{x} = [x,\bar{x}]\in\widetilde{\mathcal{P}}(H)$ the action is $\mathcal{A}_H(\tilde{x}) := -\int_{D^2}\bar{x}^*\omega + \int_0^1 H_t(x(t))\,dt$ (C7). If x is nondegenerate, a symplectic trivialization of $\bar{x}^*TM$ turns $t\mapsto d\phi_H^t(x(0))$ into a path Ψ in $\mathrm{Sp}(2n)$ from id to a matrix without eigenvalue 1. Set $\mu_{\mathrm{CZ}}(\tilde{x}) := \mu_{\mathrm{CZ}}(\Psi)$, which depends only on $\tilde{x}$, and $\deg\tilde{x} := n - \mu_{\mathrm{CZ}}(\tilde{x})$ (C8). The *action spectrum* is $\operatorname{Spec}(H) := \mathcal{A}_H(\widetilde{\mathcal{P}}(H))\subset\mathbb{R}$; it is countable if H is nondegenerate.

Proposition 7.10 (Recapping and the small Morse case)**.** *For $A \in \Gamma$: $\mathcal{A}_H(\tilde{x}\#A) = \mathcal{A}_H(\tilde{x}) - \omega(A)$, $\mu_{\mathrm{CZ}}(\tilde{x}\#A) = \mu_{\mathrm{CZ}}(\tilde{x}) + 2c_1(A)$, hence $\deg(\tilde{x}\#A) = \deg\tilde{x} - 2c_1(A)$. If f is a Morse function and $\varepsilon > 0$ is small enough, the elements of $\mathcal{P}(\varepsilon f)$ are the constant orbits at critical points p, and with the constant capping $\hat{p}$ one has $\mathcal{A}_{\varepsilon f}(\hat{p}) = \varepsilon f(p)$, $\mu_{\mathrm{CZ}}(\hat{p}) = n - \mathrm{ind}_p(f)$, $\deg\hat{p} = \mathrm{ind}_p(f)$.*

Proof idea. By C2–C3, $X_H = J\nabla H$ for the metric g_J, so at a critical point the linearized flow is e^{tJ_0S} with $S = \varepsilon\,\mathrm{Hess}_p f$, and $\mu_{\mathrm{CZ}} = \frac{1}{2}\operatorname{sign} S = n - \mathrm{ind}_p(f)$. Changing the capping by A changes the trivialization along x by a loop of Maslov index $\pm c_1(A)$, and the loop axiom of Chapter 1 gives a change of $\pm 2c_1(A)$. The sign is forced by the index formula of Convention 7.12: gluing a sphere bubble of Chern number $c_1(A)$ to a Floer cylinder raises its Fredholm index by $2c_1(A)$.

Definition 7.11 (Floer trajectories)**.** Let $J = (J_t)_{t\in S^1}$ be a smooth loop of ω-compatible almost complex structures. For $\tilde{x}_\pm \in \widetilde{\mathcal{P}}(H)$ let $\mathcal{M}(\tilde{x}_-, \tilde{x}_+) = \mathcal{M}(\tilde{x}_-, \tilde{x}_+; H, J)$ be the set of smooth $u\colon \mathbb{R}\times S^1 \to M$ with

$$\partial_s u + J_t(u)\big(\partial_t u - X_{H_t}(u)\big) = 0, \qquad \lim_{s\to\pm\infty} u(s,t) = x_\pm(t) \text{ uniformly in } t, \qquad [x_+, \bar{x}_-\#u] = \tilde{x}_+,$$

where $\bar{x}_-\#u$ is the capping of x_+ obtained by gluing u to $\bar{x}_-$. $\mathbb{R}$ acts by translation in s. Every such u satisfies $E(u) := \iint |\partial_s u|^2_{J_t}\, ds\, dt = \mathcal{A}_H(\tilde{x}_-) - \mathcal{A}_H(\tilde{x}_+)$, and if H is nondegenerate every finite-energy solution of the equation lies in some $\mathcal{M}(\tilde{x}_-, \tilde{x}_+)$ [Sal99].

Convention 7.12 (Direction of the differential; action filtration)**.** With C3, C7, C8 and Proposition 7.10, for generic J the space $\mathcal{M}(\tilde{x}_-, \tilde{x}_+)$ is a manifold of dimension $\mu_{\mathrm{CZ}}(\tilde{x}_+) - \mu_{\mathrm{CZ}}(\tilde{x}_-) = \deg\tilde{x}_- - \deg\tilde{x}_+$. (Check: for εf and t-independent J, the t-independent solutions are the negative gradient lines $u' = -\nabla f$ from p_- to p_+, a space of dimension $\mathrm{ind}_{p_-} f - \mathrm{ind}_{p_+} f$.) Hence a differential of degree +1 must count cylinders *arriving* at its input:

$$d\tilde{x}_+ := \sum_{\deg\tilde{x}_- = \deg\tilde{x}_+ + 1} \#\big(\mathcal{M}(\tilde{x}_-, \tilde{x}_+)/\mathbb{R}\big)\, \tilde{x}_-.$$

The input sits at $s = +\infty$ and the output at $s = -\infty$, and by the energy identity *d strictly increases the action*. For $\lambda \in \mathbb{R}$ the chains all of whose terms have action $> \lambda$ form a subcomplex $\mathrm{CF}^*_{>\lambda}(H)$, with quotient $\mathrm{CF}^*_{\le\lambda}(H)$, and $\mathrm{CF}^*_{(a,b]} := \mathrm{CF}^*_{>a}/\mathrm{CF}^*_{>b}$ computes filtered Floer cohomology $\mathrm{HF}^*_{(a,b]}(H)$. This agrees with the symplectic-cohomology literature [Sei08a]. The *transposed complex* $(\mathrm{CF}_*(H), \partial)$ has the same generators and grading, $\partial\tilde{x}_- := \sum \#(\mathcal{M}(\tilde{x}_-, \tilde{x}_+)/\mathbb{R})\,\tilde{x}_+$, and the opposite Novikov condition. It is homological, ∂ lowers degree and action, and $\mathrm{CF}_*^{<\lambda}(H)$ is a subcomplex. Most of the literature on spectral invariants (Oh, Usher, Entov–Polterovich) uses this complex, and Section 7.4 uses it too so that their formulas can be quoted verbatim. The cohomological unit 1 corresponds to the fundamental class $[M]$.

Warning 7.13. Salamon [Sal99] and McDuff–Salamon [MS12a] use $\iota_{X_H}\omega = +dH$, which replaces X_H by X_{-H}. Some authors also reverse the sign of the action. Each change reverses "increases" and "decreases" above or flips the sign of μ_{CZ}, and the combinations differ from source to source. Before importing an index formula, test it on Proposition 7.10.

Definition 7.14 (Floer complex)**.** Let H be nondegenerate. $\mathrm{CF}^k(H)$ is the $\mathbb{F}$-vector space of formal sums $\sum a_{\tilde{x}}\tilde{x}$ over $\tilde{x} \in \widetilde{\mathcal{P}}(H)$ with $\deg\tilde{x} = k$ such that for every $C \in \mathbb{R}$ only finitely many $\tilde{x}$ with $a_{\tilde{x}} \neq 0$ have $\mathcal{A}_H(\tilde{x}) < C$. Λ acts by $e^A \cdot \tilde{x} := \tilde{x}\#(-A)$, which raises action by $\omega(A)$ and degree by $2c_1(A)$, making $\mathrm{CF}^*(H)$ a free graded Λ-module of rank $\#\mathcal{P}(H)$. The differential is d of Convention 7.12, counted with signs from coherent orientations [FH93] when $\mathrm{char}\,\mathbb{F} \neq 2$.

Theorem 7.15 (Floer cohomology; Floer, Floer–Hofer, Hofer–Salamon, Ono)**.** *Let (M^{2n},ω) be closed and semipositive, $\mathbb{F}$ a field and H nondegenerate. For J in a comeagre subset of the space of smooth loops of ω-compatible almost complex structures: (a) if $\deg\tilde{x}_- = \deg\tilde{x}_+ + 1$, then $\mathcal{M}(\tilde{x}_-,\tilde{x}_+)/\mathbb{R}$ is a finite set, and d is a well-defined Λ-linear map of degree $+1$ with $d^2=0$; (b) for any two such pairs $(H^\pm,J^\pm)$ there are continuation isomorphisms $\mathrm{HF}^*(H^+,J^+)\to\mathrm{HF}^*(H^-,J^-)$, canonical and compatible with composition, so $\mathrm{HF}^*(H):=H^*(\mathrm{CF}^*(H),d)$ is independent of (H,J) up to canonical isomorphism; (c) $\mathrm{HF}^*(H)\cong H^*(M;\mathbb{F})\otimes_{\mathbb{F}}\Lambda$ as graded Λ-modules. If moreover $\omega|_{\pi_2(M)}=0=c_1|_{\pi_2(M)}$, then $\Gamma=0$ and (a)–(c) hold over $\mathbb{Z}$:* $\mathrm{HF}^*(H;\mathbb{Z})\cong H^*(M;\mathbb{Z})$.

Proof idea. Each hypothesis feeds one step. *Compactness*: Gromov–Floer compactness (Chapter 5) shows that bounded-energy sequences converge to broken trajectories with sphere bubbles. There are no nonconstant spheres if $\omega|_{\pi_2}=0$; under semipositivity bubbling occurs in codimension ≥ 2 for generic J, so it misses the 0- and 1-dimensional moduli spaces [HS95; Ono95]. *Transversality* for generic t-dependent J: Floer–Hofer–Salamon [FHS95]. *Gluing* identifies the boundary of compactified 1-dimensional spaces with broken trajectories, which gives $d^2=0$. *Orientations*: [FH93]. *Continuation maps* count solutions of the s-dependent equation for a homotopy H^s equal to $H^\pm$ for $\pm s\gg 0$, and satisfy $E(u)=\mathcal{A}_{H^-}(\tilde{x}_-)-\mathcal{A}_{H^+}(\tilde{x}_+)+\iint(\partial_s H_t^s)(u)\,ds\,dt$. For (c), continue to εf with t-independent J: all index-one trajectories are then t-independent, so $\mathrm{CF}^*(\varepsilon f)$ is the Morse cochain complex of f tensored with Λ [SZ92; HS95].

Remark 7.16 (General closed manifolds). Without semipositivity, multiply covered spheres of negative Chern number obstruct transversality, and d must be defined by virtual counts. This gives Theorem 7.15 for $\mathrm{char}\,\mathbb{F}=0$ [FO99; LT98c; Rua99; Par16], and over $\mathbb{Z}$ with global Kuranishi charts and FOP perturbations [BX22; Rez22; BX26] (preprints). See Chapter 5 for the status of these foundations. In my view the aspherical case is the one to learn first: it contains all the analysis and none of the virtual machinery.

7.3 PSS, products and what Floer cohomology cannot see

Let $QH^*(M)=H^*(M;\mathbb{F})\otimes_{\mathbb{F}}\Lambda$ with the quantum product $*$ of Chapter 5. For Hamiltonians H,K let $(H\#K)_t:=H_t+K_t\circ(\phi_H^t)^{-1}$, which generates $\phi_H^t\phi_K^t$. The *pair-of-pants product* $\mathrm{HF}^*(H)\otimes_\Lambda\mathrm{HF}^*(K)\to\mathrm{HF}^*(H\#K)$ counts finite-energy solutions of Floer's equation on a sphere with two positive (input) and one negative (output) cylindrical ends, with Hamiltonian perturbations interpolating between H, K and $H\#K$ ([PSS96], [MS12a, Ch. 12]).

Theorem 7.17 (PSS isomorphism; Piunikhin–Salamon–Schwarz 1996)**.** *Let (M,ω) be closed and semipositive, $\mathbb{F}$ a field and H nondegenerate. There is a degree-preserving Λ-linear isomorphism $\mathrm{PSS}_H\colon QH^*(M)\to\mathrm{HF}^*(H)$ which commutes with continuation isomorphisms and satisfies $\mathrm{PSS}_{H\#K}(a*b)=\mathrm{PSS}_H(a)*\mathrm{PSS}_K(b)$ whenever H, K, $H\#K$ are nondegenerate.*

Proof idea. PSS_H counts "spiked planes": a half-infinite Morse gradient line of an auxiliary Morse function, attached at its finite end to a map $u\colon\mathbb{C}\to M$ that is J-holomorphic near the origin and satisfies Floer's equation for H near infinity, where it converges to an orbit. Reversing the roles gives an inverse; the two composites are chain homotopic to the identity by gluing and a deformation of the domain. Complete proofs for semipositive M are in [MS12a, Ch. 12]; the integral version for general M is in [BX22] (preprint).

The theorem also gives a module structure. $QH^*(M)$ acts on $\mathrm{HF}^*(H)$, either by counting Floer cylinders through a cycle Poincaré dual to a or, equivalently, by $a\cdot\xi:=\mathrm{PSS}_K(a)*\xi$ for a C^2-small

K followed by continuation, and $\mathrm{HF}^*(H)$ is a free $QH^*(M)$-module of rank one generated by $\mathrm{PSS}_H(1)$. A loop γ in $\mathrm{Ham}(M,\omega)$ induces a filtration-shifting isomorphism of Floer complexes which, under PSS, is quantum multiplication by the Seidel element of γ (Chapter 5). This is the mechanism by which Floer theory detects $\pi_1(\mathrm{Ham})$, and the reason the invariants below live on the universal cover $\widetilde{\mathrm{Ham}}(M,\omega)$.

The moral, which I would put on the first page of any course: $\mathrm{HF}^*(H)$ *is a deformation invariant and knows nothing about* ϕ_H *except that it is Hamiltonian.* Everything quantitative comes from the filtered complex, that is, from where in the action spectrum the classes of $QH^*(M)$ are born.

7.4 Spectral invariants

Spectral invariants go back to Viterbo's generating-function minimax values for $\mathbb{R}^{2n}$ and cotangent bundles [Vit92]; Schwarz [Sch00] constructed them in Floer theory on symplectically aspherical manifolds, Oh [Oh05a; Oh05b] on general closed ones, and Usher [Ush08] supplied the clean proof of spectrality. Following Convention 7.12, I use the transposed complex, its PSS isomorphism $QH_*(M)\to \mathrm{HF}_*(H)$, and Poincaré duality $\mathrm{PD}\colon QH^k(M)\to QH_{2n-k}(M)$.

Definition 7.18 (Spectral invariants)**.** Let H be nondegenerate and $0\neq a\in QH^*(M)$. Then

$$c(a;H):=\inf\left\{\lambda\in\mathbb{R}:\ \mathrm{PSS}(\mathrm{PD}\,a)\in \mathrm{im}\left(H_*(\mathrm{CF}_*^{<\lambda}(H))\to \mathrm{HF}_*(H)\right)\right\}.$$

H is *normalized* if $\int_M H_t\,\omega^n=0$ for all t. For $\tilde\phi\in\widetilde{\mathrm{Ham}}(M,\omega)$ represented by the path $(\phi_H^t)_{t\in[0,1]}$ of a normalized H, set $c(a;\tilde\phi):=c(a;H)$.

For $H=\varepsilon f$ as in Proposition 7.10, $\mathrm{PD}(1)=[M]$ is represented by the sum of the local maxima, so $c(1;\varepsilon f)=\varepsilon\max f$; similarly $c(\mathrm{vol};\varepsilon f)=\varepsilon\min f$, where $\mathrm{vol}\in H^{2n}(M;\mathbb{F})$ is Poincaré dual to a point.

Theorem 7.19 (Properties of spectral invariants; Viterbo, Schwarz, Oh, Usher)**.** *Let (M,ω) be closed and semipositive, $0\neq a,b\in QH^*(M)$, and H,K nondegenerate.*

1. *(Spectrality) $c(a;H)\in\mathrm{Spec}(H)$.*
2. *(Continuity) $\int_0^1\min_M(H_t-K_t)\,dt\le c(a;H)-c(a;K)\le\int_0^1\max_M(H_t-K_t)\,dt$. Hence $c(a;\cdot)$ extends continuously to all $H\in C^0(S^1\times M)$.*
3. *(Shift) $c(a;H+s)=c(a;H)+\int_0^1 s(t)\,dt$ for $s\in C^\infty(S^1)$.*
4. *(Homotopy invariance) If H,K are normalized and (ϕ_H^t), (ϕ_K^t) are homotopic rel endpoints, then $c(a;H)=c(a;K)$; so $c(a;\cdot)$ is well defined on $\widetilde{\mathrm{Ham}}(M,\omega)$.*
5. *(Triangle inequality) $c(a*b;H\#K)\le c(a;H)+c(b;K)$.*
6. *(Normalization and invariance) $c(a;0)=0$ for $0\neq a\in H^*(M;\mathbb{F})$, and $c(a;H\circ\psi)=c(a;H)$ for $\psi\in\mathrm{Symp}_0(M,\omega)$.*

If $\omega|_{\pi_2(M)}=0=c_1|_{\pi_2(M)}$, then $c(a;\cdot)$ descends to $\mathrm{Ham}(M,\omega)$ [Sch00].

Definition 7.20 (Spectral norm)**.** For $\tilde\phi\in\widetilde{\mathrm{Ham}}(M,\omega)$ put $\gamma(\tilde\phi):=c(1;\tilde\phi)+c(1;\tilde\phi^{-1})$, and for $\phi\in\mathrm{Ham}(M,\omega)$ put $\gamma(\phi):=\inf\{\gamma(\tilde\phi):\tilde\phi\mapsto\phi\}$.

Theorem 7.21 (The spectral norm; Schwarz 2000, Oh 2005)**.** *For (M,ω) closed and semipositive, γ is a conjugation-invariant norm on* $\mathrm{Ham}(M,\omega)$*: $\gamma(\phi)\geq 0$ with equality iff $\phi=\mathrm{id}$, $\gamma(\phi^{-1})=\gamma(\phi)$, $\gamma(\phi\psi)\leq\gamma(\phi)+\gamma(\psi)$, $\gamma(\psi\phi\psi^{-1})=\gamma(\phi)$ for $\psi\in\mathrm{Symp}_0(M,\omega)$, and $\gamma(\phi)\leq\|\phi\|_{\mathrm{Hof}}$ (Definition 7.23).*

Proof idea. Everything except nondegeneracy follows from Theorem 7.19. Nondegeneracy comes from an energy–capacity inequality for an action selector: if ϕ displaces a ball B, then $\gamma(\phi)$ is bounded below by a positive capacity of B ([Sch00; Oh05b], [MS12a, Thm. 12.4.4], [FGS05; Ush10]).

Warning 7.22 (Signs). The *cohomological* spectral invariant $c^\flat(a;H):=\sup\{\text{min-action of }\alpha:\ \alpha\in \mathrm{CF}^*(H),\ d\alpha=0,\ [\alpha]=\mathrm{PSS}_H(a)\}$, which is common in papers written in symplectic-cohomology conventions, satisfies $c^\flat(1;\varepsilon f)=\varepsilon\min f$. Time reversal $H\mapsto -H(-t,\cdot)$ identifies $\mathrm{CF}^*(H)$ with CF_{2n-*} of a Hamiltonian generating $\tilde\phi^{-1}$ and negates actions, which gives $c^\flat(a;\tilde\phi)=-c(a;\tilde\phi^{-1})$. In particular the triangle inequality for $c^\flat$ is superadditive. When comparing papers, test them on εf.

Theorem 7.19 holds for all closed M once virtual counts are admitted (Oh, Usher; for bulk-deformed versions, [Fuk+19]). A fact about $\mathbb{CP}^n$ worth knowing: $\gamma\leq\omega([\mathbb{CP}^1])$ on all of $\mathrm{Ham}(\mathbb{CP}^n)$, by a short argument from [EP03] using the quantum relation $p^{*(n+1)}=e^L$ for the hyperplane class p and the line class L. Kislev–Shelukhin [KS21] proved sharp bounds of this type for Lagrangian spectral norms.

7.5 Hofer geometry

Definition 7.23 (Hofer norm and displacement energy)**.** For $H\in C^\infty([0,1]\times M)$, compactly supported in $[0,1]\times\mathrm{int}\,M$ if M is open, let $\|H\|:=\int_0^1(\max_M H_t-\min_M H_t)\,dt$. For $\phi\in\mathrm{Ham}(M,\omega)$ let $\|\phi\|_{\mathrm{Hof}}:=\inf\{\|H\|:\phi_H^1=\phi\}$, and let $d_{\mathrm{Hof}}(\phi,\psi):=\|\phi^{-1}\psi\|_{\mathrm{Hof}}$. The *displacement energy* of $U\subset M$ is $e(U):=\inf\{\|\phi\|_{\mathrm{Hof}}:\phi\in\mathrm{Ham}(M,\omega),\ \phi(U)\cap\overline{U}=\emptyset\}\in[0,\infty]$.

Bi-invariance ($H\circ\psi^{-1}$ generates $\psi\phi_H\psi^{-1}$ for $\psi\in\mathrm{Symp}(M,\omega)$), symmetry and the triangle inequality are formal. Nondegeneracy is not, and it is the first genuinely "hard" fact about Ham: the analogous L^p norms with $p<\infty$ are degenerate (Eliashberg–Polterovich; see [Pol01]).

Theorem 7.24 (Nondegeneracy and energy–capacity; Hofer 1990, Lalonde–McDuff 1995, Usher 2010)**.** *For every symplectic manifold (M,ω), d_{Hof} is a bi-invariant metric on* $\mathrm{Ham}(M,\omega)$*. More precisely: (a) if $\iota\colon\overline{B}^{2n}(a)\to M$ is a symplectic embedding of the closed ball of capacity a (radius $\sqrt{a/\pi}$), then $e(\iota(\overline{B}^{2n}(a)))\geq a/2$; (b) if M is closed, then $e(U)\geq c_G(U)$ for every open $U\subset M$, where c_G is the Gromov width (Chapter 6), and this is sharp.*

Hofer [Hof90] treated $\mathbb{R}^{2n}$ and Lalonde–McDuff [LM95b] proved (a) in general by a beautiful geometric argument reducing it to nonsqueezing. Usher [Ush10] proved (b), in the form $e(U)\geq c_{HZ}^\circ(U)$ for the π_1-sensitive Hofer–Zehnder capacity, using spectral invariants and Frauenfelder–Ginzburg–Schlenk [FGS05]. A nonidentity ϕ displaces a small ball, so (a) gives nondegeneracy.

Remark 7.25 (Large-scale geometry). Whether $\mathrm{Ham}(M,\omega)$ has infinite Hofer diameter for every closed M is open (Problem 7.40). It is known for surfaces of positive genus [LM95a], for $\omega|_{\pi_2}=0=c_1|_{\pi_2}$ [Sch00], for S^2 [Pol98], for $\mathbb{CP}^n$ and $S^2\times S^2$ (via Theorem 7.27 and its analogue for $S^2\times S^2$ [EP03]), whenever the asymptotic spectral invariants descend to Ham [McD10], and

whenever M carries a nonconstant autonomous Hamiltonian all of whose contractible closed orbits are constant [Ush13]. In the last case Usher even embeds $(\mathbb{R}^\infty, \ell^\infty)$ bi-Lipschitzly, via boundary depth. Much of the difficulty lies in $\pi_1(\mathrm{Ham})$: the asymptotic spectral invariants live on $\widetilde{\mathrm{Ham}}$, and whether they descend is governed by the Seidel representation [McD10]. On S^2, Cristofaro-Gardiner–Humilière–Seyfaddini [CHS24a] answered the Kapovich–Polterovich question with PFH spectral invariants: $\mathrm{Ham}(S^2)$ is not quasi-isometric to $\mathbb{R}$. The local theory of Hofer geodesics (quasi-autonomous Hamiltonians, second variation) is due to Bialy–Polterovich [BP94], Lalonde–McDuff [LM95a] and Ustilovsky [Ust96]; Polterovich's book [Pol01] is the place to read it.

7.6 Calabi quasimorphisms and symplectic quasi-states

Banyaga's theorem (Chapter 2) says that $\mathrm{Ham}(M,\omega)$ is simple for M closed, so it has no nontrivial homomorphisms to $\mathbb{R}$. Entov–Polterovich found the next best thing.

Definition 7.26 (Quasimorphisms; Calabi property)**.** A *quasimorphism* on a group G is a map $\mu\colon G \to \mathbb{R}$ with $\sup_{f,g} |\mu(fg) - \mu(f) - \mu(g)| < \infty$; it is *homogeneous* if $\mu(f^k) = k\mu(f)$ for $k \in \mathbb{Z}$. A homogeneous quasimorphism μ on $\widetilde{\mathrm{Ham}}(M,\omega)$ has the *Calabi property* if, for every displaceable open $U \subset M$ and every $\tilde\phi$ represented by a path generated by H with $\operatorname{supp} H_t \subset U$ for all t, one has $\mu(\tilde\phi) = \int_0^1 \int_M H_t\, \omega^n\, dt$.

Theorem 7.27 (Calabi quasimorphism; Entov–Polterovich 2003)**.** *Let $(M,\omega) = (\mathbb{CP}^n, \omega_{FS})$, $n \geq 1$, and $\mathrm{vol}(M) := \int_M \omega^n$. For $\phi \in \mathrm{Ham}(M,\omega)$ and any lift $\tilde\phi \in \widetilde{\mathrm{Ham}}(M,\omega)$ the limit*

$$\mu(\phi) := -\mathrm{vol}(M) \lim_{k\to\infty} \frac{c(1;\tilde\phi^k)}{k}$$

exists and is independent of the lift. The map μ is a homogeneous quasimorphism on $\mathrm{Ham}(M,\omega)$ with the Calabi property (for ϕ generated by H with $\operatorname{supp} H_t \subset U$, U displaceable, $\mu(\phi) = \int_0^1 \int_M H_t\, \omega^n\, dt$), and it is Lipschitz with respect to d_{Hof}.

Proof idea. By the triangle inequality, $\tilde\phi \mapsto c(1;\tilde\phi)$ is subadditive up to a constant. Because $QH^*(\mathbb{CP}^n)$ is a field, Poincaré duality gives the reverse inequality up to a bounded error, and homogenization produces a quasimorphism. If $\operatorname{supp} H \subset U$ and U is displaceable, the values $c(1;kH)$ stay bounded, and the shift property applied to the normalization of H produces the Calabi term. Independence of the lift uses finiteness of $\pi_1(\mathrm{Ham}(\mathbb{CP}^n))$ for $n \leq 2$ and an argument of Seidel, based on his elements, for $n \geq 3$.

In [EP03] the same holds for $S^2 \times S^2$ with the split form, with 1 replaced by an idempotent. More generally, for closed spherically monotone M whose even quantum homology algebra is semisimple, $c(e;\cdot)$ with e the unit of a field summand gives a homogeneous Calabi quasimorphism on $\widetilde{\mathrm{Ham}}(M,\omega)$; examples include $S^2 \times S^2$, the one-point blow-up of $\mathbb{CP}^2$ and complex Grassmannians [EP03, Thm. 1.5]. Non-monotone and bulk-deformed versions, which can produce infinitely many linearly independent Calabi quasimorphisms, are in [Fuk+19]; Entov's ICM survey [Ent14] maps out the field. The quasimorphism reorganizes into a functional on functions.

Definition 7.28 (Symplectic quasi-states; heaviness)**.** A *symplectic quasi-state* on (M,ω) is $\zeta\colon C(M)\to\mathbb{R}$ with: $\zeta(1)=1$; $F\le G\Rightarrow\zeta(F)\le\zeta(G)$; $\zeta(F+G)=\zeta(F)+\zeta(G)$ and $\zeta(sF)=s\zeta(F)$ for $s\in\mathbb{R}$ whenever $F,G\in C^\infty(M)$ satisfy $\{F,G\}=0$; and $\zeta(F)=0$ if $\operatorname{supp}F$ is displaceable. A closed $X\subset M$ is *ζ-heavy* if $\zeta(F)\ge\min_X F$ for all $F\in C(M)$, and *ζ-superheavy* if $\zeta(F)\le\max_X F$ for all $F\in C(M)$.

Theorem 7.29 (Entov–Polterovich 2006, 2009)**.** *Let $(M,\omega)=(\mathbb{CP}^n,\omega_{FS})$, $n\ge1$. For $F\in C^\infty(M)$, $\zeta(F):=\lim_{k\to\infty}c(1;kF)/k$ exists, extends uniquely to a 1-Lipschitz map $C(M)\to\mathbb{R}$, and this extension is a symplectic quasi-state. Moreover $\zeta(F)=\big(\int_M F\omega^n-\mu(\phi_F^1)\big)/\mathrm{vol}(M)$, and: (a) superheavy sets are heavy; (b) heavy sets are stably nondisplaceable; (c) every superheavy set meets every heavy set. The Clifford torus $\{[z_0:\cdots:z_n]:|z_0|=\cdots=|z_n|\}$ is superheavy; for $n=1$ it is the equator.*

The theorem is from [EP06; EP09], where it is proved under the semisimplicity hypotheses above. Its most useful corollary is the "stem" criterion. If $\Phi\colon M\to\mathbb{R}^k$ has Poisson-commuting components and all of its fibres but one are displaceable, then the remaining fibre is superheavy, and in particular nondisplaceable. This is how one sees at once that the Clifford torus cannot be displaced. Without semisimplicity, $e=1$ still gives a *partial* quasi-state (homogeneous only for $s\ge0$ and subadditive on commuting pairs), and (a)–(c) survive [EP09]. Two recent developments: Mak–Sun–Varolgunes [MSV24] characterized heaviness of a compact set by nonvanishing of its relative symplectic cohomology, and Buhovsky–Entov–Polterovich [BEP12] introduced Poisson-bracket invariants of quadruples of sets, positive by Floer-theoretic arguments, with applications to Hamiltonian chords and, via Berezin–Toeplitz quantization, to quantum noise [Pol14].

7.7 Barcodes

Definition 7.30 (Persistence modules and barcodes)**.** A *persistence module of finite type* over $\mathbb{F}$ is a family of finite-dimensional $\mathbb{F}$-vector spaces $(V_t)_{t\in\mathbb{R}}$ with linear maps $\pi_{s,t}\colon V_s\to V_t$ for $s\le t$, such that $\pi_{t,t}=\mathrm{id}$ and $\pi_{t,u}\pi_{s,t}=\pi_{s,u}$; $V_t=0$ for $t\ll0$; there is a finite set $S\subset\mathbb{R}$ with $\pi_{s,t}$ an isomorphism whenever $[s,t]\cap S=\emptyset$; and $\pi_{s,t}$ is an isomorphism for $s<t$ close enough to t. Such a module is isomorphic to a finite direct sum of interval modules $\mathbb{F}_{(a,b]}$ ($\mathbb{F}$ for $t\in(a,b]$, zero otherwise), with $a<b\le+\infty$ and $(a,+\infty]:=(a,+\infty)$; the multiset of intervals is its *barcode*. The *bottleneck distance* d_{bot} between two barcodes is the infimum of the $\delta\ge0$ for which there is a partial matching of their bars such that matched bars have endpoints within δ and unmatched bars have length at most 2δ.

Let $\omega|_{\pi_2(M)}=0=c_1|_{\pi_2(M)}$, let $\phi\in\mathrm{Ham}(M,\omega)$ be nondegenerate, and let H be normalized with $\phi_H^1=\phi$. Then $V_t(\phi):=H_*(\mathrm{CF}_*^{<t}(H))$ is such a persistence module, independent of H up to isomorphism; call its barcode $\mathcal{B}(\phi)$. All endpoints lie in $\mathrm{Spec}(H)$. There are exactly $b(M;\mathbb{F})$ infinite bars, whose left endpoints are spectral invariants; finite bars record pairs of orbits cancelled by the differential. The *boundary depth* $\beta(\phi)$ is the length of the longest finite bar, an invariant introduced by Usher [Ush11] at the chain level.

Theorem 7.31 (Hofer stability of barcodes; Polterovich–Shelukhin 2016, Usher–Zhang 2016)**.** *For nondegenerate $\phi,\psi\in\mathrm{Ham}(M,\omega)$ with $\omega|_{\pi_2}=0=c_1|_{\pi_2}$, $d_{\mathrm{bot}}(\mathcal{B}(\phi),\mathcal{B}(\psi))\le d_{\mathrm{Hof}}(\phi,\psi)$. Consequently $\mathcal{B}$ extends by continuity to degenerate ϕ, with values in the completion of the space of barcodes.*

The proof is one line: continuation maps are filtered up to shifts bounded by $\int\max(H-K)$ and $\int\max(K-H)$, so they interleave the two persistence modules. Usher–Zhang [UZ16] built

barcodes over Novikov fields on arbitrary closed M, where the extra information carried by the Γ-action complicates matters. Polterovich–Shelukhin [PS16] used $\mathbb{Z}/p$-equivariant versions of these barcodes for the first striking application. On $\Sigma \times N$, with Σ a closed surface of genus ≥ 4 and N closed with $\omega|_{\pi_2} = 0 = c_1|_{\pi_2}$, there are Hamiltonian diffeomorphisms arbitrarily Hofer-far from all p-th powers, and in particular from all autonomous ones. Kislev–Shelukhin [KS21] proved $\beta(\phi) \leq \gamma(\phi)$ (answering a question of Usher), together with sharp bounds on Lagrangian spectral norms. Barcodes also measure dynamical complexity (barcode entropy, Chapter 9) and drive Shelukhin's proof of the Hofer–Zehnder conjecture (Chapter 9). The textbook is Polterovich–Rosen–Samvelyan–Zhang [Pol+20].

7.8 C^0 symplectic topology

The subject starts from a rigidity theorem: being symplectic, a condition on first derivatives, survives uniform limits. For volume-preserving maps the analogous statement is trivial, because volume is a measure; for symplectic maps it is a consequence of nonsqueezing.

Definition 7.32 (Symplectic and Hamiltonian homeomorphisms)**.** $\mathrm{Sympeo}(M,\omega)$ is the closure of $\mathrm{Symp}(M,\omega)$ in $\mathrm{Homeo}(M)$ for the C^0 topology (uniform convergence on compact sets), and $\overline{\mathrm{Ham}}(M,\omega)$ is the C^0-closure of $\mathrm{Ham}(M,\omega)$. A continuous isotopy $(\phi^t)_{t\in[0,1]}$ with $\phi^0 = \mathrm{id}$ is a *continuous Hamiltonian flow* if there are smooth normalized H_k such that $\phi^t_{H_k} \to \phi^t$ uniformly in (t,x) and (H_k) is Cauchy for $\|\cdot\|$ of Definition 7.23. Its limit $H \in L^{(1,\infty)}$ is a *continuous Hamiltonian*. $\mathrm{Hameo}(M,\omega)$ is the set of time-one maps of continuous Hamiltonian flows [OM07]. Thus $\mathrm{Ham} \subset \mathrm{Hameo} \subset \overline{\mathrm{Ham}} \subset \mathrm{Sympeo}$.

Theorem 7.33 (Eliashberg–Gromov C^0 rigidity; Eliashberg 1987, Gromov 1986)**.** *Let (M,ω) be a symplectic manifold and $\phi_k \in \mathrm{Symp}(M,\omega)$ a sequence converging uniformly on compact sets to a diffeomorphism ϕ. Then $\phi \in \mathrm{Symp}(M,\omega)$. That is,* $\mathrm{Sympeo}(M,\omega) \cap \mathrm{Diff}(M) = \mathrm{Symp}(M,\omega)$.

Proof idea. A linear map that preserves the Gromov width of all ellipsoids is symplectic or anti-symplectic, and a diffeomorphism that preserves the widths of all small ellipsoids satisfies the same dichotomy pointwise. Nonsqueezing (Chapter 6) shows that capacities of images behave well under uniform limits [Eli87; Gro86].

Oh and Müller [OM07] set out to build a C^0 Hamiltonian dynamics. Its basic consistency result is that the continuous Hamiltonian of a continuous Hamiltonian flow is unique: if $\phi^t_{H_k} \to \mathrm{id}$ uniformly and $H_k \to H$ in $L^{(1,\infty)}$, then $H = 0$. Viterbo [Vit06] proved this for uniform convergence of H_k, and Buhovsky–Seyfaddini [BS13] in general. Rigidity statements of the same flavour: the Poisson bracket is lower semicontinuous in the C^0 topology, $\|\{F,G\}\|_\infty \leq \liminf \|\{F_k,G_k\}\|_\infty$ when $F_k \to F$ and $G_k \to G$ uniformly [CV08; EP10; Buh10a]; and if $\phi \in \mathrm{Sympeo}(M,\omega)$ maps a coisotropic submanifold C onto a smooth submanifold, then $\phi(C)$ is coisotropic and ϕ maps the characteristic foliation of C to that of $\phi(C)$ [HLS15]. For Lagrangians, an earlier result of this kind is due to Laudenbach–Sikorav [LS94]. Then came the counterexample.

Theorem 7.34 (C^0 failure of Arnold; Buhovsky–Humilière–Seyfaddini 2018)**.** *Every closed connected symplectic manifold of dimension at least* 4 *admits an element of* $\mathrm{Hameo}(M,\omega)$ *with exactly one fixed point.*

In dimension 2 the opposite holds: elements of $\overline{\mathrm{Ham}}(\Sigma,\omega)$ satisfy (AC-crit) (Matsumoto [Mat00], building on Franks; see [BHS18] for the history). So Theorem 7.34 [BHS18] is a genuinely

high-dimensional flexibility phenomenon. In a sequel, Buhovsky–Humilière–Seyfaddini showed that a formulation of the Arnold conjecture involving fixed points *together with spectral invariants* does survive for Hamiltonian homeomorphisms of aspherical manifolds [BHS21]. The input is the following continuity theorem.

Theorem 7.35 (C^0 continuity of the spectral norm)**.** *Equip* $\mathrm{Ham}(M,\omega)$ *with* $d_{C^0}(\phi,\psi) := \max_x \mathrm{dist}(\phi(x),\psi(x))$ *for a Riemannian distance. Then* γ *is continuous with respect to* d_{C^0}*, and hence extends continuously to* $\overline{\mathrm{Ham}}(M,\omega)$*, if* (M,ω) *is: a closed surface (Seyfaddini [Sey13]); closed with* $\omega|_{\pi_2} = 0 = c_1|_{\pi_2}$ *(Buhovsky–Humilière–Seyfaddini [BHS21]);* $(\mathbb{CP}^n, \omega_{FS})$ *(Shelukhin [She22c], Kawamoto [Kaw22]); or negative monotone (Kawamoto [Kaw22]).*

The extension uses only $|\gamma(\phi) - \gamma(\psi)| \le \gamma(\phi\psi^{-1})$ and $d_{C^0}(\phi\psi^{-1}, \mathrm{id}) = d_{C^0}(\phi,\psi)$. Combined with $\beta \le \gamma$ [KS21], it shows that barcodes of aspherical manifolds are C^0-continuous and so are defined for Hamiltonian homeomorphisms [BHS21].

The most spectacular application concerns surfaces. Fathi [Fat80] proved that for $n \ge 3$ the kernel of the mass-flow homomorphism on volume-preserving homeomorphisms of a closed n-manifold is simple (and that $\mathrm{Homeo}_c(D^n, \mathrm{vol})$ is simple), and asked about $n = 2$.

Theorem 7.36 (Simplicity conjecture; Cristofaro-Gardiner–Humilière–Seyfaddini 2024)**.** *The group of compactly supported area-preserving homeomorphisms of the open* 2*-disc is not simple; in fact it is not perfect.*

Theorem 7.37 (Cristofaro-Gardiner–Humilière–Mak–Seyfaddini–Smith 2022)**.** *For every compact connected surface* Σ*, possibly with boundary, with area form* ω*, the group* $\overline{\mathrm{Ham}}(\Sigma,\omega)$ *(equal to the kernel of the mass flow on* $\mathrm{Homeo}_0(\Sigma,\omega)$*, and to* $\mathrm{Homeo}_0(S^2,\omega)$ *for* $\Sigma = S^2$*) is not simple. If* $\partial\Sigma \neq \emptyset$*, the Calabi homomorphism extends to a homomorphism* $\mathrm{Hameo}(\Sigma,\omega) \to \mathbb{R}$*.*

The proof of Theorem 7.36 in [CHS24b] uses PFH spectral invariants. Their asymptotics recover the Calabi invariant (a conjecture of Hutchings, verified for monotone twist maps), and they are C^0-continuous. These two properties show that the normal subgroup of "finite-energy homeomorphisms" is proper, because Oh's infinite twist maps lie outside it. The genus-0 results were also obtained in [CHS24a]. The proof of Theorem 7.37 in [Cri+22] replaces PFH by Lagrangian Floer theory of links $L = L_1 \sqcup \cdots \sqcup L_k \subset \Sigma$ in $\mathrm{Sym}^k \Sigma$ (quantitative Heegaard Floer cohomology), which works in all genera. It is a good example of a theme of this chapter: a quantitative invariant whose only job is to be C^0-continuous and to see Calabi in the limit. Whether $\mathrm{Hameo}(\Sigma,\omega)$ is a proper subgroup of the finite-energy group was left open in [Cri+22].

7.9 Floer homotopy theory

Cohen–Jones–Segal [CJS95] observed that Floer moduli spaces, compactified by broken trajectories, form a *flow category*: manifolds with corners whose boundary strata are products of lower ones. If these carry coherent stable framings (more generally, orientations for a bordism theory), they assemble into a spectrum whose homology is Floer homology. For thirty years the obstructions (framings, and in the closed case transversality and isotropy) kept this programme mostly aspirational. Since 2021 it has become a working technology. Abouzaid–Blumberg [AB21] used complex-oriented Kuranishi flow categories, together with Poincaré duality for Morava K-theory of classifying spaces of finite groups, to prove (AC-hom) in characteristic p. Their foundational account [AB24] builds a stable ∞-category of flow categories and bimodules for

various bordism theories. Large [Lar21] constructed spectral Fukaya categories of Liouville manifolds, and Porcelli–Smith [PS24] built a Donaldson–Fukaya category over the sphere spectrum from framed flow modules, obtaining constraints on smooth structures of exact Lagrangians in certain plumbings (Chapter 8). The integral Arnold conjecture [BX22; Rez22] belongs to the same circle of ideas. The philosophy, which I endorse: Floer theory with coefficients in generalized cohomology theories detects tangential and smooth information (framings, exotic structures, torsion) that ordinary Floer cohomology loses. In the Hamiltonian setting of this chapter, however, it has so far been used mainly to sharpen fixed-point counts.

7.10 What to read

Salamon, Lectures on Floer homology *[Sal99]*
The fastest complete route through the construction, the index theory and the small-Hamiltonian computation. Mind Warning 7.13.

Audin–Damian, Morse Theory and Floer Homology *[AD14]*
Slow and careful, aspherical case only. Read it for the details everyone else skips: gluing, the proof that $d^2 = 0$, orientations. The Morse half is a good warm-up.

McDuff–Salamon, J-holomorphic Curves, *Ch. 12 [MS12a]*
Floer theory written from the Gromov–Witten side: semipositive case, PSS, products, Seidel elements, spectral invariants.

Hofer–Zehnder [HZ94], Chs. 5–6
Hofer's metric on $\mathbb{R}^{2n}$ (Ch. 5), and the Arnold conjecture with an early account of Floer homology (Ch. 6). Read it for the variational intuition that predates Floer.

Floer [Flo89b]; Salamon–Zehnder [SZ92]; Hofer–Salamon [HS95]
Read Floer in the original; it is clearer than its reputation. Salamon–Zehnder fix the grading and the small-H computation; Hofer–Salamon introduce Novikov rings and the semipositive case.

Oh, Symplectic Topology and Floer Homology *[Oh15]*
Encyclopedic, two volumes; the reference for Oh's approach to spectral invariants. Not for a first reading.

Polterovich, The Geometry of the Group of Symplectic Diffeomorphisms *[Pol01]*
Hofer geometry; short, elegant, still the best introduction.

Entov–Polterovich [EP03; EP06; EP09]; Entov's ICM survey [Ent14]
Read the survey first, then the 2003 paper, then "Rigid subsets".

Polterovich–Rosen–Samvelyan–Zhang [Pol+20]
Barcodes from scratch, with the Hamiltonian applications.

C^0: [BHS18; CHS24b]
The introductions of these two papers are the best available surveys of C^0 symplectic topology; then [OM07] for the definitions.

Floer homotopy: [CJS95], then the introduction of [AB21]
For the technology, [AB24] and [AMS21].

7.11 Frontier and open problems

Problem 7.38 (Degenerate Arnold conjecture)**.** *Prove (AC-cl), and ideally (AC-crit), for every closed (M,ω). Status: open beyond the cases of Remark 7.3. Theorem 7.34 shows that the proof cannot be C^0-robust.*

Problem 7.39 (Strong Arnold conjecture)**.** *Prove (AC-Morse). Status: open; it follows from Theorem 7.4 whenever $N_{\mathrm{Morse}}(M) = b(M;\mathbb{F})$ for some $\mathbb{F}$, and from Theorem 7.6 in the case of Remark 7.7.*

Problem 7.40 (Hofer diameter)**.** *Is $\sup_\phi \|\phi\|_{\mathrm{Hof}} = \infty$ on $\mathrm{Ham}(M,\omega)$ for every closed (M,ω)? Status: open; see Remark 7.25 for known cases.*

Problem 7.41 (C^0 continuity of γ)**.** *Is the spectral norm C^0-continuous on every closed symplectic manifold? Status: open beyond Theorem 7.35; partial results for rational M are in [Kaw22].*

Question 7.42 (Simplicity in higher dimensions)**.** *Is $\overline{\mathrm{Ham}}(M,\omega)$, or $\mathrm{Hameo}(M,\omega)$, simple for closed M of dimension ≥ 4? I know of no viable approach: the surface proofs use PFH or symmetric products, which have no evident higher-dimensional replacement.*

Conjecture 7.43 (Viterbo's spectral bound conjecture)**.** *For a closed Riemannian manifold Q there is $C(Q)$ such that every exact Lagrangian in the unit codisc bundle D^*Q that is Hamiltonian isotopic to the zero section has Lagrangian spectral norm at most $C(Q)$. Status: proved for $Q = T^n$ and string point-invertible manifolds [She22b], S^n, $\mathbb{RP}^n$, $\mathbb{CP}^n$ and $\mathbb{HP}^n$ [She22c], and compact homogeneous spaces (Guillermou–Vichery [GV22], Viterbo [Vit22], preprints); open in general. This is not the systolic Viterbo conjecture of Chapter 9.*

Two more questions are open. The C^0 flux conjecture asks whether $\mathrm{Ham}(M,\omega)$ is C^0-closed in $\mathrm{Symp}_0(M,\omega)$ (Chapter 2). And Floer homotopy theory has yet to find genuinely new Hamiltonian applications, as opposed to sharper counts.

7.12 Logical dependencies

To state Conjecture 7.2 and Theorems 7.4 and 7.6 one needs only Ham and Definition 7.1. Floer cohomology requires, in order, Definitions 7.8, 7.9, 7.11 and 7.14, Proposition 7.10, and Convention 7.12; stating Theorem 7.15 presupposes the Conley–Zehnder index (Chapter 1), semipositivity, and Fredholm theory with Gromov–Floer compactness (Chapter 5), and its general form presupposes virtual counts. Theorem 7.17 needs quantum cohomology. Definitions 7.18 and 7.20 and Theorems 7.19 and 7.21 need PSS and the transposed complex. Definition 7.23 and Theorem 7.24 are elementary to state (Gromov width from Chapter 6). Definitions 7.26 and 7.28 and Theorems 7.27 and 7.29 need spectral invariants and semisimplicity; Definition 7.30 and Theorem 7.31 need filtered complexes; Definition 7.32 and Theorems 7.33, 7.34, 7.36 and 7.37 need only point-set topology to state.

Chapter 8

Lagrangian Floer theory, Fukaya categories, and mirror symmetry

Floer's paper [Flo88b] did Morse theory for an action functional on the space of paths between two Lagrangians, and almost everything in this chapter grows out of it. Counting polygons instead of strips gives the Fukaya category. Wrapping noncompact Lagrangians at infinity gives the wrapped Fukaya category, whose closed-string shadow is symplectic cohomology. Mirror symmetry predicts that these are categories of coherent sheaves or matrix factorizations. For exact manifolds they are also, as we now know, categories of sheaves on the skeleton.

There are three things to take away. First, the obstruction to defining $\mathrm{HF}(L_0, L_1)$ is real: disks bubble off at the boundary in codimension one. The right response is the curved A_∞-algebra of Fukaya–Oh–Ohta–Ono with its bounding cochains, not a list of special hypotheses. Second, generation theorems make the categorical package computable, and in cotangent bundles they reduce symplectic questions to the based loop space. This is how the nearby Lagrangian conjecture was brought to within a smooth classification problem. Third, in the exact world the sheaf model does real work; for closed manifolds there is no such shortcut. Conventions C1–C10 are in force: Floer theory is cohomological and normalized as in C7–C8 (Chapter 7), and A_∞ signs follow Seidel [Sei08b].

8.1 Floer cohomology of a pair of Lagrangians

Convention 8.1 (Standing setting)**.** Unless said otherwise, (M, λ) is a *Liouville manifold*: the completion $\widehat{W} = W \cup_{\partial W} \big([1,\infty) \times \partial W\big)$ of a Liouville domain (W, λ) (Chapter 3), with $\omega = d\lambda$ (C5) and $\lambda = r\alpha$ on the end, where $\alpha = \lambda|_{\partial W}$. An *exact Lagrangian* is a closed Lagrangian $L \subset M$ together with a chosen $f_L \in C^\infty(L)$ satisfying $\lambda|_L = df_L$. Almost complex structures are ω-compatible (C2) and of contact type on the end ($\lambda \circ J = dr$ for $r \gg 1$), so that the maximum principle keeps holomorphic curves with boundary on compact Lagrangians inside a compact set. Hamiltonians on M have compact support unless said otherwise. $\mathbb{F}$ is a coefficient field. If $\mathrm{char}\,\mathbb{F} \neq 2$, Lagrangians carry spin structures, which orient the moduli spaces (Remark 8.16).

Definition 8.2 (Path space, action, chords)**.** Let $L_0, L_1 \subset M$ be exact Lagrangians and $H \in C^\infty([0,1] \times M)$. Put $\mathcal{P}(L_0, L_1) = \{x \in C^\infty([0,1], M) : x(0) \in L_0,\ x(1) \in L_1\}$ and

$$\mathcal{A}_H(x) \;=\; -\int_0^1 x^*\lambda \;+\; f_{L_1}(x(1)) - f_{L_0}(x(0)) \;+\; \int_0^1 H_t(x(t))\,dt.$$

A *Hamiltonian chord* is a path $x \in \mathcal{P}(L_0, L_1)$ with $\dot{x}(t) = X_{H_t}(x(t))$. Write $\mathcal{X}(L_0, L_1; H)$ for the set of chords. The map $x \mapsto x(1)$ identifies $\mathcal{X}(L_0, L_1; H)$ with $\phi_H^1(L_0) \cap L_1$, and x is *nondegenerate* if this intersection is transverse at $x(1)$. For $H = 0$ the chords are the constant paths at points $p \in L_0 \cap L_1$, and $\mathcal{A}_0(p) = f_{L_1}(p) - f_{L_0}(p)$.

For a vector field ξ along x with $\xi(i) \in TL_i$, Cartan's formula gives $\delta \int x^*\lambda = \int \omega(\xi, \dot{x})\, dt + \lambda(\xi)|_0^1$. The boundary terms cancel against the variation of $f_{L_1}(x(1)) - f_{L_0}(x(0))$, and $dH = \omega(\cdot, X_H)$ (C3) then yields

$$d\mathcal{A}_H(x)\xi = \int_0^1 \omega\big(\dot{x} - X_{H_t}(x), \xi\big)\, dt.$$

So $\operatorname{Crit} \mathcal{A}_H$ is the set of chords, and for the L^2 metric $\int_0^1 \omega(\cdot, J_t \cdot)\, dt$ the gradient is $\nabla \mathcal{A}_H(x) = J_t(x)(\dot{x} - X_{H_t}(x))$, as for loops in C7. Its negative gradient lines are the Floer strips.

Definition 8.3 (Floer strips)**.** Fix a family $J = (J_t)_{t \in [0,1]}$ of almost complex structures as in Convention 8.1, and let $x_\pm \in \mathcal{X}(L_0, L_1; H)$. A *Floer strip* from x_- to x_+ is a smooth map $u : \mathbb{R} \times [0,1] \to M$ with $u(s,0) \in L_0$, $u(s,1) \in L_1$,

$$\partial_s u + J_t(u)\big(\partial_t u - X_{H_t}(u)\big) = 0,$$

and $u(s,\cdot) \to x_\pm$ uniformly as $s \to \pm\infty$. Its energy is $E(u) = \iint |\partial_s u|^2_{g_{J_t}}\, ds\, dt = \mathcal{A}_H(x_-) - \mathcal{A}_H(x_+)$. We write $\widetilde{\mathcal{M}}(x_-, x_+)$ for the space of Floer strips and $\mathcal{M}(x_-, x_+) = \widetilde{\mathcal{M}}(x_-, x_+)/\mathbb{R}$, where $\mathbb{R}$ acts by translation in s.

Definition 8.4 (Gradings)**.** Let $x \in L_0 \cap L_1$ be a transverse intersection point.

(a) ($\mathbb{Z}$-gradings; Kontsevich, Seidel [Sei00].) Assume $2c_1(M) = 0$, and fix a nowhere vanishing section Θ of $(\Lambda^n_{\mathbb{C}} T^*M)^{\otimes 2}$, where the complex structure is the one given by J. For a Lagrangian subspace $\Lambda \subset T_pM$ with basis $v_1, \dots, v_n$, the *squared phase* is $\alpha_\Theta(\Lambda) = \Theta(v_1 \wedge \cdots \wedge v_n)/|\Theta(v_1 \wedge \cdots \wedge v_n)| \in S^1$; it does not depend on the basis. A *grading* of a Lagrangian L is a continuous function $\theta_L : L \to \mathbb{R}$ with $e^{2\pi i \theta_L(p)} = \alpha_\Theta(T_pL)$. A grading exists if and only if the Maslov class $\mu_L \in H^1(L; \mathbb{Z})$, the degree of $p \mapsto \alpha_\Theta(T_pL)$, vanishes. The shift is $L[k] = (L, \theta_L - k)$. Choose a unitary frame of T_xM in which $T_xL_0 = \mathbb{R}^n$ and $T_xL_1 = e^{i\pi\beta_1}\mathbb{R} \oplus \cdots \oplus e^{i\pi\beta_n}\mathbb{R}$ with $\beta_j \in (0,1)$. Then

$$\deg(x) = n + \theta_{L_1}(x) - \theta_{L_0}(x) - (\beta_1 + \cdots + \beta_n) \in \mathbb{Z}.$$

(b) ($\mathbb{Z}/2$-gradings.) If L_0, L_1 are oriented, $\deg(x) \bmod 2$ is determined by the local intersection sign, up to a universal shift.

(c) (Monotone case.) Let $L \subset (M, \omega)$ be a Lagrangian in an arbitrary symplectic manifold. Call L *monotone* if there is $\kappa > 0$ with $\omega(\beta) = \kappa\, \mu_L(\beta)$ for all $\beta \in \pi_2(M, L)$, where μ_L is the boundary Maslov index (Chapter 1). Let $N_L \in \{1, 2, \dots\} \cup \{\infty\}$ be the positive generator of $\mu_L(\pi_2(M, L)) \subset \mathbb{Z}$. On the chords in the component of $\mathcal{P}(L, L)$ containing the constant paths, the relative index of strips is well defined in $\mathbb{Z}/N_L$.

In each case the Fredholm index of the linearized operator at a strip u from x_- to x_+ satisfies $\operatorname{ind} D_u \equiv \deg(x_-) - \deg(x_+)$, as an equality in $\mathbb{Z}$, $\mathbb{Z}/2$ or $\mathbb{Z}/N_L$ respectively.

Normalization check. Let L_1 be the graph of dg, for a C^2-small Morse function g, in a Weinstein neighbourhood T^*L_0 ($\lambda = p\, dq$), with the grading induced from θ_{L_0}. Then $L_0 \cap L_1 = \operatorname{Crit}(g)$ and (a) gives $\deg(x) = \operatorname{ind}_x(g)$, consistent with C8.

Definition 8.5 (The Floer complex in the exact case)**.** Let L_0, L_1 be exact (and graded, if a $\mathbb{Z}$-grading is wanted). Let H be a Hamiltonian all of whose chords are nondegenerate and J a generic family. The *Floer complex* is $\mathrm{CF}^*(L_0, L_1; H, J) = \bigoplus_{x \in \mathcal{X}(L_0, L_1; H)} \mathbb{F} \cdot x$, graded by Definition 8.4 applied to $\phi_H^1(L_0) \cap L_1$ (the grading of $\phi_H^1(L_0)$ transported along the isotopy), with differential

$$d(x_+) \;=\; \sum_{\deg x_- = \deg x_+ + 1} \#\mathcal{M}(x_-, x_+)\; x_-.$$

The count is mod 2 if $\mathrm{char}\,\mathbb{F} = 2$, and signed using the spin structures otherwise. The differential raises degree by 1 and raises the action. Its cohomology is $\mathrm{HF}^*(L_0, L_1)$.

In the normalization check, the rigid strips are the negative gradient lines of g, so $\mathrm{CF}^*(L_0, L_1)$ is the Morse cochain complex of g.

Theorem 8.6 (Lagrangian Arnold conjecture for exact Lagrangians; Floer 1988)**.** *Let (M, λ) be a Liouville manifold, $L \subset M$ a closed exact Lagrangian, and H a compactly supported Hamiltonian with $\phi_H^1(L) \pitchfork L$. For generic J, the differential of Definition 8.5 on $\mathrm{CF}^*(L, \phi_H^1(L); \mathbb{Z}/2)$ squares to zero, its cohomology is independent of (H, J) up to canonical isomorphism, and*

$$\mathrm{HF}^*(L, \phi_H^1(L); \mathbb{Z}/2) \cong H^*(L; \mathbb{Z}/2).$$

Consequently $\#\big(L \cap \phi_H^1(L)\big) \geq \sum_i \dim_{\mathbb{Z}/2} H^i(L; \mathbb{Z}/2)$.

Proof idea. A disk v with boundary on L has area $\int_{\partial v} df_L = 0$, and M has no holomorphic spheres. So there is no bubbling, the energy identity gives compactness, the one-dimensional moduli spaces are compactified by broken strips ($d^2 = 0$), and continuation maps give invariance. For H a C^2-small Morse function in a Weinstein neighbourhood, Floer shows all strips are t-independent gradient lines.

Floer stated the theorem for closed M with $\pi_2(M, L) = 0$; the proof is the same. For zero sections of cotangent bundles, the Betti bound is due to Laudenbach–Sikorav [LS85] (generating functions) and the cuplength bound to Hofer [Hof85] (variational methods); Floer's proof is the one that generalizes. For L spin it holds over any $\mathbb{F}$, $\mathbb{Z}$-graded if $2c_1(M) = 0$ and $\mu_L = 0$.

Remark 8.7 (Why the general case is hard: disk bubbling). Sphere bubbling has codimension two and is excluded under semipositivity (Chapter 7). Strips can also lose a disk at a boundary point, in *codimension one*: on the boundary of one-dimensional moduli spaces, exactly where $d^2 = 0$ is proved. A disk bubble of Maslov index μ lowers the index of the remaining strip by μ. (i) If all nonconstant J-disks on L_0, L_1 have Maslov index ≥ 3, generic one-dimensional moduli spaces see no disk bubbling. (ii) Maslov index 2 disks bubble only off constant strips and contribute

$$d^2 = \pm\big(\mathfrak{m}_0(L_0) - \mathfrak{m}_0(L_1)\big) \cdot \mathrm{id},$$

where $\mathfrak{m}_0(L) \in \mathbb{F}$ counts Maslov index 2 J-disks on L through a generic point (the sign is convention-dependent). (iii) Disks of Maslov index ≤ 0 come in families of excess dimension; this is what Section 8.2 is for.

Theorem 8.8 (Monotone Lagrangians; Oh 1993–1996, Biran 2006)**.** *Let (M, ω) be closed or Liouville and let $L \subset M$ be a closed, connected, monotone Lagrangian with $N_L \geq 2$. Then for every $\phi \in \mathrm{Ham}(M)$ with $\phi(L) \pitchfork L$ and generic J, the Floer differential on $\mathrm{CF}(L, \phi(L); \mathbb{Z}/2)$ squares to zero, and its cohomology $\mathrm{HF}^*(L, L; \mathbb{Z}/2)$ is $\mathbb{Z}/N_L$-graded and independent of (ϕ, J) up to canonical isomorphism. Moreover, there is a spectral sequence with $E_1^{p,q} = H^{p+q-pN_L}(L; \mathbb{Z}/2)$ and differentials $d_r \colon E_r^{p,q} \to E_r^{p+r, q-r+1}$, converging to $\mathrm{HF}^{p+q \bmod N_L}(L, L; \mathbb{Z}/2)$. As a map on $H^*(L)$, d_r has degree $1 - rN_L$.*

Corollary 8.9. *Let L be as in Theorem 8.8 with $n = \dim L$. If $N_L \geq n+2$, then $\mathrm{HF}^*(L,L;\mathbb{Z}/2) \cong H^*(L;\mathbb{Z}/2)$ (degrees mod N_L) and L is nondisplaceable. If L is displaceable by a Hamiltonian isotopy (for instance, $L \subset \mathbb{C}^n$), then $N_L \leq n+1$.*

Proof idea. d_r shifts degrees by $1 - rN_L$ and $H^*(L)$ lives in degrees $0,\dots,n$, so all d_r vanish if $N_L > n+1$. If L is displaceable, $\mathrm{HF} = 0$ and $1 \in H^0(L)$ must be killed, forcing $N_L \leq n+1$.

Oh [Oh93] constructed the theory for $N_L \geq 3$; his Addendum [Oh95] treats $N_L = 2$ via Remark 8.7(ii), since $\mathfrak{m}_0(L) = \mathfrak{m}_0(\phi(L))$. The spectral sequence is Oh's [Oh96]; the indexing above is Biran's [Bir06, Thm. 5.2.A], the version I recommend reading. Biran–Cornea's *pearl complex* [BC09] computes $\mathrm{HF}(L,L)$ from a Morse function and disks, with products and the $\mathrm{QH}(M)$-module structure; it is the most efficient tool in the monotone world.

8.2 Obstruction theory: curved A_∞-algebras and bounding cochains

Fukaya–Oh–Ohta–Ono's idea [Fuk+09] is to package *all* disk counts into a structure that is always defined, and then ask algebraically whether it can be deformed to admit a differential.

Definition 8.10 (Universal Novikov ring)**.** $\Lambda_0 = \left\{ \sum_{i\geq 0} a_i T^{\lambda_i} : a_i \in \mathbb{F},\ \lambda_i \in \mathbb{R}_{\geq 0},\ \lambda_i \to \infty \right\}$, with maximal ideal Λ_+ given by the condition that all $\lambda_i > 0$, and $\Lambda = \Lambda_0[T^{-1}]$, the universal Novikov field (the version of the Novikov field of Chapter 5 used in Lagrangian Floer theory). The valuation is $\mathrm{val}(\sum a_i T^{\lambda_i}) = \min\{\lambda_i : a_i \neq 0\}$. A disk u is weighted by $T^{\omega(u)}$.

Definition 8.11 (Curved A_∞-algebra; bounding cochains)**.** A *curved A_∞-algebra* over Λ_0 is a $\mathbb{Z}/2$-graded, T-adically complete, free Λ_0-module C with Λ_0-multilinear maps $\mathfrak{m}_k \colon C^{\otimes k} \to C$ for $k \geq 0$, of degree $2-k$, such that $\mathfrak{m}_0 = \mathfrak{m}_0(1) \in \Lambda_+ C$ and, for all $k \geq 0$ and homogeneous $a_1,\dots,a_k$,

$$\sum_{0\leq i\leq i+j\leq k} (-1)^{\maltese_i}\, \mathfrak{m}_{k-j+1}\big(a_k,\dots,a_{i+j+1},\mathfrak{m}_j(a_{i+j},\dots,a_{i+1}),a_i,\dots,a_1\big) = 0, \qquad \maltese_i = \sum_{p=1}^{i} (|a_p|-1).$$

A *strict unit* is $e \in C^0$ with $\mathfrak{m}_1(e) = 0$, $\mathfrak{m}_2(a,e) = a$, $\mathfrak{m}_2(e,a) = (-1)^{|a|}a$, and $\mathfrak{m}_k(\dots,e,\dots) = 0$ for $k \geq 3$. A *weak bounding cochain* is an element $b \in \Lambda_+ C^1$ with

$$\sum_{k\geq 0} \mathfrak{m}_k(b,\dots,b) = W(b)\,e \qquad \text{for some } W(b) \in \Lambda_+.$$

It is a *bounding cochain* if $W(b) = 0$. The deformed operations are

$$\mathfrak{m}_k^b(a_k,\dots,a_1) = \sum_{l_0,\dots,l_k\geq 0} \mathfrak{m}_{k+l_0+\cdots+l_k}(b^{l_k},a_k,b^{l_{k-1}},\dots,a_1,b^{l_0}).$$

They again satisfy the A_∞ relations, with $\mathfrak{m}_0^b = W(b)e$, and therefore $(\mathfrak{m}_1^b)^2 = 0$. (Expand the relation with $k=1$ and use the unit.)

Signs are Seidel's [Sei08b] with curvature allowed. FOOO order inputs oppositely and keep a $\mathbb{Z}$-grading via a degree-2 variable (their $\Lambda_{0,\mathrm{nov}}$).

Theorem 8.12 (Fukaya–Oh–Ohta–Ono 2009)**.** *Let (M,ω) be a closed symplectic manifold and $L\subset M$ a closed, relatively spin Lagrangian ($w_2(TL)$ lies in the image of $H^2(M;\mathbb{Z}/2)\to H^2(L;\mathbb{Z}/2)$), and take $\mathbb{F}=\mathbb{Q}$. Virtual counts of J-disks with boundary on L and $k+1$ boundary marked points define a curved, homotopy-unital A_∞-algebra $(H^*(L;\Lambda_0),\{\mathfrak{m}_k\})$. Modulo Λ_+, $\mathfrak{m}_1\equiv 0$ and $\mathfrak{m}_2$ is the cup product up to sign. The structure is independent of all choices up to gapped filtered A_∞ homotopy equivalence. Hence the moduli spaces $\mathcal{M}(L)\subset\mathcal{M}_{\mathrm{weak}}(L)$ of (weak) bounding cochains up to gauge equivalence, and the* potential $\mathfrak{PO}=W\colon\mathcal{M}_{\mathrm{weak}}(L)\to\Lambda_+$*, are symplectic invariants of L. For $b\in\mathcal{M}_{\mathrm{weak}}(L)$ and $b'\in\mathcal{M}_{\mathrm{weak}}(L')$ with $W(b)=W'(b')$, Floer cohomology $\mathrm{HF}((L,b),(L',b');\Lambda)$ is defined and is invariant under Hamiltonian isotopy. If $\mathrm{HF}((L,b),(L,b);\Lambda)\neq 0$, then for every $\phi\in\mathrm{Ham}(M)$ with $\phi(L)\pitchfork L$,*

$$\#\big(L\cap\phi(L)\big)\ \geq\ \dim_\Lambda \mathrm{HF}((L,b),(L,b);\Lambda),$$

and in particular L is nondisplaceable.

The slogan: *Floer cohomology is defined for unobstructed Lagrangians* ($\mathcal{M}(L)\neq\emptyset$, or weak bounding cochains in a common fibre of W), *while the A_∞-structure is defined for all.* Exact Lagrangians are unobstructed ($b=0$). Monotone ones with $N_L\geq 2$ are weakly unobstructed, with $W(0)=\mathfrak{m}_0(L)T^{\omega(\beta)}$ for $\mu(\beta)=2$. Fixed loci of antisymplectic involutions are unobstructed under orientation, relative spin and Chern class hypotheses, e.g. $c_1(M)|_{\pi_2(M)}\equiv 0 \bmod 4$ [Fuk+17]. The construction needs virtual techniques (Chapter 5; foundations in [Fuk+20]).

The theory is computed most completely for toric manifolds. Let X be a closed toric manifold with moment polytope $P=\{u\in\mathbb{R}^n : \ell_j(u)\geq 0,\ j=1,\dots,m\}$, where $\ell_j(u)=\langle v_j,u\rangle-\lambda_j$ and the $v_j\in\mathbb{Z}^n$ are the primitive inward normals. Let $L(u)$ be the fibre over $u\in\operatorname{Int}P$. For X Fano, Cho–Oh [CO06] showed that the Maslov index 2 disks on $L(u)$ correspond to facets: one regular family in a class β_j with $\partial\beta_j=v_j$ and area $\ell_j(u)$ (with the moment map normalized for $T^n=\mathbb{R}^n/\mathbb{Z}^n$; in the normalization of Chapter 2, $T^n=\mathbb{R}^n/2\pi\mathbb{Z}^n$, the areas are $2\pi\ell_j(u)$). In variables y with $\mathrm{val}(y_i)=u_i$, recording the fibre and a weak bounding cochain in $H^1(L(u);\Lambda_0)$, the potential is the Hori–Vafa Laurent polynomial

$$W(y)=\sum_{j=1}^{m}T^{-\lambda_j}y^{v_j}.$$

Theorem 8.13 (Nondisplaceable toric fibres; Cho–Oh 2006, FOOO 2010)**.** *Let X be a closed toric Fano manifold. If $y\in(\Lambda\setminus 0)^n$ is a critical point of W with $\mathrm{val}(y)=u\in\operatorname{Int}P$, then for the corresponding weak bounding cochain b one has $\mathrm{HF}((L(u),b),(L(u),b);\Lambda)\cong H^*(T^n;\Lambda)$ (even over Λ_0). In particular $L(u)$ is nondisplaceable, and $\#\big(L(u)\cap\phi(L(u))\big)\geq 2^n$ for every $\phi\in\mathrm{Ham}(X)$ with $\phi(L(u))\pitchfork L(u)$.*

For non-Fano X, W acquires corrections from disks with sphere bubbles; FOOO [Fuk+10] prove the analogue for all closed toric manifolds, show that some fibre is always nondisplaceable, and with bulk deformations [Fuk+11] find continua of nondisplaceable fibres in some toric manifolds. Floer theory also detects Entov–Polterovich heaviness [EP09]: if an idempotent e of bulk-deformed quantum cohomology has nonzero image under the closed–open map to $\mathrm{HF}((L,b),(L,b);\Lambda)$, then L is ζ_e-heavy, and superheavy if e is the unit of a field summand [Fuk+19]. Floer-theoretic and quasi-state rigidity (Chapter 7) are two sides of one coin.

Example 8.14 (Clifford and Chekanov tori in $\mathbb{CP}^2$)**.** Equip $\mathbb{CP}^2$ with its Fubini–Study form, and let A be one third of the area of a line, which is the area of a Maslov index 2 disk on a monotone Lagrangian. The Clifford torus $T_{\mathrm{Cl}}=\{|z_0|=|z_1|=|z_2|\}$ has potential $W_{\mathrm{Cl}}=T^A\big(x+y+\frac{1}{xy}\big)$. For

the Chekanov torus T_{Ch}, Auroux [Aur07] computed that in suitable coordinates on $H^1(T_{\mathrm{Ch}}; \Lambda_0)$ (after rescaling his variables by T^A)

$$W_{\mathrm{Ch}} = T^A\Big(u + \frac{(1+w)^2}{u^2 w}\Big) = T^A\Big(u + \frac{1}{u^2 w} + \frac{2}{u^2} + \frac{w}{u^2}\Big).$$

They agree under the wall-crossing substitution $u = x + y$, $w = x/y$ [Aur07, §5], which is birational but not monomial. For monotone tori the Maslov 2 disk counts are independent of J, so W is a Hamiltonian isotopy invariant up to a monomial change of coordinates in $\mathrm{GL}_2(\mathbb{Z})$. Three monomials versus four: the tori are not Hamiltonian isotopic. Chekanov's original arguments were different [Che96; CS10], but this one generalizes: Vianna's infinitely many monotone tori [Via16] are distinguished by the Newton polytopes of their potentials (Chapter 6). Both potentials have critical values $3\zeta T^A$, $\zeta^3 = 1$, as they must by the next proposition.

Proposition 8.15 (Auroux 2007, attributed to Kontsevich and Seidel)**.** *Let M be a closed monotone symplectic manifold and $L \subset M$ a closed, oriented, spin, monotone Lagrangian with $N_L \geq 2$. If $\mathrm{HF}(L,L) \neq 0$, then $\mathfrak{m}_0(L)$ is an eigenvalue of quantum multiplication $c_1(M)\star\colon \mathrm{QH}^*(M) \to \mathrm{QH}^*(M)$.*

Proof idea. The closed–open map $\mathcal{CO}^0\colon \mathrm{QH}^*(M) \to \mathrm{HF}^*(L,L)$ is a unital ring map sending $c_1(M) = \mathrm{PD}[D]$, D anticanonical, to $\mathfrak{m}_0(L)\cdot 1$, since a Maslov 2 disk meets D once. So $c_1 \star - \mathfrak{m}_0$ maps to zero in a nonzero ring and cannot be invertible.

Remark 8.16 (Orientations). For $\mathrm{char}\,\mathbb{F} \neq 2$, moduli spaces of disks on L are oriented by a (relative) spin structure; changing it by $\epsilon \in H^1(L; \mathbb{Z}/2)$ changes the sign of a disk u by $(-1)^{\epsilon(\partial u)}$. Sources: FOOO [Fuk+09], and Seidel [Sei08b, Part II] for the exact case. My advice is to work over $\mathbb{Z}/2$ until the signs are the point.

8.3 Fukaya categories

Definition 8.17 (A_∞-categories; Seidel's conventions)**.** An *A_∞-category* $\mathcal{A}$ over $\mathbb{F}$ consists of a set of objects; $\mathbb{Z}$-graded (or $\mathbb{Z}/2$-graded) vector spaces $\hom_{\mathcal{A}}(X_0, X_1)$; and maps $\mu^d\colon \hom(X_{d-1}, X_d) \otimes \cdots \otimes \hom(X_0, X_1) \to \hom(X_0, X_d)[2-d]$ for $d \geq 1$ satisfying

$$\sum_{m,n} (-1)^{\maltese_n} \mu^{d-m+1}\big(a_d, \ldots, a_{n+m+1}, \mu^m(a_{n+m}, \ldots, a_{n+1}), a_n, \ldots, a_1\big) = 0, \qquad \maltese_n = \sum_{j=1}^{n} (|a_j| - 1).$$

The *cohomological category* $H(\mathcal{A})$ has morphism spaces $H(\hom_{\mathcal{A}}, \mu^1)$ and composition $[a_2]\cdot[a_1] = (-1)^{|a_1|}[\mu^2(a_2, a_1)]$, which is associative by the relation with $d = 3$; $\mathcal{A}$ is *c-unital* if $H(\mathcal{A})$ has identity morphisms. An A_∞-functor $\mathcal{F}\colon \mathcal{A} \to \mathcal{B}$ consists of a map on objects and multilinear maps $\mathcal{F}^d$ of degree $1-d$ satisfying the obvious quadratic relations. It is a *quasi-equivalence* if $H(\mathcal{F})$ is an equivalence of categories. *Twisted complexes* $\mathrm{Tw}\,\mathcal{A}$ are formal sums $\bigoplus_i X_i[k_i]$ with a strictly lower-triangular differential δ satisfying $\sum_r \mu^r(\delta, \ldots, \delta) = 0$; $H^0(\mathrm{Tw}\,\mathcal{A})$ is triangulated. The *split-closed derived category* $D^\pi\mathcal{A}$ is the idempotent completion of $H^0(\mathrm{Tw}\,\mathcal{A})$. A set of objects *(split-)generates* if its triangulated (and idempotent-complete) envelope is everything.

Theorem 8.18 (The exact Fukaya category; Seidel 2008)**.** *Let (M, λ) be a Liouville manifold with $2c_1(M) = 0$ and a fixed Θ as in Definition 8.4(a), and let $\mathbb{F}$ be a field. There is a $\mathbb{Z}$-graded, c-unital A_∞-category* $\mathrm{Fuk}(M)$ *with the following description. Its objects are closed exact Lagrangians*

equipped with a grading and a spin structure (the latter only if $\operatorname{char}\mathbb{F}\neq 2$*). Its morphism spaces are* $\hom(L_0,L_1)=\mathrm{CF}^*(L_0,L_1;H_{L_0L_1},J_{L_0L_1})$*. The map* μ^d *has as coefficient of* x_0 *in* $\mu^d(x_d,\dots,x_1)$ *the signed count of rigid solutions* $u\colon S\to M$ *of a perturbed Cauchy–Riemann equation* $(du-Y)^{0,1}=0$*, where* S *is a disk with* $d+1$ *boundary punctures, the boundary components are mapped to* $L_0,\dots,L_d$*,* u *converges to the inputs* $x_j\in\mathcal{X}(L_{j-1},L_j)$ *and the output* $x_0\in\mathcal{X}(L_0,L_d)$*, and* $\deg x_0=\sum_j\deg x_j+2-d$*. The perturbation data are chosen consistently over the Deligne–Mumford–Stasheff compactifications of the moduli spaces of disks with* $d+1$ *boundary points. Different choices give quasi-equivalent categories, and* $H^0(\mathrm{Fuk}(M))$ *has morphisms* $\mathrm{HF}^0(L_0,L_1)$*.*

Consistent perturbation data are the technical heart; Seidel [Sei08b, Part II] does this honestly. The idea goes back to Fukaya [Fuk93] and Kontsevich [Kon95b]. In good cases, Lagrangian surgery at an intersection point x realizes $\mathrm{Cone}(x)$ geometrically.

Remark 8.19 (Beyond the exact case). For closed monotone M and $w\in\mathbb{F}$, Oh's argument gives a $\mathbb{Z}/2$-graded A_∞-category $\mathrm{Fuk}(M)_w$ of closed, oriented, spin, monotone Lagrangians with $N_L\geq 2$ and $\mathfrak{m}_0(L)=w$ (possibly with flat line bundles). By Proposition 8.15, $\mathrm{Fuk}(M)_w\simeq 0$ unless w is an eigenvalue of $c_1\star$. Sheridan [She16] gives the definitive account, with a generation criterion. For general closed M the objects are pairs (L,b), $b\in\mathcal{M}_{\mathrm{weak}}(L)$, over Λ, split by the value of W. A complete construction with all coherences is still not in one written source (Section 8.10).

A Lagrangian sphere $V\subset M$, parametrized up to isotopy, has a *Dehn twist* $\tau_V\in\mathrm{Symp}_c(M)$ defined via the normalized geodesic flow in a Weinstein neighbourhood (Chapter 3); it is the monodromy around a Lefschetz critical point.

Theorem 8.20 (Exact triangle for Dehn twists; Seidel 2003)**.** *Let* M *be a Liouville manifold,* $V\subset M$ *an exact Lagrangian sphere and* L_0,L_1 *exact Lagrangians, with* $\mathbb{F}=\mathbb{Z}/2$ *(or with gradings and spin structures, over any* $\mathbb{F}$*). In* $D^\pi\,\mathrm{Fuk}(M)$ *there is an exact triangle*

$$\mathrm{HF}^*(V,L_0)\otimes V\xrightarrow{\mathrm{ev}}L_0\longrightarrow\tau_V(L_0)\longrightarrow\big(\mathrm{HF}^*(V,L_0)\otimes V\big)[1].$$

In particular there is a long exact sequence

$$\cdots\to\mathrm{HF}^*(V,L_0)\otimes\mathrm{HF}^*(L_1,V)\to\mathrm{HF}^*(L_1,L_0)\to\mathrm{HF}^*(L_1,\tau_V(L_0))\to\cdots.$$

The sequence is [Sei03]; the categorical form is in [Sei08b]. A first application: the Dehn twist of the zero section of T^*S^n shifts its grading, $\tau(\widetilde{S}^n)=\widetilde{S}^n[1-n]$, Floer cohomology detects the shift, and so τ has infinite order in $\pi_0\,\mathrm{Symp}_c(T^*S^n)$ for $n\geq 2$ [Sei00]. For $n=2$, τ^2 is smoothly isotopic to the identity with compact support ([Sei99], crediting Kronheimer), while $\mathrm{Symp}_c(T^*S^2)\simeq\mathbb{Z}$ weakly, generated by τ [Sei08c]. In A_m Milnor fibres, $m\geq 3$, Seidel used τ^2 to produce infinitely many smoothly isotopic, pairwise non-Lagrangian-isotopic Lagrangian spheres [Sei99]: the prototype of four-dimensional symplectic mapping class phenomena (Chapter 6).

In Lefschetz fibrations the triangle becomes a computational engine. Let $\pi\colon E\to\mathbb{C}$ be an exact Lefschetz fibration with fibre F (Chapter 3), and choose a distinguished basis of vanishing paths, with vanishing cycles $V_1,\dots,V_m\subset F$ and thimbles $\Delta_1,\dots,\Delta_m\subset E$.

Definition 8.21 (Directed category of vanishing cycles)**.** $\mathrm{Fuk}^{\rightarrow}(V_1,\dots,V_m)$ is the A_∞-category with objects $1,\dots,m$; morphism spaces $\hom(i,j)=\mathrm{CF}^*(V_i,V_j)$ for $i<j$, $\mathbb{F}\cdot e_i$ for $i=j$, and 0 for $i>j$; and μ^d inherited from $\mathrm{Fuk}(F)$ (with e_i a strict unit).

Theorem 8.22 (Seidel 2008)**.** *The thimbles $\Delta_1, \dots, \Delta_m$ form a full exceptional collection generating the derived Fukaya category $D^b \operatorname{Fuk}(\pi)$ of the Lefschetz fibration, and their endomorphism A_∞-algebra is quasi-isomorphic to $\operatorname{Fuk}^{\to}(V_1, \dots, V_m)$. Hence $D^b \operatorname{Fuk}(\pi) \simeq H^0\big(\operatorname{Tw} \operatorname{Fuk}^{\to}(V_1, \dots, V_m)\big)$. Changing the distinguished basis by a Hurwitz move acts by mutation of exceptional collections.*

This is [Sei08b, Part III], following [Sei01]: the symplectic counterpart of Beilinson's theorem, and the engine of most HMS proofs for Fano and Landau–Ginzburg mirrors.

8.4 Symplectic cohomology and wrapped Fukaya categories

Definition 8.23 (Symplectic cohomology)**.** Let (W, λ) be a Liouville domain with completion $\widehat{W}$ and $\alpha = \lambda|_{\partial W}$. A Hamiltonian $H\colon S^1 \times \widehat{W} \to \mathbb{R}$ is *admissible of slope a* if $H = ar + b$ on $\{r \geq r_0\}$, where $a > 0$ is not the period of a closed Reeb orbit of α. For such H (nondegenerate) and J of contact type at infinity, the Hamiltonian Floer complex $\mathrm{CF}^*(H)$ is defined with the conventions C7–C8, and the maximum principle gives compactness. Continuation maps $\mathrm{HF}^*(H_a) \to \mathrm{HF}^*(H_{a'})$ for $a \leq a'$ form a directed system, and

$$\mathrm{SH}^*(W) = \varinjlim_{a \to \infty} \mathrm{HF}^*(H_a).$$

Equivalently, one can use a single Hamiltonian that is quadratic in r at infinity. For a smaller than the minimal Reeb period, $\mathrm{HF}^*(H_a) \cong H^*(W)$ (by C8), and this gives a canonical map $c^*\colon H^*(W) \to \mathrm{SH}^*(W)$. The grading is in $\mathbb{Z}$ if $2c_1(W) = 0$ and a trivialization is chosen, and in $\mathbb{Z}/2$ in general.

The definition goes back to Floer–Hofer [FH94] and Viterbo [Vit99]; learn it from Seidel's "biased view" [Sei08a], whose conventions C7 records. $\mathrm{SH}^*(W)$ is invariant under exact symplectomorphisms of $\widehat{W}$; its high-slope generators are Reeb orbits of ∂W (Chapter 9).

Proposition 8.24 (Structure)**.** *Let W be a Liouville domain.*

(a) The pair-of-pants product makes $\mathrm{SH}^*(W)$ a graded-commutative unital $\mathbb{F}$-algebra, and c^* is a unital ring map from $(H^*(W), \cup)$. With the BV operator $\Delta\colon \mathrm{SH}^* \to \mathrm{SH}^{*-1}$ coming from the S^1-action on loops, $\mathrm{SH}^*(W)$ is a BV algebra.

(b) $\mathrm{SH}^*(W) = 0$ if and only if the unit vanishes, if and only if $c^*(1) = 0$.

(c) (Viterbo functoriality [Vit99].) If $V \subset W$ is a Liouville subdomain (a compact codimension-zero submanifold with boundary for which $\lambda|_V$ is, up to adding an exact form, a Liouville form with ∂V of contact type), there is a unital ring map $\mathrm{SH}^*(W) \to \mathrm{SH}^*(V)$ compatible with restriction $H^*(W) \to H^*(V)$ via c^*. In particular, $\mathrm{SH}^*(W) = 0$ implies $\mathrm{SH}^*(V) = 0$.

(d) (Vanishing.) $\mathrm{SH}^*(W) = 0$ if W is a subcritical Weinstein domain (Cieliebak [Cie02a]), or more generally a flexible one (Bourgeois–Ekholm–Eliashberg [BEE12]; see Chapter 4). Also $\mathrm{SH}^*(W; \mathbb{Z}/2) = 0$ if W embeds as a Liouville subdomain of a Liouville manifold in which ∂W is displaceable by a compactly supported Hamiltonian isotopy. This combines the vanishing of Rabinowitz Floer homology for displaceable hypersurfaces (Cieliebak–Frauenfelder [CF09]) with Ritter's theorem that this vanishing is equivalent to $\mathrm{SH} = 0$ [Rit13].

Part (c) gives Viterbo's proof of Gromov's theorem (Chapter 6): a closed exact $L \subset \mathbb{C}^n$ would make D^*L a Liouville subdomain of a ball B, but $\mathrm{SH}(B) = 0 \neq \mathrm{SH}(D^*L)$ by Theorem 8.25. The

converse of (d) fails: Murphy–Siegel's subflexible domains [MS18] have $\mathrm{SH} = 0$ but are not flexible, as twisted SH shows.

Theorem 8.25 (Viterbo's isomorphism; Viterbo 1999, Abbondandolo–Schwarz 2006, Salamon–Weber 2006, Abouzaid 2015)**.** *Let Q be a closed, oriented, spin n-manifold, and give T^*Q the $\mathbb{Z}$-grading induced by the vertical Lagrangian distribution. Then there is an isomorphism of graded algebras*

$$\mathrm{SH}^*(T^*Q;\mathbb{F}) \cong H_{n-*}(\mathcal{L}Q;\mathbb{F}),$$

where $\mathcal{L}Q = C^0(S^1, Q)$ is the free loop space and the right-hand side carries the Chas–Sullivan loop product. Under it, c^* becomes Poincaré duality $H^*(Q) \cong H_{n-*}(Q)$ followed by the inclusion of constant loops. If $\mathrm{char}\,\mathbb{F} = 2$*, no orientability or spin hypothesis is needed.*

Viterbo announced the group isomorphism [Vit99; Vit18]; complete proofs are due to Abbondandolo–Schwarz [AS06; AS14], Salamon–Weber [SW06] and Abouzaid [Abo15], and the product to Abbondandolo–Schwarz [AS10a], matching Chas–Sullivan string topology [CS99]. For oriented non-spin Q over $\mathbb{Z}$ one must twist by the local system on $\mathcal{L}Q$ transgressing $w_2(Q)$; this was overlooked for years until Kragh [Kra18], and Abouzaid [Abo15] gives the general twisted statement. In my view the conceptual content is that $\mathrm{SH}^*(T^*Q)$ is the Hochschild cohomology of $C_{-*}(\Omega Q)$, a statement about the wrapped category (Theorem 8.28).

Definition 8.26 (Wrapped Floer cohomology and the wrapped Fukaya category)**.** An *exact cylindrical Lagrangian* in $\widehat{W}$ is a properly embedded Lagrangian L with $\lambda|_L = df_L$, which on $\{r \geq 1\}$ equals $[1,\infty) \times \partial L$ for a Legendrian $\partial L \subset \partial W$, with f_L locally constant there. For such L_0, L_1 and time-independent admissible H_a, the *wrapped Floer cohomology* is $\mathrm{HW}^*(L_0, L_1) = \varinjlim_a \mathrm{HF}^*(L_0, L_1; H_a)$. The *wrapped Fukaya category* $\mathcal{W}(W)$ has closed exact and exact cylindrical Lagrangians (graded, spin) as objects and an A_∞-structure lifting these groups. It is constructed by Abouzaid–Seidel [AS10c] with a quadratic Hamiltonian and rescaling, and by Ganatra–Pardon–Shende [GPS20] as a localization at continuation morphisms.

There are closed–open and open–closed maps

$$\mathcal{CO}\colon \mathrm{SH}^*(W) \to \mathrm{HH}^*(\mathcal{W}(W)), \qquad \mathcal{OC}\colon \mathrm{HH}_{*-n}(\mathcal{W}(W)) \to \mathrm{SH}^*(W)$$

(Ganatra [Gan12]). Since each $\mathrm{HW}^*(L, L)$ is a unital $\mathrm{SH}^*(W)$-module, $\mathrm{SH}^*(W) = 0$ forces $\mathcal{W}(W) \simeq 0$.

Theorem 8.27 (Generation criterion; Abouzaid 2010)**.** *Let W be a Liouville domain, and let $\mathcal{B} \subset \mathcal{W}(W)$ be the full subcategory on a finite set of objects. If the unit $1 \in \mathrm{SH}^*(W)$ lies in the image of $\mathcal{OC}\colon \mathrm{HH}_{*-n}(\mathcal{B}) \to \mathrm{SH}^*(W)$, then every object of $\mathcal{W}(W)$ is split-generated by $\mathcal{B}$.*

Theorem 8.28 (The cotangent fibre; Abbondandolo–Schwarz 2006, Abouzaid 2011, 2012)**.** *Let Q be a closed, connected, oriented, spin manifold (or let* $\mathrm{char}\,\mathbb{F} = 2$*), $q \in Q$, and $T^*_q Q$ the cotangent fibre. Then $\mathrm{HW}^*(T^*_q Q, T^*_q Q) \cong H_{-*}(\Omega_q Q;\mathbb{F})$ as rings, where the product on the right is the Pontryagin product; at the cochain level, $CW^*(T^*_q Q, T^*_q Q)$ is A_∞-quasi-isomorphic to $C_{-*}(\Omega_q Q;\mathbb{F})$. The fibre generates $\mathcal{W}(T^*Q)$, so $D^\pi\mathcal{W}(T^*Q) \simeq \mathrm{Perf}\left(C_{-*}(\Omega_q Q)\right)$.*

The group isomorphism is [AS06], the A_∞ statement [Abo12c], and generation [Abo11], via Theorem 8.27. Without spin, both hold after twisting by the background class $\pi^* w_2(Q)$.

Theorem 8.29 (Generation by cocores; Chantraine–Dimitroglou Rizell–Ghiggini–Golovko, Ganatra–Pardon–Shende)**.** *Let W be a Weinstein domain (Chapter 3). The Lagrangian cocore disks of the index-n handles of a Weinstein handle decomposition generate $\mathcal{W}(W)$.*

This is due to Chantraine–Dimitroglou Rizell–Ghiggini–Golovko [Cha+24] and, independently and for Weinstein sectors (adding linking disks of the stops), Ganatra–Pardon–Shende [GPS24b]. GPS's *Liouville sectors* [GPS20] make $\mathcal{W}$ covariantly functorial, and *sectorial descent* [GPS24b] expresses $\mathcal{W}$ of a Weinstein manifold as a homotopy colimit over a cover by sectors; this local-to-global principle makes the comparison with sheaves (Section 8.7) possible. The closed-string counterpart is Bourgeois–Ekholm–Eliashberg's computation of SH from the Legendrian contact homology of the attaching spheres [BEE12].

8.5 The nearby Lagrangian conjecture

Conjecture 8.30 (Arnold's nearby Lagrangian conjecture)**.** *Let Q be a closed manifold. Every closed exact Lagrangian $L \subset (T^*Q, d\lambda_{\mathrm{can}})$ is Hamiltonian isotopic to the zero section.*

The isotopy may be taken compactly supported. Exactness is essential: Lagrangian tori in a Darboux ball are not exact.

Theorem 8.31 (Known cases)**.** *Conjecture 8.30 holds for $Q = S^1$ (elementary), for $Q = S^2$ (Hind [Hin12]: every Lagrangian sphere in T^*S^2 is Hamiltonian isotopic to the zero section), for $Q = \mathbb{RP}^2$ (Hind–Pinsonnault–Wu [HPW16], with a simpler proof by Adaloglou [Ada25]: every Lagrangian $\mathbb{RP}^2$ in $T^*\mathbb{RP}^2$ is Hamiltonian isotopic to the zero section), and for $Q = T^2$ (Dimitroglou Rizell–Goodman–Ivrii [DGI16]).*

For S^2 and $\mathbb{RP}^2$ one also needs $L \cong Q$, which follows from Theorem 8.32(a) and the classification of surfaces. These proofs use holomorphic foliations in a compactification, in the tradition of Gromov and McDuff (Chapter 6); in higher dimensions the tool is the wrapped category of Theorem 8.28.

Theorem 8.32 (Nearby Lagrangians are simple homotopy equivalent to the zero section)**.** *Let Q be closed and $L \subset T^*Q$ a closed exact Lagrangian. Then:*

- *(a) the Maslov class of L vanishes, and the projection $\pi|_L \colon L \to Q$ is a homotopy equivalence (Abouzaid [Abo12b] assuming $\mu_L = 0$; Kragh [Kra13] in general, with an appendix by Abouzaid proving $\mu_L = 0$);*
- *(b) $\pi|_L$ is a simple homotopy equivalence (Abouzaid–Kragh [AK18]);*
- *(c) the stable Lagrangian Gauss map $L \to U/O$ (defined using the vertical polarization) induces the zero map on all homotopy groups, and is nullhomotopic if Q is a homotopy sphere [Abo+25]; its composite with the delooped J-homomorphism $U/O \to B(\mathbb{Z} \times BG)$ is nullhomotopic (after looping, [AK16]; in general [Abo+25, Thm. B], following Jin).*

The history is instructive. Fukaya–Seidel–Smith [FSS08] (Q simply connected and spin, L spin with $\mu_L = 0$, $\operatorname{char} \mathbb{F} \neq 2$) proved that $L \to Q$ has degree ± 1 and $H^*(L;\mathbb{F}) \cong H^*(Q;\mathbb{F})$, and Nadler [Nad09] showed from the sheaf side that such L is equivalent in the Fukaya category to the zero section with a local system. The idea: $\mathrm{HF}^*(T^*_q Q, L)$ is a module over $\mathrm{HW}^*(T^*_q Q) \simeq C_{-*}(\Omega Q)$ which, by Theorem 8.28, determines L, and exactness forces it to be a rank-one local system.

Abouzaid upgraded cohomology to homotopy type; Kragh's parametrized spectra removed the hypotheses; Guillermou's sheaf quantization [Gui23] gives an independent proof. What remains is a genuinely smooth question: even for $Q = S^n$ it is unknown in general whether $L \cong S^n$, though Abouzaid [Abo12a] excludes exotic spheres not bounding parallelizable manifolds in T^*S^{4k+1}.

8.6 Homological mirror symmetry

Conjecture 8.33 (Kontsevich 1994; informal)**.** *Let X be a compact Calabi–Yau manifold with a complexified Kähler class near the large volume limit, and let $X^\vee$ be its mirror family near the large complex structure limit, regarded as a variety over the Novikov field Λ. Then there is a Λ-linear equivalence of triangulated categories*

$$D^\pi \operatorname{Fuk}(X) \;\simeq\; D^b \operatorname{Coh}(X^\vee),$$

under which the Novikov parameter corresponds to the complex parameter of the family via the mirror map, and the symmetric equivalence $D^\pi \operatorname{Fuk}(X^\vee) \simeq D^b \operatorname{Coh}(X)$ holds as well.

It is informal because "mirror" is not a definition: a precise version needs a construction of $X^\vee$ (Batyrev, Greene–Plesser, or intrinsically via SYZ or Gross–Siebert). Kontsevich proposed it in 1994 [Kon95b], before D-brane physics made it look inevitable. For Fano X the mirror is a Landau–Ginzburg model $(X^\vee, W)$, with $D^\pi \operatorname{Fuk}(X)_w \simeq \operatorname{MF}(X^\vee, W - w)$ (matrix factorizations); conversely, Fukaya–Seidel and wrapped categories are mirror to D^b Coh of possibly noncompact or singular varieties. The potential W_{Cl} of Example 8.14 is the Landau–Ginzburg mirror of $\mathbb{CP}^2$.

Theorem 8.34 (Punctured spheres; Abouzaid–Auroux–Efimov–Katzarkov–Orlov 2013)**.** *Let $\Sigma = \mathbb{CP}^1 \setminus \{p_1, \dots, p_k\}$ with $k \geq 3$. Then the derived wrapped Fukaya category $D\mathcal{W}(\Sigma)$ is equivalent to the triangulated category of singularities $D_{\mathrm{sg}}(W^{-1}(0))$ of a Landau–Ginzburg model (X_k, W). Here X_k is a noncompact toric threefold, a toric resolution of a singular affine toric threefold. For $k = 3$ (the pair of pants), $X_3 = \mathbb{C}^3$ and $W = xyz$. The equivalence lifts to a quasi-equivalence of A_∞-enhancements.*

Proven cases of Conjecture 8.33 and its variants include the following.

- *Compact Calabi–Yau:* the elliptic curve (Polishchuk–Zaslow [PZ98]); abelian varieties, partially (Fukaya [Fuk02]); the four-torus (Abouzaid–Smith [AS10d]); the quartic K3 surface (Seidel [Sei15]); Calabi–Yau hypersurfaces in $\mathbb{CP}^n$ of every dimension ≥ 3 (Sheridan [She15]); generalized Greene–Plesser mirror pairs, a class of Batyrev–Borisov pairs (Sheridan–Smith [SS21]), with the mirror map shown to have integral coefficients by Ganatra–Hanlon–Hicks–Pomerleano–Sheridan [Gan+25].
- *General type and punctured surfaces:* the genus-two curve (Seidel [Sei11]); higher-dimensional pairs of pants (Sheridan [She11]); punctured spheres (Theorem 8.34); very affine hypersurfaces (Gammage–Shende [GS22], via Theorem 8.39).
- *Toric and log Calabi–Yau:* toric varieties, from tropical Lagrangian sections (Abouzaid [Abo06; Abo09]) and from the coherent–constructible correspondence (Fang–Liu–Treumann–Zaslow [Fan+11], Kuwagaki [Kuw20]); log Calabi–Yau surfaces (Hacking–Keating [HK22]).

The proofs come in two families. Seidel's strategy for the quartic, which Sheridan pushed to all dimensions, computes a small A_∞-algebra on both sides, recognizes both as versal deformations of the same object, and concludes by generation. The other route is SYZ [SYZ96]: mirrors carry

dual special Lagrangian torus fibrations, and the mirror is the moduli space of fibres with local systems, corrected by Maslov 0 disks via wall-crossing as in Example 8.14 [Aur07; AAK16]. Its algebro-geometric shadows are the Gross–Siebert program [GS11] and Kontsevich–Soibelman's non-Archimedean version [KS06]. Fukaya's *family Floer* program builds the mirror functor from Floer cohomology with all fibres at once; Abouzaid proved it faithful [Abo17], and fully faithful when there are no singular fibres and fibres bound no holomorphic disks [Abo21] (overview: [Abo14]). Closed-string mirror symmetry (the quintic numbers of Candelas–de la Ossa–Green–Parkes [Can+91], proved by Givental [Giv96] and Lian–Liu–Yau [LLY97]) is expected to follow from HMS via Hochschild invariants; I do not pursue it.

8.7 Microlocal sheaves

Let X be a manifold, $\mathbb{F}$ a field, and $D(X)$ the bounded derived category of sheaves of $\mathbb{F}$-vector spaces on X (in Theorem 8.38, locally bounded complexes).

Definition 8.35 (Microsupport; Kashiwara–Schapira)**.** For $F \in D(X)$, a point $(x_0, \xi_0) \in T^*X$ is *not* in the microsupport $\mathrm{SS}(F)$ if it has an open neighbourhood $U \subset T^*X$ with the following property: for every $x_1 \in X$ and every C^1 function φ defined near x_1 with $\varphi(x_1) = 0$ and $(x_1, d\varphi(x_1)) \in U$, one has $\left(R\Gamma_{\{\varphi \geq 0\}} F\right)_{x_1} = 0$. Thus $\mathrm{SS}(F)$ is a closed conic subset of T^*X, and $\mathrm{SS}(F) \cap 0_X = \operatorname{supp} F$.

So F propagates in codirections outside $\mathrm{SS}(F)$. The notion is invariant under contact transformations, which is the source of its symplectic content.

Theorem 8.36 (Involutivity; Kashiwara–Schapira 1990)**.** *For every $F \in D(X)$, $\mathrm{SS}(F)$ is coisotropic in T^*X. If X is real analytic, F is $\mathbb{R}$-constructible if and only if its stalks have finite-dimensional cohomology and $\mathrm{SS}(F)$ is contained in a closed conic subanalytic Lagrangian subset of T^*X.*

Theorem 8.37 (Nadler–Zaslow 2009, Nadler 2009)**.** *Let X be a compact real analytic manifold. Let $\mathrm{Fuk}(T^*X)$ be the A_∞-category whose objects are exact, not necessarily compact, Lagrangian branes in T^*X with subanalytic closure in the fibrewise compactification, and whose morphisms are computed after a small positive perturbation by the normalized geodesic flow near infinity (what is now called the infinitesimally wrapped category). Then there is a quasi-equivalence $\mathrm{Sh}_c(X) \simeq \mathrm{Tw}\,\mathrm{Fuk}(T^*X)$, where $\mathrm{Sh}_c(X)$ is the dg-category of bounded $\mathbb{R}$-constructible complexes of sheaves on X.*

Nadler–Zaslow [NZ09] construct the quasi-embedding, sending the standard sheaf on an open U to the graph of $d\log$ of a defining function of U; Nadler [Nad09] proves essential surjectivity. Independently, Tamarkin [Tam18] proved nondisplaceability by sheaf theory alone, using sheaves on $X \times \mathbb{R}_t$ microsupported in $\{\tau \geq 0\}$: the Tamarkin category, a sheaf model of the Novikov-filtered world.

Theorem 8.38 (Sheaf quantization of Hamiltonian isotopies; Guillermou–Kashiwara–Schapira 2012)**.** *Let $\Phi = (\phi_s)_{s \in I}$, with $I \ni 0$ an open interval, be a homogeneous Hamiltonian isotopy of $T^*X \setminus 0_X$ that is the identity outside $\pi^{-1}(C)$ for a compact $C \subset X$. There is a unique $K \in D(X \times X \times I)$ with $K|_{s=0} \cong \mathbb{F}_\Delta$ whose microsupport outside the zero section is the conic graph of Φ. For each s, convolution with K_s is an autoequivalence of $D(X)$ with $\mathrm{SS}(K_s \circ F) \setminus 0 = \phi_s(\mathrm{SS}(F) \setminus 0)$.*

GKS [GKS12] apply this, via Tamarkin's trick of making compactly supported isotopies of T^*X homogeneous on $T^*(X \times \mathbb{R})$, to reprove nondisplaceability of the zero section for compact X. The same technology gives Chiu's contact nonsqueezing [Chi17], the Legendrian invariants of

Shende–Treumann–Zaslow [STZ17], and the theorem of Ng–Rutherford–Shende–Sivek–Zaslow [Ng+20] that augmentations of the Chekanov–Eliashberg DGA of a Legendrian knot in $\mathbb{R}^3$ are sheaves microsupported on the knot. Using cluster structures on such moduli, Casals–Gao [CG22] found the first Legendrian links with infinitely many exact fillings in B^4: all max-tb (n, m) torus links except $(2, m)$, $(3, 3)$, $(3, 4)$, $(3, 5)$. Casals–Weng [CW24] and Gao–Shen–Weng [GSW24] made this a systematic theory.

For a Weinstein manifold W with skeleton $\mathfrak{c}_W$ and a stable polarization (more generally, Maslov data), Shende [She21b] and Nadler–Shende [NS20] define microlocal sheaves $\mu\,\mathrm{Sh}_{\mathfrak{c}_W}(\mathfrak{c}_W)$ by embedding W in a high-dimensional cotangent bundle.

Theorem 8.39 (Microlocal Morse theory; Ganatra–Pardon–Shende 2024)**.** *Let W be a Weinstein manifold, real analytic with subanalytic core $\mathfrak{c}_W$ (this can always be arranged), and fix a stable polarization of W. Then there is an equivalence*

$$\mathrm{Perf}\,\mathcal{W}(W)^{\mathrm{op}} \;\simeq\; \mu\,\mathrm{Sh}_{\mathfrak{c}_W}(\mathfrak{c}_W)^c$$

onto the compact objects of the category of microlocal sheaves on the core. For a general Liouville manifold with subanalytic core the functor is fully faithful. For $W = T^*X$ with X connected and real analytic, the right-hand side is the category of compact locally constant sheaves, $\mathrm{Perf}\,C_{-*}(\Omega X)$*, which recovers Theorem 8.28.*

This is [GPS24a, Thms. 1.1, 1.4], which also treats stopped versions $\mathcal{W}(W, \Lambda)$. The practical consequence is large: wrapped categories of Weinstein manifolds become computable as limits of local sheaf categories attached to the singularities of the skeleton, which is how Gammage–Shende prove HMS for very affine hypersurfaces [GS22]. Quantitatively, Asano–Ike [AI20] and Zhang [Zha20] develop interleaving distances on Tamarkin categories that recover displacement energy bounds: a sheaf version of the persistence theory of Chapter 7.

8.8 Lagrangian cobordism

Definition 8.40 (Lagrangian cobordism; Arnold, Biran–Cornea)**.** Let $L_1, \dots, L_k$ and L' be closed Lagrangians in (M, ω). A *Lagrangian cobordism* $V \colon L' \rightsquigarrow (L_1, \dots, L_k)$ is a properly embedded Lagrangian $V \subset (\mathbb{C} \times M, \omega_{\mathbb{C}} \oplus \omega)$ such that, for some $R > 0$, $V \cap \big(\{\Re z \le -R\} \times M\big) = \bigsqcup_{i=1}^{k} (-\infty, -R] \times \{i\} \times L_i$, $V \cap \big(\{\Re z \ge R\} \times M\big) = [R, \infty) \times \{1\} \times L'$, and $V \cap \big(\{|\Re z| \le R\} \times M\big)$ is compact. V is *monotone* if it is monotone as a Lagrangian in $\mathbb{C} \times M$ in the sense of Definition 8.4(c).

Arnold introduced Lagrangian cobordism [Arn80a; Arn80b]; without monotonicity it is governed by the h-principle and computable by homotopy theory (Audin [Aud85], Eliashberg). Biran–Cornea showed that the monotone version is rigid and seen by Floer theory, first on complexes [BC13], then categorically [BC14].

Theorem 8.41 (Cobordisms are cone decompositions; Biran–Cornea 2014)**.** *Let M be a closed monotone symplectic manifold, and let $V \colon L' \rightsquigarrow (L_1, \dots, L_k)$ be a connected monotone cobordism with $N_V \ge 2$, all of whose ends are objects of* $\mathrm{Fuk}(M)_w$ *for a common w (Remark 8.19). Work over $\mathbb{Z}/2$, ungraded. Then there are objects $Z_1 = L_1, Z_2, \dots, Z_k \cong L'$ of $D\,\mathrm{Fuk}(M)_w$ and exact triangles $L_i \to Z_{i-1} \to Z_i \to L_i[1]$ for $2 \le i \le k$. Thus $L' \cong \mathrm{Cone}\big(L_k \to \mathrm{Cone}(L_{k-1} \to \cdots \mathrm{Cone}(L_2 \to L_1) \cdots)\big)$, and L' lies in the triangulated subcategory generated by $L_1, \dots, L_k$.*

Hence the monotone cobordism group (relations $[L'] = \sum_i [L_i]$) maps to $K_0(D^\pi \operatorname{Fuk}(M))$; Haug [Hau15] proved this an isomorphism for T^2, using HMS. For K3 surfaces X mirror to algebraic K3s, Sheridan–Smith [SS20a] used Mumford's infinite-dimensionality of zero-cycles to show that classes of Maslov-zero tori in K_0 lie in no countable subgroup, while Lagrangian spheres span finite rank (mirror to Beauville–Voisin).

8.9 What to read

Auroux, A beginner's introduction to Fukaya categories *[Aur14]*
Read this first: HF, A_∞-categories, twisted complexes, the exact triangle and the T^2 mirror, all by example. Skip nothing.

Seidel, Fukaya categories and Picard–Lefschetz theory *[Sei08b]*
The reference for the exact case. Part I (A_∞ algebra, twisted complexes, split-closure) is the cleanest treatment in print; read it all. In Part II read gradings, orientations and the construction, skimming the analysis. Part III (Lefschetz fibrations) is essential for HMS with Fano or Landau–Ginzburg mirrors.

Smith, A symplectic prolegomenon *[Smi15]*
Why categorical methods matter; read it early and again at the end.

Seidel, A biased view of symplectic cohomology *[Sei08a]*
The source of our conventions for SH. Supplement it with Oancea's survey [Oan04] for the history and the action filtration, and Ritter [Rit13] for the TQFT operations and twisted coefficients.

Abouzaid, Symplectic cohomology and Viterbo's theorem *[Abo15]*
A book-length, honest proof of Theorem 8.25, signs included; the model of a complete Floer-theoretic argument.

Ganatra's thesis [Gan12]
Open–closed maps, Hochschild invariants and the Calabi–Yau structure of wrapped categories.

Ganatra–Pardon–Shende [GPS20; GPS24b; GPS24a]
Read all three introductions, then the first paper for sectors and localization; you will use the statements more than the long proofs.

Fukaya–Oh–Ohta–Ono [Fuk+09]
Read Chapter 1 and the introduction, plus the parts on bounding cochains; use the rest as a reference. Fukaya's survey [Fuk13] is the best short account of the ideas. Their toric papers [Fuk+10; Fuk+11] are the most readable entry to the theory in action.

Biran–Cornea [BC09; BC14]
Pearl complexes and the monotone theory, done concretely; and then Lagrangian cobordism.

Kashiwara–Schapira, Sheaves on Manifolds *[KS90]*
Chapters V–VI for microsupport and involutivity; Chapter VIII for constructibility. Then Guillermou's Astérisque volume [Gui23], which is the systematic sheaf-theoretic treatment of cotangent-bundle symplectic topology, and Nadler–Zaslow [NZ09].

Original papers

Floer [Flo88b]; Oh [Oh93]; Seidel's exact triangle [Sei03]; Abouzaid's generation criterion [Abo10]; Abouzaid–Seidel [AS10c]; Auroux's anticanonical-divisor paper [Aur07], which is the best introduction to SYZ with wall-crossing; Sheridan [She15], as the model of a complete HMS proof.

8.10 Frontier and open problems

Problem 8.42 (Nearby Lagrangians)**.** *Prove Conjecture 8.30 for some Q of dimension ≥ 3, for instance $Q = S^3$ or T^3. Status: by Theorem 8.32, $L \to Q$ is a simple homotopy equivalence and the stable Gauss map is trivial on homotopy groups (nullhomotopic for homotopy spheres). Even "L is diffeomorphic to Q" is open in general. Floer theory with coefficients in spectra (Chapter 7) is the current tool.*

Problem 8.43 (HMS for compact Calabi–Yau manifolds)**.** *Prove Conjecture 8.33 for general Batyrev–Borisov mirror pairs (Calabi–Yau complete intersections in toric varieties) and identify the mirror map intrinsically. Status: known for hypersurfaces in projective space and for generalized Greene–Plesser pairs, where the mirror map is now known to be integral (Section 8.6). The SYZ/family Floer approach is complete only away from singular fibres.*

Problem 8.44 (Foundations for closed manifolds)**.** *Give a complete construction of the Fukaya category of an arbitrary closed symplectic manifold, with objects (L, b), including functoriality under Lagrangian correspondences and the open–closed maps. Status: FOOO's framework [Fuk+09; Fuk+20] covers much of this; a complete written account of the full category is missing.*

Problem 8.45 (Sheaf models beyond the exact case)**.** *Is there a sheaf-theoretic model of* $\mathrm{Fuk}(M)$ *for closed M, perhaps over the Novikov field via Tamarkin categories? Status: open. For Weinstein manifolds, Theorem 8.39 answers it; for closed manifolds even the definition of the candidate is unclear.*

Question 8.46 (Detecting flexibility)**.** *Which Floer-theoretic invariants characterize flexible Weinstein manifolds?* $\mathrm{SH} = 0$ *does not: Murphy–Siegel's subflexible domains [MS18] have* $\mathrm{SH} = 0$*, and therefore vanishing wrapped category, but nonzero twisted* SH*. Does vanishing of* SH *with all twisted coefficients (or of the wrapped category under all bulk deformations) imply flexibility? Status: open, as far as I know. Partial vanishing results, where* SH *vanishes after inverting some primes but not others (Abouzaid–Seidel [AS10b], preprint; Lazarev–Sylvan [LS23]), show how fine the question is.*

Problem 8.47 (Lagrangian cobordism groups)**.** *Compute the monotone (or unobstructed) Lagrangian cobordism group of a symplectic manifold of dimension ≥ 4 and decide whether the map to $K_0(D^\pi \mathrm{Fuk})$ is an isomorphism. Status: known for T^2 [Hau15] and some other surfaces; essentially open in higher dimension.*

Problem 8.48 (Classification of exact Lagrangians in Weinstein manifolds)**.** *Beyond cotangent bundles, classify closed exact Lagrangians up to Hamiltonian isotopy in basic Weinstein manifolds such as Milnor fibres of isolated singularities and plumbings. Status: answered only in special four-dimensional cases; generation theorems constrain such Lagrangians categorically, and passing from categorical to geometric information is the open step.*

8.11 Logical dependencies

Theorems 8.6 and 8.8 need Liouville manifolds (Chapter 3), Definitions 8.2 and 8.3, gradings and the boundary Maslov index (Definition 8.4, Chapter 1), and Definition 8.5; they presuppose Fredholm theory for strips, Gromov compactness with boundary and generic transversality (Chapter 5). Theorems 8.12 and 8.13 add Definitions 8.10 and 8.11 and virtual chains. Theorems 8.18, 8.20 and 8.22 need Definitions 8.17 and 8.21; Theorems 8.25 and 8.27 to 8.29 need Definitions 8.23 and 8.26 and Hamiltonian Floer theory (Chapter 7); Theorem 8.32 needs these plus parametrized spectra. Theorems 8.36 to 8.39 need Definition 8.35 and derived categories of sheaves; Theorem 8.41 needs Definition 8.40.

Chapter 9

Hamiltonian and Reeb dynamics

Hamiltonian dynamics is where symplectic topology came from. This chapter asks the classical questions. Does a Hamiltonian system on a compact energy surface have a periodic orbit (Section 9.1)? How many (Sections 9.2 and 9.5)? Are periodic orbits C^∞-generically dense, as Poincaré hoped (Section 9.4)? How short must the shortest one be (Section 9.6)? I also describe the holomorphic curve machinery for contact manifolds built to answer them (Section 9.3). The prototype is the Arnold conjecture (Chapter 7).

There are two things to take away. First, the three-dimensional theory has been transformed since 2007. Taubes's proof of the Weinstein conjecture and Hutchings's embedded contact homology (ECH) turned questions about Reeb flows on 3-manifolds into questions about a topologically invariant Floer theory with an action filtration. The asymptotics of its spectral invariants then gave two-or-infinity theorems and smooth closing lemmas, and broken book decompositions gave Birkhoff sections for generic flows. Second, in higher dimensions the tools are still variational methods, Hamiltonian Floer theory and symplectic cohomology, and the Weinstein conjecture in dimension ≥ 5 is open. The most dramatic recent result there is negative: the Haim-Kislev–Ostrover counterexample (2024) to Viterbo's volume–capacity conjecture.

9.1 Reeb orbits and the Weinstein conjecture

Convention 9.1 (Reeb orbits and their indices)**.** Throughout, Y is a closed connected manifold of dimension $2n-1 \geq 3$ (rather than $2n+1$ as in C6, so that hypersurfaces in $\mathbb{R}^{2n}$ have the expected n) and α is a contact form with $\xi = \ker \alpha$ and Reeb field R_α as in C6.

- A *closed Reeb orbit* is a map $\gamma\colon \mathbb{R}/T\mathbb{Z} \to Y$, $T > 0$, with $\dot\gamma = R_\alpha(\gamma)$, taken modulo translation of the parameter. Its *action* is $\mathcal{A}(\gamma) = \int_\gamma \alpha = T$.
- γ is *simple* if it is injective. Every closed orbit is an iterate γ_0^k of a unique simple one, and "number of closed orbits" always means the number of simple ones.
- γ is *nondegenerate* if $d\phi^T|_{\xi_{\gamma(0)}}$, where ϕ^t is the Reeb flow, does not have 1 as an eigenvalue. The form α is nondegenerate if all closed orbits, iterates included, are nondegenerate.
- For $n = 2$, a nondegenerate orbit is *elliptic* or *hyperbolic* according as the eigenvalues of $d\phi^T|_\xi$ lie on the unit circle or are real.
- Given a homotopy class τ of symplectic trivializations of $\gamma^*\xi$, the path $d\phi^t|_\xi$ in $\mathrm{Sp}(2n-2)$ has a Conley–Zehnder index $\mu^\tau_{\mathrm{CZ}}(\gamma)$ (Chapter 1). It is normalized so that $t \mapsto e^{2\pi i\theta t}$,

$t \in [0,1]$, has index $2\lfloor\theta\rfloor + 1$ for $\theta \notin \mathbb{Z}$. This agrees with C8 (the minimum of a C^2-small Morse Hamiltonian has $\mu_{CZ} = n$). With it, the short orbit on $\partial E(a,b) \subset \mathbb{C}^2$, $a < b$, $a/b \notin \mathbb{Q}$, has $\mu_{CZ} = 3$ in the trivialization extending over a spanning disk.

The gradings of contact homology and ECH in Section 9.3 are homological, as in their sources.

Conjecture 9.2 (Weinstein conjecture; Weinstein 1979)**.** *Let Y be a closed manifold of dimension $2n-1 \geq 3$ and α a contact form on Y. Then R_α has at least one closed orbit. Equivalently, every closed hypersurface Σ of contact type in a symplectic manifold carries a closed characteristic, i.e. a closed leaf of the characteristic line field* $\ker(\omega|_{T\Sigma})$.

The forms are equivalent: if a Liouville field Z is transverse to Σ, the Reeb field of $\iota_Z\omega|_\Sigma$ spans $\ker(\omega|_{T\Sigma})$, and $\{0\} \times Y$ is of contact type in the symplectization. On a regular level $\Sigma = H^{-1}(c)$, closed characteristics are the periodic orbits of X_H for *any* such H.

Rabinowitz [Rab78] and Weinstein [Wei78] found closed characteristics on star-shaped and convex hypersurfaces in $\mathbb{R}^{2n}$ by variational methods in 1978. Weinstein [Wei79] saw that the proofs use only contact type, a symplectically invariant property. His formulation also assumed $H^1(\Sigma;\mathbb{R}) = 0$, which none of the proofs below needs. Contact type cannot be dropped: some proper Hamiltonians on $\mathbb{R}^{2n}$ have a regular compact level with no periodic orbit. Such Hamiltonians exist in class C^∞ for $2n \geq 6$ [Gin95; Gin97; Her99] and in class C^2 for $2n = 4$ [GG03].

Theorem 9.3 (Viterbo 1987)**.** *Every closed hypersurface of contact type in $(\mathbb{R}^{2n}, \omega_0)$ carries a closed characteristic [Vit87a]. In particular, so does every closed hypersurface that is transverse to the radial vector field [Rab78].*

Theorem 9.4 (Almost existence; Hofer–Zehnder 1987, Struwe 1990)**.** *Let $H\colon \mathbb{R}^{2n} \to \mathbb{R}$ be smooth and proper. For Lebesgue-almost every $c \in \mathbb{R}$ with $H^{-1}(c) \neq \emptyset$, the level set $H^{-1}(c)$ contains a nonconstant periodic orbit of X_H [HZ87; Str90].*

Proof idea. The Hofer–Zehnder capacity (Chapter 6) of $\{H < c\}$ is finite and nondecreasing in c, so it is differentiable at almost every c. At such c, nearly optimal admissible Hamiltonians produce fast periodic orbits trapped near $H^{-1}(c)$ with bounded period, and Arzelà–Ascoli gives an orbit on the level. For contact-type Σ, the Liouville flow gives nearby hypersurfaces with conjugate characteristic foliations, which recovers Theorem 9.3. Learn the proof from [HZ94, Ch. 4].

The argument works wherever a neighbourhood of Σ has finite capacity. By Usher's sharp energy–capacity inequality [Ush10], every displaceable contact-type hypersurface in a closed symplectic manifold therefore carries a contractible closed characteristic. For closed M itself, $c_{HZ}(M)$ can be infinite: Zehnder's irrational 4-torus has a family of hypersurfaces without closed characteristics. It is finite if some Gromov–Witten invariant of M with two point constraints is nonzero [Lu06], e.g. for $\mathbb{CP}^n$ [HV92].

Theorem 9.5 (Hofer 1993)**.** *Let (Y,ξ) be a closed contact 3-manifold and α a contact form for ξ. If Y is diffeomorphic to S^3, or ξ is overtwisted, or $\pi_2(Y) \neq 0$, then R_α has a contractible closed orbit [Hof93b].*

Proof idea. Take an α-adapted J on $\mathbb{R} \times Y$ (Definition 9.16). In the overtwisted case, a Bishop family of holomorphic disks with boundary on $\{0\} \times D$ emanates from the elliptic point of an overtwisted disk D. The family cannot reach ∂D and has energy at most $\int_D d\alpha$, so gradients blow up, and rescaling gives a nonconstant finite energy plane. Hofer's key lemma: such a plane

is asymptotic, along a sequence of loops, to a closed Reeb orbit, which is contractible because it bounds the plane. This proof created the framework of Section 9.3.

Theorem 9.6 (Taubes 2007)**.** *Let Y be a closed oriented 3-manifold. Every contact form on Y has a closed Reeb orbit [Tau07].*

Proof idea. For a spin^c structure with torsion c_1, $\widehat{\mathrm{HM}}^*(Y)$ is nonzero in infinitely many degrees [KM07]. Taubes perturbs the Seiberg–Witten equations by r times a term built from α. A spectral flow estimate bounds the energy of solutions representing these classes. As $r \to \infty$, their curvature then concentrates along a nonempty finite union of Reeb orbits. ECH makes this "Seiberg–Witten solutions become orbit sets" philosophy precise (Theorem 9.19).

Planar contact structures had been treated by Abbas–Cieliebak–Hofer [ACH05]. Hutchings–Taubes [HT09b] extended Taubes's theorem to stable Hamiltonian structures (on closed 3-manifolds other than T^2-bundles over S^1), showed that nondegenerate contact forms have at least two closed orbits, and at least three unless Y is a lens space, and later proved the three-dimensional Arnold chord conjecture [HT11; HT13].

Remark 9.7 (Higher dimensions). In dimension ≥ 5, Conjecture 9.2 is open. It is known in these cases:

- contact-type hypersurfaces in $\mathbb{R}^{2n}$, and displaceable ones in closed manifolds;
- contact-type hypersurfaces in $\mathbb{CP}^n$, and more generally hypersurfaces that separate the constraints of a nonzero Gromov–Witten count, e.g. in uniruled manifolds [HV92; LT00; Lu06];
- every contact form on ∂W, for W a Liouville domain such that the canonical map $H^*(W) \to \mathrm{SH}^*(W)$ (Chapter 8) is not an isomorphism, e.g. $\mathrm{SH}^*(W) = 0$ or $W = D^*Q$ [Vit99]. Without Reeb orbits, the action filtration would force that map to be an isomorphism.
- iterated planar contact manifolds [Acu17; AM22];
- contact manifolds containing a Plastikstufe [Nie06; AH09]. As Borman–Eliashberg–Murphy observe [BEM15], this includes all overtwisted ones; see also [CMP19].

9.2 Multiplicity, surfaces of section, and two-or-infinity

The model is the boundary of an irrational ellipsoid $E(a_1, \dots, a_n)$, with $a_i/a_j \notin \mathbb{Q}$ for $i \neq j$. It has exactly n closed orbits, the coordinate circles.

Theorem 9.8 (Closed characteristics on convex hypersurfaces)**.** *Let $\Sigma \subset \mathbb{R}^{2n}$, $n \geq 2$, be a compact C^2 hypersurface bounding a strictly convex domain. Then Σ carries:*

(a) at least 2 closed characteristics [EH87] (also, independently, Ekeland–Lassoued and Szulkin);

(b) at least $\lfloor n/2 \rfloor + 1$, and at least n if all are nondegenerate [LZ02];

(c) at least n if $n \in \{3, 4\}$ [WHL07; Wan16a].

Ekeland–Lasry [EL80] had obtained n for Σ pinched between concentric balls of radii r and $r\sqrt{2}$. W. Wang [Wan16b] proved $\lfloor (n+1)/2 \rfloor + 1$, and the bound n is now claimed in general (Remark 9.15). The tools are Clarke duality, Ekeland's index theory, Long's iteration theory for Maslov indices [Lon02], and resonance identities.

Definition 9.9 (Dynamical convexity; Hofer–Wysocki–Zehnder 1998)**.** A contact form α on S^{2n-1} with $\ker\alpha = \xi_{\mathrm{st}}$ is *dynamically convex* if every closed Reeb orbit γ (iterates included) has $\mu_{\mathrm{CZ}}(\gamma) \geq n+1$. Here μ_{CZ} is computed in the global trivialization of ξ_{st}, which is unique up to homotopy. For degenerate γ, μ_{CZ} means its lower semicontinuous extension, the minimum of μ_{CZ} over nondegenerate paths C^0-close to the linearized flow. For $n = 2$ the condition is $\mu_{\mathrm{CZ}} \geq 3$.

The boundary of a strictly convex domain $K \ni 0$, with λ_0, is dynamically convex [HWZ98]. Unlike convexity, dynamical convexity is invariant under strict contactomorphisms. Hofer–Wysocki–Zehnder asked whether every dynamically convex form comes from a convex domain; Chaidez–Edtmair [CE22] showed it does not. For convex $X \subset \mathbb{R}^4$ with boundary Y, they proved $c < \mathrm{Ru}(Y)^2\rho(Y)/\mathrm{vol}(X) < C$ for universal constants $C > c > 0$, where Ru is the Ruelle invariant (mean rotation of the linearized flow) and ρ the systolic ratio (Definition 9.31). Then they built dynamically convex forms violating these bounds. Dynamical convexity captures what holomorphic curves see, not convex geometry.

Definition 9.10 (Birkhoff section)**.** Let X be a nowhere vanishing vector field on a closed 3-manifold Y. A *Birkhoff section* (global surface of section) for X is an immersed compact surface $S \to Y$ such that:

- its interior is embedded and transverse to X;
- its boundary is a finite union of covers of closed orbits of X (the *binding*);
- for some $T > 0$, every orbit segment of X of time-length T meets S.

It is *disk-like* if S is an embedded disk.

Periodic points of the first-return map ψ of $\operatorname{int} S$ are the closed orbits off the binding. For a Reeb field, ψ preserves $d\alpha|_{\operatorname{int} S}$, of total area the action of the binding: Poincaré's route from flows to area-preserving maps.

Theorem 9.11 (Hofer–Wysocki–Zehnder 1998)**.** *Let α be a dynamically convex contact form on (S^3, ξ_{st}). Then there is a closed Reeb orbit P, unknotted, with self-linking number -1 and $\mu_{\mathrm{CZ}}(P) = 3$, that binds an open book decomposition of S^3 whose pages are all disk-like Birkhoff sections. Consequently, R_α has exactly two or infinitely many closed orbits. This applies in particular to every strictly convex energy surface in $\mathbb{R}^4$ [HWZ98].*

Proof idea. The pages are projections of an $\mathbb{R}$-family of embedded index-2 finite energy planes asymptotic to P. Dynamical convexity rules out breaking of this family, and intersection theory makes the projections disjoint. The return map on a page has a fixed point by Brouwer, and Theorem 9.12 on the complementary open annulus then gives the dichotomy. The argument is done first for nondegenerate α and extended by approximation.

Theorem 9.12 (Franks 1992, 1996)**.** *(a) Let f be either an area-preserving homeomorphism of the closed annulus, or a homeomorphism of the open annulus preserving a finite measure homeomorphic to Lebesgue measure; in both cases assume f isotopic to the identity. If f has a periodic point, then it has infinitely many interior periodic points [Fra92; Fra96]. (b) An area-preserving homeomorphism of S^2 with at least three fixed points has infinitely many periodic points [Fra92; Fra96].*

The 1996 paper repairs a gap for the open annulus. With Bangert's variational argument for the remaining case [Ban93], Theorem 9.12 proves that *every Riemannian metric on S^2 has infinitely many closed geodesics*. Franks–Handel proved that a nontrivial Hamiltonian diffeomorphism of a

closed surface has periodic points of arbitrarily high period, provided it has at least three fixed points if the surface is S^2 [FH03]. Le Calvez extended this to homeomorphisms [Le 06]. These are the two-dimensional cases of the conjectures of Section 9.5.

Theorem 9.13 (Two or infinitely many)**.** *Let λ be a contact form on a closed connected 3-manifold Y, and $\xi = \ker \lambda$.*

(a) R_λ has at least two closed orbits [CH16].

(b) If $c_1(\xi)$ is torsion, R_λ has exactly two or infinitely many closed orbits [Cri+26]. The nondegenerate case is [CHP19].

(c) If λ is nondegenerate, R_λ has exactly two or infinitely many closed orbits, and infinitely many unless Y is S^3 or a lens space [CDR23].

(d) If R_λ has exactly two closed orbits, then λ is nondegenerate, Y is S^3 or a lens space, and the orbits are irrationally elliptic cores of a genus-one Heegaard splitting [Cri+23].

By (b), every compact regular energy level in $\mathbb{R}^4$ transverse to the radial field has two or infinitely many closed orbits. Hofer–Wysocki–Zehnder conjectured this in 2003 and proved it for generic such levels [HWZ03]. Likewise, every Finsler metric on S^2 has two or infinitely many prime closed geodesics.

Proof idea. For (a): if γ were the only simple orbit, all orbit sets would have action in $\mathcal{A}(\gamma)\mathbb{Z}$. Spectral invariants of classes with $U\sigma_{k+1} = \sigma_k$ increase strictly (an extra argument if λ is degenerate), so $c_{\sigma_k} \geq k\mathcal{A}(\gamma)$, contradicting $c_{\sigma_k} \sim \sqrt{2k \operatorname{vol}(Y, \lambda)}$ (Theorem 9.21). Part (b) refines this counting. For (c), Colin–Dehornoy–Rechtman build from ECH curves a *broken book decomposition*, an open book whose pages are transverse to the flow away from finitely many "broken" binding orbits, and argue on the pages as Franks does.

Theorem 9.14 (Generic Reeb flows in dimension three)**.** *Let (Y, ξ) be a closed cooriented contact 3-manifold.*

(a) Every Kupka–Smale Reeb flow of a contact form for ξ admits a Birkhoff section [CM22]. Kupka–Smale means that all closed orbits are nondegenerate and that the stable and unstable manifolds of hyperbolic orbits are transverse.

(b) Among contact forms for ξ, two properties each hold on a C^∞-open dense set: the flow has a Birkhoff section with nondegenerate binding; the flow has positive topological entropy [Col+24].

Remark 9.15 (Higher-dimensional spheres). On $(S^{2n-1}, \xi_{\text{st}})$ the main results are these.

- Dynamically convex flows on S^{2n-1} have at least $\lceil n/2 \rceil + 1$ closed orbits, and at least n if nondegenerate [GG20].
- A dynamically convex flow with a hyperbolic closed orbit has infinitely many [Çin+26].
- A 2024 preprint of Çineli–Ginzburg–Gürel, announced as accepted in the *Annals* [ÇGG24a], proves at least n closed orbits for every dynamically convex flow on the boundary of a star-shaped domain in $\mathbb{R}^{2n}$. Since strictly convex implies dynamically convex, this would settle the Ekeland–Hofer question behind Theorem 9.8. The same preprint proves "exactly n or infinitely many" for centrally symmetric domains with nondegenerate flow.
- For elliptic orbits and prequantization bundles see [AM17; GGM18].

9.3 Holomorphic curves in symplectizations: SFT and ECH

Definition 9.16 (Adapted almost complex structures; finite energy curves)**.** Let (Y, λ) be a closed contact manifold of dimension $2n-1$. An almost complex structure J on $\mathbb{R} \times Y$ is *λ-adapted* if it is invariant under translation in r, $J\partial_r = R_\lambda$, $J\xi = \xi$, and $d\lambda(\cdot, J\cdot)|_\xi$ is a metric on ξ. Such J is compatible with $d(e^r\lambda)$ (C2, C6). Let (Σ, j) be a closed Riemann surface and $\Gamma \subset \Sigma$ finite. A *finite energy curve* is a map $u\colon \Sigma \setminus \Gamma \to \mathbb{R} \times Y$ with $du \circ j = J \circ du$ and Hofer energy

$$E(u) = \sup_{\varphi} \int_{\Sigma\setminus\Gamma} u^* d\big(\varphi(r)\lambda\big) < \infty,$$

where the supremum is over smooth nondecreasing $\varphi\colon \mathbb{R} \to [0,1]$. Suppose λ is nondegenerate, and let $z \in \Gamma$ be a non-removable puncture. Then $r \circ u \to \pm\infty$ near z, and u is exponentially asymptotic to the *trivial cylinder* $\mathbb{R}\times\gamma_z$ over a closed Reeb orbit γ_z [Hof93b; HWZ96]. Accordingly z is a positive or negative puncture. Curves in completed cobordisms with cylindrical ends are defined in the same way.

The Fredholm index of such a curve, i.e. the expected dimension of the moduli space of unparametrized curves near u with j varying, is

$$\operatorname{ind}(u) = (n-3)\chi(\Sigma\setminus\Gamma) + 2c_1^\tau(u^*\xi) + \sum_{z\in\Gamma^+} \mu^\tau_{\mathrm{CZ}}(\gamma_z) - \sum_{z\in\Gamma^-} \mu^\tau_{\mathrm{CZ}}(\gamma_z).$$

Here $\Gamma^\pm$ are the positive and negative punctures, τ trivializes ξ along the asymptotic orbits, and c_1^τ is the relative first Chern number. By SFT compactness [Bou+03], sequences of bounded energy converge to *holomorphic buildings*: stacks of curves matched along Reeb orbits.

Symplectic field theory [EGH00] packages counts of such curves into algebra. The full theory counts all genera and puncture data; rational SFT counts genus 0; *contact homology* counts genus 0 with one positive puncture. Informally, the contact homology algebra of nondegenerate λ is free graded-commutative on the "good" closed orbits, with $|\gamma| = \mu^\tau_{\mathrm{CZ}}(\gamma) + n - 3$, and its differential counts rigid curves with one positive puncture modulo $\mathbb{R}$. Legendrian contact homology is the relative version: Chekanov's DGA in $\mathbb{R}^3$ [Che02], and [EES05b; EES07; EHK16]; see Chapters 4 and 8.

Warning 9.17 (Foundations). Transversality in SFT is harder than in Gromov–Witten theory. Branched covers of trivial cylinders have negative index, yet they appear in boundary strata, and no choice of J makes them transverse. As I read the status in 2026:

- Cylindrical contact homology is defined without virtual techniques for dynamically convex forms in dimension 3 [HN16], and for hypertight ones (no contractible orbits) [BH18; HN22].
- Contact homology is constructed with virtual techniques by Pardon [Par19] and by Bao–Honda [BH23].
- For the full SFT, polyfold theory supplies the abstract framework [HWZ21], with an outline of its application [FH18], and Ishikawa's preprint [Ish18] constructs Kuranishi structures in all genera. There is still no complete, refereed construction of the full theory with its algebra; see the overview [HS24a].

Many arguments that "use SFT" need only somewhere injective curves, where classical transversality works; check which kind you are reading.

ECH avoids all of this in dimension three by exploiting positivity of intersections and an index inequality that forces the relevant curves to be embedded.

Definition 9.18 (Embedded contact homology; Hutchings)**.** Let Y be a closed oriented 3-manifold and λ a nondegenerate contact form.

- An *orbit set* is a finite set $\alpha = \{(\alpha_i, m_i)\}$ of distinct simple orbits α_i with multiplicities $m_i \in \mathbb{Z}_{>0}$. Its class is $[\alpha] = \sum_i m_i[\alpha_i] \in H_1(Y)$ and its action is $\mathcal{A}(\alpha) = \sum_i m_i \mathcal{A}(\alpha_i)$. It is an *ECH generator* if $m_i = 1$ whenever α_i is hyperbolic.
- Let $\alpha, \beta = \{(\beta_j, n_j)\}$ be orbit sets with $[\alpha] = [\beta]$, and $Z \in H_2(Y, \alpha, \beta)$. The *ECH index* is

$$I(\alpha, \beta, Z) = c_\tau(Z) + Q_\tau(Z) + \sum_i \sum_{k=1}^{m_i} \mu^\tau_{\mathrm{CZ}}(\alpha_i^k) - \sum_j \sum_{k=1}^{n_j} \mu^\tau_{\mathrm{CZ}}(\beta_j^k).$$

  Here τ trivializes ξ over the orbits, c_τ is the relative first Chern class of ξ over Z, and Q_τ the relative self-intersection number. I does not depend on τ.
- For generic λ-adapted J, the differential on the $\mathbb{Z}/2$-vector space spanned by the ECH generators in class $\Gamma \in H_1(Y)$ is $\partial\alpha = \sum_\beta \#_2\big(\mathcal{M}_1(\alpha, \beta)/\mathbb{R}\big)\,\beta$. Here $\mathcal{M}_1(\alpha, \beta)$ is the set of J-holomorphic currents in $\mathbb{R} \times Y$ from α to β of ECH index 1. A current is a finite set $\{(C_k, d_k)\}$ of somewhere injective finite energy curves with positive multiplicities, asymptotic to α at $+\infty$ and to β at $-\infty$.

The homology is $\mathrm{ECH}_*(Y, \lambda, \Gamma)$ [Hut02; Hut14]. Coefficients in $\mathbb{Z}$ need coherent orientations.

The key is the *index inequality* $\mathrm{ind}(C) \le I(C) - 2\delta(C)$ for somewhere injective C, where $\delta(C) \ge 0$ counts singularities [Hut02]. It forces a generic ECH index 1 current to be one embedded index-1 curve plus trivial cylinders. The proof that $\partial^2 = 0$ is a hard gluing theorem [HT07; HT09a].

Theorem 9.19 (ECH = monopole Floer cohomology; Taubes 2010)**.** *Let Y be a closed connected oriented 3-manifold, λ a nondegenerate contact form with $\xi = \ker\lambda$, and $\Gamma \in H_1(Y)$. There is a canonical isomorphism of relatively graded modules, with $\mathbb{Z}/2$ or $\mathbb{Z}$ coefficients,*

$$\mathrm{ECH}_*(Y, \lambda, \Gamma) \;\cong\; \widehat{\mathrm{HM}}^{-*}\big(Y, \mathfrak{s}_\xi + \mathrm{PD}(\Gamma)\big)$$

[Tau10]. Here $\mathfrak{s}_\xi$ is the spin^c structure of ξ and $\widehat{\mathrm{HM}}$ is the "from" version of monopole Floer cohomology. The isomorphism intertwines the U-maps and sends the class of the empty orbit set (the ECH contact invariant*) to the Kronheimer–Mrowka contact invariant. In particular $\mathrm{ECH}_*(Y, \lambda, \Gamma)$ depends only on Y, Γ and $\mathfrak{s}_\xi$.*

With Kutluhan–Lee–Taubes [KLT20], or directly by Colin–Ghiggini–Honda [CGH24], one gets, over $\mathbb{Z}/2$,

$$\bigoplus\nolimits_\Gamma \mathrm{ECH}_*(Y, \lambda, \Gamma) \cong HF^+(-Y)\,:$$

three Floer theories agree, one of them built from Reeb dynamics.

Definition 9.20 (ECH spectral invariants)**.** For nondegenerate λ and $L > 0$, generators of action $< L$ span a subcomplex, because ∂ decreases action. Let $\mathrm{ECH}^L_*(Y, \lambda, \Gamma)$ be its homology. For $0 \ne \sigma \in \mathrm{ECH}_*(Y, \xi, \Gamma)$, the common homology of Theorem 9.19, set $c_\sigma(Y, \lambda) = \inf\{L : \sigma \in \mathrm{im}(\mathrm{ECH}^L_* \to \mathrm{ECH}_*)\}$. This extends by continuity to degenerate λ, and is always the action of some orbit set.

The ECH capacities of a 4-dimensional Liouville domain (Chapter 6) are the spectral invariants of its boundary for the classes σ_k singled out by the U-map.

Theorem 9.21 (Volume property; Cristofaro-Gardiner–Hutchings–Ramos 2015)**.** *Let (Y, λ) be a closed connected contact 3-manifold, where λ may be degenerate. Let $\Gamma \in H_1(Y)$ with $c_1(\xi) + 2\,\mathrm{PD}(\Gamma)$ torsion, so that $\mathrm{ECH}_*(Y, \xi, \Gamma)$ has a relative $\mathbb{Z}$-grading, and fix an absolute grading I. If (σ_k) is a sequence of nonzero homogeneous classes with $I(\sigma_k) \to \infty$, then*

$$\lim_{k\to\infty} \frac{c_{\sigma_k}(Y,\lambda)^2}{I(\sigma_k)} = \mathrm{vol}(Y,\lambda) := \int_Y \lambda \wedge d\lambda.$$

For a 4-dimensional Liouville domain (X, ω) with all $c_k(X) < \infty$ this gives $\lim_{k\to\infty} c_k(X)^2/k = 4\,\mathrm{vol}(X)$, where $\mathrm{vol}(X) = \int_X \omega^2/2$ [CHR15].

Check on the ball: $c_k(B^4(a)) = da$ with $d^2 + d \le 2k \le d^2 + 3d$, so $c_k^2/k \to 2a^2 = 4\,\mathrm{vol}(B^4(a))$.

Remark 9.22 (Periodic Floer homology). ECH of the mapping torus of an area-preserving surface diffeomorphism ϕ, whose closed orbits are the periodic orbits of ϕ, is *periodic Floer homology* $\mathrm{PFH}_*(\phi)$ [Hut02]. Lee–Taubes [LT12] identified it, for monotone ϕ, with Seiberg–Witten Floer cohomology of the mapping torus. PFH spectral invariants drive the proof of the simplicity conjecture (Chapter 7) and the surface closing lemmas of Section 9.4.

9.4 Closing lemmas

Are periodic orbits of a generic Hamiltonian system dense? In the C^1 topology, yes: this is Pugh's closing lemma [Pug67], extended to Hamiltonian systems by Pugh–Robinson [PR83]. For C^r with $r \ge 2$ the closing lemma is open in general; it is Problem 10 on Smale's list [Sma98]. All the C^∞ results come from symplectic topology, in low dimensions.

Theorem 9.23 (C^∞ closing lemma for Reeb flows; Irie 2015)**.** *Let (Y, λ) be a closed contact 3-manifold. For a C^∞-residual set of $f \in C^\infty(Y, \mathbb{R}_{>0})$, the union of the closed Reeb orbits of $f\lambda$ is dense in Y [Iri15].*

Proof idea. Fix a nonempty open U and $h \ge 0$ supported in U with $h \not\equiv 0$. Suppose no $(1 + th)\lambda$, $t \in [0, 1]$, has a closed orbit meeting U. Then the orbit sets and their actions do not depend on t. Each $c_\sigma((1 + th)\lambda)$ is continuous in t with values in a fixed null set, so it is constant. But the volume increases with t, which contradicts Theorem 9.21. A Baire argument finishes the proof.

Theorem 9.24 (C^∞ closing lemma for Hamiltonian surface maps; Asaoka–Irie 2016)**.** *Let (Σ, ω) be a closed oriented surface with an area form, $\phi \in \mathrm{Ham}(\Sigma, \omega)$, and $U \subset \Sigma$ nonempty and open. Then there are $\phi_j \in \mathrm{Ham}(\Sigma, \omega)$, each with a periodic point in U, such that $\phi_j \to \phi$ in C^∞. Consequently, a C^∞-generic element of $\mathrm{Ham}(\Sigma, \omega)$ has dense periodic points [AI16].*

For all area-preserving diffeomorphisms of closed surfaces, the C^∞ closing lemma is proved in a preprint of Cristofaro-Gardiner–Prasad–Zhang [CPZ21]. A quantitative version, for rational Hamiltonian isotopy classes with the "U-cycle property", is proved in a preprint of Edtmair–Hutchings [EH21]. That property is now known in all genera [Cri+25b]. Both preprints use PFH spectral invariants and a Weyl law for them. Moreover, C^∞-generic 3-dimensional Reeb flows and area-preserving surface maps have *equidistributed* periodic orbits [Iri21; Pra24].

9.5 Hamiltonian diffeomorphisms: Conley, Hofer–Zehnder, pseudo-rotations

Let (M,ω) be closed and $\phi = \phi_H^1 \in \mathrm{Ham}(M,\omega)$. A point x is *k-periodic* if $\phi^k(x) = x$. It is *contractible* if the loop $\phi_H^t(x)$, $t \in [0,k]$, is contractible; by Floer theory this does not depend on H. A *simple* periodic orbit is the orbit of a k-periodic point of minimal period k. The Arnold conjecture (Chapter 7) counts fixed points. The question here is when there must be infinitely many periodic points.

Theorem 9.25 (Conley conjecture)**.** *Let (M,ω) be a closed symplectic manifold.*

(a) (Hingston 2009 for tori; Ginzburg 2010.) Suppose $\omega|_{\pi_2(M)} = 0$ and $c_1(TM)|_{\pi_2(M)} = 0$. If the fixed points of $\phi \in \mathrm{Ham}(M,\omega)$ are isolated, then ϕ has simple contractible periodic orbits of arbitrarily large period. In particular, every $\phi \in \mathrm{Ham}(M,\omega)$ has infinitely many simple contractible periodic orbits [Hin09; Gin10].

(b) Suppose there is no $A \in \pi_2(M)$ with $\omega(A) > 0$ and $\langle c_1(TM), A\rangle > 0$. This holds for instance if $\omega|_{\pi_2(M)} = 0$, or $c_1(TM)|_{\pi_2(M)} = 0$, or M is negative monotone. Then every $\phi \in \mathrm{Ham}(M,\omega)$ has infinitely many periodic points [GG19; Çin18].

Part (b) caps the cases $c_1|_{\pi_2} = 0$ [GG09; Hei12], negative monotone [GG12] and $\omega|_{\pi_2} = 0$ [Çin18]; in the first two one again gets arbitrarily large periods [GG15].

Proof idea. Assume the fixed points are isolated, so that each has local Floer homology. Suppose none is a *symplectically degenerate maximum*, i.e. a fixed point of mean index 0 with local Floer homology in the extremal degree. Then index bookkeeping in the Floer homology of the iterates produces new orbits, as in Salamon–Zehnder's nondegenerate case [SZ92]. Hingston and Ginzburg showed that a symplectically degenerate maximum forces simple periodic orbits of all large prime periods. Their proof controls the filtered Floer homology of carefully built Hamiltonians.

Example 9.26 (Irrational rotations)**.** The map $\phi[z_0 : \cdots : z_n] = [e^{2\pi i\theta_0} z_0 : \cdots : e^{2\pi i\theta_n} z_n]$ is a Hamiltonian diffeomorphism of $(\mathbb{CP}^n, \omega_{FS})$. If $1, \theta_1 - \theta_0, \dots, \theta_n - \theta_0$ are linearly independent over $\mathbb{Q}$, its only periodic points are the $n+1$ coordinate points. So the Conley conjecture fails on $\mathbb{CP}^n$; for $n = 1$, ϕ is an irrational rotation of S^2.

Conjecture 9.27 (Hofer–Zehnder conjecture, homological form)**.** *Let (M,ω) be a closed symplectic manifold and $\mathbb{F}$ a field. Let $\phi \in \mathrm{Ham}(M,\omega)$ have finitely many contractible fixed points x, with $\sum_x \dim_{\mathbb{F}} \mathrm{HF}^{\mathrm{loc}}(\phi, x; \mathbb{F}) > \dim_{\mathbb{F}} H_*(M;\mathbb{F})$. Then ϕ has infinitely many contractible periodic points.*

Hofer–Zehnder [HZ94, p. 263] conjectured that a Hamiltonian diffeomorphism with more fixed points than the Arnold conjecture requires has infinitely many periodic points. The homological count is Shelukhin's sharpening. On S^2 the geometric version is Theorem 9.12(b).

Theorem 9.28 (Shelukhin 2022)**.** *Let (M,ω) be a closed monotone symplectic manifold and $\mathbb{F}$ a field such that the even quantum homology of M over $\mathbb{F}$ (with Novikov coefficients as in [She22a]) is semisimple. Then Conjecture 9.27 holds for (M,ω) and $\mathbb{F}$. If $\operatorname{char}\mathbb{F} = 0$, such a ϕ moreover has a simple contractible p-periodic point for every sufficiently large prime p [She22a].*

Proof idea. Semisimplicity gives a uniform bound $\beta \le 8n$ on the boundary depth of the Floer barcode of every Hamiltonian diffeomorphism. A Smith-type inequality from the $\mathbb{Z}/p$-equivariant pair-of-pants product gives $p\,\beta_{\mathrm{tot}}(\phi) \le \beta_{\mathrm{tot}}(\phi^p)$ for large primes p, where β_{tot} is the

total length of the finite bars, and the fixed point hypothesis makes $\beta_{\mathrm{tot}}(\phi) > 0$. So ϕ^p has at least linearly many finite bars, which is impossible if ϕ has finitely many periodic points.

Allais reproved the case of $\mathbb{CP}^n$ with generating functions [All22], and Atallah–Lou extended the theorem to semipositive manifolds [AL25].

Definition 9.29 (Pseudo-rotation)**.** A *Hamiltonian pseudo-rotation* of $\mathbb{CP}^n$ is a $\phi \in \mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$ with exactly $n+1$ periodic points, the minimum allowed by Fortune's theorem [For85; GG18]. More generally, for a field $\mathbb{F}$, an $\mathbb{F}$*-pseudo-rotation* of (M, ω) is a $\phi \in \mathrm{Ham}(M, \omega)$ such that every iterate ϕ^k, $k \geq 1$, is nondegenerate with vanishing Floer differential over $\mathbb{F}$ [ÇGG22].

The rotations of Example 9.26 are pseudo-rotations. The Anosov–Katok method [AK70] gives wilder ones. For example, a closed toric manifold whose torus action has ℓ fixed points carries pseudo-rotations with exactly $\ell + 1$ ergodic measures [LS22]. Pseudo-rotations are the extremal case of the Hofer–Zehnder conjecture, and they turn out to be rigid:

- an irrational pseudo-rotation of the disk is a C^0-limit of periodic maps, and is C^0-rigid if its rotation number is very Liouvillean [Bra15a; Bra15b];
- pseudo-rotations of $\mathbb{CP}^n$ are C^0-rigid under a Liouville condition on the mean indices [GG18];
- a closed monotone manifold with a $\mathbb{Z}/2$-pseudo-rotation has a nonconstant holomorphic sphere through every point (proved with quantum Steenrod squares) [She20; She21a; ÇGG22].

Remark 9.30 (Barcode entropy). The *barcode entropy* of ϕ [ÇGG24b] is the limit as $\varepsilon \to 0$ of the exponential growth rate in k of the number of bars of length $\geq \varepsilon$ in the barcode of $\mathrm{HF}(\phi^k)$. On closed monotone manifolds it is at most the topological entropy, and at least the entropy on any compact hyperbolic invariant set; with Katok's horseshoes, the two coincide for surfaces. In higher dimensions they can differ [Çin25]. There are analogues for Reeb and geodesic flows [Çin+25; GGM24].

9.6 Systolic inequalities and Viterbo's conjecture

Definition 9.31 (Systolic ratio; Zoll)**.** Let α be a contact form on a closed $(2n-1)$-manifold Y, and $T_{\min}(\alpha)$ the minimal action of a closed Reeb orbit. The *systolic ratio* is $\rho(\alpha) = T_{\min}(\alpha)^n / \mathrm{vol}(Y, \alpha)$, where $\mathrm{vol}(Y, \alpha) = \int_Y \alpha \wedge (d\alpha)^{n-1}$. The form α is *Zoll* if every Reeb orbit is closed, with a common minimal period.

For $\partial B^{2n}(a)$ with λ_0, $T_{\min} = a$ and $\mathrm{vol} = \int_{B^{2n}(a)} \omega_0^n = a^n$. So the round sphere, which is Zoll, has $\rho = 1$.

Proposition 9.32 (The EHZ capacity of convex domains)**.** *Let $K \subset \mathbb{R}^{2n}$ be a compact convex domain with smooth boundary and $0 \in \mathrm{int}\, K$. Then the first Ekeland–Hofer capacity, the Hofer–Zehnder capacity and the symplectic homology capacity of K coincide. They equal $T_{\min}(\lambda_0|_{\partial K})$, the minimal action of a closed characteristic on ∂K [HZ94; AK22; Iri22; GHR22]. The common value $c_{EHZ}(K)$ extends continuously, in the Hausdorff topology, to all convex bodies.*

Since $\mathrm{vol}(\partial K, \lambda_0) = \int_K \omega_0^n = n!\,\mathrm{vol}(K)$, we get $\rho(\lambda_0|_{\partial K}) = c_{EHZ}(K)^n / (n!\,\mathrm{vol}(K))$. So the following is a systolic inequality.

Conjecture 9.33 (Viterbo 2000; false)**.** *For every convex body $K \subset \mathbb{R}^{2n}$ and every normalized symplectic capacity c, $c(K)^n \le n!\,\mathrm{vol}(K)$, i.e.* $\mathrm{vol}(K) \ge \mathrm{vol}(B^{2n}(c(K)))$*. Equality holds only if* $\operatorname{int} K$ *is symplectomorphic to a ball [Vit00].*

For the Gromov width this is the volume bound. Artstein-Avidan–Milman–Ostrover [AMO08] bound the cylindrical capacity, which dominates every normalized capacity, of a convex body K by A_0 times the capacity of the ball of the same volume, with A_0 universal. So $\sup_K \rho \le A_0^n$ over convex $K \subset \mathbb{R}^{2n}$, and the supremum is attained [HO26]. The *strong Viterbo conjecture* says that all normalized capacities agree on convex domains. It holds for monotone toric domains in $\mathbb{R}^4$ [GHR22], for convex domains in $\mathbb{R}^4$ that are C^3-close to the ball [Edt24], and for domains in $\mathbb{C}^n$ that are C^2-close to the ball [ABE25].

Theorem 9.34 (Abbondandolo–Bramham–Hryniewicz–Salomão)**.** *(a) There is a C^3-neighbourhood $\mathcal{U}$ of the set of Zoll contact forms on S^3 such that $\rho(\alpha) \le 1$ for every $\alpha \in \mathcal{U}$, with equality if and only if α is Zoll [Abb+18a]. (b) Every cooriented contact structure on a closed 3-manifold is defined by contact forms of arbitrarily large systolic ratio [Abb+19].*

Part (a), proved with a disk-like Birkhoff section and a generating function for its return map, gives Viterbo's inequality for c_{EH} and c_{HZ} on convex domains in $\mathbb{R}^4$ C^3-close to a ball. Zoll forms are local maximizers in all dimensions [BK21; AB23], after the one-parameter version of Álvarez Paiva–Balacheff [ÁB14]. Part (b), extended to all dimensions by Sağlam [Sağ21], shows that only convexity could give a global bound; dynamical convexity is not enough, since ρ can approach 2 on dynamically convex forms on S^3 [Abb+18b].

Theorem 9.35 (Haim-Kislev–Ostrover 2024)**.** *Let $K \subset \mathbb{R}^2_q$ be the regular pentagon with vertices $(\cos\frac{2\pi k}{5}, \sin\frac{2\pi k}{5})$, $k = 0, \dots, 4$, and $T \subset \mathbb{R}^2_p$ its rotation by $-\pi/2$, with vertices $(\cos(\frac{2\pi k}{5} - \frac{\pi}{2}), \sin(\frac{2\pi k}{5} - \frac{\pi}{2}))$. The Lagrangian product $K \times T \subset \mathbb{R}^4$ satisfies*

$$c_{EHZ}(K \times T) = 2\cos\tfrac{\pi}{10}\big(1 + \cos\tfrac{\pi}{5}\big), \qquad \frac{c_{EHZ}(K \times T)^2}{2\,\mathrm{vol}(K \times T)} = \frac{\sqrt{5}+3}{5} > 1.$$

Hence Conjecture 9.33 fails for c_{EHZ} in $\mathbb{R}^4$, and, by taking products, in every $\mathbb{R}^{2n}$ with $n \ge 2$. Since $c_{\mathrm{Gr}}(K \times T)^2 \le 2\,\mathrm{vol}(K \times T)$, the strong Viterbo conjecture fails too [HO26].

Proof idea. For convex $K \subset \mathbb{R}^n_q$ and $T \subset \mathbb{R}^n_p$, closed characteristics on $\partial(K \times T)$ are closed Minkowski billiard trajectories in K. So $c_{EHZ}(K \times T)$ is the minimal length of such a trajectory, measured by the support function of T [AO14; Rud24]. For polytopes, Haim-Kislev's combinatorial formula [Hai19] turns this into a finite optimization, which for the pentagons can be done by hand. The systolic ratio is multiplicative under symplectic ℓ^2-products of bodies with equal capacity [HO23]. Taking such products gives $\rho \ge ((\sqrt{5}+3)/5)^{\lfloor n/2 \rfloor}$ in $\mathbb{R}^{2n}$.

Conjecture 9.36 (Mahler 1939)**.** *For every centrally symmetric convex body $K \subset \mathbb{R}^m$, $\mathrm{vol}(K)\,\mathrm{vol}(K^\circ) \ge 4^m/m!$, where K° is the polar body.*

Mahler posed this [Mah39b] and proved it for $m = 2$ [Mah39a]. Iriyeh–Shibata proved $m = 3$ [IS20]. It is open for $m \ge 4$.

Theorem 9.37 (Artstein-Avidan–Karasev–Ostrover 2014)**.** *For every centrally symmetric convex body $K \subset \mathbb{R}^m$, $c_{HZ}(K \times K^\circ) = 4$, where $K \times K^\circ \subset \mathbb{R}^m_q \times \mathbb{R}^m_p = \mathbb{R}^{2m}$. Consequently Conjecture 9.33 for the bodies $K \times K^\circ$ and $c = c_{HZ}$ implies Conjecture 9.36 [AKO14].*

Indeed, Viterbo would give $4^m \leq m! \operatorname{vol}(K) \operatorname{vol}(K^\circ)$. A pentagon is not centrally symmetric, so Theorem 9.35 does not touch Mahler. But the symplectic route to Mahler now needs Viterbo's inequality for centrally symmetric bodies, for which the evidence is thin, although Haim-Kislev–Ostrover note that it holds for sufficiently symmetric domains. The honest questions now are: for which convex bodies does $\rho \leq 1$ hold, and what is $\sup \rho$ in $\mathbb{R}^4$? A relevant new fact: on convex bodies in $\mathbb{R}^4$ the cylindrical capacity equals c_{EHZ}, by a preprint of Abbondandolo–Edtmair–Kang [AEK24]. So the pentagon example separates the Gromov width from $c_{EHZ} = c_Z$.

9.7 What to read

Hofer–Zehnder, Symplectic Invariants and Hamiltonian Dynamics *[HZ94]*
Still the best introduction to the dynamical side. The core is Chapters 1–4: the action principle, capacities, the construction of c_{HZ}, almost existence, and the Weinstein conjecture in $\mathbb{R}^{2n}$. The Floer theory chapter is dated in technique but not in ideas.

Ekeland [Eke90]; Abbondandolo [Abb01]; Long [Lon02]
For convex multiplicity: Clarke duality and index theory (Ekeland), Morse theory of the action (Abbondandolo). Read Long only if you plan to work on the problem.

Hutchings, Lecture notes on ECH *[Hut14]*
The entry point to ECH, and a model of exposition. Read it before anything by Taubes. Then read [Hut11a] for capacities and the short paper [CH16] as the model ECH dynamics argument.

Wendl, Lectures on Symplectic Field Theory *[Wen16]*
Punctured curves done carefully, with an honest account of transversality. Then the introduction of EGH [EGH00] for the vision, BEHWZ [Bou+03] for compactness, Bourgeois [Bou09] for computations, Hind–Siegel [HS24a] for the state of the art, and Pardon [Par19] for contact homology done completely.

Hofer [Hof93b]; Hofer–Wysocki–Zehnder [HWZ98; HWZ03]
Hofer's overtwisted argument is the model of how curves find orbits. For HWZ read the introductions, then Salomão–Hryniewicz [SH18] for the modern view of surfaces of section.

Taubes [Tau07]
Read the introduction and outline for the mechanism; the analysis is for specialists.

Ginzburg–Gürel, The Conley conjecture and beyond *[GG15]*
The survey for Section 9.5. Then read Ginzburg [Gin10], Shelukhin [She22a], and [ÇGG24b] for barcodes.

Ostrover, ICM 2014 [Ost14]
Where convex geometry meets capacities. Then read [AKO14], [Abb+18a] and [HO26]. The last is short and elementary once you accept billiards.

9.8 Frontier and open problems

Problem 9.38 (Weinstein conjecture in higher dimensions)**.** *Prove Conjecture 9.2 in dimension* ≥ 5*, even just for Weinstein-fillable contact manifolds. This is open, and there is no higher-dimensional substitute for Taubes's Seiberg–Witten input.*

Conjecture 9.39 (n or infinitely many; Gürel 2015)**.** *Every Reeb flow on* $(S^{2n-1}, \xi_{\mathrm{st}})$ *has either exactly* n *or infinitely many closed orbits [Gür15].*

For $n = 2$ this is Theorem 9.13(b); for $n \geq 3$ see Remark 9.15. Without dynamical convexity even the lower bound n is open.

Problem 9.40 (Two or infinity without torsion)**.** *Remove the torsion hypothesis from Theorem 9.13(b): does every contact form on a closed* 3*-manifold have two or infinitely many closed Reeb orbits? The nondegenerate case is Theorem 9.13(c).*

Problem 9.41 (Hofer–Zehnder and Chance–McDuff)**.** *Prove Conjecture 9.27 beyond semisimple quantum homology. Prove the Chance–McDuff conjecture, as recorded in [GG15]: a closed symplectic manifold admitting a Hamiltonian diffeomorphism with finitely many periodic points has a nonvanishing Gromov–Witten invariant. Only the pseudo-rotation cases listed after Definition 9.29 are known.*

Problem 9.42 (Higher-dimensional closing lemmas)**.** *Is there a* C^∞ *closing lemma for Reeb flows in dimension* ≥ 5*, or for Hamiltonian diffeomorphisms in dimension* ≥ 4*?*

Problem 9.43 (After Viterbo)**.** *Does* $c(K)^n \leq n!\,\mathrm{vol}(K)$ *hold for centrally symmetric convex* $K \subset \mathbb{R}^{2n}$*? A yes would prove Mahler's conjecture (Theorem 9.37). What is* $\sup \rho$ *over convex bodies in* $\mathbb{R}^4$*? It is finite and at least* $(\sqrt{5}+3)/5$*. How many distinct normalized capacities are there on convex domains?*

Problem 9.44 (Full SFT)**.** *Construct full symplectic field theory, with its algebraic structure and invariance (Warning 9.17).*

9.9 Logical dependencies

To state the main results, introduce in order: contact forms, Reeb orbits, nondegeneracy and μ_{CZ} (Convention 9.1), enough for Conjecture 9.2; dynamical convexity and Birkhoff sections (Definitions 9.9 and 9.10), for Theorems 9.11 and 9.13; adapted J and finite energy curves (Definition 9.16), then the ECH complex and spectral invariants (Definitions 9.18 and 9.20), for Theorems 9.19 and 9.21. Theorems 9.25 and 9.28 need local Floer homology and quantum homology (Chapter 7); Section 9.6 needs only capacities (Chapter 6). The statements presuppose Fredholm theory for punctured curves with exponential weights, SFT compactness, and, for Theorem 9.19, monopole Floer homology [KM07].

Chapter 10

Frontiers, reading paths, and a map of the subject

This chapter is the one to read first if you are a colleague and last if you are a student. It compresses the subject into a map, gives reading paths that I have actually seen work, collects the open problems that I think matter in one place with their current status, and ends with a few paragraphs of opinion about where things are going.

10.1 A map of the subject

The cleanest way to hold the subject in one's head is by the *rigidity–flexibility dichotomy*: for each kind of symplectic object, ask whether its classification reduces to algebraic topology (an h-principle holds) or whether it does not (there is a holomorphic-curve obstruction). Table 10.1 is my summary; every row is a theorem or a cluster of theorems discussed in the chapter indicated.

The second organizing principle is by *language*. There are three, and a working symplectic topologist today needs all of them: the *variational* language (action functionals, critical point theory, Conley index; Chapter 9 and the first half of Chapter 7), the *holomorphic-curve* language (moduli spaces, compactness, transversality, virtual cycles; Chapter 5 and everything downstream), and the *categorical and sheaf-theoretic* language (A_∞-categories, wrapped Fukaya categories, microlocal sheaves; Chapter 8). Historically each language subsumed the previous one without replacing it; the sheaf-theoretic language currently covers only the Weinstein and cotangent-bundle part of the subject, which is why closed symplectic manifolds still require curves.

The third organizing principle is by *class of manifold*, in increasing order of difficulty: cotangent bundles T^*Q (where everything is governed by the loop space of Q: Viterbo's theorem, Nadler–Zaslow, the nearby Lagrangian conjecture), Weinstein and Liouville manifolds (where the Weinstein skeleton and SH govern), contact manifolds (the boundaries of the previous class, and the setting of dynamics), monotone and semipositive closed manifolds (where classical Floer theory applies), and general closed symplectic manifolds (where virtual techniques are unavoidable).

Object	Flexible regime (h-principle holds)	Rigid regime (curves obstruct)
Symplectic structures	open manifolds (Gromov; Chapter 4)	closed manifolds: SW = Gr obstructions in dim 4 (Chapter 6); dim ≥ 6 unknown (Chapter 3)
Symplectic embeddings	high codimension; Lagrangian "caps"	balls, ellipsoids, polydisks: nonsqueezing, capacities, staircases (Chapter 6)
Lagrangians	immersions (Gromov–Lees); embeddings with loose Legendrian boundary (Eliashberg–Murphy)	closed exact embeddings (Gromov); nearby Lagrangians; tori up to isotopy (Chapters 6 and 8)
Legendrians	loose Legendrians in dim ≥ 5 (Murphy)	Legendrian knots in dim 3; nonloose Legendrians: contact homology, sheaves (Chapters 4 and 8)
Contact structures	overtwisted (Eliashberg; Borman–Eliashberg–Murphy)	tight: fillability hierarchy, Floer-type invariants (Chapter 4)
Weinstein structures	flexible (Cieliebak–Eliashberg)	$SH \neq 0$, nontrivial wrapped Fukaya category (Chapter 8)
Hamiltonian dynamics	smooth conjugacy classes, topological entropy	fixed points (Arnold), periodic orbits (Conley, Weinstein), closing lemmas (Chapters 7 and 9)
Diffeomorphism groups	Diff, Homeo	Symp, Ham: Hofer geometry, quasimorphisms, simplicity (Chapter 7)

Table 10.1: The rigidity–flexibility dichotomy by object.

10.2 Reading paths

What follows assumes a student who has finished a first-year graduate sequence in differential topology and algebraic topology. The times are what I have seen work for strong students who are doing nothing else.

The core (one year)

1. *Linear algebra and basic theory* (6 weeks). McDuff–Salamon [MS17], Chapters 1–4 and the first half of Chapter 5, supplemented by Cannas da Silva [Can01] where McDuff–Salamon is terse. Do the exercises on Moser's trick; they are the ones that matter. Read Arnold [Arn78], Chapters 7–8, for where the subject came from.

2. *Pseudoholomorphic curves* (3 months). Wendl's lecture notes [Wen14] first, then McDuff–Salamon [MS12a], Chapters 2–4 and 6, with Appendices B and C open beside them. Read Gromov's paper [Gro85], sections 0 and 1, after the compactness theorem and not before. Prove nonsqueezing yourself from the ingredients.

3. *Floer theory* (3 months). Salamon's Park City lectures [Sal99] straight through, then Audin–Damian [AD14] for the details that Salamon elides (gluing, orientations). Read Floer's 1989 paper [Flo89b]; it is clearer than its reputation. Then Seidel's "biased view" [Sei08a]

for symplectic cohomology and Auroux's beginner's introduction [Aur14] for Fukaya categories.

4. *Contact geometry and flexibility* (6 weeks, can be done in parallel with the above). Geiges [Gei08], Chapters 1–4 and 6; Eliashberg–Mishachev [EM02], Part I and the chapter on symplectic and contact h-principles; Eliashberg's survey [Eli15].
5. *Dynamics* (6 weeks). Hofer–Zehnder [HZ94], Chapters 1–4; Hutchings's ECH lecture notes [Hut14].

After this a student can read current papers in any part of the subject with effort. What to read next depends on the destination.

Destination: Fukaya categories and mirror symmetry

Seidel's book [Sei08b], Part I in full and Part II selectively; Abouzaid's generation criterion [Abo10]; Ganatra's thesis [Gan12]; Ganatra–Pardon–Shende [GPS20; GPS24b; GPS24a], introductions first; Sheridan's CY hypersurfaces paper [She15] as the model of a complete HMS proof; Nadler–Zaslow [NZ09] and Guillermou's Astérisque [Gui23] for sheaves. See Chapter 8.

Destination: four-manifolds

Gompf–Stipsicz [GS99]; McDuff's rational and ruled paper [McD90]; Taubes's collected volume [Tau00], read with McDuff's Gromov-invariant lectures; Donaldson [Don96; Don99]; Auroux's surveys on Lefschetz fibrations; T.-J. Li's Kodaira-dimension surveys. See Chapters 3 and 6.

Destination: dynamics and contact topology

Hofer–Zehnder [HZ94] in full; Hofer's 1993 paper [Hof93b]; Hofer–Wysocki–Zehnder on convex energy surfaces [HWZ98]; Hutchings [Hut14] then Taubes [Tau07]; Ginzburg–Gürel's Conley conjecture survey [GG15]; Wendl's SFT lectures; the closing-lemma and Birkhoff-section papers of the 2020s. See Chapters 4 and 9.

Destination: quantitative and C^0 symplectic topology

Polterovich's book [Pol01]; Oh's spectral invariants [Oh05a] with Usher's clarifications; Entov–Polterovich [EP03]; Polterovich–Rosen–Samvelyan–Zhang on barcodes [Pol+20]; Schlenk's embedding survey [Sch18]; the simplicity conjecture paper [CHS24b]. See Chapters 6 and 7.

Destination: foundations

McDuff–Salamon [MS12a] completely; Pardon [Par16]; Abouzaid–McLean–Smith on global Kuranishi charts [AMS21]; Fukaya–Oh–Ohta–Ono's Kuranishi monograph [Fuk+20]; Hofer–Wysocki–Zehnder's polyfold book [HWZ21]; Abouzaid–Blumberg [AB21] for the homotopy-theoretic future. See Chapters 5 and 7.

10.3 Open problems

The following list is personal. Each problem is stated in the form I believe is both open and worth working on, with its status as of late 2026 and a pointer to the chapter where the context lives. I have tried to avoid problems that are open only because nobody has written down a known argument.

Existence and classification of symplectic manifolds

Problem 10.1 (Existence in dimension at least six)**.** *Let M be a closed manifold of dimension $2n \geq 6$ admitting an almost complex structure and a class $a \in H^2(M;\mathbb{R})$ with $a^n \neq 0$. Must M admit a symplectic form? No obstruction beyond these two is known in dimension ≥ 6, and no general existence theorem either. In dimension 4 the answer is no, by Taubes (Chapters 3 and 6).*

Problem 10.2 (Symplectic homotopy projective planes)**.** *Is every closed symplectic 4-manifold homeomorphic to $\mathbb{CP}^2$ symplectomorphic to $(\mathbb{CP}^2, \lambda\omega_{FS})$? Known when $K\cdot[\omega] < 0$; the case $K\cdot[\omega] > 0$ is open (Chapter 3).*

Problem 10.3 (Symplectic Calabi–Yau surfaces)**.** *Classify closed symplectic 4-manifolds with $c_1 = 0$ up to diffeomorphism. Their rational homology is known to be that of a $K3$ surface, an Enriques surface or a torus bundle over a torus; whether they are diffeomorphic to the known examples is open (Chapter 3).*

Problem 10.4 (Uniqueness up to deformation)**.** *Does a closed 4-manifold with $b^+ > 1$ admit finitely many symplectic forms up to deformation and diffeomorphism? For the $K3$ surface, is the symplectic form unique up to deformation and diffeomorphism? Both open (Chapter 3).*

Lagrangian submanifolds

Problem 10.5 (Nearby Lagrangian conjecture)**.** *Every closed exact Lagrangian in T^*Q, Q closed, is Hamiltonian isotopic to the zero section. Known for $Q = S^1$, S^2, $\mathbb{RP}^2$ and T^2; in general the projection to Q is known to be a simple homotopy equivalence, with vanishing Maslov class and with stable Lagrangian Gauss map trivial on homotopy groups, but even the diffeomorphism type of L is open (Chapter 8).*

Problem 10.6 (Lagrangian tori in the projective plane)**.** *Classify monotone Lagrangian tori in $\mathbb{CP}^2$ up to Hamiltonian isotopy. Vianna's construction gives infinitely many, indexed by Markov triples; whether these are all is open (Chapters 6 and 8).*

Problem 10.7 (Lagrangian embeddings into $\mathbb{C}^n$)**.** *Which closed n-manifolds admit Lagrangian embeddings into $\mathbb{C}^n$, and how many Lagrangian isotopy and Hamiltonian isotopy classes does each have? Gromov excludes exact ones, Audin's conjecture for tori is settled (Cieliebak–Mohnke), and the Klein bottle is excluded in dimension four (Nemirovski, Shevchishin); beyond that the existence question is open, and the classification question is open even for monotone tori in $\mathbb{C}^n$, $n \geq 3$ (Chapter 6).*

Fixed points and periodic orbits

Problem 10.8 (Arnold conjecture, Morse-number and cuplength forms)**.** *For a Hamiltonian diffeomorphism ϕ of a closed symplectic manifold M, is $\#\operatorname{Fix}(\phi)$ bounded below by the minimal number of critical points of a smooth function on M, and, without nondegeneracy, by the cuplength plus one? The Betti-number bound for nondegenerate ϕ is a theorem over every field; the degenerate bound is known for symplectically aspherical M and a few other cases (Chapter 7).*

Problem 10.9 (Weinstein conjecture in higher dimensions)**.** *Does every Reeb vector field on a closed contact manifold of dimension ≥ 5 have a closed orbit? Known in dimension 3 (Taubes), for overtwisted structures, and for many fillable cases (Chapter 9).*

Problem 10.10 (Conley and Hofer–Zehnder conjectures)**.** *For which closed symplectic manifolds does every Hamiltonian diffeomorphism have infinitely many periodic points? The Conley conjecture holds for symplectically aspherical, negative monotone and Calabi–Yau manifolds; the Hofer–Zehnder conjecture (more fixed points than the homological minimum forces infinitely many periodic points) is known for monotone manifolds with semisimple quantum cohomology (Shelukhin, extended to the semipositive case by Atallah–Lou) and open in general (Chapter 9).*

Problem 10.11 (Multiplicity on convex hypersurfaces)**.** *Does every closed convex hypersurface in $\mathbb{R}^{2n}$ carry at least n closed characteristics, and either exactly n or infinitely many? Known: at least two (Ekeland–Hofer), at least $\lfloor n/2 \rfloor + 1$ (Long–Zhu), at least n for $n \leq 4$; either two or infinitely many when $n = 2$ (Hofer–Wysocki–Zehnder); and a 2024 preprint of Çineli–Ginzburg–Gürel claims the bound n for all dynamically convex flows (Chapter 9).*

Problem 10.12 (Closing lemmas beyond dimension three)**.** *Does the C^∞ closing lemma hold for Reeb flows on closed contact manifolds of dimension ≥ 5, or for Hamiltonian diffeomorphisms of closed symplectic manifolds of dimension ≥ 4? Known in dimension 3 for Reeb flows (Irie) and for area-preserving surface maps (Chapter 9).*

Quantitative and C^0 questions

Problem 10.13 (Hofer diameter)**.** *Is the Hofer diameter of $\mathrm{Ham}(M, \omega)$ infinite for every closed symplectic manifold? Known for surfaces, for manifolds with semisimple quantum cohomology via quasimorphisms, and in several other cases (Chapter 7).*

Problem 10.14 (C^0 flux conjecture)**.** *Is $\mathrm{Ham}(M, \omega)$ closed in $\mathrm{Symp}(M, \omega)$ in the C^0 topology? The C^1 version is Ono's theorem (Chapters 2 and 7).*

Problem 10.15 (Simplicity in higher dimensions)**.** *Is the group of Hamiltonian homeomorphisms of a closed symplectic manifold of dimension ≥ 4 simple? The two-dimensional case is settled negatively through PFH and quantitative Heegaard Floer theory (Chapter 7).*

Embeddings and capacities

Problem 10.16 (Ellipsoids in higher dimensions)**.** *Determine when $E(a_1, \ldots, a_n)$ embeds symplectically into $E(b_1, \ldots, b_n)$ for $n \geq 3$. In dimension 4 this is McDuff's theorem via ECH capacities; in higher dimensions no complete set of obstructions is known (Chapter 6).*

Problem 10.17 (After Viterbo)**.** *Now that Viterbo's conjecture is false (Haim-Kislev–Ostrover 2024), what is the sharp upper bound for the systolic ratio of convex bodies in $\mathbb{R}^{2n}$, and for which classes of convex bodies (centrally symmetric, Lagrangian products, monotone toric domains) does the inequality $c(K)^n \leq n!\,\mathrm{vol}(K)$ survive? Is Mahler's conjecture true in dimension ≥ 4? (Chapter 9)*

Problem 10.18 (Packing stability in higher dimensions)**.** *Compute the packing numbers of balls and ellipsoids into balls in dimension ≥ 6. The four-dimensional case is governed by McDuff–Polterovich, Biran and ECH capacities; in higher dimensions Gromov's two-ball theorem and packing stability (Biran, Buse–Hind) are known, but there is no complete table even for small numbers of balls (Chapter 6).*

Categorical and sheaf-theoretic questions

Problem 10.19 (Homological mirror symmetry for compact Calabi–Yau manifolds)**.** *Prove Kontsevich's equivalence $D^{\pi}\,\mathrm{Fuk}(X) \simeq D^{b}\,\mathrm{Coh}(X^{\vee})$ for compact Calabi–Yau manifolds beyond hypersurfaces in projective space and the toric-complete-intersection cases, and identify the mirror map intrinsically (Chapter 8).*

Problem 10.20 (Detecting flexibility)**.** *Characterize flexible Weinstein manifolds of dimension ≥ 6 by Floer-theoretic invariants. Vanishing of symplectic cohomology is necessary but not sufficient: Murphy–Siegel's subflexible domains have* $\mathrm{SH} = 0$ *and are not flexible, as twisted symplectic cohomology detects. Whether vanishing of* SH *with all twisted coefficients, or of the wrapped Fukaya category under all bulk deformations, forces flexibility is open (Chapters 4 and 8).*

Problem 10.21 (Fukaya categories of closed manifolds)**.** *Find a sheaf-theoretic or otherwise curve-free model of the Fukaya category of a closed symplectic manifold, or prove that none exists. For Weinstein manifolds the wrapped Fukaya category is equivalent to microlocal sheaves on the skeleton (Ganatra–Pardon–Shende); nothing comparable is known in the closed case (Chapter 8).*

Foundations

Problem 10.22 (A unified virtual framework with all structures)**.** *Construct virtual fundamental cycles for all moduli spaces of the subject (closed curves, strips, punctured curves in SFT) in a single framework carrying orientations, integer and equivariant structures, and all the operations (products, S^1-actions, coherent gluing), with the full algebraic structure of symplectic field theory as a theorem. Each piece exists in some framework; the full SFT package is still being written (Chapters 5 and 9).*

10.4 Where the subject is going

Four directions seem to me to carry most of the energy. First, *Floer theory with coefficients in spectra*. The integral Arnold conjecture was the proof of concept; the real prize is that generalized cohomology detects more than ordinary cohomology does, as Abouzaid–Blumberg and Porcelli–Smith have begun to show for nearby Lagrangians. I expect the next decade to be the one in which the Cohen–Jones–Segal program becomes routine. Second, *sheaves*. Ganatra–Pardon–Shende's theorem that wrapped Fukaya categories of Weinstein manifolds are microlocal sheaf categories has turned a large part of the subject into something computable by people who have never written a Sobolev space, and the cluster-algebraic structures found on Legendrian fillings by Casals, Gao, Shen, Weng and others show that the computations are rich. The closed case remains the obstacle. Third, *quantitative dynamics through ECH and PFH*. Spectral invariants from embedded contact homology and periodic Floer homology have resolved in five years questions (closing lemmas, the simplicity conjecture, Birkhoff sections) that had stood for forty, and their higher-dimensional analogues are the obvious target. Fourth, the *flexible–rigid boundary*: Murphy's loose Legendrians and Cieliebak–Eliashberg's flexible Weinstein manifolds showed that flexibility is far more pervasive than anyone expected in 2010, and the question of exactly which structures are flexible, and whether vanishing of the known invariants characterizes flexibility, is the right question to ask of every object in the table above.

Two areas where I expect less movement, said plainly. The classification of symplectic four-manifolds has been stuck for twenty years: the tools (Seiberg–Witten, Lefschetz fibrations, knot surgery) have produced an enormous zoo and no organizing principle, and I do not see

one coming. And the existence problem in dimension six and above (Problem 10.1) will need an idea that is not currently on the table, since neither curves nor h-principles seem to reach it.

10.5 Logical dependencies

This chapter introduces no new definitions. Its open problems are stated in terms of the definitions of Chapter 2 (symplectic manifold, Hamiltonian diffeomorphism), Chapter 3 (deformation equivalence, symplectic Calabi–Yau), Chapter 4 (Reeb vector field, flexible Weinstein manifold), Chapter 6 (capacities, systolic ratio, ellipsoids), Chapter 7 (Hofer metric, Hamiltonian homeomorphism), Chapter 8 (Fukaya category, symplectic cohomology) and Chapter 9 (closed characteristics, closing lemma).

References

[ACH05] C. Abbas, K. Cieliebak, and H. Hofer. "The Weinstein conjecture for planar contact structures in dimension three". In: *Commentarii Mathematici Helvetici* 80.4 (2005), pp. 771–793.

[Abb01] A. Abbondandolo. *Morse Theory for Hamiltonian Systems*. Vol. 425. Chapman & Hall/CRC Research Notes in Mathematics. Boca Raton, FL: Chapman & Hall/CRC, 2001.

[AB23] A. Abbondandolo and G. Benedetti. "On the local systolic optimality of Zoll contact forms". In: *Geometric and Functional Analysis* 33.2 (2023), pp. 299–363.

[ABE25] A. Abbondandolo, G. Benedetti, and O. Edtmair. "Symplectic capacities of domains close to the ball and Banach–Mazur geodesics in the space of contact forms". In: *Duke Mathematical Journal* 174.8 (2025), pp. 1567–1646.

[Abb+18a] A. Abbondandolo, B. Bramham, U. L. Hryniewicz, and P. A. S. Salomão. "Sharp systolic inequalities for Reeb flows on the three-sphere". In: *Inventiones Mathematicae* 211.2 (2018), pp. 687–778.

[Abb+18b] A. Abbondandolo, B. Bramham, U. L. Hryniewicz, and P. A. S. Salomão. "Systolic ratio, index of closed orbits and convexity for tight contact forms on the three-sphere". In: *Compositio Mathematica* 154.12 (2018), pp. 2643–2680.

[Abb+19] A. Abbondandolo, B. Bramham, U. L. Hryniewicz, and P. A. S. Salomão. "Contact forms with large systolic ratio in dimension three". In: *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (5)* 19.4 (2019), pp. 1561–1582.

[AEK24] A. Abbondandolo, O. Edtmair, and J. Kang. *On closed characteristics of minimal action on a convex three-sphere*. Preprint. 2024. arXiv: 2412.01777.

[AK22] A. Abbondandolo and J. Kang. "Symplectic homology of convex domains and Clarke's duality". In: *Duke Mathematical Journal* 171.3 (2022), pp. 739–830.

[AS06] A. Abbondandolo and M. Schwarz. "On the Floer homology of cotangent bundles". In: *Comm. Pure Appl. Math.* 59.2 (2006), pp. 254–316.

[AS10a] A. Abbondandolo and M. Schwarz. "Floer homology of cotangent bundles and the loop product". In: *Geom. Topol.* 14.3 (2010), pp. 1569–1722.

[AS14] A. Abbondandolo and M. Schwarz. "Corrigendum: On the Floer homology of cotangent bundles". In: *Comm. Pure Appl. Math.* 67.4 (2014), pp. 670–691.

[Abo06] M. Abouzaid. "Homogeneous coordinate rings and mirror symmetry for toric varieties". In: *Geom. Topol.* 10 (2006), pp. 1097–1156. arXiv: math/0511644.

[Abo09] M. Abouzaid. "Morse homology, tropical geometry, and homological mirror symmetry for toric varieties". In: *Selecta Math. (N.S.)* 15.2 (2009), pp. 189–270. arXiv: math/0610004.

[Abo10] M. Abouzaid. "A geometric criterion for generating the Fukaya category". In: *Publ. Math. Inst. Hautes Études Sci.* 112 (2010), pp. 191–240. arXiv: 1001.4593.

[Abo11] M. Abouzaid. "A cotangent fibre generates the Fukaya category". In: *Adv. Math.* 228.2 (2011), pp. 894–939. arXiv: 1003.4449.

[Abo12a] M. Abouzaid. "Framed bordism and Lagrangian embeddings of exotic spheres". In: *Ann. of Math. (2)* 175.1 (2012), pp. 71–185. arXiv: 0812.4781.

[Abo12b] M. Abouzaid. "Nearby Lagrangians with vanishing Maslov class are homotopy equivalent". In: *Invent. Math.* 189.2 (2012), pp. 251–313. arXiv: 1005.0358.

[Abo12c] M. Abouzaid. "On the wrapped Fukaya category and based loops". In: *J. Symplectic Geom.* 10.1 (2012), pp. 27–79. arXiv: 0907.5606.

[Abo14] M. Abouzaid. "Family Floer cohomology and mirror symmetry". In: *Proceedings of the International Congress of Mathematicians—Seoul 2014. Vol. II*. Seoul: Kyung Moon Sa, 2014, pp. 813–836. arXiv: 1404.2659.

[Abo15] M. Abouzaid. "Symplectic cohomology and Viterbo's theorem". In: *Free Loop Spaces in Geometry and Topology*. Vol. 24. IRMA Lect. Math. Theor. Phys. Zürich: European Mathematical Society, 2015, pp. 271–485. arXiv: 1312.3354.

[Abo17] M. Abouzaid. "The family Floer functor is faithful". In: *J. Eur. Math. Soc. (JEMS)* 19.7 (2017), pp. 2139–2217. arXiv: 1408.6794.

[Abo21] M. Abouzaid. "Homological mirror symmetry without correction". In: *J. Amer. Math. Soc.* 34.4 (2021), pp. 1059–1173. arXiv: 1703.07898.

[AAK16] M. Abouzaid, D. Auroux, and L. Katzarkov. "Lagrangian fibrations on blowups of toric varieties and mirror symmetry for hypersurfaces". In: *Publ. Math. Inst. Hautes Études Sci.* 123 (2016), pp. 199–282. arXiv: 1205.0053.

[AB21] M. Abouzaid and A. J. Blumberg. *Arnold conjecture and Morava K-theory*. Preprint. 2021. arXiv: 2103.01507.

[AB24] M. Abouzaid and A. J. Blumberg. *Foundation of Floer homotopy theory I: flow categories*. Preprint. 2024. arXiv: 2404.03193.

[Abo+25] M. Abouzaid, S. Courte, S. Guillermou, and T. Kragh. "Twisted generating functions and the nearby Lagrangian conjecture". In: *Duke Math. J.* 174.5 (2025), pp. 949–1011. arXiv: 2011.13178.

[AK16] M. Abouzaid and T. Kragh. "On the immersion classes of nearby Lagrangians". In: *J. Topol.* 9.1 (2016), pp. 232–244. arXiv: 1305.6810.

[AK18] M. Abouzaid and T. Kragh. "Simple homotopy equivalence of nearby Lagrangians". In: *Acta Math.* 220.2 (2018), pp. 207–237. arXiv: 1603.05431.

[AMS21] M. Abouzaid, M. McLean, and I. Smith. *Complex cobordism, Hamiltonian loops and global Kuranishi charts*. Preprint. 2021. arXiv: 2110.14320.

[AMS24] M. Abouzaid, M. McLean, and I. Smith. "Gromov–Witten invariants in complex and Morava-local K-theories". In: *Geometric and Functional Analysis* 34.6 (2024), pp. 1647–1733.

[AS10b] M. Abouzaid and P. Seidel. *Altering symplectic manifolds by homologous recombination*. Preprint. 2010. arXiv: 1007.3281.

[AS10c] M. Abouzaid and P. Seidel. "An open string analogue of Viterbo functoriality". In: *Geom. Topol.* 14.2 (2010), pp. 627–718.

[AS10d] M. Abouzaid and I. Smith. "Homological mirror symmetry for the 4-torus". In: *Duke Math. J.* 152.3 (2010), pp. 373–440. arXiv: 0903.3065.

[Abr98] M. Abreu. "Topology of symplectomorphism groups of $S^2 \times S^2$". In: *Invent. Math.* 131.1 (1998), pp. 1–23.

[AM17] M. Abreu and L. Macarini. "Dynamical convexity and elliptic periodic orbits for Reeb flows". In: *Mathematische Annalen* 369.1–2 (2017), pp. 331–386.

[AM00] M. Abreu and D. McDuff. "Topology of symplectomorphism groups of rational ruled surfaces". In: *J. Amer. Math. Soc.* 13.4 (2000), pp. 971–1009.

[Acu17] B. Acu. *The Weinstein conjecture for iterated planar contact structures*. Preprint. 2017. arXiv: 1710.07724.

[AM22] B. Acu and A. Moreno. "Planarity in higher-dimensional contact manifolds". In: *International Mathematics Research Notices* 6 (2022), pp. 4222–4258.

[Ada25] N. Adaloglou. "Uniqueness of Lagrangians in $T^*\mathbb{R}P^2$". In: *Ann. Math. Qué.* 49.1 (2025), pp. 215–222. arXiv: 2201.09299.

[AO01] S. Akbulut and B. Ozbagci. "Lefschetz fibrations on compact Stein surfaces". In: *Geometry & Topology* 5 (2001). See also the erratum in the same volume, pp. 319–334.

[AP08] A. Akhmedov and B. D. Park. "Exotic smooth structures on small 4-manifolds". In: *Inventiones Mathematicae* 173.1 (2008), pp. 209–223. arXiv: math/0701664.

[AP10a] A. Akhmedov and B. D. Park. *Exotic smooth structures on $S^2 \times S^2$*. Preprint (v5, July 2010); not published. 2010. arXiv: 1005.3346.

[AP10b] A. Akhmedov and B. D. Park. "Exotic smooth structures on small 4-manifolds with odd signatures". In: *Inventiones Mathematicae* 181.3 (2010), pp. 577–603. arXiv: math/0701829.

[AH09] P. Albers and H. Hofer. "On the Weinstein conjecture in higher dimensions". In: *Comment. Math. Helv.* 84 (2009), pp. 429–436.

[All22] S. Allais. "On the Hofer–Zehnder conjecture on $\mathbb{CP}^d$ via generating functions". In: *International Journal of Mathematics* 33.10–11 (2022). With an appendix by Egor Shelukhin, Paper No. 2250072.

[ÁB14] J. C. Álvarez Paiva and F. Balacheff. "Contact geometry and isosystolic inequalities". In: *Geometric and Functional Analysis* 24.2 (2014), pp. 648–669.

[ÁEN20] D. Álvarez-Gavela, Y. Eliashberg, and D. Nadler. *Positive arborealization of polarized Weinstein manifolds*. Preprint. 2020. arXiv: 2011.08962.

[Amo+96] J. Amorós, M. Burger, K. Corlette, D. Kotschick, and D. Toledo. *Fundamental Groups of Compact Kähler Manifolds*. Vol. 44. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 1996.

[Anj02] S. Anjos. "Homotopy type of symplectomorphism groups of $S^2 \times S^2$". In: *Geom. Topol.* 6 (2002), pp. 195–218.

[AG04] S. Anjos and G. Granja. "Homotopy decomposition of a group of symplectomorphisms of $S^2 \times S^2$". In: *Topology* 43.3 (2004), pp. 599–618.

[Anj+24] S. Anjos, J. Li, T.-J. Li, and M. Pinsonnault. "Stability of the symplectomorphism groups of rational surfaces". In: *Math. Ann.* 389.1 (2024), pp. 85–119.

[AP13] S. Anjos and M. Pinsonnault. "The homotopy Lie algebra of symplectomorphism groups of 3-fold blow-ups of the projective plane". In: *Math. Z.* 275.1-2 (2013), pp. 245–292.

[AK70] D. V. Anosov and A. B. Katok. "New examples in smooth ergodic theory. Ergodic diffeomorphisms". In: *Trudy Moskovskogo Matematičeskogo Obščestva* 23 (1970). English translation: Trans. Moscow Math. Soc. 23 (1970), 1–35, pp. 3–36.

[Arn67] V. I. Arnol'd. "Characteristic class entering in quantization conditions". In: *Functional Analysis and Its Applications* 1.1 (1967), pp. 1–13. Translated from Funktsional. Anal. i Prilozhen. **1**:1 (1967), 1–14.

[Arn80a] V. I. Arnol'd. "Lagrange and Legendre cobordisms. I". In: *Funct. Anal. Appl.* 14.3 (1980). Translated from Funktsional. Anal. i Prilozhen. 14 (1980), no. 3, 1–13, pp. 167–177.

[Arn80b] V. I. Arnol'd. "Lagrange and Legendre cobordisms. II". In: *Funct. Anal. Appl.* 14.4 (1980). Translated from Funktsional. Anal. i Prilozhen. 14 (1980), no. 4, 8–17, pp. 252–260.

[Arn78] V. I. Arnold. *Mathematical Methods of Classical Mechanics*. Vol. 60. Graduate Texts in Mathematics. New York: Springer, 1978.

[Arn65] V. I. Arnold. "Sur une propriété topologique des applications globalement canoniques de la mécanique classique". In: *C. R. Acad. Sci. Paris* 261 (1965), pp. 3719–3722.

[AKO14] S. Artstein-Avidan, R. Karasev, and Y. Ostrover. "From symplectic measurements to the Mahler conjecture". In: *Duke Mathematical Journal* 163.11 (2014), pp. 2003–2022.

[AMO08] S. Artstein-Avidan, V. Milman, and Y. Ostrover. "The M-ellipsoid, symplectic capacities and volume". In: *Commentarii Mathematici Helvetici* 83.2 (2008), pp. 359–369.

[AO14] S. Artstein-Avidan and Y. Ostrover. "Bounds for Minkowski billiard trajectories in convex bodies". In: *International Mathematics Research Notices* 1 (2014), pp. 165–193.

[AI20] T. Asano and Y. Ike. "Persistence-like distance on Tamarkin's category and symplectic displacement energy". In: *J. Symplectic Geom.* 18.3 (2020), pp. 613–649. arXiv: 1712.06847.

[AI16] M. Asaoka and K. Irie. "A C^∞ closing lemma for Hamiltonian diffeomorphisms of closed surfaces". In: *Geometric and Functional Analysis* 26.5 (2016), pp. 1245–1254.

[AM93] P. S. Aspinwall and D. R. Morrison. "Topological field theory and rational curves". In: *Communications in Mathematical Physics* 151.2 (1993), pp. 245–262.

[AL25] M. S. Atallah and H. Lou. "On the Hofer–Zehnder conjecture for semipositive symplectic manifolds". In: *Journal of Modern Dynamics* 21 (2025), pp. 755–804.

[Ati82] M. F. Atiyah. "Convexity and commuting Hamiltonians". In: *Bull. London Math. Soc.* 14.1 (1982), pp. 1–15.

[AB84] M. F. Atiyah and R. Bott. "The moment map and equivariant cohomology". In: *Topology* 23.1 (1984), pp. 1–28.

[Aud85] M. Audin. "Quelques calculs en cobordisme lagrangien". In: *Ann. Inst. Fourier (Grenoble)* 35.3 (1985), pp. 159–194.

[Aud04] M. Audin. *Torus Actions on Symplectic Manifolds*. 2nd ed. Vol. 93. Progress in Mathematics. Basel: Birkhäuser, 2004.

[AD14] M. Audin and M. Damian. *Morse Theory and Floer Homology*. Universitext. London: Springer, 2014.

[AL94] M. Audin and J. Lafontaine, eds. *Holomorphic Curves in Symplectic Geometry*. Vol. 117. Progress in Mathematics. Basel: Birkhäuser, 1994.

[ALP94] M. Audin, F. Lalonde, and L. Polterovich. "Symplectic rigidity: Lagrangian submanifolds". In: *Holomorphic Curves in Symplectic Geometry*. Ed. by M. Audin and J. Lafontaine. Vol. 117. Progress in Mathematics. Basel: Birkhäuser, 1994, pp. 271–321.

[Aur97] D. Auroux. "Asymptotically holomorphic families of symplectic submanifolds". In: *Geometric and Functional Analysis* 7.6 (1997), pp. 971–995.

[Aur00] D. Auroux. "Symplectic 4-manifolds as branched coverings of $\mathbb{CP}^2$". In: *Inventiones Mathematicae* 139.3 (2000), pp. 551–602.

[Aur05] D. Auroux. "A stable classification of Lefschetz fibrations". In: *Geometry & Topology* 9 (2005), pp. 203–217.

[Aur06] D. Auroux. "Mapping class group factorizations and symplectic 4-manifolds: some open problems". In: *Problems on Mapping Class Groups and Related Topics*. Ed. by B. Farb. Vol. 74. Proceedings of Symposia in Pure Mathematics. Providence, RI: American Mathematical Society, 2006, pp. 123–132.

[Aur07] D. Auroux. "Mirror symmetry and T-duality in the complement of an anticanonical divisor". In: *J. Gökova Geom. Topol. GGT* 1 (2007), pp. 51–91. arXiv: 0706.3207.

[Aur14] D. Auroux. "A beginner's introduction to Fukaya categories". In: *Contact and Symplectic Topology*. Vol. 26. Bolyai Soc. Math. Stud. Cham: Springer, 2014, pp. 85–136.

[Aur15] D. Auroux. "Infinitely many monotone Lagrangian tori in $\mathbb{R}^6$". In: *Invent. Math.* 201.3 (2015), pp. 909–924.

[ADK03] D. Auroux, S. K. Donaldson, and L. Katzarkov. "Luttinger surgery along Lagrangian tori and non-isotopy for singular symplectic plane curves". In: *Mathematische Annalen* 326.1 (2003), pp. 185–203.

[AS08] D. Auroux and I. Smith. "Lefschetz pencils, branched covers and symplectic invariants". In: *Symplectic 4-Manifolds and Algebraic Surfaces (Cetraro, 2003)*. Ed. by F. Catanese and G. Tian. Vol. 1938. Lecture Notes in Mathematics. Berlin: Springer, 2008, pp. 1–53.

[Bai+26] S. Bai, E. Shelukhin, Y. Wang, and G. Xu. *A proof of the Arnold–Givental conjecture*. Preprint. 2026. arXiv: 2608.27242.

[BX22] S. Bai and G. Xu. *Arnold conjecture over integers*. Preprint. 2022. arXiv: 2209.08599.

[BX26] S. Bai and G. Xu. *Integral Hamiltonian Floer theory: foundations*. Preprint. 2026. arXiv: 2606.15847.

[BK08] S. Baldridge and P. Kirk. "A symplectic manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2\#3\overline{\mathbb{CP}}^2$". In: *Geometry & Topology* 12.2 (2008), pp. 919–940.

[Ban93] V. Bangert. "On the existence of closed geodesics on two-spheres". In: *International Journal of Mathematics* 4.1 (1993), pp. 1–10.

[Ban78] A. Banyaga. "Sur la structure du groupe des difféomorphismes qui préservent une forme symplectique". In: *Comment. Math. Helv.* 53.2 (1978), pp. 174–227.

[Ban97] A. Banyaga. *The Structure of Classical Diffeomorphism Groups*. Vol. 400. Mathematics and its Applications. Dordrecht: Kluwer Academic Publishers, 1997.

[BH18] E. Bao and K. Honda. "Definition of cylindrical contact homology in dimension three". In: *Journal of Topology* 11.4 (2018), pp. 1002–1053.

[BH23] E. Bao and K. Honda. "Semi-global Kuranishi charts and the definition of contact homology". In: *Advances in Mathematics* 414 (2023), Paper No. 108864.

[Bau08] S. Bauer. "Almost complex 4-manifolds with vanishing first Chern class". In: *Journal of Differential Geometry* 79.1 (2008), pp. 25–32.

[BF97] K. Behrend and B. Fantechi. "The intrinsic normal cone". In: *Inventiones Mathematicae* 128.1 (1997), pp. 45–88.

[BK21] G. Benedetti and J. Kang. "A local contact systolic inequality in dimension three". In: *Journal of the European Mathematical Society* 23.3 (2021), pp. 721–764.

[Ben83] D. Bennequin. "Entrelacements et équations de Pfaff". In: *Astérisque* 107–108 (1983), pp. 87–161.

[BV82] N. Berline and M. Vergne. "Classes caractéristiques équivariantes. Formule de localisation en cohomologie équivariante". In: *C. R. Acad. Sci. Paris Sér. I Math.* 295.9 (1982), pp. 539–541.

[Ber+21] M. Bertozzi, T. S. Holm, E. Maw, D. McDuff, G. T. Mwakyoma, A. R. Pires, and M. Weiler. "Infinite staircases for Hirzebruch surfaces". In: *Research Directions in Symplectic and Contact Geometry and Topology*. Ed. by B. Acu, C. Cannizzo, D. McDuff, Z. Myer, Y. Pan, and L. Traynor. Vol. 27. Association for Women in Mathematics Series. Cham: Springer, 2021, pp. 47–157.

[BP94] M. Bialy and L. Polterovich. "Geodesics of Hofer's metric on the group of Hamiltonian diffeomorphisms". In: *Duke Math. J.* 76.1 (1994), pp. 273–292.

[Bir97] P. Biran. "Symplectic packing in dimension 4". In: *Geom. Funct. Anal.* 7.3 (1997), pp. 420–437.

[Bir99] P. Biran. "A stability property of symplectic packing". In: *Invent. Math.* 136.1 (1999), pp. 123–155.

[Bir01] P. Biran. "From symplectic packing to algebraic geometry and back". In: *European Congress of Mathematics, Vol. II (Barcelona, 2000)*. Vol. 202. Progr. Math. Basel: Birkhäuser, 2001, pp. 507–524.

[Bir06] P. Biran. "Lagrangian non-intersections". In: *Geom. Funct. Anal.* 16.2 (2006), pp. 279–326.

[BC09] P. Biran and O. Cornea. "Rigidity and uniruling for Lagrangian submanifolds". In: *Geom. Topol.* 13.5 (2009), pp. 2881–2989.

[BC13] P. Biran and O. Cornea. "Lagrangian cobordism. I". In: *J. Amer. Math. Soc.* 26.2 (2013), pp. 295–340. arXiv: 1109.4984.

[BC14] P. Biran and O. Cornea. "Lagrangian cobordism and Fukaya categories". In: *Geom. Funct. Anal.* 24.6 (2014), pp. 1731–1830. arXiv: 1304.6032.

[BEM15] M. S. Borman, Y. Eliashberg, and E. Murphy. "Existence and classification of overtwisted contact structures in all dimensions". In: *Acta Math.* 215 (2015), pp. 281–361.

[Bou09] F. Bourgeois. "A survey of contact homology". In: *New Perspectives and Challenges in Symplectic Field Theory*. Vol. 49. CRM Proceedings and Lecture Notes. Providence, RI: American Mathematical Society, 2009, pp. 45–71.

[BEE12] F. Bourgeois, T. Ekholm, and Y. Eliashberg. "Effect of Legendrian surgery". In: *Geom. Topol.* 16 (2012), pp. 301–389.

[Bou+03] F. Bourgeois, Y. Eliashberg, H. Hofer, K. Wysocki, and E. Zehnder. "Compactness results in symplectic field theory". In: *Geometry & Topology* 7.2 (2003), pp. 799–888.

[BK10] F. Bourgeois and O. van Koert. "Contact homology of left-handed stabilizations and plumbing of open books". In: *Commun. Contemp. Math.* 12 (2010), pp. 223–263.

[Bow12] J. Bowden. "Exactly fillable contact structures without Stein fillings". In: *Algebr. Geom. Topol.* 12.3 (2012), pp. 1807–1814.

[Bow+26] J. Bowden, F. Gironella, A. Moreno, and Z. Zhou. "Tight contact structures without symplectic fillings are everywhere". In: *J. Eur. Math. Soc. (JEMS)* (2026). Published online first, 2026. arXiv: 2211.03680.

[Bra15a] B. Bramham. "Periodic approximations of irrational pseudo-rotations using pseudoholomorphic curves". In: *Annals of Mathematics* 181.3 (2015), pp. 1033–1086.

[Bra15b] B. Bramham. "Pseudo-rotations with sufficiently Liouvillean rotation number are C^0-rigid". In: *Inventiones Mathematicae* 199.2 (2015), pp. 561–580.

[Bre+19] J. Breen, A. Christian, K. Honda, and Y. Huang. *Convex hypersurface theory in contact topology*. Preprint; first versions by Honda and Huang. 2019. arXiv: 1907.06025.

[BHH23] J. Breen, K. Honda, and Y. Huang. *The Giroux correspondence in arbitrary dimensions*. Preprint. 2023. arXiv: 2307.02317.

[Buh10a] L. Buhovsky. "The 2/3-convergence rate for the Poisson bracket". In: *Geom. Funct. Anal.* 19.6 (2010), pp. 1620–1649.

[Buh10b] L. Buhovsky. "The Maslov class of Lagrangian tori and quantum products in Floer cohomology". In: *J. Topol. Anal.* 2.1 (2010), pp. 57–75.

[Buh14] L. Buhovsky. "Towards the C^0 flux conjecture". In: *Algebr. Geom. Topol.* 14.6 (2014), pp. 3493–3508. arXiv: 1309.1325.

[BEP12] L. Buhovsky, M. Entov, and L. Polterovich. "Poisson brackets and symplectic invariants". In: *Selecta Math. (N.S.)* 18.1 (2012), pp. 89–157.

[BHS18] L. Buhovsky, V. Humilière, and S. Seyfaddini. "A C^0 counterexample to the Arnold conjecture". In: *Invent. Math.* 213.2 (2018), pp. 759–809.

[BHS21] L. Buhovsky, V. Humilière, and S. Seyfaddini. "The action spectrum and C^0 symplectic topology". In: *Math. Ann.* 380.1–2 (2021), pp. 293–316.

[BS13] L. Buhovsky and S. Seyfaddini. "Uniqueness of generating Hamiltonians for topological Hamiltonian flows". In: *J. Symplectic Geom.* 11.1 (2013), pp. 37–52.

[BH13] O. Buse and R. Hind. "Ellipsoid embeddings and symplectic packing stability". In: *Compos. Math.* 149.5 (2013), pp. 889–902.

[Cal70] E. Calabi. "On the group of automorphisms of a symplectic manifold". In: *Problems in Analysis (Lectures at the Symposium in honor of Salomon Bochner, Princeton, 1969)*. Ed. by R. C. Gunning. Princeton, NJ: Princeton University Press, 1970, pp. 1–26.

[Can+91] P. Candelas, X. C. de la Ossa, P. S. Green, and L. Parkes. "A pair of Calabi–Yau manifolds as an exactly soluble superconformal theory". In: *Nuclear Phys. B* 359.1 (1991), pp. 21–74.

[Can01] A. Cannas da Silva. *Lectures on Symplectic Geometry*. Vol. 1764. Lecture Notes in Mathematics. Berlin: Springer, 2001.

[CLM94] S. E. Cappell, R. Lee, and E. Y. Miller. "On the Maslov index". In: *Communications on Pure and Applied Mathematics* 47.2 (1994), pp. 121–186.

[CV08] F. Cardin and C. Viterbo. "Commuting Hamiltonians and Hamilton–Jacobi multi-time equations". In: *Duke Math. J.* 144.2 (2008), pp. 235–284.

[CG22] R. Casals and H. Gao. "Infinitely many Lagrangian fillings". In: *Ann. of Math. (2)* 195.1 (2022), pp. 207–249. arXiv: 2001.01334.

[CMP19] R. Casals, E. Murphy, and F. Presas. "Geometric criteria for overtwistedness". In: *J. Amer. Math. Soc.* 32 (2019), pp. 563–604.

[CPP15] R. Casals, D. M. Pancholi, and F. Presas. "Almost contact 5-manifolds are contact". In: *Ann. of Math. (2)* 182 (2015), pp. 429–490.

[CW24] R. Casals and D. Weng. "Microlocal theory of Legendrian links and cluster algebras". In: *Geom. Topol.* 28.2 (2024), pp. 901–1000. arXiv: 2204.13244.

[CE22] J. Chaidez and O. Edtmair. "3D convex contact forms and the Ruelle invariant". In: *Inventiones Mathematicae* 229.1 (2022), pp. 243–301.

[Cha+24] B. Chantraine, G. Dimitroglou Rizell, P. Ghiggini, and R. Golovko. "Geometric generation of the wrapped Fukaya category of Weinstein manifolds and sectors". In: *Ann. Sci. Éc. Norm. Supér. (4)* 57.1 (2024), pp. 1–85. arXiv: 1712.09126.

[CS99] M. Chas and D. Sullivan. *String topology*. Preprint. 1999. arXiv: math/9911159.

[Che02] Y. Chekanov. "Differential algebra of Legendrian links". In: *Invent. Math.* 150.3 (2002), pp. 441–483.

[CS10] Y. Chekanov and F. Schlenk. "Notes on monotone Lagrangian twist tori". In: *Electron. Res. Announc. Math. Sci.* 17 (2010), pp. 104–121.

[Che96] Y. V. Chekanov. "Lagrangian tori in a symplectic vector space and global symplectomorphisms". In: *Math. Z.* 223.4 (1996), pp. 547–559.

[Che98] Y. V. Chekanov. "Lagrangian intersections, symplectic energy, and areas of holomorphic curves". In: *Duke Math. J.* 95.1 (1998), pp. 213–226.

[Chi17] S.-F. Chiu. "Nonsqueezing property of contact balls". In: *Duke Math. J.* 166.4 (2017), pp. 605–655. arXiv: 1405.1178.

[CO06] C.-H. Cho and Y.-G. Oh. "Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds". In: *Asian J. Math.* 10.4 (2006), pp. 773–814.

[Cie02a] K. Cieliebak. "Handle attaching in symplectic homology and the chord conjecture". In: *Journal of the European Mathematical Society* 4.2 (2002), pp. 115–142.

[Cie02b] K. Cieliebak. *Subcritical Stein manifolds are split*. Preprint. 2002. arXiv: `math/0204351`.

[CE12] K. Cieliebak and Y. Eliashberg. *From Stein to Weinstein and Back: Symplectic Geometry of Affine Complex Manifolds*. Vol. 59. American Mathematical Society Colloquium Publications. Providence, RI: American Mathematical Society, 2012.

[CE14] K. Cieliebak and Y. Eliashberg. "Flexible Weinstein manifolds". In: *Symplectic, Poisson, and Noncommutative Geometry*. Ed. by T. Eguchi, Y. Eliashberg, and Y. Maeda. Vol. 62. Mathematical Sciences Research Institute Publications. New York: Cambridge University Press, 2014, pp. 1–42. arXiv: `1305.1635`.

[CF09] K. Cieliebak and U. Frauenfelder. "A Floer homology for exact contact embeddings". In: *Pacific J. Math.* 239.2 (2009), pp. 251–316. arXiv: `0710.0972`.

[Cie+07] K. Cieliebak, H. Hofer, J. Latschev, and F. Schlenk. "Quantitative symplectic geometry". In: *Dynamics, Ergodic Theory, and Geometry*. Vol. 54. Math. Sci. Res. Inst. Publ. Cambridge: Cambridge University Press, 2007, pp. 1–44.

[CM07] K. Cieliebak and K. Mohnke. "Symplectic hypersurfaces and transversality in Gromov–Witten theory". In: *Journal of Symplectic Geometry* 5.3 (2007), pp. 281–356.

[CM18] K. Cieliebak and K. Mohnke. "Punctured holomorphic curves and Lagrangian embeddings". In: *Invent. Math.* 212.1 (2018), pp. 213–295.

[Çin18] E. Çineli. "Conley conjecture and local Floer homology". In: *Archiv der Mathematik* 111 (2018), pp. 647–656.

[Çin25] E. Çineli. "A generalized pseudo-rotation with positive topological entropy". In: *Bulletin of the London Mathematical Society* 57.4 (2025), pp. 1140–1149.

[ÇGG22] E. Çineli, V. L. Ginzburg, and B. Z. Gürel. "From pseudo-rotations to holomorphic curves via quantum Steenrod squares". In: *International Mathematics Research Notices* 3 (2022), pp. 2274–2297.

[ÇGG24a] E. Çineli, V. L. Ginzburg, and B. Z. Gürel. *Closed orbits of dynamically convex Reeb flows: towards the HZ- and multiplicity conjectures*. Preprint; announced as to appear in Annals of Mathematics. 2024. arXiv: `2410.13093`.

[ÇGG24b] E. Çineli, V. L. Ginzburg, and B. Z. Gürel. "Topological entropy of Hamiltonian diffeomorphisms: a persistence homology and Floer theory perspective". In: *Mathematische Zeitschrift* 308.4 (2024), Paper No. 73.

[Çin+25] E. Çineli, V. L. Ginzburg, B. Z. Gürel, and M. Mazzucchelli. "On the barcode entropy of Reeb flows". In: *Selecta Mathematica (N.S.)* 31.4 (2025), Paper No. 64.

[Çin+26] E. Çineli, V. L. Ginzburg, B. Z. Gürel, and M. Mazzucchelli. "Invariant sets and hyperbolic closed Reeb orbits". In: *Advances in Mathematics* 484 (2026), Paper No. 110709.

[CJS95] R. L. Cohen, J. D. S. Jones, and G. B. Segal. "Floer's infinite-dimensional Morse theory and homotopy theory". In: *The Floer Memorial Volume*. Vol. 133. Progr. Math. Basel: Birkhäuser, 1995, pp. 297–325.

[Col01] V. Colin. "Une infinité de structures de contact tendues sur les variétés toroïdales". In: *Comment. Math. Helv.* 76.2 (2001), pp. 353–372.

[Col+24] V. Colin, P. Dehornoy, U. Hryniewicz, and A. Rechtman. "Generic properties of 3-dimensional Reeb flows: Birkhoff sections and entropy". In: *Commentarii Mathematici Helvetici* 99.3 (2024), pp. 557–611.

[CDR23] V. Colin, P. Dehornoy, and A. Rechtman. "On the existence of supporting broken book decompositions for contact forms in dimension 3". In: *Inventiones Mathematicae* 231.3 (2023), pp. 1489–1539.

[CGH24] V. Colin, P. Ghiggini, and K. Honda. "The equivalence of Heegaard Floer homology and embedded contact homology via open book decompositions I, II; III: from hat to plus". In: *Publications Mathématiques de l'IHÉS* 139 (2024), pp. 13–385.

[CGH09] V. Colin, E. Giroux, and K. Honda. "Finitude homotopique et isotopique des structures de contact tendues". In: *Publ. Math. Inst. Hautes Études Sci.* 109 (2009), pp. 245–293.

[CZ83] C. C. Conley and E. Zehnder. "The Birkhoff–Lewis fixed point theorem and a conjecture of V. I. Arnold". In: *Inventiones Mathematicae* 73.1 (1983), pp. 33–49.

[CZ84] C. C. Conley and E. Zehnder. "Morse-type index theory for flows and periodic solutions for Hamiltonian equations". In: *Communications on Pure and Applied Mathematics* 37.2 (1984), pp. 207–253.

[CM22] G. Contreras and M. Mazzucchelli. "Existence of Birkhoff sections for Kupka–Smale Reeb flows of closed contact 3-manifolds". In: *Geometric and Functional Analysis* 32.5 (2022), pp. 951–979.

[CK99] D. A. Cox and S. Katz. *Mirror Symmetry and Algebraic Geometry*. Vol. 68. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 1999.

[CHS24a] D. Cristofaro-Gardiner, V. Humilière, and S. Seyfaddini. "PFH spectral invariants on the two-sphere and the large scale geometry of Hofer's metric". In: *J. Eur. Math. Soc. (JEMS)* 26.12 (2024), pp. 4537–4584. arXiv: 2102.04404.

[CHS24b] D. Cristofaro-Gardiner, V. Humilière, and S. Seyfaddini. "Proof of the simplicity conjecture". In: *Ann. of Math. (2)* 199.1 (2024), pp. 181–257.

[Cri19] D. Cristofaro-Gardiner. "Symplectic embeddings from concave toric domains into convex ones". In: *J. Differential Geom.* 112.2 (2019), pp. 199–232.

[CH18] D. Cristofaro-Gardiner and R. Hind. "Symplectic embeddings of products". In: *Comment. Math. Helv.* 93.1 (2018), pp. 1–32.

[CHM18] D. Cristofaro-Gardiner, R. Hind, and D. McDuff. "The ghost stairs stabilize to sharp symplectic embedding obstructions". In: *J. Topol.* 11.2 (2018), pp. 309–378.

[Cri+25a] D. Cristofaro-Gardiner, T. S. Holm, A. Mandini, and A. R. Pires. "On infinite staircases in toric symplectic four-manifolds". In: *J. Differential Geom.* 129.2 (2025), pp. 335–413.

[Cri+23] D. Cristofaro-Gardiner, U. Hryniewicz, M. Hutchings, and H. Liu. "Contact three-manifolds with exactly two simple Reeb orbits". In: *Geometry & Topology* 27.9 (2023), pp. 3801–3831.

[Cri+26] D. Cristofaro-Gardiner, U. Hryniewicz, M. Hutchings, and H. Liu. "Proof of Hofer–Wysocki–Zehnder's two or infinity conjecture". In: *Journal of the American Mathematical Society* 39.4 (2026), pp. 915–983.

[Cri+22] D. Cristofaro-Gardiner, V. Humilière, C. Y. Mak, S. Seyfaddini, and I. Smith. "Quantitative Heegaard Floer cohomology and the Calabi invariant". In: *Forum Math. Pi* 10 (2022), Paper No. e27.

[CH16] D. Cristofaro-Gardiner and M. Hutchings. "From one Reeb orbit to two". In: *Journal of Differential Geometry* 102.1 (2016), pp. 25–36.

[CHP19] D. Cristofaro-Gardiner, M. Hutchings, and D. Pomerleano. "Torsion contact forms in three dimensions have two or infinitely many Reeb orbits". In: *Geometry & Topology* 23.7 (2019), pp. 3601–3645.

[CHR15] D. Cristofaro-Gardiner, M. Hutchings, and V. G. B. Ramos. "The asymptotics of ECH capacities". In: *Invent. Math.* 199.1 (2015), pp. 187–214.

[Cri+25b] D. Cristofaro-Gardiner, D. Pomerleano, R. Prasad, and B. Zhang. "A note on U-cyclic elements in monopole Floer homology". In: *Proceedings of the American Mathematical Society, Series B* 12 (2025), pp. 218–228. arXiv: 2110.13844.

[CPZ21] D. Cristofaro-Gardiner, R. Prasad, and B. Zhang. *Periodic Floer homology and the smooth closing lemma for area-preserving surface diffeomorphisms*. Preprint. 2021. arXiv: 2110.02925.

[Dam12] M. Damian. "Floer homology on the universal cover, Audin's conjecture and other constraints on Lagrangian submanifolds". In: *Comment. Math. Helv.* 87.2 (2012), pp. 433–463.

[Dar82] G. Darboux. "Sur le problème de Pfaff". In: *Bull. Sci. Math. (2)* 6 (1882), pp. 14–36, 49–68.

[Del88] T. Delzant. "Hamiltoniens périodiques et images convexes de l'application moment". In: *Bull. Soc. Math. France* 116.3 (1988), pp. 315–339.

[DE14] G. Dimitroglou Rizell and J. D. Evans. "Unlinking and unknottedness of monotone Lagrangian submanifolds". In: *Geom. Topol.* 18.2 (2014), pp. 997–1034.

[DGI16] G. Dimitroglou Rizell, E. Goodman, and A. Ivrii. "Lagrangian isotopy of tori in $S^2 \times S^2$ and $\mathbb{C}P^2$". In: *Geom. Funct. Anal.* 26.5 (2016), pp. 1297–1358.

[DG04] F. Ding and H. Geiges. "A Legendrian surgery presentation of contact 3-manifolds". In: *Math. Proc. Cambridge Philos. Soc.* 136 (2004), pp. 583–598.

[DIW21] A. Doan, E.-N. Ionel, and T. Walpuski. *The Gopakumar–Vafa finiteness conjecture*. Accepted for publication in Annals of Mathematics (2025). 2021. arXiv: 2103.08221.

[DW21] A. Doan and T. Walpuski. "Castelnuovo's bound and rigidity in almost complex geometry". In: *Advances in Mathematics* 379 (2021), Paper No. 107550.

[DW23] A. Doan and T. Walpuski. "Equivariant Brill–Noether theory for elliptic operators and superrigidity of J-holomorphic maps". In: *Forum Math. Sigma* 11, e3 (2023).

[Don96] S. K. Donaldson. "Symplectic submanifolds and almost-complex geometry". In: *Journal of Differential Geometry* 44.4 (1996), pp. 666–705.

[Don99] S. K. Donaldson. "Lefschetz pencils on symplectic manifolds". In: *Journal of Differential Geometry* 53.2 (1999), pp. 205–236.

[Dui80] J. J. Duistermaat. "On global action-angle coordinates". In: *Comm. Pure Appl. Math.* 33.6 (1980), pp. 687–706.

[DH82] J. J. Duistermaat and G. J. Heckman. "On the variation in the cohomology of the symplectic form of the reduced phase space". In: *Invent. Math.* 69.2 (1982), pp. 259–268.

[Edt24] O. Edtmair. "Disk-like surfaces of section and symplectic capacities". In: *Geometric and Functional Analysis* 34.5 (2024), pp. 1399–1459.

[EH21] O. Edtmair and M. Hutchings. *PFH spectral invariants and C^∞ closing lemmas*. Preprint. 2021. arXiv: 2110.02463.

[Eke90] I. Ekeland. *Convexity Methods in Hamiltonian Mechanics*. Vol. 19. Ergebnisse der Mathematik und ihrer Grenzgebiete (3). Berlin: Springer-Verlag, 1990.

[EH87] I. Ekeland and H. Hofer. "Convex Hamiltonian energy surfaces and their periodic trajectories". In: *Communications in Mathematical Physics* 113.3 (1987), pp. 419–469.

[EH89] I. Ekeland and H. Hofer. "Symplectic topology and Hamiltonian dynamics". In: *Math. Z.* 200.3 (1989), pp. 355–378.

[EH90] I. Ekeland and H. Hofer. "Symplectic topology and Hamiltonian dynamics. II". In: *Math. Z.* 203.4 (1990), pp. 553–567.

[EL80] I. Ekeland and J.-M. Lasry. "On the number of periodic trajectories for a Hamiltonian flow on a convex energy surface". In: *Annals of Mathematics* 112.2 (1980), pp. 283–319.

[Ekh+13] T. Ekholm, Y. Eliashberg, E. Murphy, and I. Smith. "Constructing exact Lagrangian immersions with few double points". In: *Geom. Funct. Anal.* 23 (2013), pp. 1772–1803.

[EES05a] T. Ekholm, J. Etnyre, and M. Sullivan. "Non-isotopic Legendrian submanifolds in $\mathbb{R}^{2n+1}$". In: *J. Differential Geom.* 71 (2005), pp. 85–128.

[EES05b] T. Ekholm, J. Etnyre, and M. Sullivan. "The contact homology of Legendrian submanifolds in $\mathbb{R}^{2n+1}$". In: *J. Differential Geom.* 71 (2005), pp. 177–305.

[EES07] T. Ekholm, J. Etnyre, and M. Sullivan. "Legendrian contact homology in $P \times \mathbb{R}$". In: *Trans. Amer. Math. Soc.* 359 (2007), pp. 3301–3335.

[EHK16] T. Ekholm, K. Honda, and T. Kálmán. "Legendrian knots and exact Lagrangian cobordisms". In: *Journal of the European Mathematical Society* 18.11 (2016), pp. 2627–2689.

[Eli79] Y. Eliashberg. *Estimates on the number of fixed points of area preserving transformations*. Preprint. Syktyvkar University, 1979.

[Eli87] Y. Eliashberg. "A theorem on the structure of wave fronts and its applications in symplectic topology". In: *Funct. Anal. Appl.* 21.3 (1987), pp. 227–232. Translated from Funktsional. Anal. i Prilozhen. 21 (1987), no. 3, 65–72.

[Eli89] Y. Eliashberg. "Classification of overtwisted contact structures on 3-manifolds". In: *Invent. Math.* 98 (1989), pp. 623–637.

[Eli90a] Y. Eliashberg. "Filling by holomorphic discs and its applications". In: *Geometry of Low-Dimensional Manifolds, 2 (Durham, 1989)*. Vol. 151. London Math. Soc. Lecture Note Ser. Cambridge Univ. Press, 1990, pp. 45–67.

[Eli90b] Y. Eliashberg. "Topological characterization of Stein manifolds of dimension > 2". In: *International Journal of Mathematics* 1.1 (1990), pp. 29–46.

[Eli92] Y. Eliashberg. "Contact 3-manifolds twenty years since J. Martinet's work". In: *Ann. Inst. Fourier (Grenoble)* 42 (1992), pp. 165–192.

[Eli96] Y. Eliashberg. "Unique holomorphically fillable contact structure on the 3-torus". In: *Int. Math. Res. Not.* 1996.2 (1996), pp. 77–82.

[Eli98] Y. Eliashberg. "Invariants in contact topology". In: *Doc. Math.* Extra Vol. II (Proceedings of the ICM, Berlin 1998) (1998), pp. 327–338.

[Eli15] Y. Eliashberg. "Recent advances in symplectic flexibility". In: *Bull. Amer. Math. Soc. (N.S.)* 52 (2015), pp. 1–26.

[EF09] Y. Eliashberg and M. Fraser. "Topologically trivial Legendrian knots". In: *J. Symplectic Geom.* 7.2 (2009), pp. 77–127.

[EGL20] Y. Eliashberg, S. Ganatra, and O. Lazarev. "Flexible Lagrangians". In: *Int. Math. Res. Not.* 2020.8 (2020), pp. 2408–2435.

[EGH00] Y. Eliashberg, A. Givental, and H. Hofer. "Introduction to symplectic field theory". In: *Geometric and Functional Analysis* Special Volume, Part II (2000). GAFA 2000 (Tel Aviv, 1999), pp. 560–673.

[EG91] Y. Eliashberg and M. Gromov. "Convex symplectic manifolds". In: *Several Complex Variables and Complex Geometry, Part 2 (Santa Cruz, CA, 1989)*. Vol. 52. Proceedings of Symposia in Pure Mathematics. Providence, RI: American Mathematical Society, 1991, pp. 135–162.

[EM02] Y. Eliashberg and N. Mishachev. *Introduction to the h-Principle*. Vol. 48. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2002.

[EM13] Y. Eliashberg and E. Murphy. "Lagrangian caps". In: *Geom. Funct. Anal.* 23 (2013), pp. 1483–1514.

[EP93] Y. Eliashberg and L. Polterovich. "Unknottedness of Lagrangian surfaces in symplectic 4-manifolds". In: *Internat. Math. Res. Notices* 1993.11 (1993), pp. 295–301.

[EP96] Y. Eliashberg and L. Polterovich. "Local Lagrangian 2-knots are trivial". In: *Ann. of Math. (2)* 144.1 (1996), pp. 61–76.

[EP97] Y. Eliashberg and L. Polterovich. "The problem of Lagrangian knots in four-manifolds". In: *Geometric Topology (Athens, GA, 1993)*. Ed. by W. H. Kazez. Vol. 2. AMS/IP Stud. Adv. Math. Providence, RI: American Mathematical Society, 1997, pp. 313–327.

[ET98] Y. Eliashberg and W. P. Thurston. *Confoliations*. Vol. 13. University Lecture Series. Amer. Math. Soc., 1998.

[Ent14] M. Entov. "Quasi-morphisms and quasi-states in symplectic topology". In: *Proceedings of the International Congress of Mathematicians (Seoul 2014), Vol. II*. Seoul: Kyung Moon Sa, 2014, pp. 1147–1171.

[EP03] M. Entov and L. Polterovich. "Calabi quasimorphism and quantum homology". In: *Int. Math. Res. Not.* 2003.30 (2003), pp. 1635–1676.

[EP06] M. Entov and L. Polterovich. "Quasi-states and symplectic intersections". In: *Comment. Math. Helv.* 81.1 (2006), pp. 75–99.

[EP09] M. Entov and L. Polterovich. "Rigid subsets of symplectic manifolds". In: *Compos. Math.* 145.3 (2009), pp. 773–826.

[EP10] M. Entov and L. Polterovich. "C^0-rigidity of Poisson brackets". In: *Symplectic Topology and Measure Preserving Dynamical Systems*. Vol. 512. Contemp. Math. Providence, RI: Amer. Math. Soc., 2010, pp. 25–32.

[Etn03] J. B. Etnyre. "Introductory lectures on contact geometry". In: *Topology and Geometry of Manifolds (Athens, GA, 2001)*. Vol. 71. Proc. Sympos. Pure Math. Amer. Math. Soc., 2003, pp. 81–107.

[Etn05] J. B. Etnyre. "Legendrian and transversal knots". In: *Handbook of Knot Theory*. Elsevier, 2005, pp. 105–185.

[Etn06] J. B. Etnyre. "Lectures on open book decompositions and contact structures". In: *Floer Homology, Gauge Theory, and Low-Dimensional Topology*. Vol. 5. Clay Math. Proc. Amer. Math. Soc., 2006, pp. 103–141.

[EH01a] J. B. Etnyre and K. Honda. "Knots and contact geometry I: torus knots and the figure eight knot". In: *J. Symplectic Geom.* 1.1 (2001), pp. 63–120.

[EH01b] J. B. Etnyre and K. Honda. "On the nonexistence of tight contact structures". In: *Ann. of Math. (2)* 153 (2001), pp. 749–766.

[EH02] J. B. Etnyre and K. Honda. "Tight contact structures with no symplectic fillings". In: *Invent. Math.* 148 (2002), pp. 609–626.

[EN20] J. B. Etnyre and L. L. Ng. "Legendrian contact homology in $\mathbb{R}^3$". In: *Surv. Differ. Geom.* 25 (2020), pp. 103–161.

[Eva10] J. D. Evans. "Lagrangian spheres in del Pezzo surfaces". In: *J. Topol.* 3.1 (2010), pp. 181–227.

[Eva11] J. D. Evans. "Symplectic mapping class groups of some Stein and rational surfaces". In: *J. Symplectic Geom.* 9.1 (2011), pp. 45–82.

[Fan+11] B. Fang, C.-C. M. Liu, D. Treumann, and E. Zaslow. "A categorification of Morelli's theorem". In: *Invent. Math.* 186.1 (2011), pp. 79–114. arXiv: 1007.0053.

[Fat80] A. Fathi. "Structure of the group of homeomorphisms preserving a good measure on a compact manifold". In: *Ann. Sci. École Norm. Sup. (4)* 13.1 (1980), pp. 45–93.

[FW22] B. Filippenko and K. Wehrheim. "A polyfold proof of the Arnold conjecture". In: *Selecta Math. (N.S.)* 28.1 (2022), Paper No. 11.

[FS98] R. Fintushel and R. J. Stern. "Knots, links, and 4-manifolds". In: *Inventiones Mathematicae* 134.2 (1998), pp. 363–400.

[FS09] R. Fintushel and R. J. Stern. "Six lectures on four 4-manifolds". In: *Low Dimensional Topology*. Ed. by T. S. Mrowka and P. S. Ozsváth. Vol. 15. IAS/Park City Mathematics Series. Providence, RI: American Mathematical Society, 2009, pp. 265–315. arXiv: math/0610700.

[FH18] J. W. Fish and H. Hofer. *Lectures on polyfolds and symplectic field theory*. Preprint. 2018. arXiv: 1808.07147.

[Flo86] A. Floer. "Proof of the Arnold conjecture for surfaces and generalizations to certain Kähler manifolds". In: *Duke Math. J.* 53.1 (1986), pp. 1–32.

[Flo88a] A. Floer. "A relative Morse index for the symplectic action". In: *Communications on Pure and Applied Mathematics* 41.4 (1988), pp. 393–407.

[Flo88b] A. Floer. "Morse theory for Lagrangian intersections". In: *J. Differential Geom.* 28.3 (1988), pp. 513–547.

[Flo88c] A. Floer. "The unregularized gradient flow of the symplectic action". In: *Communications on Pure and Applied Mathematics* 41.6 (1988), pp. 775–813.

[Flo89a] A. Floer. "Cuplength estimates on Lagrangian intersections". In: *Comm. Pure Appl. Math.* 42.4 (1989), pp. 335–356.

[Flo89b] A. Floer. "Symplectic fixed points and holomorphic spheres". In: *Comm. Math. Phys.* 120.4 (1989), pp. 575–611.

[FH93] A. Floer and H. Hofer. "Coherent orientations for periodic orbit problems in symplectic geometry". In: *Math. Z.* 212.1 (1993), pp. 13–38.

[FH94] A. Floer and H. Hofer. "Symplectic homology. I. Open sets in $\mathbb{C}^n$". In: *Math. Z.* 215.1 (1994), pp. 37–88.

[FHS95] A. Floer, H. Hofer, and D. Salamon. "Transversality in elliptic Morse theory for the symplectic action". In: *Duke Math. J.* 80.1 (1995), pp. 251–292.

[For85] B. Fortune. "A symplectic fixed point theorem for $\mathbb{CP}^n$". In: *Invent. Math.* 81.1 (1985), pp. 29–46.

[FW85] B. Fortune and A. Weinstein. "A symplectic fixed point theorem for complex projective spaces". In: *Bull. Amer. Math. Soc. (N.S.)* 12.1 (1985), pp. 128–130.

[Fra59] T. Frankel. "Fixed points and torsion on Kähler manifolds". In: *Ann. of Math. (2)* 70 (1959), pp. 1–8.

[Fra92] J. Franks. "Geodesics on S^2 and periodic points of annulus homeomorphisms". In: *Inventiones Mathematicae* 108.2 (1992), pp. 403–418.

[Fra96] J. Franks. "Area preserving homeomorphisms of open surfaces of genus zero". In: *New York Journal of Mathematics* 2 (1996), pp. 1–19.

[FH03] J. Franks and M. Handel. "Periodic points of Hamiltonian surface diffeomorphisms". In: *Geometry & Topology* 7 (2003), pp. 713–756.

[FGS05] U. Frauenfelder, V. L. Ginzburg, and F. Schlenk. "Energy capacity inequalities via an action selector". In: *Geometry, Spectral Theory, Groups, and Dynamics*. Vol. 387. Contemp. Math. Providence, RI: Amer. Math. Soc., 2005, pp. 129–152.

[FM15] D. Frenkel and D. Müller. "Symplectic embeddings of 4-dimensional ellipsoids into cubes". In: *J. Symplectic Geom.* 13.4 (2015), pp. 765–847.

[FV11] S. Friedl and S. Vidussi. "Twisted Alexander polynomials detect fibered 3-manifolds". In: *Annals of Mathematics (2)* 173.3 (2011), pp. 1587–1643.

[Fuk93] K. Fukaya. "Morse homotopy, A^∞-category, and Floer homologies". In: *Proceedings of GARC Workshop on Geometry and Topology '93 (Seoul, 1993)*. Vol. 18. Lecture Notes Series. Seoul: Seoul National University, 1993, pp. 1–102.

[Fuk02] K. Fukaya. "Mirror symmetry of abelian varieties and multi-theta functions". In: *J. Algebraic Geom.* 11.3 (2002), pp. 393–512.

[Fuk06] K. Fukaya. "Application of Floer homology of Lagrangian submanifolds to symplectic topology". In: *Morse Theoretic Methods in Nonlinear Analysis and in Symplectic Topology*. Vol. 217. NATO Sci. Ser. II Math. Phys. Chem. Dordrecht: Springer, 2006, pp. 231–276.

[Fuk13] K. Fukaya. "Floer homology of Lagrangian submanifolds". In: *Sugaku Expositions* 26.1 (2013). Translation of Sūgaku **63** (2011), no. 1, 43–66, pp. 99–127. arXiv: 1106.4882.

[Fuk+09] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. *Lagrangian Intersection Floer Theory: Anomaly and Obstruction. Parts I and II*. Vol. 46. AMS/IP Studies in Advanced Mathematics. Providence, RI: American Mathematical Society and International Press, 2009.

[Fuk+10] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. "Lagrangian Floer theory on compact toric manifolds. I". In: *Duke Math. J.* 151.1 (2010), pp. 23–175.

[Fuk+11] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. "Lagrangian Floer theory on compact toric manifolds II: bulk deformations". In: *Selecta Math. (N.S.)* 17.3 (2011), pp. 609–711.

[Fuk+12] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. *Technical details on Kuranishi structure and virtual fundamental chain*. Preprint. 2012. arXiv: 1209.4410.

[Fuk+17] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. "Antisymplectic involution and Floer cohomology". In: *Geom. Topol.* 21.1 (2017), pp. 1–106.

[Fuk+19] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. *Spectral Invariants with Bulk, Quasimorphisms and Lagrangian Floer Theory*. Vol. 260. Mem. Amer. Math. Soc. 1254. Providence, RI: Amer. Math. Soc., 2019.

[Fuk+20] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono. *Kuranishi Structures and Virtual Fundamental Chains*. Springer Monographs in Mathematics. Singapore: Springer, 2020.

[FO99] K. Fukaya and K. Ono. "Arnold conjecture and Gromov–Witten invariant". In: *Topology* 38.5 (1999), pp. 933–1048.

[FSS08] K. Fukaya, P. Seidel, and I. Smith. "Exact Lagrangian submanifolds in simply-connected cotangent bundles". In: *Invent. Math.* 172.1 (2008), pp. 1–27. arXiv: math/0701783.

[FP97] W. Fulton and R. Pandharipande. "Notes on stable maps and quantum cohomology". In: *Algebraic Geometry—Santa Cruz 1995*. Vol. 62, Part 2. Proceedings of Symposia in Pure Mathematics. Providence, RI: American Mathematical Society, 1997, pp. 45–96.

[GS22] B. Gammage and V. Shende. "Mirror symmetry for very affine hypersurfaces". In: *Acta Math.* 229.2 (2022), pp. 287–346. arXiv: 1707.02959.

[Gan12] S. Ganatra. "Symplectic cohomology and duality for the wrapped Fukaya category". PhD thesis. Cambridge, MA: Massachusetts Institute of Technology, 2012. arXiv: 1304.7312.

[Gan+25] S. Ganatra, A. Hanlon, J. Hicks, D. Pomerleano, and N. Sheridan. "Integrality of mirror maps and arithmetic homological mirror symmetry for Greene–Plesser mirrors". In: *Adv. Math.* 481 (2025), Paper No. 110535. arXiv: 2312.01949.

[GPS20] S. Ganatra, J. Pardon, and V. Shende. "Covariantly functorial wrapped Floer theory on Liouville sectors". In: *Publ. Math. Inst. Hautes Études Sci.* 131 (2020), pp. 73–200.

[GPS24a] S. Ganatra, J. Pardon, and V. Shende. "Microlocal Morse theory of wrapped Fukaya categories". In: *Ann. of Math. (2)* 199.3 (2024), pp. 943–1042. arXiv: 1809.08807.

[GPS24b] S. Ganatra, J. Pardon, and V. Shende. "Sectorial descent for wrapped Fukaya categories". In: *J. Amer. Math. Soc.* 37.2 (2024), pp. 499–635. arXiv: 1809.03427.

[GSW24] H. Gao, L. Shen, and D. Weng. "Augmentations, fillings, and clusters". In: *Geom. Funct. Anal.* 34.3 (2024), pp. 798–867. arXiv: 2008.10793.

[Gei91] H. Geiges. "Contact structures on 1-connected 5-manifolds". In: *Mathematika* 38.2 (1991), pp. 303–311.

[Gei92] H. Geiges. "Symplectic structures on T^2-bundles over T^2". In: *Duke Mathematical Journal* 67.3 (1992), pp. 539–555.

[Gei08] H. Geiges. *An Introduction to Contact Topology*. Vol. 109. Cambridge Studies in Advanced Mathematics. Cambridge Univ. Press, 2008.

[Ghi05] P. Ghiggini. "Strongly fillable contact 3-manifolds without Stein fillings". In: *Geom. Topol.* 9 (2005), pp. 1677–1687.

[Gin95] V. L. Ginzburg. "An embedding $S^{2n-1} \to \mathbb{R}^{2n}$, $2n - 1 \geq 7$, whose Hamiltonian flow has no periodic trajectories". In: *International Mathematics Research Notices* 2 (1995), pp. 83–98.

[Gin97] V. L. Ginzburg. "A smooth counterexample to the Hamiltonian Seifert conjecture in $\mathbb{R}^6$". In: *International Mathematics Research Notices* 13 (1997), pp. 641–650.

[Gin10] V. L. Ginzburg. "The Conley conjecture". In: *Annals of Mathematics* 172.2 (2010), pp. 1127–1180.

[GG03] V. L. Ginzburg and B. Z. Gürel. "A C^2-smooth counterexample to the Hamiltonian Seifert conjecture in $\mathbb{R}^4$". In: *Annals of Mathematics* 158.3 (2003), pp. 953–976.

[GG09] V. L. Ginzburg and B. Z. Gürel. "Action and index spectra and periodic orbits in Hamiltonian dynamics". In: *Geometry & Topology* 13.5 (2009), pp. 2745–2805.

[GG12] V. L. Ginzburg and B. Z. Gürel. "Conley conjecture for negative monotone symplectic manifolds". In: *International Mathematics Research Notices* 8 (2012), pp. 1748–1767.

[GG15] V. L. Ginzburg and B. Z. Gürel. "The Conley conjecture and beyond". In: *Arnold Mathematical Journal* 1.3 (2015), pp. 299–337.

[GG18] V. L. Ginzburg and B. Z. Gürel. "Hamiltonian pseudo-rotations of projective spaces". In: *Inventiones Mathematicae* 214.3 (2018), pp. 1081–1130.

[GG19] V. L. Ginzburg and B. Z. Gürel. "Conley conjecture revisited". In: *International Mathematics Research Notices* 3 (2019), pp. 761–798.

[GG20] V. L. Ginzburg and B. Z. Gürel. "Lusternik–Schnirelmann theory and closed Reeb orbits". In: *Mathematische Zeitschrift* 295.1–2 (2020), pp. 515–582.

[GGM18] V. L. Ginzburg, B. Z. Gürel, and L. Macarini. "Multiplicity of closed Reeb orbits on prequantization bundles". In: *Israel Journal of Mathematics* 228.1 (2018), pp. 407–453.

[GGM24] V. L. Ginzburg, B. Z. Gürel, and M. Mazzucchelli. "Barcode entropy of geodesic flows". In: *Journal of the European Mathematical Society* (2024). Online first. arXiv: 2212.00943.

[Gir00] E. Giroux. “Structures de contact en dimension trois et bifurcations des feuilletages de surfaces”. In: *Invent. Math.* 141 (2000), pp. 615–689.

[Gir02] E. Giroux. “Géométrie de contact: de la dimension trois vers les dimensions supérieures”. In: *Proceedings of the International Congress of Mathematicians, Vol. II (Beijing, 2002)*. Beijing: Higher Ed. Press, 2002, pp. 405–414.

[GP17] E. Giroux and J. Pardon. “Existence of Lefschetz fibrations on Stein and Weinstein domains”. In: *Geom. Topol.* 21 (2017), pp. 963–997.

[Giv96] A. B. Givental. “Equivariant Gromov–Witten invariants”. In: *Internat. Math. Res. Notices* 1996.13 (1996), pp. 613–663. arXiv: `alg-geom/9603021`.

[GKZ20] O. Goertsches, P. Konstantis, and L. Zoller. “GKM theory and Hamiltonian non-Kähler actions in dimension 6”. In: *Adv. Math.* 368 (2020), p. 107141. arXiv: `1903.11684`.

[Gom95] R. E. Gompf. “A new construction of symplectic manifolds”. In: *Ann. of Math. (2)* 142.3 (1995), pp. 527–595.

[Gom98] R. E. Gompf. “Handlebody construction of Stein surfaces”. In: *Ann. of Math. (2)* 148 (1998), pp. 619–693.

[Gom04] R. E. Gompf. “Toward a topological characterization of symplectic manifolds”. In: *Journal of Symplectic Geometry* 2.2 (2004), pp. 177–206.

[GS99] R. E. Gompf and A. I. Stipsicz. *4-Manifolds and Kirby Calculus*. Vol. 20. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 1999.

[Got82] M. J. Gotay. “On coisotropic imbeddings of presymplectic manifolds”. In: *Proc. Amer. Math. Soc.* 84.1 (1982), pp. 111–114.

[Gra59] J. W. Gray. “Some global properties of contact structures”. In: *Ann. of Math. (2)* 69 (1959), pp. 421–450.

[Gro85] M. Gromov. “Pseudo holomorphic curves in symplectic manifolds”. In: *Invent. Math.* 82.2 (1985), pp. 307–347.

[Gro69] M. Gromov. “Stable mappings of foliations into manifolds”. In: *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* 33 (1969), pp. 707–734. English translation: Math. USSR-Izv. **3** (1969), 671–694.

[Gro86] M. Gromov. *Partial Differential Relations*. Vol. 9. Ergebnisse der Mathematik und ihrer Grenzgebiete (3). Berlin: Springer-Verlag, 1986.

[GS11] M. Gross and B. Siebert. “From real affine geometry to complex geometry”. In: *Ann. of Math. (2)* 174.3 (2011), pp. 1301–1428. arXiv: `math/0703822`.

[GS82a] V. Guillemin and S. Sternberg. “Convexity properties of the moment mapping”. In: *Invent. Math.* 67.3 (1982), pp. 491–513.

[GS82b] V. Guillemin and S. Sternberg. “Geometric quantization and multiplicities of group representations”. In: *Invent. Math.* 67.3 (1982), pp. 515–538.

[Gui94] V. Guillemin. *Moment Maps and Combinatorial Invariants of Hamiltonian T^n-spaces*. Vol. 122. Progress in Mathematics. Boston: Birkhäuser, 1994.

[GS89] V. Guillemin and S. Sternberg. “Birational equivalence in the symplectic category”. In: *Inventiones Mathematicae* 97.3 (1989), pp. 485–522.

[Gui23] S. Guillermou. *Sheaves and Symplectic Geometry of Cotangent Bundles*. Vol. 440. Astérisque. Paris: Société Mathématique de France, 2023. arXiv: 1905.07341.

[GKS12] S. Guillermou, M. Kashiwara, and P. Schapira. "Sheaf quantization of Hamiltonian isotopies and applications to nondisplaceability problems". In: *Duke Math. J.* 161.2 (2012), pp. 201–245. arXiv: 1005.1517.

[GV22] S. Guillermou and N. Vichery. *Viterbo's spectral bound conjecture for homogeneous spaces*. Preprint. 2022. arXiv: 2203.13700.

[Gür15] B. Z. Gürel. "Perfect Reeb flows and action-index relations". In: *Geometriae Dedicata* 174 (2015), pp. 105–120.

[Gut08] L. Guth. "Symplectic embeddings of polydisks". In: *Invent. Math.* 172.3 (2008), pp. 477–489.

[Gut14] J. Gutt. "Generalized Conley–Zehnder index". In: *Ann. Fac. Sci. Toulouse Math. (6)* 23.4 (2014), pp. 907–932.

[GH18] J. Gutt and M. Hutchings. "Symplectic capacities from positive S^1-equivariant symplectic homology". In: *Algebr. Geom. Topol.* 18.6 (2018), pp. 3537–3600.

[GHR22] J. Gutt, M. Hutchings, and V. G. B. Ramos. "Examples around the strong Viterbo conjecture". In: *Journal of Fixed Point Theory and Applications* 24.2 (2022), Paper No. 41.

[GR24] J. Gutt and V. G. B. Ramos. *The equivalence of Ekeland–Hofer and equivariant symplectic homology capacities*. Preprint. 2024. arXiv: 2412.09555.

[HK22] P. Hacking and A. Keating. "Homological mirror symmetry for log Calabi–Yau surfaces". In: *Geom. Topol.* 26.8 (2022). With an appendix by Wendelin Lutz, pp. 3747–3833. arXiv: 2005.05010.

[Hai19] P. Haim-Kislev. "On the symplectic size of convex polytopes". In: *Geometric and Functional Analysis* 29.2 (2019), pp. 440–463.

[HO23] P. Haim-Kislev and Y. Ostrover. "Remarks on symplectic capacities of p-products". In: *International Journal of Mathematics* 34.4 (2023), Paper No. 2350021.

[HO26] P. Haim-Kislev and Y. Ostrover. "A counterexample to Viterbo's conjecture". In: *Ann. of Math. (2)* 203.2 (2026), pp. 603–622. arXiv: 2405.16513.

[Hau15] L. Haug. "The Lagrangian cobordism group of T^2". In: *Selecta Math. (N.S.)* 21.3 (2015), pp. 1021–1069. arXiv: 1310.8056.

[Hei12] D. Hein. "The Conley conjecture for irrational symplectic manifolds". In: *Journal of Symplectic Geometry* 10.2 (2012), pp. 183–202.

[Her99] M.-R. Herman. "Examples of compact hypersurfaces in $\mathbb{R}^{2p}$, $2p \geq 6$, with no periodic orbits". In: *Hamiltonian Systems with Three or More Degrees of Freedom*. Ed. by C. Simó. Vol. 533. NATO Advanced Science Institutes Series C: Mathematical and Physical Sciences. Dordrecht: Kluwer Academic Publishers, 1999.

[Hin04] R. Hind. "Lagrangian spheres in $S^2 \times S^2$". In: *Geom. Funct. Anal.* 14.2 (2004), pp. 303–318.

[Hin12] R. Hind. "Lagrangian unknottedness in Stein surfaces". In: *Asian J. Math.* 16.1 (2012), pp. 1–36.

[Hin15] R. Hind. "Some optimal embeddings of symplectic ellipsoids". In: *J. Topol.* 8.3 (2015), pp. 871–883.

[HK14] R. Hind and E. Kerman. "New obstructions to symplectic embeddings". In: *Invent. Math.* 196.2 (2014), pp. 383–452. Correction: Invent. Math. 214 (2018), no. 2, 1023–1029.

[HPW16] R. Hind, M. Pinsonnault, and W. Wu. "Symplectormophism groups of non-compact manifolds, orbifold balls, and a space of Lagrangians". In: *J. Symplectic Geom.* 14.1 (2016), pp. 203–226. arXiv: 1305.7291.

[HS24a] R. Hind and K. Siegel. *Symplectic field theory: an overview*. Preprint. 2024. arXiv: 2410.19936.

[Hin09] N. Hingston. "Subharmonic solutions of Hamiltonian equations on tori". In: *Annals of Mathematics* 170.2 (2009), pp. 529–560.

[HS24b] A. Hirschi and M. Swaminathan. "Global Kuranishi charts and a product formula in symplectic Gromov–Witten theory". In: *Selecta Mathematica (N.S.)* 30.5 (2024), Paper No. 87.

[Hof85] H. Hofer. "Lagrangian embeddings and critical point theory". In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 2.6 (1985), pp. 407–462.

[Hof88] H. Hofer. "Lusternik–Schnirelman-theory for Lagrangian intersections". In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 5.5 (1988), pp. 465–499.

[Hof90] H. Hofer. "On the topological properties of symplectic maps". In: *Proc. Roy. Soc. Edinburgh Sect. A* 115.1-2 (1990), pp. 25–38.

[Hof93a] H. Hofer. "Estimates for the energy of a symplectic map". In: *Comment. Math. Helv.* 68.1 (1993), pp. 48–72.

[Hof93b] H. Hofer. "Pseudoholomorphic curves in symplectizations with applications to the Weinstein conjecture in dimension three". In: *Invent. Math.* 114 (1993), pp. 515–563.

[HLS97] H. Hofer, V. Lizan, and J.-C. Sikorav. "On genericity for holomorphic curves in four-dimensional almost-complex manifolds". In: *Journal of Geometric Analysis* 7.1 (1997), pp. 149–159.

[HS95] H. Hofer and D. Salamon. "Floer homology and Novikov rings". In: *The Floer Memorial Volume*. Vol. 133. Progress in Mathematics. Basel: Birkhäuser, 1995, pp. 483–524.

[HV92] H. Hofer and C. Viterbo. "The Weinstein conjecture in the presence of holomorphic spheres". In: *Communications on Pure and Applied Mathematics* 45.5 (1992), pp. 583–622.

[HWZ96] H. Hofer, K. Wysocki, and E. Zehnder. "Properties of pseudoholomorphic curves in symplectisations. I. Asymptotics". In: *Annales de l'Institut Henri Poincaré C, Analyse Non Linéaire* 13.3 (1996), pp. 337–379.

[HWZ98] H. Hofer, K. Wysocki, and E. Zehnder. "The dynamics on three-dimensional strictly convex energy surfaces". In: *Annals of Mathematics* 148.1 (1998), pp. 197–289.

[HWZ03] H. Hofer, K. Wysocki, and E. Zehnder. "Finite energy foliations of tight three-spheres and Hamiltonian dynamics". In: *Annals of Mathematics* 157.1 (2003), pp. 125–255.

[HWZ17] H. Hofer, K. Wysocki, and E. Zehnder. "Applications of polyfold theory I: the polyfolds of Gromov–Witten theory". In: *Memoirs of the American Mathematical Society* 248.1179 (2017).

[HWZ21] H. Hofer, K. Wysocki, and E. Zehnder. *Polyfold and Fredholm Theory*. Vol. 72. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Cham: Springer, 2021.

[HZ87] H. Hofer and E. Zehnder. "Periodic solutions on hypersurfaces and a result by C. Viterbo". In: *Inventiones Mathematicae* 90.1 (1987), pp. 1–9.

[HZ90] H. Hofer and E. Zehnder. "A new capacity for symplectic manifolds". In: *Analysis, et cetera*. Ed. by P. H. Rabinowitz and E. Zehnder. Boston, MA: Academic Press, 1990, pp. 405–427.

[HZ94] H. Hofer and E. Zehnder. *Symplectic Invariants and Hamiltonian Dynamics*. Birkhäuser Advanced Texts. Basel: Birkhäuser, 1994.

[Hon00] K. Honda. "On the classification of tight contact structures I". In: *Geom. Topol.* 4 (2000), pp. 309–368.

[HKM02] K. Honda, W. H. Kazez, and G. Matić. "Convex decomposition theory". In: *Int. Math. Res. Not.* 2002.2 (2002), pp. 55–88.

[Hör71] L. Hörmander. "Fourier integral operators. I". In: *Acta Mathematica* 127 (1971), pp. 79–183.

[HLS15] V. Humilière, R. Leclercq, and S. Seyfaddini. "Coisotropic rigidity and C^0-symplectic geometry". In: *Duke Math. J.* 164.4 (2015), pp. 767–799.

[Hum97] C. Hummel. *Gromov's Compactness Theorem for Pseudo-holomorphic Curves*. Vol. 151. Progress in Mathematics. Basel: Birkhäuser, 1997.

[Hut02] M. Hutchings. "An index inequality for embedded pseudoholomorphic curves in symplectizations". In: *Journal of the European Mathematical Society* 4.4 (2002), pp. 313–361.

[Hut11a] M. Hutchings. "Quantitative embedded contact homology". In: *J. Differential Geom.* 88.2 (2011), pp. 231–266.

[Hut11b] M. Hutchings. "Recent progress on symplectic embedding problems in four dimensions". In: *Proc. Natl. Acad. Sci. USA* 108.20 (2011), pp. 8093–8099.

[Hut14] M. Hutchings. "Lecture notes on embedded contact homology". In: *Contact and Symplectic Topology*. Ed. by F. Bourgeois, V. Colin, and A. Stipsicz. Vol. 26. Bolyai Society Mathematical Studies. Cham: Springer, 2014, pp. 389–484. arXiv: 1303.5789.

[Hut16] M. Hutchings. "Beyond ECH capacities". In: *Geom. Topol.* 20.2 (2016), pp. 1085–1126.

[HN16] M. Hutchings and J. Nelson. "Cylindrical contact homology for dynamically convex contact forms in three dimensions". In: *Journal of Symplectic Geometry* 14.4 (2016), pp. 983–1012.

[HN22] M. Hutchings and J. Nelson. "S^1-equivariant contact homology for hypertight contact forms". In: *Journal of Topology* 15.3 (2022), pp. 1455–1539.

[HT07] M. Hutchings and C. H. Taubes. "Gluing pseudoholomorphic curves along branched covered cylinders. I". In: *Journal of Symplectic Geometry* 5.1 (2007), pp. 43–137.

[HT09a] M. Hutchings and C. H. Taubes. "Gluing pseudoholomorphic curves along branched covered cylinders. II". In: *Journal of Symplectic Geometry* 7.1 (2009), pp. 29–133.

[HT09b] M. Hutchings and C. H. Taubes. "The Weinstein conjecture for stable Hamiltonian structures". In: *Geometry & Topology* 13 (2009), pp. 901–941.

[HT11] M. Hutchings and C. H. Taubes. "Proof of the Arnold chord conjecture in three dimensions, I". In: *Mathematical Research Letters* 18.2 (2011), pp. 295–313.

[HT13] M. Hutchings and C. H. Taubes. "Proof of the Arnold chord conjecture in three dimensions, II". In: *Geometry & Topology* 17.5 (2013), pp. 2601–2688.

[IP04] E.-N. Ionel and T. H. Parker. "The symplectic sum formula for Gromov–Witten invariants". In: *Annals of Mathematics (2)* 159.3 (2004), pp. 935–1025.

[IP18] E.-N. Ionel and T. H. Parker. "The Gopakumar–Vafa formula for symplectic manifolds". In: *Annals of Mathematics (2)* 187.1 (2018), pp. 1–64.

[IP19] E.-N. Ionel and T. H. Parker. "Thin compactifications and relative fundamental classes". In: *Journal of Symplectic Geometry* 17.3 (2019), pp. 703–752.

[Iri15] K. Irie. "Dense existence of periodic Reeb orbits and ECH spectral invariants". In: *Journal of Modern Dynamics* 9 (2015), pp. 357–363.

[Iri21] K. Irie. "Equidistributed periodic orbits of C^∞-generic three-dimensional Reeb flows". In: *Journal of Symplectic Geometry* 19.3 (2021), pp. 531–566.

[Iri22] K. Irie. "Symplectic homology of fiberwise convex sets and homology of loop spaces". In: *Journal of Symplectic Geometry* 20.2 (2022), pp. 417–470.

[IS20] H. Iriyeh and M. Shibata. "Symmetric Mahler's conjecture for the volume product in the 3-dimensional case". In: *Duke Mathematical Journal* 169.6 (2020), pp. 1077–1134.

[Ish18] S. Ishikawa. *Construction of general symplectic field theory*. Preprint. 2018. arXiv: 1807.09455.

[Kan97] Y. Kanda. "The classification of tight contact structures on the 3-torus". In: *Comm. Anal. Geom.* 5.3 (1997), pp. 413–438.

[Kar99] Y. Karshon. *Periodic Hamiltonian Flows on Four Dimensional Manifolds*. Vol. 141. Memoirs of the American Mathematical Society 672. Providence, RI: American Mathematical Society, 1999.

[KT03] Y. Karshon and S. Tolman. "Complete invariants for Hamiltonian torus actions with two dimensional quotients". In: *J. Symplectic Geom.* 2.1 (2003), pp. 25–82.

[KT14] Y. Karshon and S. Tolman. "Classification of Hamiltonian torus actions with two-dimensional quotients". In: *Geom. Topol.* 18.2 (2014), pp. 669–716.

[Kas80] A. Kas. "On the handlebody decomposition associated to a Lefschetz fibration". In: *Pacific Journal of Mathematics* 89.1 (1980), pp. 89–104.

[KS90] M. Kashiwara and P. Schapira. *Sheaves on Manifolds*. Vol. 292. Grundlehren der Mathematischen Wissenschaften. Berlin: Springer-Verlag, 1990.

[Kaw22] Y. Kawamoto. "On C^0-continuity of the spectral norm for symplectically non-aspherical manifolds". In: *Int. Math. Res. Not.* 2022.21 (2022), pp. 17187–17230.

[KS02] M. Khovanov and P. Seidel. "Quivers, Floer cohomology, and braid group actions". In: *J. Amer. Math. Soc.* 15.1 (2002), pp. 203–271.

[Kir84a] F. Kirwan. "Convexity properties of the moment mapping. III". In: *Invent. Math.* 77.3 (1984), pp. 547–552.

[Kir84b] F. C. Kirwan. *Cohomology of Quotients in Symplectic and Algebraic Geometry*. Vol. 31. Mathematical Notes. Princeton, NJ: Princeton University Press, 1984.

[KS21] A. Kislev and E. Shelukhin. "Bounds on spectral norms and barcodes". In: *Geom. Topol.* 25.7 (2021), pp. 3257–3350.

[Kod64] K. Kodaira. "On the structure of compact complex analytic surfaces. I". In: *Amer. J. Math.* 86 (1964), pp. 751–798.

[Kon95a] M. Kontsevich. "Enumeration of rational curves via torus actions". In: *The Moduli Space of Curves (Texel Island, 1994)*. Ed. by R. Dijkgraaf, C. Faber, and G. van der Geer. Vol. 129. Progress in Mathematics. Boston, MA: Birkhäuser, 1995, pp. 335–368.

[Kon95b] M. Kontsevich. "Homological algebra of mirror symmetry". In: *Proceedings of the International Congress of Mathematicians (Zürich, 1994), Vol. I*. Basel: Birkhäuser, 1995, pp. 120–139. arXiv: `alg-geom/9411018`.

[KM94a] M. Kontsevich and Y. Manin. "Gromov–Witten classes, quantum cohomology, and enumerative geometry". In: *Communications in Mathematical Physics* 164.3 (1994), pp. 525–562.

[KS06] M. Kontsevich and Y. Soibelman. "Affine structures and non-Archimedean analytic spaces". In: *The Unity of Mathematics*. Vol. 244. Progr. Math. Boston, MA: Birkhäuser, 2006, pp. 321–385. arXiv: `math/0406564`.

[Kra13] T. Kragh. "Parametrized ring-spectra and the nearby Lagrangian conjecture". In: *Geom. Topol.* 17.2 (2013). With an appendix by Mohammed Abouzaid, pp. 639–731. arXiv: `1107.4674`.

[Kra18] T. Kragh. "The Viterbo transfer as a map of spectra". In: *J. Symplectic Geom.* 16.1 (2018), pp. 85–226. arXiv: `0712.2533`.

[KM07] P. Kronheimer and T. Mrowka. *Monopoles and Three-Manifolds*. Vol. 10. New Mathematical Monographs. Cambridge: Cambridge University Press, 2007.

[KM94b] P. B. Kronheimer and T. S. Mrowka. "The genus of embedded surfaces in the projective plane". In: *Mathematical Research Letters* 1.6 (1994), pp. 797–808.

[KM97] P. B. Kronheimer and T. S. Mrowka. "Monopoles and contact structures". In: *Invent. Math.* 130 (1997), pp. 209–255.

[KLT20] Ç. Kutluhan, Y.-J. Lee, and C. H. Taubes. "HF=HM I–V". In: *Geometry & Topology* 24.6–7 (2020), pp. 2829–3748.

[Kuw20] T. Kuwagaki. "The nonequivariant coherent-constructible correspondence for toric stacks". In: *Duke Math. J.* 169.11 (2020), pp. 2125–2197. arXiv: `1610.03214`.

[KO00] D. Kwon and Y.-G. Oh. "Structure of the image of (pseudo)-holomorphic discs with totally real boundary condition". In: *Communications in Analysis and Geometry* 8.1 (2000), pp. 31–82.

[LM95a] F. Lalonde and D. McDuff. "Hofer's L^∞-geometry: energy and stability of Hamiltonian flows. I, II". In: *Invent. Math.* 122.1 (1995), pp. 1–33, 35–69.

[LM95b] F. Lalonde and D. McDuff. "The geometry of symplectic energy". In: *Ann. of Math. (2)* 141.2 (1995), pp. 349–371.

[LM96a] F. Lalonde and D. McDuff. "J-curves and the classification of rational and ruled symplectic 4-manifolds". In: *Contact and Symplectic Geometry (Cambridge, 1994)*. Ed. by C. B. Thomas. Vol. 8. Publications of the Newton Institute. Cambridge: Cambridge University Press, 1996, pp. 3–42.

[LM96b] F. Lalonde and D. McDuff. "The classification of ruled symplectic 4-manifolds". In: *Mathematical Research Letters* 3.6 (1996), pp. 769–778.

[LMP98] F. Lalonde, D. McDuff, and L. Polterovich. "On the flux conjectures". In: *Geometry, Topology, and Dynamics (Montreal, 1995)*. Vol. 15. CRM Proceedings and Lecture Notes. Providence, RI: American Mathematical Society, 1998, pp. 69–85. arXiv: dg-ga/9706015.

[LMP99] F. Lalonde, D. McDuff, and L. Polterovich. "Topological rigidity of Hamiltonian loops and quantum homology". In: *Inventiones Mathematicae* 135.2 (1999), pp. 369–385.

[LP04] F. Lalonde and M. Pinsonnault. "The topology of the space of symplectic balls in rational 4-manifolds". In: *Duke Math. J.* 122.2 (2004), pp. 347–397.

[Lar21] T. Large. "Spectral Fukaya categories for Liouville manifolds". PhD thesis. Massachusetts Institute of Technology, 2021.

[LS85] F. Laudenbach and J.-C. Sikorav. "Persistance d'intersection avec la section nulle au cours d'une isotopie hamiltonienne dans un fibré cotangent". In: *Invent. Math.* 82.2 (1985), pp. 349–357.

[LS94] F. Laudenbach and J.-C. Sikorav. "Hamiltonian disjunction and limits of Lagrangian submanifolds". In: *Int. Math. Res. Not.* 1994.4 (1994), pp. 161–168.

[Laz20a] O. Lazarev. "Contact manifolds with flexible fillings". In: *Geom. Funct. Anal.* 30 (2020), pp. 188–254.

[Laz20b] O. Lazarev. "Maximal contact and symplectic structures". In: *J. Topol.* 13 (2020), pp. 1058–1083.

[Laz22] O. Lazarev. "Symplectic flexibility and the Grothendieck group of the Fukaya category". In: *J. Topol.* 15.1 (2022). arXiv title: Geometric and algebraic presentations of Weinstein domains, pp. 204–237. arXiv: 1910.01101.

[LS23] O. Lazarev and Z. Sylvan. "Prime-localized Weinstein subdomains". In: *Geom. Topol.* 27.2 (2023), pp. 699–737.

[Laz00] L. Lazzarini. "Existence of a somewhere injective pseudo-holomorphic disc". In: *Geometric and Functional Analysis* 10.4 (2000), pp. 829–862.

[Le 06] P. Le Calvez. "Periodic orbits of Hamiltonian homeomorphisms of surfaces". In: *Duke Mathematical Journal* 133.1 (2006), pp. 125–184.

[LS22] F. Le Roux and S. Seyfaddini. "The Anosov–Katok method and pseudo-rotations in symplectic dynamics". In: *Journal of Fixed Point Theory and Applications* 24.2 (2022), Paper No. 36.

[LT12] Y.-J. Lee and C. H. Taubes. "Periodic Floer homology and Seiberg–Witten–Floer cohomology". In: *Journal of Symplectic Geometry* 10.1 (2012), pp. 81–164.

[Lee76] J. A. Lees. "On the classification of Lagrange immersions". In: *Duke Math. J.* 43 (1976), pp. 217–224.

[Ler95] E. Lerman. "Symplectic cuts". In: *Math. Res. Lett.* 2.3 (1995), pp. 247–258.

[LS10] N. C. Leung and M. Symington. "Almost toric symplectic four-manifolds". In: *J. Symplectic Geom.* 8.2 (2010), pp. 143–187. arXiv: `math/0312165`.

[LLW22] J. Li, T.-J. Li, and W. Wu. *Symplectic Torelli groups of rational surfaces*. Preprint. 2022. arXiv: `2212.01873`.

[LT98a] J. Li and G. Tian. "Virtual moduli cycles and Gromov–Witten invariants of algebraic varieties". In: *Journal of the American Mathematical Society* 11.1 (1998), pp. 119–174.

[LT98b] J. Li and G. Tian. "Virtual moduli cycles and Gromov–Witten invariants of general symplectic manifolds". In: *Topics in Symplectic 4-Manifolds (Irvine, CA, 1996)*. Ed. by R. J. Stern. Vol. I. First International Press Lecture Series. Cambridge, MA: International Press, 1998, pp. 47–83.

[LT99] J. Li and G. Tian. "Comparison of algebraic and symplectic Gromov–Witten invariants". In: *Asian Journal of Mathematics* 3.3 (1999), pp. 689–728.

[LR01] A.-M. Li and Y. Ruan. "Symplectic surgery and Gromov–Witten invariants of Calabi–Yau 3-folds". In: *Inventiones Mathematicae* 145.1 (2001), pp. 151–218.

[Li06a] T.-J. Li. "Quaternionic bundles and Betti numbers of symplectic 4-manifolds with Kodaira dimension zero". In: *International Mathematics Research Notices*, 37385 (2006), 28 pp.

[Li06b] T.-J. Li. "Symplectic 4-manifolds with Kodaira dimension zero". In: *Journal of Differential Geometry* 74.2 (2006), pp. 321–352.

[Li06c] T.-J. Li. "The Kodaira dimension of symplectic 4-manifolds". In: *Floer Homology, Gauge Theory, and Low-Dimensional Topology*. Vol. 5. Clay Mathematics Proceedings. Providence, RI: American Mathematical Society, 2006, pp. 249–261.

[Li19] T.-J. Li. "Kodaira dimension in low dimensional topology". In: *Tsinghua Lectures in Mathematics*. Ed. by L. Ji et al. Vol. 45. Advanced Lectures in Mathematics. Survey; arXiv version 2015. Somerville, MA and Beijing: International Press and Higher Education Press, 2019, pp. 265–291. arXiv: `1511.04831`.

[LL95a] T.-J. Li and A.-K. Liu. "General wall crossing formula". In: *Mathematical Research Letters* 2.6 (1995), pp. 797–810.

[LL95b] T.-J. Li and A.-K. Liu. "Symplectic structure on ruled surfaces and a generalized adjunction formula". In: *Mathematical Research Letters* 2.4 (1995), pp. 453–471.

[LL01] T.-J. Li and A.-K. Liu. "Uniqueness of symplectic canonical class, surface cone and symplectic cone of 4-manifolds with $b^+ = 1$". In: *Journal of Differential Geometry* 58.2 (2001), pp. 331–370.

[LLY97] B. H. Lian, K. Liu, and S.-T. Yau. "Mirror principle. I". In: *Asian J. Math.* 1.4 (1997), pp. 729–763. arXiv: `alg-geom/9712011`.

[LSV24] J. Licata, M. Scharitzer, and V. Vértesi. *The Giroux correspondence in dimension 3*. Preprint. 2024. arXiv: `2408.01079`.

[LV80] G. Lion and M. Vergne. *The Weil Representation, Maslov Index and Theta Series*. Vol. 6. Progress in Mathematics. Boston: Birkhäuser, 1980.

[LM97] P. Lisca and G. Matić. "Tight contact structures and Seiberg–Witten invariants". In: *Invent. Math.* 129 (1997), pp. 509–525.

[LS09] P. Lisca and A. I. Stipsicz. "On the existence of tight contact structures on Seifert fibered 3-manifolds". In: *Duke Math. J.* 148.2 (2009), pp. 175–209.

[LT98c] G. Liu and G. Tian. "Floer homology and Arnold conjecture". In: *J. Differential Geom.* 49.1 (1998), pp. 1–74.

[LT00] G. Liu and G. Tian. "Weinstein conjecture and GW-invariants". In: *Communications in Contemporary Mathematics* 2 (2000), pp. 405–459.

[Liu96] A.-K. Liu. "Some new applications of general wall crossing formula, Gompf's conjecture and its applications". In: *Mathematical Research Letters* 3.5 (1996), pp. 569–585.

[LP01] A. Loi and R. Piergallini. "Compact Stein surfaces with boundary as branched covers of B^4". In: *Inventiones Mathematicae* 143.2 (2001), pp. 325–348.

[Lon02] Y. Long. *Index Theory for Symplectic Paths with Applications*. Vol. 207. Progress in Mathematics. Basel: Birkhäuser, 2002.

[LZ02] Y. Long and C. Zhu. "Closed characteristics on compact convex hypersurfaces in $\mathbb{R}^{2n}$". In: *Annals of Mathematics* 155.2 (2002), pp. 317–368.

[Lu06] G. Lu. "Gromov–Witten invariants and pseudo symplectic capacities". In: *Israel Journal of Mathematics* 156 (2006), pp. 1–63.

[Lut95] K. M. Luttinger. "Lagrangian tori in $\mathbb{R}^4$". In: *Journal of Differential Geometry* 42.2 (1995), pp. 220–228.

[MM23] N. Magill and D. McDuff. "Staircase symmetries in Hirzebruch surfaces". In: *Algebr. Geom. Topol.* 23.9 (2023), pp. 4235–4307.

[MMW24] N. Magill, D. McDuff, and M. Weiler. "Staircase patterns in Hirzebruch surfaces". In: *Comment. Math. Helv.* 99.3 (2024), pp. 437–555.

[MPW25] N. Magill, A. R. Pires, and M. Weiler. "A classification of infinite staircases for Hirzebruch surfaces". In: *J. Topol.* 18.1 (2025), e70017.

[Mah39a] K. Mahler. "Ein Minimalproblem für konvexe Polygone". In: *Mathematica (Zutphen) B* 7 (1939), pp. 118–127.

[Mah39b] K. Mahler. "Ein Übertragungsprinzip für konvexe Körper". In: *Časopis pro Pěstování Matematiky a Fysiky* 68 (1939), pp. 93–102.

[MSV24] C. Y. Mak, Y. Sun, and U. Varolgunes. "A characterization of heaviness in terms of relative symplectic cohomology". In: *J. Topol.* 17.1 (2024), Paper No. e12327.

[MW74] J. Marsden and A. Weinstein. "Reduction of symplectic manifolds with symmetry". In: *Rep. Math. Phys.* 5.1 (1974), pp. 121–130.

[Mar71] J. Martinet. "Formes de contact sur les variétés de dimension 3". In: *Proceedings of Liverpool Singularities Symposium II*. Vol. 209. Lecture Notes in Math. Springer, 1971, pp. 142–163.

[Mas65] V. P. Maslov. *Theory of Perturbations and Asymptotic Methods*. In Russian. Moscow: Izdatel'stvo Moskovskogo Universiteta, 1965. French translation: *Théorie des perturbations et méthodes asymptotiques*, Dunod, Paris, 1972.

[MNW13] P. Massot, K. Niederkrüger, and C. Wendl. "Weak and strong fillability of higher dimensional contact manifolds". In: *Invent. Math.* 192 (2013), pp. 287–373.

[Mat00] S. Matsumoto. "Arnold conjecture for surface homeomorphisms". In: *Topology Appl.* 104.1–3 (2000). Proceedings of the French–Japanese Conference "Hyperspace Topologies and Applications" (La Bussière, 1997), pp. 191–214.

[Mau+06] D. Maulik, N. Nekrasov, A. Okounkov, and R. Pandharipande. "Gromov–Witten theory and Donaldson–Thomas theory, I". In: *Compositio Mathematica* 142.5 (2006), pp. 1263–1285.

[MW94] J. D. McCarthy and J. G. Wolfson. "Symplectic normal connect sum". In: *Topology* 33.4 (1994), pp. 729–764.

[McD84] D. McDuff. "Examples of simply-connected symplectic non-Kählerian manifolds". In: *J. Differential Geom.* 20.1 (1984), pp. 267–277.

[McD87] D. McDuff. "Examples of symplectic structures". In: *Invent. Math.* 89.1 (1987), pp. 13–36.

[McD88] D. McDuff. "The moment map for circle actions on symplectic manifolds". In: *J. Geom. Phys.* 5.2 (1988), pp. 149–160.

[McD90] D. McDuff. "The structure of rational and ruled symplectic 4-manifolds". In: *Journal of the American Mathematical Society* 3.3 (1990). Erratum: J. Amer. Math. Soc. 5 (1992), 987–988, pp. 679–712.

[McD91a] D. McDuff. "Blow ups and symplectic embeddings in dimension 4". In: *Topology* 30.3 (1991), pp. 409–421.

[McD91b] D. McDuff. "Symplectic manifolds with contact type boundaries". In: *Inventiones Mathematicae* 103 (1991), pp. 651–671.

[McD91c] D. McDuff. "The local behaviour of holomorphic curves in almost complex 4-manifolds". In: *Journal of Differential Geometry* 34.1 (1991), pp. 143–164.

[McD93] D. McDuff. "Remarks on the uniqueness of symplectic blowing up". In: *Symplectic Geometry*. Ed. by D. Salamon. Vol. 192. London Mathematical Society Lecture Note Series. Cambridge: Cambridge University Press, 1993, pp. 157–168.

[McD94] D. McDuff. "Singularities and positivity of intersections of J-holomorphic curves". In: *Holomorphic Curves in Symplectic Geometry*. Ed. by M. Audin and J. Lafontaine. Vol. 117. Progress in Mathematics. Basel: Birkhäuser, 1994, pp. 191–215.

[McD97] D. McDuff. "Lectures on Gromov invariants for symplectic 4-manifolds". In: *Gauge Theory and Symplectic Geometry (Montreal, 1995)*. Ed. by J. Hurtubise and F. Lalonde. Vol. 488. NATO ASI Series C: Mathematical and Physical Sciences. Dordrecht: Kluwer, 1997, pp. 175–210.

[McD98] D. McDuff. "From symplectic deformation to isotopy". In: *Topics in Symplectic 4-Manifolds (Irvine, CA, 1996)*. Ed. by R. J. Stern. Vol. 1. First International Press Lecture Series. Cambridge, MA: International Press, 1998, pp. 85–99. arXiv: dg-ga/9606004.

[McD00] D. McDuff. "Quantum homology of fibrations over S^2". In: *International Journal of Mathematics* 11.5 (2000), pp. 665–721.

[McD04] D. McDuff. "Lectures on groups of symplectomorphisms". In: *Rend. Circ. Mat. Palermo (2) Suppl.* 72 (2004), pp. 43–78.

[McD09a] D. McDuff. "Hamiltonian S^1-manifolds are uniruled". In: *Duke Math. J.* 146.3 (2009), pp. 449–507.

[McD09b] D. McDuff. "Some 6-dimensional Hamiltonian S^1-manifolds". In: *J. Topol.* 2.3 (2009), pp. 589–623. arXiv: 0808.3549.

[McD09c] D. McDuff. "Symplectic embeddings and continued fractions: a survey". In: *Jpn. J. Math.* 4.2 (2009), pp. 121–139.

[McD09d] D. McDuff. "Symplectic embeddings of 4-dimensional ellipsoids". In: *J. Topol.* 2.1 (2009), pp. 1–22.

[McD10] D. McDuff. "Monodromy in Hamiltonian Floer theory". In: *Comment. Math. Helv.* 85.1 (2010), pp. 95–133.

[McD11a] D. McDuff. "The Hofer conjecture on embedding symplectic ellipsoids". In: *J. Differential Geom.* 88.3 (2011), pp. 519–532.

[McD11b] D. McDuff. "The topology of toric symplectic manifolds". In: *Geom. Topol.* 15.1 (2011), pp. 145–190.

[MP94] D. McDuff and L. Polterovich. "Symplectic packings and algebraic geometry". In: *Invent. Math.* 115.3 (1994), pp. 405–429.

[MS12a] D. McDuff and D. Salamon. *J-holomorphic Curves and Symplectic Topology*. 2nd ed. Vol. 52. American Mathematical Society Colloquium Publications. Providence, RI: American Mathematical Society, 2012.

[MS17] D. McDuff and D. Salamon. *Introduction to Symplectic Topology*. 3rd ed. Oxford Graduate Texts in Mathematics. Oxford: Oxford University Press, 2017.

[MS12b] D. McDuff and F. Schlenk. "The embedding capacity of 4-dimensional symplectic ellipsoids". In: *Ann. of Math. (2)* 175.3 (2012), pp. 1191–1282.

[MS24] D. McDuff and K. Siegel. "Symplectic capacities, unperturbed curves and convex toric domains". In: *Geom. Topol.* 28.3 (2024), pp. 1213–1285.

[MT06] D. McDuff and S. Tolman. "Topological properties of Hamiltonian circle actions". In: *International Mathematics Research Papers* 2006.4 (2006), Art. ID 72826, 77 pp.

[MW17a] D. McDuff and K. Wehrheim. "Smooth Kuranishi atlases with isotropy". In: *Geometry & Topology* 21.5 (2017), pp. 2725–2809.

[MW17b] D. McDuff and K. Wehrheim. "The topology of Kuranishi atlases". In: *Proceedings of the London Mathematical Society (3)* 115.2 (2017), pp. 221–292.

[MW18] D. McDuff and K. Wehrheim. "The fundamental class of smooth Kuranishi atlases with trivial isotropy". In: *Journal of Topology and Analysis* 10.1 (2018), pp. 71–243.

[McL09] M. McLean. "Lefschetz fibrations and symplectic homology". In: *Geometry & Topology* 13.4 (2009), pp. 1877–1944.

[MT99] C. T. McMullen and C. H. Taubes. "4-manifolds with inequivalent symplectic forms and 3-manifolds with inequivalent fibrations". In: *Mathematical Research Letters* 6.5-6 (1999), pp. 681–696.

[Mei98] E. Meinrenken. "Symplectic surgery and the Spin^c-Dirac operator". In: *Adv. Math.* 134.2 (1998), pp. 240–277.

[Mey73] K. R. Meyer. "Symmetries and integrals in mechanics". In: *Dynamical Systems (Proc. Sympos., Univ. Bahia, Salvador, 1971)*. Ed. by M. M. Peixoto. New York: Academic Press, 1973, pp. 259–272.

[MW95] M. J. Micallef and B. White. "The structure of branch points in minimal surfaces and in pseudoholomorphic curves". In: *Annals of Mathematics (2)* 141.1 (1995), pp. 35–85.

[MST96] J. W. Morgan, Z. Szabó, and C. H. Taubes. "A product formula for the Seiberg–Witten invariants and the generalized Thom conjecture". In: *J. Differential Geom.* 44.4 (1996), pp. 706–788.

[Mos65] J. Moser. "On the volume elements on a manifold". In: *Trans. Amer. Math. Soc.* 120 (1965), pp. 286–294.

[Mur12] E. Murphy. *Loose Legendrian embeddings in high dimensional contact manifolds*. Preprint. 2012. arXiv: 1201.2245.

[MS18] E. Murphy and K. Siegel. "Subflexible symplectic manifolds". In: *Geom. Topol.* 22 (2018), pp. 2367–2401.

[Nad09] D. Nadler. "Microlocal branes are constructible sheaves". In: *Selecta Math. (N.S.)* 15.4 (2009), pp. 563–619. arXiv: math/0612399.

[NS20] D. Nadler and V. Shende. *Sheaf quantization in Weinstein symplectic manifolds*. Preprint; arXiv v4 (2026) is marked as the accepted version. 2020. arXiv: 2007.10154.

[NZ09] D. Nadler and E. Zaslow. "Constructible sheaves and the Fukaya category". In: *J. Amer. Math. Soc.* 22.1 (2009), pp. 233–286. arXiv: math/0604379.

[Nem09] S. Nemirovski. "Lagrangian Klein bottles in $\mathbb{R}^{2n}$". In: *Geom. Funct. Anal.* 19.3 (2009), pp. 902–909.

[Ng+20] L. Ng, D. Rutherford, V. Shende, S. Sivek, and E. Zaslow. "Augmentations are sheaves". In: *Geom. Topol.* 24.5 (2020), pp. 2149–2286. arXiv: 1502.04939.

[Nie06] K. Niederkrüger. "The plastikstufe—a generalization of the overtwisted disk to higher dimensions". In: *Algebr. Geom. Topol.* 6 (2006), pp. 2473–2508.

[Oan04] A. Oancea. "A survey of Floer homology for manifolds with contact type boundary or symplectic homology". In: *Ensaios Mat.* 7 (2004), pp. 51–91.

[Oh93] Y.-G. Oh. "Floer cohomology of Lagrangian intersections and pseudo-holomorphic disks. I". In: *Comm. Pure Appl. Math.* 46.7 (1993), pp. 949–993.

[Oh95] Y.-G. Oh. "Addendum to "Floer cohomology of Lagrangian intersections and pseudo-holomorphic discs. I"". In: *Comm. Pure Appl. Math.* 48.11 (1995), pp. 1299–1302.

[Oh96] Y.-G. Oh. "Floer cohomology, spectral sequences, and the Maslov class of Lagrangian embeddings". In: *Internat. Math. Res. Notices* 1996.7 (1996), pp. 305–346.

[Oh05a] Y.-G. Oh. "Construction of spectral invariants of Hamiltonian paths on closed symplectic manifolds". In: *The Breadth of Symplectic and Poisson Geometry*. Vol. 232. Progr. Math. Boston, MA: Birkhäuser, 2005, pp. 525–570.

[Oh05b] Y.-G. Oh. "Spectral invariants, analysis of the Floer moduli space, and geometry of the Hamiltonian diffeomorphism group". In: *Duke Math. J.* 130.2 (2005), pp. 199–295.

[Oh15] Y.-G. Oh. *Symplectic Topology and Floer Homology. Vols. 1 and 2*. Vol. 28–29. New Mathematical Monographs. Cambridge: Cambridge University Press, 2015.

[OM07] Y.-G. Oh and S. Müller. "The group of Hamiltonian homeomorphisms and C^0-symplectic topology". In: *J. Symplectic Geom.* 5.2 (2007), pp. 167–219.

[OO96] H. Ohta and K. Ono. "Notes on symplectic 4-manifolds with $b_2^+ = 1$. II". In: *International Journal of Mathematics* 7.6 (1996), pp. 755–770.

[Ono95] K. Ono. "On the Arnold conjecture for weakly monotone symplectic manifolds". In: *Invent. Math.* 119.3 (1995), pp. 519–537.

[Ono06] K. Ono. "Floer–Novikov cohomology and the flux conjecture". In: *Geom. Funct. Anal.* 16.5 (2006), pp. 981–1020.

[Ops07] E. Opshtein. "Maximal symplectic packings in $\mathbb{P}^2$". In: *Compos. Math.* 143.6 (2007), pp. 1558–1575.

[Ost14] Y. Ostrover. "When symplectic topology meets Banach space geometry". In: *Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II*. Seoul: Kyung Moon Sa, 2014, pp. 959–981.

[OS04] B. Ozbagci and A. I. Stipsicz. *Surgery on Contact 3-Manifolds and Stein Surfaces*. Vol. 13. Bolyai Society Mathematical Studies. Springer, 2004.

[OS00] P. Ozsváth and Z. Szabó. "The symplectic Thom conjecture". In: *Annals of Mathematics (2)* 151.1 (2000), pp. 93–124.

[OS05] P. Ozsváth and Z. Szabó. "Heegaard Floer homology and contact structures". In: *Duke Math. J.* 129 (2005), pp. 39–61.

[Pan94] P. Pansu. "Compactness". In: *Holomorphic Curves in Symplectic Geometry*. Ed. by M. Audin and J. Lafontaine. Vol. 117. Progress in Mathematics. Basel: Birkhäuser, 1994, pp. 233–249.

[Par16] J. Pardon. "An algebraic approach to virtual fundamental cycles on moduli spaces of pseudo-holomorphic curves". In: *Geometry & Topology* 20.2 (2016), pp. 779–1034.

[Par19] J. Pardon. "Contact homology and virtual fundamental cycles". In: *Journal of the American Mathematical Society* 32.3 (2019), pp. 825–919.

[Par23] J. Pardon. *Universally counting curves in Calabi–Yau threefolds*. Preprint. 2023. arXiv: 2308.02948.

[Par05] J. Park. "Simply connected symplectic 4-manifolds with $b_2^+ = 1$ and $c_1^2 = 2$". In: *Inventiones Mathematicae* 159.3 (2005), pp. 657–667.

[PW93] T. H. Parker and J. G. Wolfson. "Pseudo-holomorphic maps and bubble trees". In: *Journal of Geometric Analysis* 3.1 (1993), pp. 63–98.

[PV15] Á. Pelayo and S. Vũ Ngọc. "Hofer's question on intermediate symplectic capacities". In: *Proc. Lond. Math. Soc. (3)* 110.4 (2015), pp. 787–804.

[PV16] Á. Pelayo and S. Vũ Ngọc. "Sharp symplectic embeddings of cylinders". In: *Indag. Math. (N.S.)* 27.1 (2016), pp. 307–317. arXiv: 1304.5250.

[PSS96] S. Piunikhin, D. Salamon, and M. Schwarz. "Symplectic Floer–Donaldson theory and quantum cohomology". In: *Contact and Symplectic Geometry (Cambridge, 1994)*. Vol. 8. Publ. Newton Inst. Cambridge: Cambridge University Press, 1996, pp. 171–200.

[PZ98] A. Polishchuk and E. Zaslow. "Categorical mirror symmetry: the elliptic curve". In: *Adv. Theor. Math. Phys.* 2.2 (1998), pp. 443–470. arXiv: math/9801119.

[Pol91] L. Polterovich. "The Maslov class of the Lagrange surfaces and Gromov's pseudo-holomorphic curves". In: *Trans. Amer. Math. Soc.* 325.1 (1991), pp. 241–248.

[Pol98] L. Polterovich. "Hofer's diameter and Lagrangian intersections". In: *Int. Math. Res. Not.* 1998.4 (1998), pp. 217–223.

[Pol01] L. Polterovich. *The Geometry of the Group of Symplectic Diffeomorphisms*. Lectures in Mathematics ETH Zürich. Basel: Birkhäuser, 2001.

[Pol14] L. Polterovich. "Symplectic geometry of quantum noise". In: *Comm. Math. Phys.* 327.2 (2014), pp. 481–519.

[Pol+20] L. Polterovich, D. Rosen, K. Samvelyan, and J. Zhang. *Topological Persistence in Geometry and Analysis*. Vol. 74. University Lecture Series. Providence, RI: Amer. Math. Soc., 2020.

[PS16] L. Polterovich and E. Shelukhin. "Autonomous Hamiltonian flows, Hofer's geometry and persistence modules". In: *Selecta Math. (N.S.)* 22.1 (2016), pp. 227–296.

[PS24] N. Porcelli and I. Smith. *Bordism of flow modules and exact Lagrangians*. Preprint. 2024. arXiv: 2401.11766.

[Pra24] R. Prasad. "Generic equidistribution of periodic orbits for area-preserving surface maps". In: *International Mathematics Research Notices* 24 (2024), pp. 14802–14834.

[Pug67] C. C. Pugh. "The closing lemma". In: *American Journal of Mathematics* 89 (1967), pp. 956–1009.

[PR83] C. C. Pugh and C. Robinson. "The C^1 closing lemma, including Hamiltonians". In: *Ergodic Theory and Dynamical Systems* 3.2 (1983), pp. 261–313.

[Rab78] P. H. Rabinowitz. "Periodic solutions of Hamiltonian systems". In: *Communications on Pure and Applied Mathematics* 31.2 (1978), pp. 157–184.

[Rez22] S. Rezchikov. *Integral Arnol'd conjecture*. Preprint. 2022. arXiv: 2209.11165.

[Rit13] A. F. Ritter. "Topological quantum field theory structure on symplectic cohomology". In: *J. Topol.* 6.2 (2013), pp. 391–489. arXiv: 1003.1781.

[RS93] J. Robbin and D. Salamon. "The Maslov index for paths". In: *Topology* 32.4 (1993), pp. 827–844.

[Rua94] Y. Ruan. "Symplectic topology on algebraic 3-folds". In: *Journal of Differential Geometry* 39.1 (1994), pp. 215–227.

[Rua99] Y. Ruan. "Virtual neighborhoods and pseudo-holomorphic curves". In: *Turkish Journal of Mathematics* 23.1 (1999), pp. 161–231.

[RT95] Y. Ruan and G. Tian. "A mathematical theory of quantum cohomology". In: *Journal of Differential Geometry* 42.2 (1995), pp. 259–367.

[RT97] Y. Ruan and G. Tian. "Higher genus symplectic invariants and sigma models coupled with gravity". In: *Inventiones Mathematicae* 130.3 (1997), pp. 455–516.

[Rud24] D. Rudolf. "The Minkowski billiard characterization of the EHZ-capacity of convex Lagrangian products". In: *Journal of Dynamics and Differential Equations* 36.3 (2024), pp. 2773–2791.

[Rud99] Y. B. Rudyak. "On analytical applications of stable homotopy (the Arnold conjecture, critical points)". In: *Math. Z.* 230.4 (1999), pp. 659–672.

[RO99] Y. B. Rudyak and J. Oprea. "On the Lusternik–Schnirelmann category of symplectic manifolds and the Arnold conjecture". In: *Math. Z.* 230.4 (1999), pp. 673–678.

[Sağ21] M. Sağlam. "Contact forms with large systolic ratio in arbitrary dimensions". In: *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (5)* 22.3 (2021), pp. 1265–1308.

[Sal99] D. Salamon. "Lectures on Floer homology". In: *Symplectic Geometry and Topology (Park City, UT, 1997)*. Ed. by Y. Eliashberg and L. Traynor. Vol. 7. IAS/Park City Mathematics Series. Providence, RI: American Mathematical Society, 1999, pp. 143–229.

[Sal13] D. Salamon. "Uniqueness of symplectic structures". In: *Acta Mathematica Vietnamica* 38.1 (2013), pp. 123–144. arXiv: 1211.2940.

[SZ92] D. Salamon and E. Zehnder. "Morse theory for periodic solutions of Hamiltonian systems and the Maslov index". In: *Communications on Pure and Applied Mathematics* 45.10 (1992), pp. 1303–1360.

[SW06] D. A. Salamon and J. Weber. "Floer homology and the heat flow". In: *Geom. Funct. Anal.* 16.5 (2006), pp. 1050–1138.

[SH18] P. A. S. Salomão and U. L. Hryniewicz. "Global surfaces of section for Reeb flows in dimension three and beyond". In: *Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018, Vol. II*. Hackensack, NJ: World Scientific, 2018, pp. 941–967.

[Sch05] F. Schlenk. *Embedding Problems in Symplectic Geometry*. Vol. 40. de Gruyter Expositions in Mathematics. Berlin: Walter de Gruyter, 2005.

[Sch18] F. Schlenk. "Symplectic embedding problems, old and new". In: *Bull. Amer. Math. Soc. (N.S.)* 55.2 (2018), pp. 139–182.

[Sch98] M. Schwarz. "A quantum cup-length estimate for symplectic fixed points". In: *Invent. Math.* 133.2 (1998), pp. 353–397.

[Sch00] M. Schwarz. "On the action spectrum for closed symplectically aspherical manifolds". In: *Pacific J. Math.* 193.2 (2000), pp. 419–461.

[Sei97] P. Seidel. "π_1 of symplectic automorphism groups and invertibles in quantum homology rings". In: *Geometric and Functional Analysis* 7.6 (1997), pp. 1046–1095.

[Sei99] P. Seidel. "Lagrangian two-spheres can be symplectically knotted". In: *J. Differential Geom.* 52.1 (1999), pp. 145–171.

[Sei00] P. Seidel. "Graded Lagrangian submanifolds". In: *Bulletin de la Société Mathématique de France* 128.1 (2000), pp. 103–149.

[Sei01] P. Seidel. "Vanishing cycles and mutation". In: *European Congress of Mathematics, Vol. II (Barcelona, 2000)*. Vol. 202. Progr. Math. Basel: Birkhäuser, 2001, pp. 65–85.

[Sei03] P. Seidel. "A long exact sequence for symplectic Floer cohomology". In: *Topology* 42.5 (2003), pp. 1003–1063.

[Sei08a] P. Seidel. "A biased view of symplectic cohomology". In: *Current Developments in Mathematics, 2006*. Somerville, MA: International Press, 2008, pp. 211–253.

[Sei08b] P. Seidel. *Fukaya Categories and Picard–Lefschetz Theory*. Zurich Lectures in Advanced Mathematics. Zürich: European Mathematical Society, 2008.

[Sei08c] P. Seidel. "Lectures on four-dimensional Dehn twists". In: *Symplectic 4-Manifolds and Algebraic Surfaces*. Vol. 1938. Lecture Notes in Math. Berlin: Springer, 2008, pp. 231–267. arXiv: `math/0309012`.

[Sei11] P. Seidel. "Homological mirror symmetry for the genus two curve". In: *J. Algebraic Geom.* 20.4 (2011), pp. 727–769. arXiv: `0812.1171`.

[Sei15] P. Seidel. *Homological Mirror Symmetry for the Quartic Surface*. Vol. 236. Mem. Amer. Math. Soc. 1116. Providence, RI: American Mathematical Society, 2015. arXiv: `math/0310414`.

[SS05] P. Seidel and I. Smith. "The symplectic topology of Ramanujam's surface". In: *Comment. Math. Helv.* 80 (2005), pp. 859–881.

[Sey13] S. Seyfaddini. "C^0-limits of Hamiltonian paths and the Oh–Schwarz spectral invariants". In: *Int. Math. Res. Not.* 2013.21 (2013), pp. 4920–4960.

[She20] E. Shelukhin. "Pseudo-rotations and Steenrod squares". In: *Journal of Modern Dynamics* 16 (2020), pp. 289–304.

[She21a] E. Shelukhin. "Pseudo-rotations and Steenrod squares revisited". In: *Mathematical Research Letters* 28.4 (2021), pp. 1255–1261.

[She22a] E. Shelukhin. "On the Hofer–Zehnder conjecture". In: *Annals of Mathematics* 195.3 (2022), pp. 775–839.

[She22b] E. Shelukhin. "Symplectic cohomology and a conjecture of Viterbo". In: *Geom. Funct. Anal.* 32.6 (2022), pp. 1514–1543.

[She22c] E. Shelukhin. "Viterbo conjecture for Zoll symmetric spaces". In: *Invent. Math.* 230.1 (2022), pp. 321–373.

[She21b] V. Shende. "Microlocal category for Weinstein manifolds via the h-principle". In: *Publ. Res. Inst. Math. Sci.* 57.3-4 (2021), pp. 1041–1048. arXiv: `1707.07663`.

[STZ17] V. Shende, D. Treumann, and E. Zaslow. "Legendrian knots and constructible sheaves". In: *Invent. Math.* 207.3 (2017), pp. 1031–1133. arXiv: `1402.0490`.

[She11] N. Sheridan. "On the homological mirror symmetry conjecture for pairs of pants". In: *J. Differential Geom.* 89.2 (2011), pp. 271–367. arXiv: `1012.3238`.

[She15] N. Sheridan. "Homological mirror symmetry for Calabi–Yau hypersurfaces in projective space". In: *Invent. Math.* 199.1 (2015), pp. 1–186. arXiv: `1111.0632`.

[She16] N. Sheridan. "On the Fukaya category of a Fano hypersurface in projective space". In: *Publ. Math. Inst. Hautes Études Sci.* 124 (2016), pp. 165–317.

[SS20a] N. Sheridan and I. Smith. "Rational equivalence and Lagrangian tori on K3 surfaces". In: *Comment. Math. Helv.* 95.2 (2020), pp. 301–337. arXiv: `1809.03892`.

[SS20b] N. Sheridan and I. Smith. "Symplectic topology of K3 surfaces via mirror symmetry". In: *J. Amer. Math. Soc.* 33.3 (2020), pp. 875–915.

[SS21] N. Sheridan and I. Smith. "Homological mirror symmetry for generalized Greene–Plesser mirrors". In: *Invent. Math.* 224.2 (2021), pp. 627–682. arXiv: `1709.08937`.

[She09] V. V. Shevchishin. "Lagrangian embeddings of the Klein bottle and combinatorial properties of mapping class groups". In: *Izv. Math.* 73.4 (2009), pp. 797–859.

[Sie96] B. Siebert. *Gromov–Witten invariants of general symplectic manifolds*. Preprint. 1996. arXiv: dg-ga/9608005.

[Sie99] B. Siebert. "Algebraic and symplectic Gromov–Witten invariants coincide". In: *Annales de l'Institut Fourier (Grenoble)* 49.6 (1999), pp. 1743–1795.

[ST05] B. Siebert and G. Tian. "On the holomorphicity of genus two Lefschetz fibrations". In: *Annals of Mathematics (2)* 161.2 (2005), pp. 959–1020.

[Sie25] K. Siegel. "Higher symplectic capacities". In: *Algebr. Geom. Topol.* 25.9 (2025), pp. 5205–5278.

[Sik94] J.-C. Sikorav. "Some properties of holomorphic curves in almost complex manifolds". In: *Holomorphic Curves in Symplectic Geometry*. Ed. by M. Audin and J. Lafontaine. Vol. 117. Progress in Mathematics. Basel: Birkhäuser, 1994, pp. 165–189.

[SL91] R. Sjamaar and E. Lerman. "Stratified symplectic spaces and reduction". In: *Ann. of Math. (2)* 134.2 (1991), pp. 375–422.

[Sma62] S. Smale. "On the structure of manifolds". In: *Amer. J. Math.* 84 (1962), pp. 387–399.

[Sma65] S. Smale. "An infinite dimensional version of Sard's theorem". In: *American Journal of Mathematics* 87.4 (1965), pp. 861–866.

[Sma98] S. Smale. "Mathematical problems for the next century". In: *The Mathematical Intelligencer* 20.2 (1998), pp. 7–15.

[Smi22] G. Smirnov. "Symplectic mapping class groups of K3 surfaces and Seiberg–Witten invariants". In: *Geom. Funct. Anal.* 32.2 (2022), pp. 280–301.

[Smi00] I. Smith. "On moduli spaces of symplectic forms". In: *Mathematical Research Letters* 7.6 (2000), pp. 779–788.

[Smi15] I. Smith. "A symplectic prolegomenon". In: *Bull. Amer. Math. Soc. (N.S.)* 52.3 (2015), pp. 415–464.

[Sol06] J. P. Solomon. "Intersection theory on the moduli space of holomorphic curves with Lagrangian boundary conditions". PhD thesis. Massachusetts Institute of Technology, 2006. arXiv: math/0606429.

[SYZ96] A. Strominger, S.-T. Yau, and E. Zaslow. "Mirror symmetry is T-duality". In: *Nuclear Phys. B* 479.1–2 (1996), pp. 243–259. arXiv: hep-th/9606040.

[Str90] M. Struwe. "Existence of periodic solutions of Hamiltonian systems on almost every energy surface". In: *Boletim da Sociedade Brasileira de Matemática (N.S.)* 20.2 (1990), pp. 49–58.

[Sym03] M. Symington. "Four dimensions from two in symplectic topology". In: *Topology and Geometry of Manifolds (Athens, GA, 2001)*. Vol. 71. Proc. Sympos. Pure Math. Providence, RI: American Mathematical Society, 2003, pp. 153–208.

[Tak68] F. Takens. "The minimal number of critical points of a function on a compact manifold and the Lusternik–Schnirelman category". In: *Invent. Math.* 6 (1968), pp. 197–244.

[Tam18] D. Tamarkin. "Microlocal condition for non-displaceability". In: *Algebraic and Analytic Microlocal Analysis (AAMA, Evanston, 2012 and 2013)*. Vol. 269. Springer Proc. Math. Stat. Cham: Springer, 2018, pp. 99–223. arXiv: 0809.1584.

[Tau94] C. H. Taubes. "The Seiberg–Witten invariants and symplectic forms". In: *Mathematical Research Letters* 1.6 (1994), pp. 809–822.

[Tau95a] C. H. Taubes. "More constraints on symplectic forms from Seiberg–Witten invariants". In: *Mathematical Research Letters* 2.1 (1995), pp. 9–13.

[Tau95b] C. H. Taubes. "The Seiberg–Witten and Gromov invariants". In: *Mathematical Research Letters* 2.2 (1995), pp. 221–238.

[Tau96a] C. H. Taubes. "SW ⇒ Gr: from the Seiberg–Witten equations to pseudo-holomorphic curves". In: *Journal of the American Mathematical Society* 9.3 (1996), pp. 845–918.

[Tau96b] C. H. Taubes. "Counting pseudo-holomorphic submanifolds in dimension 4". In: *J. Differential Geom.* 44.4 (1996), pp. 818–893.

[Tau99] C. H. Taubes. "Gr ⇒ SW: from pseudo-holomorphic curves to Seiberg–Witten solutions". In: *J. Differential Geom.* 51.2 (1999), pp. 203–334.

[Tau00] C. H. Taubes. *Seiberg–Witten and Gromov Invariants for Symplectic 4-Manifolds*. Vol. 2. First International Press Lecture Series. Somerville, MA: International Press, 2000.

[Tau07] C. H. Taubes. "The Seiberg–Witten equations and the Weinstein conjecture". In: *Geometry & Topology* 11 (2007), pp. 2117–2202.

[Tau10] C. H. Taubes. "Embedded contact homology and Seiberg–Witten Floer cohomology I–V". In: *Geometry & Topology* 14.5 (2010), pp. 2497–3000.

[Thu76] W. P. Thurston. "Some simple examples of symplectic manifolds". In: *Proc. Amer. Math. Soc.* 55.2 (1976), pp. 467–468.

[TW75] W. P. Thurston and H. E. Winkelnkemper. "On the existence of contact forms". In: *Proc. Amer. Math. Soc.* 52 (1975), pp. 345–347.

[Tol98] S. Tolman. "Examples of non-Kähler Hamiltonian torus actions". In: *Invent. Math.* 131.2 (1998), pp. 299–310.

[Tol17] S. Tolman. "Non-Hamiltonian actions with isolated fixed points". In: *Invent. Math.* 210.3 (2017), pp. 877–910. arXiv: 1510.02829.

[TW00] S. Tolman and J. Weitsman. "On semifree symplectic circle actions with isolated fixed points". In: *Topology* 39.2 (2000), pp. 299–309.

[Tra95] L. Traynor. "Symplectic packing constructions". In: *J. Differential Geom.* 41.3 (1995), pp. 735–751. Reprinted with figures in J. Differential Geom. 42 (1995), no. 2, 411–429.

[Ush08] M. Usher. "Spectral numbers in Floer theories". In: *Compos. Math.* 144.6 (2008), pp. 1581–1592.

[Ush10] M. Usher. "The sharp energy-capacity inequality". In: *Commun. Contemp. Math.* 12.3 (2010), pp. 457–473.

[Ush11] M. Usher. "Boundary depth in Floer theory and its applications to Hamiltonian dynamics and coisotropic submanifolds". In: *Israel J. Math.* 184 (2011), pp. 1–57.

[Ush13] M. Usher. "Hofer's metrics and boundary depth". In: *Ann. Sci. Éc. Norm. Supér. (4)* 46.1 (2013), pp. 57–128.

[Ush19] M. Usher. "Infinite staircases in the symplectic embedding problem for four-dimensional ellipsoids into polydisks". In: *Algebr. Geom. Topol.* 19.4 (2019), pp. 1935–2022.

[UZ16] M. Usher and J. Zhang. "Persistent homology and Floer–Novikov theory". In: *Geom. Topol.* 20.6 (2016), pp. 3333–3430.

[Ust96] I. Ustilovsky. "Conjugate points on geodesics of Hofer's metric". In: *Differential Geom. Appl.* 6.4 (1996), pp. 327–342.

[Via14] R. Vianna. "On exotic Lagrangian tori in $\mathbb{CP}^2$". In: *Geom. Topol.* 18.4 (2014), pp. 2419–2476.

[Via16] R. Vianna. "Infinitely many exotic monotone Lagrangian tori in $\mathbb{CP}^2$". In: *J. Topol.* 9.2 (2016), pp. 535–551.

[Via17] R. Vianna. "Infinitely many monotone Lagrangian tori in del Pezzo surfaces". In: *Selecta Math. (N.S.)* 23.3 (2017), pp. 1955–1996.

[Vid01] S. Vidussi. "Homotopy K3's with several symplectic structures". In: *Geometry & Topology* 5 (2001), pp. 267–285.

[Vid06] S. Vidussi. "Lagrangian surfaces in a fixed homology class: existence of knotted Lagrangian tori". In: *J. Differential Geom.* 74.3 (2006), pp. 507–522.

[Vit87a] C. Viterbo. "A proof of Weinstein's conjecture in $\mathbb{R}^{2n}$". In: *Annales de l'Institut Henri Poincaré C, Analyse Non Linéaire* 4.4 (1987), pp. 337–356.

[Vit87b] C. Viterbo. "Intersection de sous-variétés lagrangiennes, fonctionnelles d'action et indice des systèmes hamiltoniens". In: *Bulletin de la Société Mathématique de France* 115 (1987), pp. 361–390.

[Vit90] C. Viterbo. "A new obstruction to embedding Lagrangian tori". In: *Invent. Math.* 100.2 (1990), pp. 301–320.

[Vit92] C. Viterbo. "Symplectic topology as the geometry of generating functions". In: *Math. Ann.* 292.4 (1992), pp. 685–710.

[Vit99] C. Viterbo. "Functors and computations in Floer homology with applications. I". In: *Geom. Funct. Anal.* 9.5 (1999), pp. 985–1033.

[Vit00] C. Viterbo. "Metric and isoperimetric problems in symplectic geometry". In: *Journal of the American Mathematical Society* 13.2 (2000), pp. 411–431.

[Vit06] C. Viterbo. "On the uniqueness of generating Hamiltonian for continuous limits of Hamiltonians flows". In: *Int. Math. Res. Not.* 2006 (2006), Art. ID 34028.

[Vit18] C. Viterbo. *Functors and computations in Floer homology with applications. Part II*. Preprint. 2018. arXiv: 1805.01316.

[Vit22] C. Viterbo. *Inverse reduction inequalities for spectral numbers and applications*. Preprint. 2022. arXiv: 2203.13172.

[Vog18] T. Vogel. "Non-loose unknots, overtwisted discs, and the contact mapping class group of S^3". In: *Geom. Funct. Anal.* 28.1 (2018), pp. 228–288. arXiv: 1612.06557.

[Wal69] C. T. C. Wall. "Non-additivity of the signature". In: *Inventiones Mathematicae* 7.3 (1969), pp. 269–274.

[Wan15] A. Wand. "Tightness is preserved by Legendrian surgery". In: *Ann. of Math. (2)* 182.2 (2015), pp. 723–738.

[Wan16a] W. Wang. "Closed characteristics on compact convex hypersurfaces in $\mathbb{R}^8$". In: *Advances in Mathematics* 297 (2016), pp. 93–148.

[Wan16b] W. Wang. "Existence of closed characteristics on compact convex hypersurfaces in $\mathbb{R}^{2n}$". In: *Calculus of Variations and Partial Differential Equations* 55.1 (2016), Paper No. 2. arXiv: 1112.5501.

[WHL07] W. Wang, X. Hu, and Y. Long. "Resonance identity, stability, and multiplicity of closed characteristics on compact convex hypersurfaces". In: *Duke Mathematical Journal* 139.3 (2007), pp. 411–462.

[Wei71] A. Weinstein. "Symplectic manifolds and their Lagrangian submanifolds". In: *Adv. Math.* 6 (1971), pp. 329–346.

[Wei77] A. Weinstein. *Lectures on Symplectic Manifolds*. Vol. 29. CBMS Regional Conference Series in Mathematics. Providence, RI: American Mathematical Society, 1977.

[Wei78] A. Weinstein. "Periodic orbits for convex Hamiltonian systems". In: *Annals of Mathematics* 108.3 (1978), pp. 507–518.

[Wei79] A. Weinstein. "On the hypotheses of Rabinowitz' periodic orbit theorems". In: *Journal of Differential Equations* 33.3 (1979), pp. 353–358.

[Wei81a] A. Weinstein. "Neighborhood classification of isotropic embeddings". In: *J. Differential Geom.* 16.1 (1981), pp. 125–128.

[Wei81b] A. Weinstein. "Symplectic geometry". In: *Bull. Amer. Math. Soc. (N.S.)* 5.1 (1981), pp. 1–13.

[Wei91] A. Weinstein. "Contact surgery and symplectic handlebodies". In: *Hokkaido Mathematical Journal* 20.2 (1991), pp. 241–251.

[Wel05] J.-Y. Welschinger. "Invariants of real symplectic 4-manifolds and lower bounds in real enumerative geometry". In: *Inventiones Mathematicae* 162.1 (2005), pp. 195–234.

[Wen10] C. Wendl. "Automatic transversality and orbifolds of punctured holomorphic curves in dimension four". In: *Commentarii Mathematici Helvetici* 85.2 (2010), pp. 347–407.

[Wen14] C. Wendl. *Lectures on Holomorphic Curves in Symplectic and Contact Geometry*. Lecture notes, version 3.2 (first version 2010). 2014. arXiv: 1011.1690.

[Wen16] C. Wendl. *Lectures on symplectic field theory*. Lecture notes, to appear in the EMS Series of Lectures in Mathematics. 2016. arXiv: 1612.01009.

[Wen18] C. Wendl. *Holomorphic Curves in Low Dimensions: From Symplectic Ruled Surfaces to Planar Contact Manifolds*. Vol. 2216. Lecture Notes in Mathematics. Cham: Springer, 2018.

[Wen23] C. Wendl. "Transversality and super-rigidity for multiply covered holomorphic curves". In: *Annals of Mathematics (2)* 198.1 (2023), pp. 93–230.

[Woo98] C. Woodward. "Multiplicity-free Hamiltonian actions need not be Kähler". In: *Invent. Math.* 131.2 (1998), pp. 311–319.

[Ye94] R. Ye. "Gromov's compactness theorem for pseudo holomorphic curves". In: *Transactions of the American Mathematical Society* 342.2 (1994), pp. 671–694.

[Zha20] J. Zhang. *Quantitative Tamarkin Theory*. CRM Short Courses. Cham: Springer, 2020. arXiv: 1807.09878.

[Zin08] A. Zinger. "Pseudocycles and integral homology". In: *Transactions of the American Mathematical Society* 360.5 (2008), pp. 2741–2765.